

Formal Verification of BayesianNetworks.jl

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BayesianNetworksProofs

```
import Mathlib.CategoryTheory.MarkovCategory.Basic
```

Lean 4 / Mathlib formalisation accompanying `BayesianNetworks.jl` (toolchain `leanprover/lean4:v4.30.0`, Mathlib tag `v4.30.0`; ADR 0005). This document is generated from the Lean sources by `mdgen`: the prose is the module docstrings and the code blocks are the verbatim, machine-checked sources. Every library module is built by `lake build --wfail`; `Audit.lean` prints the axioms of the headline results (only `propext`, `Classical.choice`, `Quot.sound`). The Roadmap has no remaining proof holes.

What is formalised

`BayesianNetworks.jl` represents a Bayesian network as an ACSet on the schema `SchBayesNet` (SPEC §8) whose semantics is a finite stochastic state $p_G : I \rightarrow \otimes_v X_v$ computed by `joint_distribution` (SPEC §5, §10), with hard interventions `do_intervention` (SPEC §21–§22) and the tensor otimes of open networks (SPEC §13). Three layers are formalised:

- **Schema layer** (`Schema/`, Mathlib-free): the ACSet schemas as Lean terms, their well-formedness, and the JSON writer behind `lake exe emit_schema`. The Julia test compares `generate_json_acset_schema(SchBayesNet)` with the committed `schemas/*.schema.json`. The

Lean terms and the Julia ACSets BasicSchema values are two independent definitions that emit_schema --check and a Julia test check to agree; neither is generated from the other.

- **Finite model** (Finite/): FinBayesNet (finite variables and mechanisms, finite state spaces, target, parents), kernel families over an arbitrary commutative semiring R (instantiated at $R \geq 0$ in Finite/Probability.lean), Propositions 1, 2, 3 and 4 of SPEC §61, and the open-network closure theorem of SPEC §13 (Finite/Open.lean). Ordered-parent refinement and exact bucket variable elimination now connect further parts of the finite model, but neither is a theorem about Julia arrays or ACSet attribute operations.
- **Abstract layer** (Markov/): the generic consequences of Mathlib's MarkovCategory / CopyDiscardCategory axioms that the Julia tests check on the finite-stochastic instance.
- Schema/Desc.lean: SchemaDesc, decidable WF, Sub, NoOutgoing, extend, and toACSetsJson in ACSets.jl's schema format.
- Schema/BayesNet.lean: the three schemas and their checked structural properties.
- Finite/BayesNet.lean: the finite shape, assignments, local/normalised kernels, closedness, topological order and joint.
- Finite/Evaluation.lean: Proposition 1, through fibres and marginalisation.
- Finite/OrderedParents.lean: contiguous positions, ordered CPT/local-kernel equivalence, coherent axis reindexing and joint normalisation.
- Finite/VariableElimination.lean: scoped factors, bucket elimination, exact joint marginalisation and elimination-order independence.
- Finite/Intervention.lean and Finite/Tensor.lean: Propositions 4 and 2.
- Finite/Open.lean: the typed-interface rule, open-network closure and split composition.
- Finite/OpenSemantics.lean: own-variable open semantics under partial gluing; the original osem_compose, a stronger theorem without locality, normalisation or total matching, and osem_compose_passthrough for arbitrary pass-through, with no hypotheses.
- Finite/BoundaryCases.lean: the pass-through counterexample to the glue formula, the pass-through formula checked on the same wire, and the mechanism-free copying limitation of subset feet.
- Markov/Basic.lean: consequences of Mathlib's abstract Markov/copy-discard classes.
- Finite/Probability.lean: Propositions 1 and 4 at $R := \mathbb{R}_{\geq 0}$.
- Roadmap.lean: remaining categorical and representation questions; no unproved declarations.

New implementation-facing mathematics

- Finite/RawRecords.lean and ReferenceTables.lean: concrete finite records, references and checked positions, with repeated slots interpreted diagonally. Boolean checks imply the compiled local/normalized model; a Julia/JSON translation proof is not asserted.
- Finite/Assignments.lean and Posterior.lean: explicit conditioning equals indicator likelihood, genuine query distributions, positive normalization and exact zero feasibility.
- Finite/FactorMarginal.lean and JunctionTree.lean: actual collect/distribute messages, side-of-cut projections, calibrated beliefs and posterior agreement with VE. Structural running intersection and exact factor assignment are the hypotheses, not correct messages. Arbitrary branching and disconnected components have proved working encodings.
- Finite/ConditionalIndependence.lean and DSeparation.lean: soundness of graph separation in the moralized ancestral DAG. No path separation is defined in terms of numerical independence. The boundary to a concrete Bayes-ball implementation remains explicit.
- Finite/NumericalContracts.lean: finite product and posterior error bounds, with a positive evidence-mass floor, and rounded products under local arithmetic contracts.
- Finite/RefinementExamples.lean: nonvacuous repeated-slot, graph and disconnected zero-support fixtures, checked without native proof shortcuts.

SPEC §61 propositions

SPEC §61	Lean	Julia (BayesianNetworks.jl)
Prop 1 — BN evaluation, $\llbracket G \rrbracket(x) = \prod_v \kappa_v(x_v \mid x_{\text{pa}(v)})$	<code>sum_joint_eq_one</code> (the product is a distribution), <code>evalSeq_eq_joint</code> (the sequential evaluator computes it)	<code>joint_distribution(bn)</code> , <code>validate(bn; closed=true)</code> , <code>topological_order</code>
Prop 2 — tensor compositionality, $\llbracket A \otimes B \rrbracket = \llbracket A \rrbracket \otimes \llbracket B \rrbracket$	<code>joint_tensor</code> , <code>closed_tensor</code> , <code>normalised_tensor</code> , <code>local_tensor</code> , <code>TopoOrder.tensor</code>	<code>otimes(A, B)</code> of <code>OpenBayesNets</code>
Prop 3 — sequential compositionality, $\llbracket B \circ A \rrbracket = \llbracket B \rrbracket \circ \llbracket A \rrbracket$	<code>marg_joint_compose_split</code> , <code>osem_compose</code> , <code>osem_compose_glue</code> (disjoint inputs/outputs), <code>osem_compose_passthrough</code> (arbitrary pass-through)	<code>compose(A, B)</code> , <code>glue(A, B; along)</code> in the categorical package
SPEC §13 revision note — pushout composition preserves “at most one mechanism per variable”	<code>composeNet_target_injective</code> , <code>compose_input_exogenous</code> , <code>compose_exogenous_input</code> , <code>composeTopo</code> , <code>compose</code>	<code>Open(bn; inputs, outputs)</code> , <code>validate(::OpenBayesNet)</code> , <code>compose</code> , <code>glue</code>
Prop 4 — hard intervention, truncated factorisation of $\text{do}(x = x)$	<code>joint_intervene</code> , <code>normalised_intervene</code> , <code>local_cut</code> , <code>sum_joint_intervene_eq_one</code>	<code>do_intervention(bn, :X ⇒ x)</code>
Props 5–7 — influence diagrams	<code>InfluenceDiagrams.jl/proofs</code> (depends on this project by path)	<code>InfluenceDiagrams.jl</code>

The Markov-category laws of the kernel calculus itself (`compose`, `otimes`, `mcopy`, `delete`) and the concrete Mathlib `FinStoch` instances are proved in `FiniteKernels.jl/proofs`.

Over a general semiring, the sum-to-one results establish normalisation, not positivity. The probability interpretation additionally requires nonnegative entries (automatic over $\mathbb{R}_{\geq 0}$). Inference results here are unnormalised marginal identities and do not authorise conditioning on zero-probability evidence.

```
namespace BayesianNetworksProofs
```

```
/-- Smoke lemma so that an empty project builds and the axiom audit has something to print. -/
theorem smoke : (1 : Nat) + 1 = 2 := rfl
```

```
end BayesianNetworksProofs
```

BayesianNetworksProofs.Schema.Desc

```
import Lean.Data.Json
```

Schema descriptions — a small, Mathlib-free data type describing the *presentation* of an ACSet schema (objects, homs, attribute types, attributes), together with the decidable well-formedness / inclusion / “no outgoing arrow” predicates that the Julia layer relies on and the JSON writer that reproduces `ACSets.jl`’s `generate_json_acset_schema` format.

The Lean terms here and the `@present` schemas in `BayesianNetworks.jl` are two independent hand-written definitions, checked to agree rather than generated from one another: `lake exe emit_schema` writes `proofs/schemas/*.schema.json` from the terms in `BayesianNetworksProofs.Schema.BayesNet`, and a Julia test asserts that the `@present` schema matches (ADR 0005).

This module deliberately imports only `Lean.Data.Json` so that the `emit_schema` executable links against Lean core alone (no Mathlib object code is required).

```
namespace BayesianNetworksProofs
```

```
open Lean (Json ToJson FromJson)
```

```
/-- A generator of arrow shape: a hom 'name : dom → cod' between objects, or an attribute 'name : dom → cod' from an object to an attribute type. -/
structure Arrow where
```

```

name : String
dom : String
cod : String
deriving ToJson, FromJson, DecidableEq, Repr

/-- A presentation of an ACSet schema: object generators, hom generators, attribute-type
generators and attribute generators. Equations are not modelled (the schemas of
'BayesianNetworks.jl' have none). -/
structure SchemaDesc where
  obs : List String
  homs : List Arrow
  attrtypes : List String
  attrs : List Arrow
  deriving ToJson, FromJson, DecidableEq, Repr

namespace SchemaDesc

/-- All generator names of a schema, in the order 'obs, homs, attrtypes, attrs'. -/
def names (S : SchemaDesc) : List String :=
  S.obs ++ S.homs.map Arrow.name ++ S.attrtypes ++ S.attrs.map Arrow.name

/-- Well-formedness: every generator name is distinct; every hom has an object as domain and
codomain; every attribute has an object as domain and an attribute type as codomain. -/
def WF (S : SchemaDesc) : Prop :=
  S.names.Nodup ∧
  (∀ h ∈ S.homs, h.dom ∈ S.obs ∧ h.cod ∈ S.obs) ∧
  (∀ a ∈ S.attrs, a.dom ∈ S.obs ∧ a.cod ∈ S.attrtypes)

instance (S : SchemaDesc) : Decidable S.WF := by unfold WF; infer_instance

/-- 'Sub S₀ S': every generator of 'S₀' is a generator of 'S' (generator-wise inclusion, the
shape of a Catlab '@present Sch <: Sch₀'). -/
def Sub (S₀ S : SchemaDesc) : Prop :=
  (∀ x ∈ S₀.obs, x ∈ S.obs) ∧ (∀ h ∈ S₀.homs, h ∈ S.homs) ∧
  (∀ x ∈ S₀.attrtypes, x ∈ S.attrtypes) ∧ (∀ a ∈ S₀.attrs, a ∈ S.attrs)

instance (S₀ S : SchemaDesc) : Decidable (Sub S₀ S) := by unfold Sub; infer_instance

/-- 'NoOutgoing S₀ S': no hom or attribute of 'S' whose domain is an object of 'S₀' has a
codomain outside 'S₀'. This is the precondition of Catlab's multi-object 'OpenACSetTypes':
the sub-schema 'S₀' can then serve as the interface (foot) type of open 'S'-ACSets. -/
def NoOutgoing (S₀ S : SchemaDesc) : Prop :=
  (∀ h ∈ S.homs, h.dom ∈ S₀.obs → h.cod ∈ S₀.obs) ∧
  (∀ a ∈ S.attrs, a.dom ∈ S₀.obs → a.cod ∈ S₀.attrtypes)

instance (S₀ S : SchemaDesc) : Decidable (NoOutgoing S₀ S) := by unfold NoOutgoing; infer_instance

/-- Extend a schema by appending generators (the Lean counterpart of '@present Sch <: Sch₀'). -/
def extend (S : SchemaDesc) (obs : List String := []) (homs : List Arrow := [])
  (attrtypes : List String := []) (attrs : List Arrow := []) : SchemaDesc where
  obs := S.obs ++ obs
  homs := S.homs ++ homs
  attrtypes := S.attrtypes ++ attrtypes
  attrs := S.attrs ++ attrs

/-- An extension always contains the schema it extends. -/
theorem sub_extend (S : SchemaDesc) (obs homs attrtypes attrs) :
  Sub S (S.extend obs homs attrtypes attrs) := by
  refine {?_, ?_, ?_, ?_} <=> intro x hx <=> simp [extend, hx]

```

ACSets.jl schema JSON

generate_json_acset_schema in ACSets/src/serialization/JSONACSets.jl emits

```

{ "version": {"ACSetSchema": "0.0.1", "ACSets": <pkg version>},
  "Ob": [{"name": ..}], "Hom": [{"name", "dom", "codom"}],

```

```
"AttrType": [{"name"}], "Attr": [{"name","dom","codom"}],
"equations": [] }
```

(codomains are spelled `codom`; `Ob/AttrType` entries are objects with a single name). JSON objects are unordered; Lean's `Json.obj` stores keys in an RB-map, so `Json.pretty` prints keys in a fixed (reverse-alphabetical) order. Consumers must compare structurally.

```
-- The 'ACSetSchema' format version written by ACSets.jl. -/
def acsetSchemaVersion : String := "0.0.1"

-- The ACSets.jl package version recorded in emitted files (informational; Julia compares
modulo 'version'). -/
def acsetsPkgVersion : String := "0.2.29"

private def arrowJson (a : Arrow) : Json :=
  Json.mkObj [{"name", a.name}, {"dom", a.dom}, {"codom", a.cod}]

private def namedJson (x : String) : Json := Json.mkObj [{"name", x}]

-- Render a schema description in the ACSets.jl 'generate_json_acset_schema' format. -/
def toACSetsJson (S : SchemaDesc) : Json :=
  Json.mkObj
    [ ("version", Json.mkObj [{"ACSetSchema", acsetSchemaVersion}, {"ACSets", acsetsPkgVersion}])
    , ("Ob", Json.arr (S.obs.map namedJson).toArray)
    , ("Hom", Json.arr (S.homs.map arrowJson).toArray)
    , ("AttrType", Json.arr (S.attrtypes.map namedJson).toArray)
    , ("Attr", Json.arr (S.attrs.map arrowJson).toArray)
    , ("equations", Json.arr #[]) ]

-- Pretty-printed, deterministic text of 'toACSetsJson' (trailing newline included). -/
def toACSetsJsonString (S : SchemaDesc) : String :=
  (S.toACSetsJson.pretty 100) ++ "\n"

end SchemaDesc

end BayesianNetworksProofs
```

BayesianNetworksProofs.Schema.BayesNet

```
import BayesianNetworksProofs.Schema.Desc
```

The three ACSet schemas of the ecosystem, as `SchemaDesc` terms:

- `schVariableSpace` — the interface sub-schema (variables and their states);
- `schBayesNet` — adds mechanisms and their ordered inputs (SPEC §8);
- `schInfluenceDiagram` — adds decisions, information arcs, utilities and decision precedence (SPEC §24).

The theorems (all by `decide`) are what the Julia layer assumes: each schema is well formed, each extension contains its base, and `schVariableSpace` has *no outgoing arrows* in `schBayesNet` / `schInfluenceDiagram`, which is the precondition of Catlab's multi-object `OpenACSetTypes`(`BayesNet`, `VariableSpace`).

```
namespace BayesianNetworksProofs

open SchemaDesc

-- 'SchVariableSpace': objects 'Variable', 'State'; hom 'state_variable'; attribute types
'Label', 'Position', 'Ref'; attributes naming variables and states, positioning states and
referencing the semantic state space. -/
def schVariableSpace : SchemaDesc where
  obs := ["Variable", "State"]
  homs := [{"state_variable", "State", "Variable"}]
  attrtypes := ["Label", "Position", "Ref"]
  attrs :=
    [ <"variable_name", "Variable", "Label">
    , <"space_ref", "Variable", "Ref">
```

```

    , <"state_name", "State", "Label">
    , <"state_position", "State", "Position"> ]

/-- 'SchBayesNet <: SchVariableSpace': mechanisms with a 'target' variable and ordered inputs. -/
def schBayesNet : SchemaDesc :=
  schVariableSpace.extend
    (obs := ["Mechanism", "Input"])
    (homs :=
      [ <"target", "Mechanism", "Variable">
        , <"input_mechanism", "Input", "Mechanism">
        , <"input_variable", "Input", "Variable"> ] )
    (attrs :=
      [ <"mechanism_name", "Mechanism", "Label">
        , <"kernel_ref", "Mechanism", "Ref">
        , <"input_position", "Input", "Position"> ] )

/-- 'SchInfluenceDiagram <: SchBayesNet': decisions, information inputs, utilities with their
inputs, and decision precedence. -/
def schInfluenceDiagram : SchemaDesc :=
  schBayesNet.extend
    (obs := ["Decision", "InformationInput", "Utility", "UtilityInput", "DecisionPrecedence"])
    (homs :=
      [ <"decision_variable", "Decision", "Variable">
        , <"information_decision", "InformationInput", "Decision">
        , <"information_variable", "InformationInput", "Variable">
        , <"utility_node", "UtilityInput", "Utility">
        , <"utility_variable", "UtilityInput", "Variable">
        , <"earlier", "DecisionPrecedence", "Decision">
        , <"later", "DecisionPrecedence", "Decision"> ] )
    (attrs :=
      [ <"decision_name", "Decision", "Label">
        , <"utility_name", "Utility", "Label">
        , <"utility_ref", "Utility", "Ref">
        , <"information_position", "InformationInput", "Position">
        , <"utility_position", "UtilityInput", "Position"> ] )

```

Theorems

```

theorem schVariableSpace_wf : schVariableSpace.WF := by decide

theorem schBayesNet_wf : schBayesNet.WF := by decide

theorem schInfluenceDiagram_wf : schInfluenceDiagram.WF := by decide

theorem sub_variableSpace_bayesNet : Sub schVariableSpace schBayesNet := by decide

/-- No hom or attribute of 'SchBayesNet' leaves '{Variable, State}': 'VariableSpace' is a valid
interface type for 'OpenACSetTypes(BayesNet, VariableSpace)'. -/
theorem noOutgoing_variableSpace_bayesNet : NoOutgoing schVariableSpace schBayesNet := by decide

theorem sub_bayesNet_influenceDiagram : Sub schBayesNet schInfluenceDiagram := by decide

theorem noOutgoing_variableSpace_influenceDiagram :
  NoOutgoing schVariableSpace schInfluenceDiagram := by decide

end BayesianNetworksProofs

```

BayesianNetworksProofs.Finite.BayesNet

```

import Mathlib.Data.Fintype.Pi
import Mathlib.Algebra.BigOperators.Group.Finset.Basic

```

Concrete finite Bayesian networks — the semantic model behind `BayesianNetworks.jl`'s finite stochastic evaluator (SPEC §5, §8, §10, §14).

A `FinBayesNet` is a finite type V of variables, a finite type M of mechanisms, a finite non-empty state space $\text{states } v$ for each variable, a target variable for each mechanism and a finite set of parents for each mechanism. This is the shape of the `SchBayesNet ACSet` (`Mechanism --target--> Variable`, Input rows collected into parents) with the attribute data erased; kernels live outside the structure, exactly as `kernel_ref` points outside the `ACSet` in Julia.

Kernels are valued in an arbitrary commutative semiring R (instantiate with $\mathbb{R}_{\geq 0}$ or $\mathbb{R}$): $\kappa \ m \ x \ y$ is the weight the mechanism m assigns to the value y of its target given the assignment x . Two properties matter:

- Local $\kappa \ m \ x$ depends on x only through the parents of m (the mechanism is a kernel $\otimes \text{parents} \rightarrow \text{target}$);
- Normalised $\kappa \ m \ x \ y = 1$ (the kernel is a stochastic map).

Closed `bn` (exactly one mechanism per variable, target bijective) is `validate(bn; closed=true)` and `TopoOrder bn` is a topological order of the derived variable graph (SPEC §11 item 7).

The joint $\kappa \ x = \prod m, \kappa \ m \ x \ (x \ (\text{target } m))$ is Proposition 1's right-hand side.

```
namespace BayesianNetworksProofs

/-- A finite Bayesian-network *shape*: variables, mechanisms, state spaces, targets, parents. -/
structure FinBayesNet where
  /-- Variables. -/
  V : Type
  /-- Mechanisms (one per 'Mechanism' row of the ACSet). -/
  M : Type
  [fintypeV : Fintype V]
  [decV : DecidableEq V]
  [fintypeM : Fintype M]
  [decM : DecidableEq M]
  /-- The finite state space of each variable. -/
  states : V → Type
  [fintypeS : ∀ v, Fintype (states v)]
  [decS : ∀ v, DecidableEq (states v)]
  [nonemptyS : ∀ v, Nonempty (states v)]
  /-- The variable generated by each mechanism ('target : Mechanism → Variable'). -/
  target : M → V
  /-- The input variables of each mechanism (the 'Input' rows of the ACSet). -/
  parents : M → Finset V

attribute [instance] FinBayesNet.fintypeV FinBayesNet.decV FinBayesNet.fintypeM FinBayesNet.decM
  FinBayesNet.fintypeS FinBayesNet.decS FinBayesNet.nonemptyS

namespace FinBayesNet

variable (bn : FinBayesNet)

/-- A joint assignment of a state to every variable. -/
abbrev Assignment := ∀ v, bn.states v

/-- A family of conditional kernels, one per mechanism: 'κ m x y' is the weight of the target
value 'y' given the assignment 'x'. -/
abbrev Kernel (R : Type) := (m : bn.M) → bn.Assignment → bn.states (bn.target m) → R

variable {bn} {R : Type}

/-- 'k' depends on the assignment only through the variables in 'P'. -/
def LocalOn (P : Finset bn.V) {v : bn.V} (k : bn.Assignment → bn.states v → R) : Prop :=
  ∀ x x' : bn.Assignment, (∀ p ∈ P, x p = x' p) → k x = k x'

/-- The kernel of 'm' reads only the parents of 'm'. -/
def Local (κ : bn.Kernel R) (m : bn.M) : Prop := LocalOn (bn.parents m) (κ m)

/-- The kernel of 'm' is a stochastic map: its weights sum to one for every assignment. -/
```

```

def Normalised [AddCommMonoid R] [One R] (κ : bn.Kernel R) (m : bn.M) : Prop :=
  ∀ x, ∑ y, κ m x y = 1

/-- Closed network: every variable has exactly one generating mechanism
('validate(bn; closed=true)'). -/
def Closed (bn : FinBayesNet) : Prop := Function.Bijective bn.target

/-- A topological order of the variables: a duplicate-free, complete list in which every parent
of a mechanism precedes the mechanism's target, and no mechanism reads its own target. -/
structure TopoOrder (bn : FinBayesNet) where
  order : List bn.V
  nodup : order.Nodup
  complete : ∀ v, v ∈ order
  /- For 'a' before 'b' in the order, 'b' is not a parent of the mechanism generating 'a'. -/
  parents_before : order.Pairwise (fun a b => ∀ m, bn.target m = a → b ∉ bn.parents m)
  no_self : ∀ m, bn.target m ∉ bn.parents m

/-- The joint weight of an assignment: the product of all mechanism kernels evaluated at it
(Proposition 1, right-hand side). -/
def joint [CommMonoid R] (κ : bn.Kernel R) (x : bn.Assignment) : R :=
  ∏ m, κ m x (x (bn.target m))

end FinBayesNet

end BayesianNetworksProofs

```

BayesianNetworksProofs.Finite.Evaluation

```

import BayesianNetworksProofs.Finite.BayesNet
import Mathlib.Algebra.BigOperators.Ring.Finset

```

Proposition 1 (BN evaluation) for the concrete finite model.

- `sum_joint_eq_one` (Prop 1a): for a closed network with local, normalised kernels the joint $\prod m, \kappa m x (x (target\ m))$ sums to one over all assignments, i.e. it is a probability distribution — the state $p_G : I \rightarrow \otimes X_v$ of SPEC §5.
- `evalSeq_eq_joint` (Prop 1b): the sequential evaluator that walks a topological order and multiplies in the kernels of the mechanisms generating each variable computes the same product.

The proof of 1a integrates the variables out one at a time along the topological order. To stay free of dependent tuples, “integrating out S ” is expressed with `fibre S x₀`, the finite set of assignments agreeing with a base assignment x_0 outside S ; `marg S F x₀` sums F over that fibre. Adding a variable to S splits the fibre by the value of that variable (`marg_insert`), and locality lets the kernel of the newest variable be pulled out of the sum.

```

namespace BayesianNetworksProofs

namespace FinBayesNet

variable {bn : FinBayesNet} {R : Type}

```

Fibres and marginalisation

```

/-- Assignments agreeing with 'x₀' outside 'S'. -/
def fibre (S : Finset bn.V) (x₀ : bn.Assignment) : Finset bn.Assignment :=
  Finset.univ.filter (fun y => ∀ v, v ∉ S → y v = x₀ v)

theorem mem_fibre {S : Finset bn.V} {x₀ y : bn.Assignment} :
  y ∈ fibre S x₀ ↔ ∀ v, v ∉ S → y v = x₀ v := by
  simp [fibre]

theorem fibre_empty (x₀ : bn.Assignment) : fibre (∅ : Finset bn.V) x₀ = {x₀} := by
  ext y

```

```

simp [mem_fibre, funext_iff]

theorem fibre_univ (x₀ : bn.Assignment) : fibre (Finset.univ : Finset bn.V) x₀ = Finset.univ := by
  ext y
  simp [mem_fibre]

/-- Splitting the fibre of 'insert v₀ S' by the value at 'v₀'. -/
theorem fibre_insert_filter {S : Finset bn.V} {v₀ : bn.V} (hv : v₀ ∉ S) (x₀ : bn.Assignment)
  (z : bn.states v₀) :
  (fibre (insert v₀ S) x₀).filter (fun y ⇒ y v₀ = z) = fibre S (Function.update x₀ v₀ z) := by
  ext y
  simp only [Finset.mem_filter, mem_fibre, Finset.mem_insert, not_or]
  constructor
  · rintro ⟨h, hz⟩ v hvS
    by_cases hvv : v = v₀
    · subst hvv
      simp [hz]
    · rw [Function.update_of_ne hvv]
      exact h v ⟨hvv, hvS⟩
  · intro h
    refine ⟨fun v ⟨hvv, hvS⟩ ⇒ ?_, ?_⟩
    · have := h v hvS
      rwa [Function.update_of_ne hvv] at this
    · have := h v₀ hv
      simp using this

variable [CommSemiring R]

/-- Sum of 'F' over the assignments agreeing with 'x₀' outside 'S'. -/
def marg (S : Finset bn.V) (F : bn.Assignment → R) (x₀ : bn.Assignment) : R :=
  ∑ y ∈ fibre S x₀, F y

theorem marg_empty (F : bn.Assignment → R) (x₀ : bn.Assignment) : marg ∅ F x₀ = F x₀ := by
  simp [marg, fibre_empty]

theorem marg_univ (F : bn.Assignment → R) (x₀ : bn.Assignment) :
  marg Finset.univ F x₀ = ∑ y, F y := by
  simp [marg, fibre_univ]

theorem marg_insert {S : Finset bn.V} {v₀ : bn.V} (hv : v₀ ∉ S) (F : bn.Assignment → R)
  (x₀ : bn.Assignment) :
  marg (insert v₀ S) F x₀ = ∑ z, marg S F (Function.update x₀ v₀ z) := by
  unfold marg
  rw [← Finset.sum_fiberwise (fibre (insert v₀ S) x₀) (fun y ⇒ y v₀) F]
  refine Finset.sum_congr rfl fun z _ => ?_
  rw [fibre_insert_filter hv]

/-- Integrating out 'S ∪ T' (disjoint) is integrating out 'S' and then, inside, 'T': the finite
Fubini theorem for fibres. -/
theorem marg_union_disjoint {S T : Finset bn.V} (hd : Disjoint S T) (F : bn.Assignment → R)
  (x₀ : bn.Assignment) : marg (S ∪ T) F x₀ = marg S (fun y ⇒ marg T F y) x₀ := by
  induction S using Finset.induction_on generalizing x₀ with
  | empty => simp [marg_empty]
  | insert v₀ S hv ih =>
    have hvT : v₀ ∉ T := fun h => (Finset.disjoint_left.1 hd (Finset.mem_insert_self v₀ S)) h
    have hd' : Disjoint S T := hd.mono_left (Finset.subset_insert _ _)
    rw [Finset.insert_union, marg_insert (by simp [hv, hvT]), marg_insert hv]
    exact Finset.sum_congr rfl fun z _ => ih hd' _

/-- A factor that is constant on the fibre comes out of the sum. -/
theorem marg_mul_left {T : Finset bn.V} {F G : bn.Assignment → R} {x₀ : bn.Assignment}
  (hF : ∀ y ∈ fibre T x₀, F y = F x₀) :
  marg T (fun y ⇒ F y * G y) x₀ = F x₀ * marg T G x₀ := by
  unfold marg
  rw [Finset.mul_sum]
  exact Finset.sum_congr rfl fun y hy => by show F y * G y = F x₀ * G y; rw [hF y hy]

```

Partial joints along a list of variables

```

/-- The product of the kernels of the mechanisms whose target lies in 'l'. -/
def partialJoint (κ : bn.Kernel R) (l : List bn.V) (x : bn.Assignment) : R :=
  ∏ m ∈ Finset.univ.filter (fun m ⇒ bn.target m ∈ l), κ m x (x (bn.target m))

theorem filter_target_cons (hij : Function.Injective bn.target) {l : List bn.V} (m₀ : bn.M) :
  Finset.univ.filter (fun m ⇒ bn.target m ∈ bn.target m₀ :: l) =
    insert m₀ (Finset.univ.filter (fun m ⇒ bn.target m ∈ l)) := by
  ext m
  simp only [Finset.mem_filter, Finset.mem_univ, true_and, List.mem_cons, Finset.mem_insert]
  constructor
  · rintro (h | h)
    · exact Or.inl (hij h)
    · exact Or.inr h
  · rintro (rfl | h)
    · exact Or.inl rfl
    · exact Or.inr h

/-- The core induction: integrating out the variables of a suffix 'l' of a topological order
(every variable in 'l' has a mechanism, every parent of a mechanism targeting 'l' that lies in
'l' comes earlier) gives total weight one, whatever the values of the other variables. -/
theorem marg_partialJoint_eq_one (κ : bn.Kernel R) (hij : Function.Injective bn.target)
  (hloc : ∀ m, Local κ m)
  (hself : ∀ m, bn.target m ∉ bn.parents m) (l : List bn.V) (hnd : l.Nodup)
  (hnorm : ∀ m, bn.target m ∈ l → Normalised κ m)
  (hpw : l.Pairwise (fun a b ⇒ ∀ m, bn.target m = a → b ∉ bn.parents m))
  (hsurj : ∀ v ∈ l, ∃ m, bn.target m = v) (x₀ : bn.Assignment) :
  marg l.toFinset (partialJoint κ l) x₀ = 1 := by
  induction l generalizing x₀ with
  | nil ⇒ simp [marg_empty, partialJoint]
  | cons v₀ l ih ⇒
    obtain ⟨m₀, hm₀⟩ := hsurj v₀ (List.mem_cons_self ..)
    subst hm₀
    rw [List.nodup_cons] at hnd
    rw [List.pairwise_cons] at hpw
    have hnotl : bn.target m₀ ∉ l.toFinset := by simp using hnd.1
    rw [List.toFinset_cons, marg_insert hnotl]
    have key : ∀ z, marg l.toFinset (partialJoint κ (bn.target m₀ :: l))
      (Function.update x₀ (bn.target m₀) z) = κ m₀ x₀ z := by
      intro z
      have hfac : ∀ y ∈ fibre l.toFinset (Function.update x₀ (bn.target m₀) z),
        partialJoint κ (bn.target m₀ :: l) y = κ m₀ x₀ z * partialJoint κ l y := by
        intro y hy
        rw [mem_fibre] at hy
        unfold partialJoint
        rw [filter_target_cons hij, Finset.prod_insert (by simp using hnd.1)]
        congr 1
        have hy0 : y (bn.target m₀) = z := by
          rw [hy _ hnotl, Function.update_self]
        have hk : κ m₀ y = κ m₀ x₀ := by
          refine hloc m₀ y x₀ fun p hp ⇒ ?_
          have hpl : p ∉ l := fun hpl ⇒ hpw.1 p hpl m₀ rfl hp
          have hne : p ≠ bn.target m₀ := fun h ⇒ hself m₀ (h ▸ hp)
          rw [hy p (by simp using hpl), Function.update_of_ne hne]
          rw [hk, hy0]
        unfold marg
        rw [Finset.sum_congr rfl hfac, ← Finset.mul_sum]
      have := ih hnd.2 (fun m hm ⇒ hnorm m (List.mem_cons_of_mem _ hm)) hpw.2
        (fun v hv ⇒ hsurj v (List.mem_cons_of_mem _ hv))
        (Function.update x₀ (bn.target m₀) z)
      unfold marg at this
      rw [this, mul_one]
    rw [Finset.sum_congr rfl fun z _ ⇒ key z]
    exact hnorm m₀ (List.mem_cons_self ..) x₀

```

Proposition 1

```

/-- **Proposition 1a.** In a closed network with a topological order, local and normalised
kernels make the joint sum to one. A probability interpretation additionally requires
nonnegative entries; this is automatic over  $\mathbb{R}_{\geq 0}$ , but not over a general semiring. -/
theorem sum_joint_eq_one (κ : bn.Kernel R) (hclosed : bn.Closed) (ord : bn.TopoOrder)
  (hloc : ∀ m, Local κ m) (hnorm : ∀ m, Normalised κ m) :
  ∑ x, joint κ x = 1 := by
  obtain ⟨x₀⟩ : Nonempty bn.Assignment := inferInstance
  have h := marg_partialJoint_eq_one κ hclosed.1 hloc ord.no_self ord.order ord.nodup
  (fun m _ => hnorm m) ord.parents_before (fun v _ => hclosed.2 v) x₀
  have hu : ord.order.toFinset = Finset.univ :=
    Finset.eq_univ_of_forall fun v => by simp using ord.complete v
  rw [hu, marg_univ] at h
  rw [← h]
  refine Finset.sum_congr rfl fun x _ => ?_
  unfold joint partialJoint
  rw [Finset.filter_true_of_mem fun m _ => ord.complete _]

/-- The sequential evaluator: walk the variables in the given order and multiply in the kernels
of the mechanisms generating each of them. This is the loop 'for v in order' of
'joint_distribution' in Julia. -/
def evalSeq (κ : bn.Kernel R) (l : List bn.V) (x : bn.Assignment) : R :=
  (l.map fun v => ∏ m ∈ Finset.univ.filter (fun m => bn.target m = v), κ m x (x (bn.target m))).prod

/-- **Proposition 1b.** The sequential evaluator along any complete duplicate-free order of the
variables computes the joint product (no acyclicity or closedness is needed for this identity). -/
theorem evalSeq_eq_joint (κ : bn.Kernel R) (l : List bn.V) (hnd : l.Nodup) (hcomp : ∀ v, v ∈ l)
  (x : bn.Assignment) : evalSeq κ l x = joint κ x := by
  unfold evalSeq joint
  have hu : l.toFinset = Finset.univ := Finset.eq_univ_of_forall fun v => by simp using hcomp v
  rw [← List.prod_toFinset _ hnd, hu]
  exact Finset.prod_fiberwise Finset.univ bn.target _

/-- Proposition 1b specialised to a topological order. -/
theorem evalSeq_topo_eq_joint (κ : bn.Kernel R) (ord : bn.TopoOrder) (x : bn.Assignment) :
  evalSeq κ ord.order x = joint κ x :=
  evalSeq_eq_joint κ ord.order ord.nodup ord.complete x

```

Marginalising the downstream variables out (Proposition 3, closed-world shadow)

```

/-- A set of "upstream" variables 'U' is *closed under parents*: every mechanism generating a
variable of 'U' reads only variables of 'U'. -/
def UpstreamClosed (U : Finset bn.V) : Prop :=
  ∀ m, bn.target m ∈ U → bn.parents m ⊆ U

/-- **Integrating the downstream variables out.** If 'U' is closed under parents, every variable
outside 'U' is generated by some mechanism, and the kernels of those mechanisms are normalised,
then summing the joint over all assignments that agree with 'x₀' on 'U' leaves exactly the
product of the kernels of the mechanisms targeting 'U'. Only the *downstream* kernels have to be
normalised; the upstream ones merely have to be local.

This is the closed-world shadow of **Proposition 3** (SPEC §61): the downstream half of a
sequential composite integrates out to one, so the marginal of '[B ∘ A]' on 'A's variables is
'[A]'. See 'BayesianNetworksProofs.Finite.Open' for the open-network form
('OpenFinBayesNet.Composable.marg_joint_compose'). -/
theorem marg_joint_downstream (κ : bn.Kernel R) (hinj : Function.Injective bn.target)
  (ord : bn.TopoOrder) (hloc : ∀ m, Local κ m) (U : Finset bn.V)
  (hnorm : ∀ m, bn.target m ∉ U → Normalised κ m)
  (hU : UpstreamClosed U) (hmech : ∀ v ∉ U, ∃ m, bn.target m = v) (x₀ : bn.Assignment) :
  marg Uᶜ (joint κ) x₀ =
  ∏ m ∈ Finset.univ.filter (fun m => bn.target m ∈ U), κ m x₀ (x₀ (bn.target m)) := by
  set l := ord.order.filter (fun v => decide (v ∉ U)) with hl
  have hmemL : ∀ v, v ∈ l ↔ v ∉ U := by
    intro v

```

```

    rw [hl, List.mem_filter]
    simp [ord.complete v]
  have hlt : l.toFinset = Uc := by
    ext v
    simp [hmemL v]
  have key := marg_partialJoint_eq_one κ hinj hloc ord.no_self l (ord.nodup.filter _)
    (fun m hm ⇒ hnorm m ((hmemL _).1 hm)) (ord.parents_before.filter _)
    (fun v hv ⇒ hmech v ((hmemL v).1 hv)) x0
  rw [← hlt]
  unfold marg at key ⊢
  have hfac : ∀ y ∈ fibre l.toFinset x0, joint κ y =
    (∏ m ∈ Finset.univ.filter (fun m ⇒ bn.target m ∈ U), κ m x0 (x0 (bn.target m)))
    * partialJoint κ l y := by
    intro y hy
    rw [mem_fibre] at hy
    have hagree : ∀ v ∈ U, y v = x0 v := fun v hv ⇒
      hy v (by simp [List.mem_toFinset, hmemL v, hv])
    unfold joint partialJoint
    rw [← Finset.prod_filter_mul_prod_filter_not Finset.univ (fun m ⇒ bn.target m ∈ U)]
    congr 1
    · refine Finset.prod_congr rfl fun m hm ⇒ ?_
      have hmU : bn.target m ∈ U := (Finset.mem_filter.1 hm).2
      rw [hloc m y x0 fun p hp ⇒ hagree p (hU m hmU hp), hagree _ hmU]
    · exact Finset.prod_congr (by ext m; simp [hmemL]) fun m _ ⇒ rfl
  rw [Finset.sum_congr rfl hfac, ← Finset.mul_sum, key, mul_one]

/-- Proposition 3 (closed-world shadow). In a closed network with a topological order and
local, normalised kernels, integrating the downstream variables out of the full joint leaves the
joint of the upstream mechanisms. -/
theorem marg_joint_upstream (κ : bn.Kernel R) (hclosed : bn.Closed) (ord : bn.TopoOrder)
  (hloc : ∀ m, Local κ m) (hnorm : ∀ m, Normalised κ m) (U : Finset bn.V)
  (hU : UpstreamClosed U) (x0 : bn.Assignment) :
  marg Uc (joint κ) x0 =
  ∏ m ∈ Finset.univ.filter (fun m ⇒ bn.target m ∈ U), κ m x0 (x0 (bn.target m)) :=
  marg_joint_downstream κ hclosed.1 ord hloc U (fun m _ ⇒ hnorm m) hU
  (fun v _ ⇒ hclosed.2 v) x0

end FinBayesNet

end BayesianNetworksProofs

```

BayesianNetworksProofs.Finite.OrderedParents

```

import BayesianNetworksProofs.Finite.Evaluation
import Mathlib.Logic.Equiv.Basic

```

An exact ordered-parent refinement of the finite kernel model (SPEC §8.2, §10.2, §11 and the §61 revision-note qualification).

ParentOrder equips every mechanism with a bijection from contiguous positions $\text{Fin } n$ to its parent variables. Thus each parent occurs exactly once and every position is used. An ordered CPT takes a dependent tuple of states in this order, followed by its child's state. Gathering an assignment into that tuple gives a local kernel. Conversely, every local kernel has an ordered CPT, and the two constructions are inverse.

Changing parent order must be accompanied by a corresponding change in the CPT axes: `toKernel_reindex` proves the precise transport law, including heterogeneous parent state types. `sum_joint_ordered` transfers the existing normalisation theorem to ordered CPTs.

This closes the unordered-versus-ordered *finite function* gap, not the entire ACSet gap. The bijection is supplied as data; this file does not parse Input rows, verify Julia's sorting routine, model `state_position`, resolve `KernelRefs`, or relate real arithmetic to floating-point arrays. Those correspondence claims remain unproved.

```

set_option autoImplicit false

```

```

namespace BayesianNetworksProofs.FinBayesNet

variable (bn : FinBayesNet)

/-- Contiguous, total and duplicate-free positions for each mechanism's parents.
Julia uses one-based positions; 'Fin n' is their zero-based mathematical index. -/
structure ParentOrder where
  count : bn.M → ℕ
  atPosition : (m : bn.M) → Fin (count m) ≈ {v : bn.V // v ∈ bn.parents m}

namespace ParentOrder

variable {bn} (o : ParentOrder bn) {R : Type}

/-- A tuple of parent states in explicit positional order. -/
abbrev Values (m : bn.M) :=
  (i : Fin (o.count m)) → bn.states (o.atPosition m i).val

/-- The equivalent parent-indexed view of a positional tuple. -/
def valuesEquiv (m : bn.M) :
  o.Values m ≈ ((v : {v : bn.V // v ∈ bn.parents m}) → bn.states v.val) :=
  Equiv.piCongrLeft (fun v : {v : bn.V // v ∈ bn.parents m} ⇒ bn.states v.val) (o.atPosition m)

/-- Gather a joint assignment according to the declared input positions. -/
def gather (m : bn.M) (x : bn.Assignment) : o.Values m :=
  fun i ⇒ x (o.atPosition m i).val

theorem valuesEquiv_gather (m : bn.M) (x : bn.Assignment) :
  o.valuesEquiv m (o.gather m x) = fun v : {v : bn.V // v ∈ bn.parents m} ⇒ x v.val := by
  apply (o.valuesEquiv m).symm.injective
  rw [Equiv.symm_apply_apply]
  rfl

/-- Extend parent values to all variables, using the model's nonempty state spaces off-scope. -/
noncomputable def extend (m : bn.M)
  (p : (v : {v : bn.V // v ∈ bn.parents m}) → bn.states v.val) : bn.Assignment :=
  fun v ⇒ if hv : v ∈ bn.parents m then p ⟨v, hv⟩ else Classical.choice (bn.nonemptyS v)

theorem gather_extend (m : bn.M) (z : o.Values m) :
  o.gather m (extend m (o.valuesEquiv m z)) = z := by
  have hg : o.gather m (extend m (o.valuesEquiv m z)) =
    (o.valuesEquiv m).symm (o.valuesEquiv m z) := by
    funext i
    simp [gather, extend, (o.atPosition m i).property, valuesEquiv]
  rw [hg, Equiv.symm_apply_apply]

/-- User-facing CPT convention: ordered parent tuple first, child's state last. -/
abbrev CPT (R : Type) :=
  (m : bn.M) → o.Values m → bn.states (bn.target m) → R

/-- Interpret an ordered CPT in the abstract assignment-based model. -/
def toKernel (t : o.CPT R) : bn.Kernel R :=
  fun m x ⇒ t m (o.gather m x)

theorem toKernel_local (t : o.CPT R) : ∀ m, Local (o.toKernel t) m := by
  intro m x y h
  apply congrArg (t m)
  funext i
  exact h (o.atPosition m i).val (o.atPosition m i).property

/-- Read a kernel on all possible positional tuples, completing the unused variables. -/
noncomputable def ofKernel (κ : bn.Kernel R) : o.CPT R :=
  fun m z ⇒ κ m (extend m (o.valuesEquiv m z))

/-- Locality makes the arbitrary off-parent completion unobservable. -/
theorem toKernel_ofKernel (κ : bn.Kernel R) (hloc : ∀ m, Local κ m) :

```

```

    o.toKernel (o.ofKernel κ) = κ := by
  funext m x
  apply hloc m
  intro v hv
  simp [extend, valuesEquiv_gather, hv]

/-- Ordered CPTs are recovered exactly, not only on the assignments used by an evaluator. -/
theorem ofKernel_toKernel (t : o.CPT R) : o.ofKernel (o.toKernel t) = t := by
  funext m z
  change t m (o.gather m (extend m (o.valuesEquiv m z))) = t m z
  rw [gather_extend]

/-- The unordered locality condition is precisely representability by an ordered CPT. -/
theorem local_iff_ordered (κ : bn.Kernel R) :
  (∀ m, Local κ m) ↔ ∃ t : o.CPT R, o.toKernel t = κ := by
  constructor
  · intro h
    exact ⟨o.ofKernel κ, o.toKernel_ofKernel κ h⟩
  · rintro ⟨t, rfl⟩
    exact o.toKernel_local t

/-- Transport new-order tuples into the old order through their common parent-indexed view. -/
def reindexValues (o' : ParentOrder bn) (m : bn.M) : o'.Values m ≈ o.Values m :=
  (o'.valuesEquiv m).trans (o.valuesEquiv m).symm

theorem reindexValues_gather (o' : ParentOrder bn) (m : bn.M) (x : bn.Assignment) :
  o.reindexValues o' m (o'.gather m x) = o.gather m x := by
  change (o.valuesEquiv m).symm (o'.valuesEquiv m (o'.gather m x)) = _
  rw [valuesEquiv_gather]
  rfl

/-- Permute the CPT's parent axes along with their declared order, leaving the child axis last. -/
def reindexCPT (o' : ParentOrder bn) (t : o.CPT R) : o'.CPT R :=
  fun m z ⇒ t m (o.reindexValues o' m z)

/-- **Ordered-parent fidelity:** coherent reindexing of parent positions and CPT axes preserves
every kernel entry, and hence every joint, marginal and intervention computed from the kernel. -/
theorem toKernel_reindex (o' : ParentOrder bn) (t : o.CPT R) :
  o'.toKernel (o.reindexCPT o' t) = o.toKernel t := by
  funext m x
  change t m (o.reindexValues o' m (o'.gather m x)) = t m (o.gather m x)
  rw [reindexValues_gather]

theorem toKernel_normalised [CommSemiring R] (t : o.CPT R)
  (ht : ∀ m z, ∑ a, t m z a = 1) : ∀ m, Normalised (o.toKernel t) m :=
  fun m x ⇒ ht m (o.gather m x)

/-- **Proposition 1 for genuinely ordered CPTs.** Normalising the child-last axis produces a
normalised joint on a closed acyclic network, independently of the chosen parent ordering. -/
theorem sum_joint_ordered [CommSemiring R] (t : o.CPT R) (hclosed : bn.Closed)
  (ord : bn.TopoOrder) (ht : ∀ m z, ∑ a, t m z a = 1) :
  ∑ x, joint (o.toKernel t) x = 1 :=
  sum_joint_eq_one _ hclosed ord (o.toKernel_local t) (o.toKernel_normalised t ht)

end ParentOrder

end BayesianNetworksProofs.FinBayesNet

```

BayesianNetworksProofs.Finite.VariableElimination

```

import BayesianNetworksProofs.Finite.Evaluation
import Mathlib.Data.List.Pairwise

```

Bucket variable elimination preserves the exact marginal (SPEC §15–§17, §55.3).

A factor has a finite scope and a value depending only on that scope. Eliminating a variable multiplies only the factors whose scopes contain it, sums that bucket over the variable, and retains all other factors. We prove this actual bucket operation equals the full-enumeration one-variable marginal. Iteration along a duplicate-free list equals the marginal over the whole list, so elimination order does not change the result.

The arithmetic is over any commutative semiring. No positivity or division is used. The result is an **unnormalised** marginal: normalising evidence of zero total weight remains undefined in probability theory and is not justified by this theorem.

This is a scoped finite-function algorithm and an oracle theorem, not a verified Julia array implementation, not a junction-tree theorem, and not decision-variable elimination.

```
set_option autoImplicit false

namespace BayesianNetworksProofs.FinBayesNet

variable {bn : FinBayesNet} {R : Type} [CommSemiring R]

/-- A scalar factor together with a sufficient (not necessarily minimal) dependency scope. -/
structure Factor (bn : FinBayesNet) (R : Type) where
  scope : Finset bn.V
  value : bn.Assignment → R
  dependsOn : ∀ x y, (∀ v ∈ scope, x v = y v) → value x = value y

namespace Factor

/-- Product of a list of factors, keeping repeated factors and their multiplicities. -/
def product (fs : List (Factor bn R)) (x : bn.Assignment) : R :=
  (fs.map fun f => f.value x).prod

theorem product_nil (x : bn.Assignment) : product ([] : List (Factor bn R)) x = 1 := rfl

theorem product_cons (f : Factor bn R) (fs : List (Factor bn R)) (x : bn.Assignment) :
  product (f :: fs) x = f.value x * product fs x := rfl

/-- The union of all factor scopes. -/
def jointScope (fs : List (Factor bn R)) : Finset bn.V :=
  fs.foldl (fun f S => f.scope ∪ S) ∅

theorem product_local (fs : List (Factor bn R)) (x y : bn.Assignment)
  (h : ∀ v ∈ jointScope fs, x v = y v) : product fs x = product fs y := by
  induction fs with
  | nil => rfl
  | cons f fs ih =>
    rw [product_cons, product_cons, f.dependsOn x y (fun v hv => h v (Finset.mem_union_left _ hv)),
      ih (fun v hv => h v (Finset.mem_union_right _ hv))]

/-- Multiply an entire bucket and sum its distinguished variable out. -/
def sumBucket (v : bn.V) (fs : List (Factor bn R)) : Factor bn R where
  scope := (jointScope fs).erase v
  value x := ∑ a : bn.states v, product fs (Function.update x v a)
  dependsOn x y h := by
    refine Finset.sum_congr rfl fun a _ => ?_
    apply product_local
    intro w hw
    by_cases he : w = v
    · subst w
      simp
    · simp only [Function.update_of_ne he]
      exact h w (Finset.mem_erase.2 ⟨he, hw⟩)

/-- The bucket is selected by membership in a factor's scope, not by factor position. -/
def bucket (v : bn.V) (fs : List (Factor bn R)) : List (Factor bn R) :=
  fs.filter (fun f => decide (v ∈ f.scope))

/-- Factors not containing the eliminated variable. -/
def outside (v : bn.V) (fs : List (Factor bn R)) : List (Factor bn R) :=
```

```

fs.filter (fun f => decide (v ∉ f.scope))

theorem product_partition (v : bn.V) (fs : List (Factor bn R)) (x : bn.Assignment) :
  product fs x = product (outside v fs) x * product (bucket v fs) x := by
  induction fs with
  | nil => simp [outside, bucket, product]
  | cons f fs ih =>
    by_cases hv : v ∈ f.scope <=>
      simp [outside, bucket, product, hv] at * <=> rw [ih] <=> ac rfl

theorem outside_update (v : bn.V) (fs : List (Factor bn R))
  (x : bn.Assignment) (a : bn.states v) :
  product (outside v fs) (Function.update x v a) = product (outside v fs) x := by
  unfold product
  apply congrArg List.prod
  apply List.map_congr_left
  intro f hf
  have hv : v ∉ f.scope := by simp [outside] using (List.mem_filter.1 hf).2
  apply f.dependsOn
  intro w hw
  have hne : w ≠ v := fun he => hv (he ▸ hw)
  rw [Function.update_of_ne hne]

/-- Replace one bucket by its marginal, preserving every factor outside it. -/
def eliminate (v : bn.V) (fs : List (Factor bn R)) : List (Factor bn R) :=
  sumBucket v (bucket v fs) :: outside v fs

/-- **One elimination step equals brute-force marginalisation**, even with overlapping scopes,
repeated factors, and variables absent from all scopes. -/
theorem eliminate_correct (v : bn.V) (fs : List (Factor bn R)) (x : bn.Assignment) :
  product (eliminate v fs) x = marg {v} (product fs) x := by
  rw [show ({v} : Finset bn.V) = insert v ∅ from rfl, marg_insert (by simp)]
  simp only [marg_empty]
  change (∑ a : bn.states v, product (bucket v fs) (Function.update x v a)) *
    product (outside v fs) x = _
  rw [Finset.sum_mul]
  refine Finset.sum_congr rfl fun a _ => ?_
  rw [product_partition v fs, outside_update, mul_comm]

/-- Successive bucket eliminations. -/
def eliminateAll (fs : List (Factor bn R)) : List bn.V → List (Factor bn R)
| [] => fs
| v :: vs => eliminateAll (eliminate v fs) vs

/-- **Variable-elimination oracle theorem.** A duplicate-free elimination order computes exactly
the full joint product summed over those variables. -/
theorem eliminateAll_correct (fs : List (Factor bn R)) (vs : List bn.V) (hvs : vs.Nodup)
  (x : bn.Assignment) :
  product (eliminateAll fs vs) x = marg vs.toFinset (product fs) x := by
  induction vs generalizing fs with
  | nil => simp [eliminateAll, marg_empty]
  | cons v vs ih =>
    have hv : v ∉ vs.toFinset := by simp using (List.nodup_cons.1 hvs).1
    rw [eliminateAll, ih _ (List.nodup_cons.1 hvs).2]
    have he : product (eliminate v fs) = marg {v} (product fs) :=
      funext (eliminate_correct v fs)
    rw [he, ← marg_union_disjoint (Finset.disjoint_singleton_right.2 hv)]
    simp

/-- Any two duplicate-free orders eliminating the same variables have the same marginal. -/
theorem eliminateAll_order_independent (fs : List (Factor bn R)) (vs ws : List bn.V)
  (hvs : vs.Nodup) (hws : ws.Nodup) (hset : vs.toFinset = ws.toFinset) :
  product (eliminateAll fs vs) = product (eliminateAll fs ws) := by
  funext x
  rw [eliminateAll_correct fs vs hvs, eliminateAll_correct fs ws hws, hset]

```

```

/-- Compile one local mechanism into a factor on its parents and child. -/
def mechanism (κ : bn.Kernel R) (hloc : ∀ m, Local κ m) (m : bn.M) : Factor bn R where
  scope := insert (bn.target m) (bn.parents m)
  value x := κ m x (x (bn.target m))
  dependsOn x y h := by
    rw [hloc m x y (fun v hv => h v (Finset.mem_insert_of_mem hv)),
        h (bn.target m) (Finset.mem_insert_self _ _)]

/-- Compile all mechanisms, retaining each mechanism exactly once. -/
noncomputable def ofKernel (κ : bn.Kernel R) (hloc : ∀ m, Local κ m) : List (Factor bn R) :=
  Finset.univ.toList.map (mechanism κ hloc)

theorem product_ofKernel (κ : bn.Kernel R) (hloc : ∀ m, Local κ m) :
  product (ofKernel κ hloc) = joint κ := by
  funext x
  simp [product, ofKernel, mechanism, joint, List.map_map]

/-- The bucket algorithm on compiled Bayesian-network factors computes the exact joint
marginal. Closedness and stochasticity are needed for a probability interpretation, not for
this semiring identity. -/
theorem eliminateAll_joint (κ : bn.Kernel R) (hloc : ∀ m, Local κ m)
  (vs : List bn.V) (hvs : vs.Nodup) (x : bn.Assignment) :
  product (eliminateAll (ofKernel κ hloc) vs) x = marg vs.toFinset (joint κ) x := by
  rw [eliminateAll_correct _ vs hvs, product_ofKernel]

end Factor

end BayesianNetworksProofs.FinBayesNet

```

BayesianNetworksProofs.Finite.Intervention

```
import BayesianNetworksProofs.Finite.Evaluation
```

Proposition 4 (hard intervention) for the concrete finite model (SPEC §21.2, §22).

do($X = a$) replaces the mechanism m_θ generating X by the point mass δ_a , ignoring its former parents. Syntactically this is cut bn m_θ (the input rows of m_θ are deleted, i.e. parents $m_\theta := \emptyset$); semantically it is intervene $\kappa m_\theta a$, which overwrites the kernel of m_θ with the indicator of a .

- joint_intervene — the truncated factorisation: the intervened joint is the indicator $[x \ X = a]$ times the product of the *other* mechanisms' kernels, which are untouched.
- normalised_intervene, local_cut — the intervened family is again normalised and local for the rewritten network, so Proposition 1 applies to it (sum_joint_intervene_eq_one).

```
namespace BayesianNetworksProofs
```

```
namespace FinBayesNet
```

```
variable {bn : FinBayesNet} {R : Type}
```

```
/-- The syntactic rewrite 'do': the mechanism 'm_θ' keeps its target but loses its inputs. -/
```

```
def cut (bn : FinBayesNet) (m_θ : bn.M) : FinBayesNet :=
  { bn with parents := Function.update bn.parents m_θ ∅ }
```

```
@[simp] theorem cut_parents (m_θ m : bn.M) :
  (bn.cut m_θ).parents m = Function.update bn.parents m_θ ∅ m := rfl
```

```
/-- A topological order of 'bn' is one of 'bn.cut m_θ' (parents only shrink). -/
```

```
def TopoOrder.cut (ord : bn.TopoOrder) (m_θ : bn.M) : (bn.cut m_θ).TopoOrder where
  order := ord.order
  nodup := ord.nodup
  complete := ord.complete
  parents_before := ord.parents_before.imp fun {a b} h (m : bn.M) hm => by
    show b ∉ Function.update bn.parents m_θ ∅ m
```

```

by_cases h m : m = m₀
• subst h m
  simp
• rw [Function.update_of_ne h m]
  exact h m h m
no_self := fun (m : bn.M) => by
  show bn.target m ∉ Function.update bn.parents m₀ ∅ m
  by_cases h m : m = m₀
  • subst h m
    simp
  • rw [Function.update_of_ne h m]
    exact ord.no_self m

theorem closed_cut (hclosed : bn.Closed) (m₀ : bn.M) : (bn.cut m₀).Closed := hclosed

variable [CommSemiring R]

/-- The semantic rewrite: replace the kernel of 'm₀' by the point mass at 'a'. -/
def intervene (κ : bn.Kernel R) (m₀ : bn.M) (a : bn.states (bn.target m₀)) : bn.Kernel R :=
  Function.update κ m₀ (fun _ y => if y = a then 1 else 0)

@[simp] theorem intervene_self (κ : bn.Kernel R) (m₀ : bn.M) (a : bn.states (bn.target m₀))
  (x : bn.Assignment) (y : bn.states (bn.target m₀)) :
  intervene κ m₀ a m₀ x y = if y = a then 1 else 0 := by
  simp [intervene]

theorem intervene_of_ne (κ : bn.Kernel R) {m₀ m : bn.M} (h : m ≠ m₀)
  (a : bn.states (bn.target m₀)) : intervene κ m₀ a m = κ m := by
  simp [intervene, Function.update_of_ne h]

/-- **Proposition 4** (truncated factorisation). -/
theorem joint_intervene (κ : bn.Kernel R) (m₀ : bn.M) (a : bn.states (bn.target m₀))
  (x : bn.Assignment) :
  joint (intervene κ m₀ a) x =
    (if x (bn.target m₀) = a then 1 else 0) *
    ∏ m ∈ Finset.univ.erase m₀, κ m x (x (bn.target m)) := by
  unfold joint
  rw [← Finset.mul_prod_erase Finset.univ _ (Finset.mem_univ m₀)]
  congr 1
  • simp
  • exact Finset.prod_congr rfl fun m hm => by
    rw [intervene_of_ne κ (Finset.ne_of_mem_erase hm)]

/-- The point-mass kernel is normalised. -/
theorem normalised_intervene_self (κ : bn.Kernel R) (m₀ : bn.M) (a : bn.states (bn.target m₀)) :
  Normalised (intervene κ m₀ a) m₀ := by
  intro x
  simp

/-- Intervening preserves normalisation of every mechanism. -/
theorem normalised_intervene (κ : bn.Kernel R) (hnorm : ∀ m, Normalised κ m) (m₀ : bn.M)
  (a : bn.states (bn.target m₀)) : ∀ m, Normalised (intervene κ m₀ a) m := by
  intro m
  by_cases h : m = m₀
  • subst h
    exact normalised_intervene_self κ m a
  • intro x
    rw [intervene_of_ne κ h]
    exact hnorm m x

/-- The point-mass kernel reads no parents at all. -/
theorem localOn_empty_intervene (κ : bn.Kernel R) (m₀ : bn.M) (a : bn.states (bn.target m₀)) :
  LocalOn ∅ (intervene κ m₀ a m₀) := by
  intro x x' _
  funext y
  simp

```

```

/-- The intervened family is local for the rewritten network 'bn.cut m₀'. -/
theorem local_cut (κ : bn.Kernel R) (hloc : ∀ m, Local κ m) (m₀ : bn.M)
  (a : bn.states (bn.target m₀)) :
  ∀ m, Local (bn := bn.cut m₀) (intervene κ m₀ a) m := by
refine fun (m : bn.M) (x x' : bn.Assignment) hx ⇒ ?_
by_cases h : m = m₀
• subst h
  exact localOn_empty_intervene κ m a x x' (fun p hp ⇒ absurd hp (Finset.notMem_empty p))
• show intervene κ m₀ a m x = intervene κ m₀ a m x'
  rw [intervene_of_ne κ h]
  refine hloc m x x' fun p hp ⇒ hx p ?_
  show p ∈ Function.update bn.parents m₀ ∅ m
  rw [Function.update_of_ne h]
  exact hp

/-- The intervened joint is again a probability distribution (Proposition 1 applied to the
rewritten network). -/
theorem sum_joint_intervene_eq_one (κ : bn.Kernel R) (hclosed : bn.Closed) (ord : bn.TopoOrder)
  (hloc : ∀ m, Local κ m) (hnorm : ∀ m, Normalised κ m) (m₀ : bn.M)
  (a : bn.states (bn.target m₀)) :
  ∑ x, joint (intervene κ m₀ a) x = 1 :=
sum_joint_eq_one (bn := bn.cut m₀) (intervene κ m₀ a) (closed_cut hclosed m₀) (ord.cut m₀)
(local_cut κ hloc m₀ a) (normalised_intervene κ hnorm m₀ a)

end FinBayesNet

end BayesianNetworksProofs

```

BayesianNetworksProofs.Finite.Tensor

```

import BayesianNetworksProofs.Finite.Evaluation
import Mathlib.Data.Fintype.Sum
import Mathlib.Data.Fintype.BigOperators
import Mathlib.Data.Finset.Sum

```

Proposition 2 (tensor compositionality) for the concrete finite model (SPEC §13.1).

The tensor $bn_1.tensor\ bn_2$ of two networks has variables $V_1 \oplus V_2$ and mechanisms $M_1 \oplus M_2$ (the disjoint union of the two ACSets, $\otimes$ of OpenBayesNets with no shared feet). Its kernels are the two families side by side, and `joint_tensor` shows that the joint of the tensor is the product of the two joints — $\llbracket A \otimes B \rrbracket = \llbracket A \rrbracket \otimes \llbracket B \rrbracket$ in the finite stochastic semantics. The tensor of closed / local / normalised / topologically ordered networks is again so.

```

namespace BayesianNetworksProofs

namespace FinBayesNet

/-- Disjoint union of two networks. Reducible so that 'simp'/'rw' see 'V = V₁ ⊕ V₂' etc. -/
@[reducible] def tensor (bn₁ bn₂ : FinBayesNet) : FinBayesNet where
  V := bn₁.V ⊕ bn₂.V
  M := bn₁.M ⊕ bn₂.M
  states := Sum.elim bn₁.states bn₂.states
  fintypeS := fun
    | .inl v ⇒ bn₁.fintypeS v
    | .inr v ⇒ bn₂.fintypeS v
  decS := fun
    | .inl v ⇒ bn₁.decS v
    | .inr v ⇒ bn₂.decS v
  nonemptyS := fun
    | .inl v ⇒ bn₁.nonemptyS v
    | .inr v ⇒ bn₂.nonemptyS v
  target := Sum.map bn₁.target bn₂.target
  parents := fun
    | .inl m ⇒ (bn₁.parents m).map Function.Embedding.inl

```

```

| .inr m ⇒ (bn₂.parents m).map Function.Embedding.inr

variable {bn₁ bn₂ : FinBayesNet} {R : Type}

@[simp] theorem tensor_target_inl (m : bn₁.M) :
  (bn₁.tensor bn₂).target (.inl m) = .inl (bn₁.target m) := rfl

@[simp] theorem tensor_target_inr (m : bn₂.M) :
  (bn₁.tensor bn₂).target (.inr m) = .inr (bn₂.target m) := rfl

@[simp] theorem tensor_parents_inl (m : bn₁.M) :
  (bn₁.tensor bn₂).parents (.inl m) = (bn₁.parents m).map Function.Embedding.inl := rfl

@[simp] theorem tensor_parents_inr (m : bn₂.M) :
  (bn₁.tensor bn₂).parents (.inr m) = (bn₂.parents m).map Function.Embedding.inr := rfl

@[simp] theorem inl_mem_map_inl {α β : Type} {s : Finset α} {a : α} :
  Sum.inl a ∈ s.map (Function.Embedding.inl (β := β)) ↔ a ∈ s := by
  simp [Finset.mem_map]

@[simp] theorem inr_mem_map_inr {α β : Type} {s : Finset β} {b : β} :
  Sum.inr b ∈ s.map (Function.Embedding.inr (α := α)) ↔ b ∈ s := by
  simp [Finset.mem_map]

@[simp] theorem inl_notMem_map_inl {α β : Type} {s : Finset α} {b : β} :
  Sum.inr b ∉ s.map (Function.Embedding.inl (β := β)) := by
  simp [Finset.mem_map]

@[simp] theorem inl_notMem_map_inr {α β : Type} {s : Finset β} {a : α} :
  Sum.inl a ∉ s.map (Function.Embedding.inr (α := α)) := by
  simp [Finset.mem_map]

/-- Restrict an assignment of the tensor to the left factor. -/
def Assignment.left (x : (bn₁.tensor bn₂).Assignment) : bn₁.Assignment := fun v ⇒ x (.inl v)

/-- Restrict an assignment of the tensor to the right factor. -/
def Assignment.right (x : (bn₁.tensor bn₂).Assignment) : bn₂.Assignment := fun v ⇒ x (.inr v)

/-- Two kernel families side by side. -/
def tensorKernel (κ₁ : bn₁.Kernel R) (κ₂ : bn₂.Kernel R) : (bn₁.tensor bn₂).Kernel R := fun
| .inl m ⇒ fun x y ⇒ κ₁ m (Assignment.left x) y
| .inr m ⇒ fun x y ⇒ κ₂ m (Assignment.right x) y

/-- Proposition 2. The joint of the tensor is the product of the joints. -/
theorem joint_tensor [CommMonoid R] (κ₁ : bn₁.Kernel R) (κ₂ : bn₂.Kernel R)
  (x : (bn₁.tensor bn₂).Assignment) :
  joint (tensorKernel κ₁ κ₂) x = joint κ₁ (Assignment.left x) * joint κ₂ (Assignment.right x) := by
  unfold joint
  exact Fintype.prod_sum_type _

theorem closed_tensor (h₁ : bn₁.Closed) (h₂ : bn₂.Closed) : (bn₁.tensor bn₂).Closed :=
  h₁.sumMap h₂

theorem normalised_tensor [AddCommMonoid R] [One R] (κ₁ : bn₁.Kernel R) (κ₂ : bn₂.Kernel R)
  (h₁ : ∀ m, Normalised κ₁ m) (h₂ : ∀ m, Normalised κ₂ m) :
  ∀ m, Normalised (tensorKernel κ₁ κ₂) m
| .inl m ⇒ fun x ⇒ h₁ m (Assignment.left x)
| .inr m ⇒ fun x ⇒ h₂ m (Assignment.right x)

theorem local_tensor (κ₁ : bn₁.Kernel R) (κ₂ : bn₂.Kernel R)
  (h₁ : ∀ m, Local κ₁ m) (h₂ : ∀ m, Local κ₂ m) :
  ∀ m, Local (tensorKernel κ₁ κ₂) m
| .inl m ⇒ fun x x' hx ⇒
  h₁ m (Assignment.left x) (Assignment.left x') fun p hp ⇒
  hx (.inl p) (Finset.mem_map_of_mem _ hp)
| .inr m ⇒ fun x x' hx ⇒

```

```

h2 m (Assignment.right x) (Assignment.right x') fun p hp =>
  hx (.inr p) (Finset.mem_map_of_mem _ hp)

/-- Concatenating topological orders gives a topological order of the tensor. -/
def TopoOrder.tensor (o1 : bn1.TopoOrder) (o2 : bn2.TopoOrder) : (bn1.tensor bn2).TopoOrder where
  order := o1.order.map Sum.inl ++ o2.order.map Sum.inr
  nodup := by
    refine List.Nodup.append (o1.nodup.map Sum.inl_injective) (o2.nodup.map Sum.inr_injective) ?_
    intro a h1 h2
    obtain ⟨v, _, rfl⟩ := List.mem_map.1 h1
    obtain ⟨w, _, h⟩ := List.mem_map.1 h2
    exact Sum.inr_ne_inl h
  complete := fun
    | .inl v => List.mem_append_left _ (List.mem_map_of_mem (o1.complete v))
    | .inr v => List.mem_append_right _ (List.mem_map_of_mem (o2.complete v))
  parents_before := by
    refine List.pairwise_append.2 (List.pairwise_map.2 (o1.parents_before.imp ?_),
      List.pairwise_map.2 (o2.parents_before.imp ?_), ?_)
    • rintro a b h (m | m) hm
      • simp using h m (Sum.inl_injective hm)
      • simp at hm
    • rintro a b h (m | m) hm
      • simp at hm
      • simp using h m (Sum.inr_injective hm)
    • rintro a ha b hb (m | m) hm
      • obtain ⟨w, _, rfl⟩ := List.mem_map.1 hb
        simp
      • obtain ⟨v, _, rfl⟩ := List.mem_map.1 ha
        simp at hm
  no_self := fun
    | .inl m => by simp using o1.no_self m
    | .inr m => by simp using o2.no_self m

end FinBayesNet

end BayesianNetworksProofs

```

BayesianNetworksProofs.Finite.Open

```
import BayesianNetworksProofs.Finite.Tensor
```

Open networks and the closure theorem (SPEC §13 and its revision note, §55.5, §61 Propositions 2 and 3), for the concrete finite model — no category theory, no quotient types.

An `OpenFinBayesNet` is a `FinBayesNet` with two distinguished sets of variables, inputs and outputs, satisfying the finite-model part of the *typed-interface rule* of `CategoricalBayesianNetworks.jl`'s `Open` (`src/open.jl`, `validation_errors (::OpenBayesNet)`):

Julia rule	Lean field
at most one mechanism per variable (<code>validate(bn; closed = false)</code>)	<code>target_inj</code>
rule 1: input-foot variables have no mechanism	<code>input_exogenous</code>
rule 3: every mechanism-free apex variable is an input	<code>exogenous_input</code>
rule 4: the derived graph is acyclic	<code>topo</code> (a <code>TopoOrder</code>)

Rule 2 (the input leg is injective) is encoded by taking feet to be *subsets* of apex variables. Rule 5's name-, reference- and position-preserving attribute conditions are **not** modelled: those attributes are erased. The interface match of `compose` is the injection $\imath$ below; `states_equiv` transports abstract state types, but does not assert preservation of a `state_position` attribute. Noninjective output legs are also absent from this representation.

Composition without quotients

Composable $A \circ B$ is an injection ι from B 's inputs into A 's outputs together with the matching of their state spaces. The pushout of the two structured cospans identifies each input of B with its partner output of A ; rather than quotient $A.V \oplus B.V$, the composite takes

- $V := A.V \oplus \text{Priv } B$ where $\text{Priv } B = \{v : B.V \mid v \notin B.\text{inputs}\}$ — one representative per class, the glued variables being represented on the A side;
- $M := A.M \oplus B.M$ — mechanisms are never identified;
- $\text{tr} : B.V \rightarrow A.V \oplus \text{Priv } B$, injective, sending an input of B to $\text{Sum.inl } (\iota \dots)$ and any other variable to Sum.inr — the right-hand pushout leg.

`composeNet` is the resulting shape and `compose` bundles it back into an `OpenFinBayesNet`, which is the closure theorem: producing the four fields is the proof that the composite is again valid. The individual statements are

- `composeNet_target_injective` — **at most one mechanism per variable is preserved**. The key step is that a mechanism of B never targets a glued variable, because B 's inputs are exogenous (`target_notMem_inputs`); so the two mechanism families cannot collide;
- `compose_input_exogenous`, `compose_exogenous_input` — the composite's inputs are exactly $A.\text{inputs}$, and every variable contributed by B keeps its mechanism;
- `composeTopo` — **acyclicity is preserved**: A 's order followed by B 's order restricted to $\text{Priv } B$ is a topological order of the composite. It is *derived*, not assumed.

Outputs are $(A.\text{outputs} \setminus \text{glued}) \cup \text{tr } (B.\text{outputs})$, exactly as in Julia's `glue`.

Semantics (Proposition 3)

`composeKernel` puts two kernel families side by side; `restrictA` and `restrictB` read an assignment of the composite as an assignment of A and of B (a glued variable is read on the A side and transported by `states_equiv`).

- `joint_compose` — $\llbracket B \circ A \rrbracket = \llbracket A \rrbracket \cdot \llbracket B \rrbracket$ pointwise: the joint of the composite factors as the product of the two joints. This is the sequential analogue of `Finite/Tensor.lean`'s `joint_tensor`, and the algebraic heart of Proposition 3;
- `local_compose`, `normalised_compose`, `closed_compose`, `sum_joint_compose_eq_one` — the composite of local / normalised kernels is local / normalised, and if A has no inputs the composite is closed, so its joint is a probability distribution;
- `marg_joint_compose` — summing the composite joint over all values of B 's private variables returns A 's joint: $\llbracket B \circ A \rrbracket$ marginalised back onto A is $\llbracket A \rrbracket$. Only B 's kernels need to be normalised;
- `marg_joint_compose_split` — **Proposition 3**. Integrating out $S \cup T$, with S on the A side and T on the B side, is $\sum_S \llbracket A \rrbracket \cdot (\sum_T \llbracket B \rrbracket)$: the composite semantics is the sum over the interface of the product of the two open semantics. Taking S to be A 's hidden variables together with the glued interface and T to be B 's hidden variables gives $\llbracket B \circ A \rrbracket = \llbracket B \rrbracket \circ \llbracket A \rrbracket$ when neither network has pass-through variables. The transfer to the open semantics of A and B on their own variable types is proved in `Finite/OpenSemantics.lean`, with partial matching and, in `osem_compose_passthrough`, arbitrary pass-through on either side.

```
namespace BayesianNetworksProofs
```

```
structure OpenFinBayesNet extends FinBayesNet where
  inputs : Finset V
  outputs : Finset V
  target_inj : Function.Injective target
  input_exogenous : ∀ v ∈ inputs, ∀ m, target m ≠ v
  exogenous_input : ∀ v, (∀ m, target m ≠ v) → v ∈ inputs
  topo : FinBayesNet.TopoOrder toFinBayesNet
```

```
namespace OpenFinBayesNet
```

```
open FinBayesNet
```

```
variable {A B : OpenFinBayesNet}
```

```
/-- Rule 1 restated: a mechanism never targets an input. -/
```

```
theorem target_notMem_inputs (O : OpenFinBayesNet) (m : O.M) : O.target m ∉ O.inputs :=
  fun h => O.input_exogenous _ h m rfl
```

```

/-- Rule 3 restated: a non-input variable is generated by some mechanism. -/
theorem exists_mechanism (O : OpenFinBayesNet) {v : O.V} (h : v ∉ O.inputs) :
  ∃ m, O.target m = v := by
  by_contra hc
  exact h (O.exogenous_input v (fun m hm => hc ⟨m, hm⟩))

/-- The hidden (non-interface) variables of an open network. -/
def hidden (O : OpenFinBayesNet) : Finset O.V := (O.inputs ∪ O.outputs)ᶜ

/-- The open semantics '[0]' of an open network: the joint with the hidden variables summed
out, a function of the interface (input and output) variables alone. -/
def osem {R : Type} [CommSemiring R] (O : OpenFinBayesNet) (κ : O.Kernel R)
  (x : O.Assignment) : R := marg O.hidden (joint κ) x

/-- The variables of 'B' that are not inputs – one representative per pushout class on the
'B' side. -/
abbrev Priv (B : OpenFinBayesNet) := {v : B.V // v ∉ B.inputs}

/-- 'some' on the non-input variables of 'B', 'none' on the inputs. -/
def privOf (v : B.V) : Option (Priv B) := if h : v ∈ B.inputs then none else some ⟨v, h⟩

/-- 'privOf' picks out exactly the non-input variables. -/
theorem privOf_eq_some {v : B.V} {w : Priv B} : privOf v = some w ↔ v = w.val := by
  unfold privOf
  by_cases h : v ∈ B.inputs
  · simp only [dif_pos h, reduceCtorEq, false_iff]
    rintro rfl
    exact w.property h
  · simp [dif_neg h, Subtype.ext_iff]

/-- An interface match: an injection of 'B's inputs into 'A's outputs (Catlab's gluing map,
validated by 'interface_matches' in Julia) together with the identification of the state spaces
it pairs up. -/
structure Composable (A B : OpenFinBayesNet) where
  ι : {v : B.V // v ∈ B.inputs} → {v : A.V // v ∈ A.outputs}
  ι_inj : Function.Injective ι
  states_equiv : ∀ w : {v : B.V // v ∈ B.inputs}, A.states (ι w).val ≈ B.states w.val

namespace Composable

/-- The right-hand pushout leg: 'B's variables inside the composite. -/
def tr (c : Composable A B) (v : B.V) : A.V ⊕ Priv B :=
  if h : v ∈ B.inputs then Sum.inl (c.ι ⟨v, h⟩).val else Sum.inr ⟨v, h⟩

variable {c : Composable A B}

theorem tr_of_mem {v : B.V} (h : v ∈ B.inputs) : c.tr v = Sum.inl (c.ι ⟨v, h⟩).val := dif_pos h

theorem tr_of_notMem {v : B.V} (h : v ∉ B.inputs) : c.tr v = Sum.inr ⟨v, h⟩ := dif_neg h

/-- The right leg is injective: two 'B' variables are identified in the composite only if they
were equal. -/
theorem tr_injective : Function.Injective c.tr := by
  intro v w h
  by_cases hv : v ∈ B.inputs <|> by_cases hw : w ∈ B.inputs
  · rw [tr_of_mem hv, tr_of_mem hw, Sum.inl.injEq] at h
    exact congrArg Subtype.val (c.ι_inj (Subtype.ext h))
  · rw [tr_of_mem hv, tr_of_notMem hw] at h
    exact absurd h (by simp)
  · rw [tr_of_notMem hv, tr_of_mem hw] at h
    exact absurd h (by simp)
  · rw [tr_of_notMem hv, tr_of_notMem hw, Sum.inr.injEq] at h
    exact congrArg Subtype.val h

variable (c) in
/-- 'tr' as an embedding, for 'Finset.map'. -/

```

```

def trEmb : B.V  $\hookleftarrow$  A.V  $\otimes$  Priv B := (c.tr, tr_injective)

@[simp] theorem trEmb_apply (v : B.V) : c.trEmb v = c.tr v := rfl

@[simp] theorem inr_mem_map_trEmb {s : Finset B.V} {w : Priv B} :
  Sum.inr w  $\in$  s.map c.trEmb  $\leftrightarrow$  w.val  $\in$  s := by
  simp only [Finset.mem_map, trEmb_apply]
  constructor
  · rintro (p, hp, h)
    by_cases hpi : p  $\in$  B.inputs
    · rw [tr_of_mem hpi] at h
      exact absurd h (by simp)
    · rw [tr_of_notMem hpi, Sum.inr.injEq] at h
      rw [← h]
      exact hp
  · exact fun h  $\Rightarrow$  (w.val, h, tr_of_notMem w.property)

variable (c) in
/-- The 'A'-outputs that are glued to inputs of 'B'. -/
def glued : Finset A.V := Finset.univ.image (fun w  $\Rightarrow$  {v : B.V // v  $\in$  B.inputs}  $\Rightarrow$  (c.i w).val)

variable (c) in
/-- The underlying shape of the composite. -/
@[reducible] def composeNet : FinBayesNet where
  V := A.V  $\otimes$  Priv B
  M := A.M  $\otimes$  B.M
  states := Sum.elim A.states (fun w  $\Rightarrow$  B.states w.val)
  fintypeS := fun
    | .inl v  $\Rightarrow$  A.fintypeS v
    | .inr w  $\Rightarrow$  B.fintypeS w.val
  decS := fun
    | .inl v  $\Rightarrow$  A.decS v
    | .inr w  $\Rightarrow$  B.decS w.val
  nonemptyS := fun
    | .inl v  $\Rightarrow$  A.nonemptyS v
    | .inr w  $\Rightarrow$  B.nonemptyS w.val
  target := Sum.elim (fun m  $\Rightarrow$  Sum.inl (A.target m))
    (fun m  $\Rightarrow$  Sum.inr (B.target m, B.target_notMem_inputs m))
  parents := Sum.elim (fun m  $\Rightarrow$  (A.parents m).map Function.Embedding.inl)
    (fun m  $\Rightarrow$  (B.parents m).map c.trEmb)

-- 'composeNet' is reducible, so instance search sees through '(composeNet c).states v' to the
-- 'Sum.elim' and no longer matches 'FinBayesNet.fintypeS'; re-expose the instances by hand.
instance instFintypeComposeStates (v : (composeNet c).V) : Fintype ((composeNet c).states v) :=
  (composeNet c).fintypeS v

instance instDecidableEqComposeStates (v : (composeNet c).V) :
  DecidableEq ((composeNet c).states v) := (composeNet c).decS v

@[simp] theorem composeNet_target_inl (m : A.M) :
  (composeNet c).target (.inl m) = .inl (A.target m) := rfl

@[simp] theorem composeNet_target_inr (m : B.M) :
  (composeNet c).target (.inr m) = .inr (B.target m, B.target_notMem_inputs m) := rfl

@[simp] theorem composeNet_parents_inl (m : A.M) :
  (composeNet c).parents (.inl m) = (A.parents m).map Function.Embedding.inl := rfl

@[simp] theorem composeNet_parents_inr (m : B.M) :
  (composeNet c).parents (.inr m) = (B.parents m).map c.trEmb := rfl

variable (c) in
/-- Appending 'B''s topological order (on its non-input variables) after 'A''s. -/
def composeTopo : (composeNet c).TopoOrder where
  order := A.topo.order.map Sum.inl ++ (B.topo.order.filterMap privOf).map Sum.inr
  nodup := by

```

```

refine List.Nodup.append (A.topo.nodup.map Sum.inl_injective)
(List.Nodup.map Sum.inr_injective ?_) ?_
• refine List.Nodup.filterMap ?_ B.topo.nodup
  intro a a' b hb hb'
  rw [Option.mem_def, privOf_eq_some] at hb hb'
  exact hb.trans hb'.symm
• rintro x hx hx'
  obtain ⟨a, _, rfl⟩ := List.mem_map.1 hx
  obtain ⟨w, _, h⟩ := List.mem_map.1 hx'
  exact Sum.inr_ne_inl h
complete := fun
| .inl a ⇒ List.mem_append_left _ (List.mem_map_of_mem (A.topo.complete a))
| .inr w ⇒ List.mem_append_right _
  (List.mem_map_of_mem (List.mem_filterMap.2 ⟨w.val, B.topo.complete w.val,
    privOf_eq_some.2 rfl⟩))
parents_before := by
refine List.pairwise_append.2 {?_, ?_, ?_}
• refine List.pairwise_map.2 (A.topo.parents_before.imp ?_)
  rintro a b hab (m | m) hm
  • rw [composeNet_target_inl, Sum.inl.injEq] at hm
    simpa using hab m hm
  • exact absurd hm (by simp)
• have hB : ((B.topo.order.filterMap privOf).Pairwise
  fun w w' : Priv B ⇒ ∀ m, B.target m = w.val → w'.val ∉ B.parents m) := by
  refine List.Pairwise.filterMap privOf ?_ B.topo.parents_before
  intro a a' hR b hb b' hb'
  rw [privOf_eq_some] at hb hb'
  subst hb; subst hb'
  exact hR
refine List.pairwise_map.2 (hB.imp ?_)
• rintro w w' hww' (m | m) hm
  • exact absurd hm (by simp)
  • rw [composeNet_target_inr, Sum.inr.injEq] at hm
    rw [composeNet_parents_inr, inr_mem_map_trEmb]
    exact hww' m (congrArg Subtype.val hm)
• rintro x hx y hy (m | m) hm
  • obtain ⟨w, _, rfl⟩ := List.mem_map.1 hy
    simp
  • obtain ⟨a, _, rfl⟩ := List.mem_map.1 hx
    exact absurd hm (by simp)
no_self := fun
| .inl m ⇒ by simpa using A.topo.no_self m
| .inr m ⇒ by
  rw [composeNet_target_inr, composeNet_parents_inr, inr_mem_map_trEmb]
  exact B.topo.no_self m

variable (c) in
/-- **The closure theorem, rule 1 ("at most one mechanism per variable").** The target map of
the composite is injective whenever those of 'A' and 'B' are. -/
theorem composeNet_target_injective : Function.Injective (composeNet c).target := by
  rintro (m | m) (m' | m') h
  • exact congrArg Sum.inl (A.target_inj (Sum.inl.inj h))
  • exact absurd h (by simp)
  • exact absurd h (by simp)
  • exact congrArg Sum.inr (B.target_inj (congrArg Subtype.val (Sum.inr.inj h)))

variable (c) in
/-- **The closure theorem, rule 1' (inputs are exogenous).** No composite mechanism targets a
variable of 'A.inputs'. -/
theorem compose_input_exogenous :
  ∀ v ∈ A.inputs.map Function.Embedding.inl, ∀ m, (composeNet c).target m ≠ v := by
  intro v hv m
  obtain ⟨a, ha, rfl⟩ := Finset.mem_map.1 hv
  cases m with
| inl m ⇒ exact fun h ⇒ A.input_exogenous a ha m (Sum.inl.inj h)
| inr m ⇒ exact fun h ⇒ by simp at h

```

```

variable (c) in
/-- **The closure theorem, rule 3 (every mechanism-free variable is an input).** A composite
variable with no generating mechanism comes from 'A' and is an input of 'A'; the variables
contributed by 'B' all keep their mechanism. -/
theorem compose_exogenous_input :
  ∀ v, (∀ m, (composeNet c).target m ≠ v) → v ∈ A.inputs.map Function.Embedding.inl := by
rintro (a | w) h
• exact Finset.mem_map_of_mem _
  (A.exogenous_input a fun m hm ⇒ h (.inl m) (congrArg Sum.inl hm))
• exact absurd (B.exogenous_input w.val fun m hm ⇒
  h (.inr m) (congrArg Sum.inr (Subtype.ext hm))) w.property

variable (c) in
/-- **The open-network closure theorem (SPEC §13, revision note).** The pushout composite of two
valid open networks along an injective interface match is again a valid open network: its inputs
are 'A''s inputs, its outputs are 'A''s unglued outputs together with 'B''s outputs, and the four
typed-interface rules hold. -/
@[reducible] def compose : OpenFinBayesNet where
  toFinBayesNet := composeNet c
  inputs := A.inputs.map Function.Embedding.inl
  outputs := (A.outputs \ c.glued).map Function.Embedding.inl ∪ B.outputs.map c.trEmb
  target_inj := composeNet_target_injective c
  input_exogenous := compose_input_exogenous c
  exogenous_input := compose_exogenous_input c
  topo := composeTopo c

```

Semantics

```

variable (c) in
/-- The 'A'-part of a composite assignment. -/
def restrictA (x : (composeNet c).Assignment) : A.Assignment := fun a ⇒ x (Sum.inl a)

variable (c) in
/-- The 'B'-part of a composite assignment: glued variables are read off the 'A' side
through the interface equivalence. -/
def restrictB (x : (composeNet c).Assignment) : B.Assignment := fun v ⇒
  if h : v ∈ B.inputs then c.states_equiv ⟨v, h⟩ (x (Sum.inl (c.ι ⟨v, h⟩).val))
  else x (Sum.inr ⟨v, h⟩)

variable (c) in
/-- On a variable that is not an input of 'B', 'restrictB' just reads the 'Sum.inr' component. -/
theorem restrictB_of_notMem (x : (composeNet c).Assignment) {v : B.V} (h : v ∉ B.inputs) :
  restrictB c x v = x (Sum.inr ⟨v, h⟩) := dif_neg h

variable (c) in
/-- Two kernel families side by side on the composite. -/
def composeKernel {R : Type} (κA : A.Kernel R) (κB : B.Kernel R) : (composeNet c).Kernel R
| .inl m ⇒ fun x y ⇒ κA m (restrictA c x) y
| .inr m ⇒ fun x y ⇒ κB m (restrictB c x) y

section Semantics

variable {R : Type}

/-- **Proposition 2 for sequential composition.** The joint of the composite is the product of
the two joints, each read off the corresponding part of the composite assignment. -/
theorem joint_compose [CommMonoid R] (κA : A.Kernel R) (κB : B.Kernel R)
  (x : (composeNet c).Assignment) :
  joint (composeKernel c κA κB) x = joint κA (restrictA c x) * joint κB (restrictB c x) := by
have h2 : ∀ m : B.M, composeKernel c κA κB (.inr m) x (x ((composeNet c).target (.inr m)))
  = κB m (restrictB c x) (restrictB c x (B.target m)) := fun m ⇒
  congrArg _ (restrictB_of_notMem c x (B.target_notMem_inputs m)).symm
unfold joint

```

```

rw [Fintype.prod_sum_type, Finset.prod_congr rfl fun m _ => h2 m]
rfl

theorem normalised_compose [AddCommMonoid R] [One R] (κA : A.Kernel R) (κB : B.Kernel R)
  (hA : ∀ m, Normalised κA m) (hB : ∀ m, Normalised κB m) :
  ∀ m, Normalised (composeKernel c κA κB) m
| .inl m => fun x => hA m (restrictA c x)
| .inr m => fun x => hB m (restrictB c x)

theorem local_compose (κA : A.Kernel R) (κB : B.Kernel R)
  (hA : ∀ m, Local κA m) (hB : ∀ m, Local κB m) :
  ∀ m, Local (composeKernel c κA κB) m
| .inl m => fun x x' hx => hA m (restrictA c x) (restrictA c x') fun p hp =>
  hx (Sum.inl p) (Finset.mem_map_of_mem _ hp)
| .inr m => fun x x' hx => hB m (restrictB c x) (restrictB c x') fun p hp => by
  have hxx := hx (c.tr p) (Finset.mem_map_of_mem c.trEmb hp)
  by_cases h : p ∈ B.inputs
  · rw [tr_of_mem h] at hxx
    simp only [restrictB, dif_pos h]
    exact congrArg _ hxx
  · rw [tr_of_notMem h] at hxx
    rw [restrictB_of_notMem c x h, restrictB_of_notMem c x' h]
    exact hxx

variable (c) in
/-- The composite variables contributed by 'A' (its own variables and the glued ones). -/
def varsA : Finset (composeNet c).V := Finset.univ.map Function.Embedding.inl

variable (c) in
/-- The composite variables contributed by 'B' alone: its non-input ("private") variables. -/
def varsB : Finset (composeNet c).V := Finset.univ.map Function.Embedding.inr

@[simp] theorem inl_mem_varsA (a : A.V) : (Sum.inl a : (composeNet c).V) ∈ varsA c :=
  Finset.mem_map_of_mem _ (Finset.mem_univ a)

@[simp] theorem inr_notMem_varsA (w : Priv B) : (Sum.inr w : (composeNet c).V) ∉ varsA c := by
  simp [varsA]

@[simp] theorem inr_mem_varsB (w : Priv B) : (Sum.inr w : (composeNet c).V) ∈ varsB c :=
  Finset.mem_map_of_mem _ (Finset.mem_univ w)

@[simp] theorem inl_notMem_varsB (a : A.V) : (Sum.inl a : (composeNet c).V) ∉ varsB c := by
  simp [varsB]

/-- The two halves are disjoint. -/
theorem disjoint_varsA_varsB : Disjoint (varsA c) (varsB c) := by
  rw [Finset.disjoint_left]
  rintro (a | w) hA hB
  · exact inl_notMem_varsB a hB
  · exact inr_notMem_varsA w hA

/-- The two halves of the composite's variables partition it. -/
theorem compl_varsA : (varsA c)c = varsB c := by
  ext v; cases v <|> simp [varsA, varsB]

/-- The composite mechanisms targeting the 'A' side are exactly 'A's. -/
theorem filter_target_mem_varsA :
  (Finset.univ.filter fun m => (composeNet c).target m ∈ varsA c)
  = Finset.univ.map Function.Embedding.inl := by
  ext m; cases m <|> simp [varsA]

/-- Proposition 3, downstream half, for open networks. Summing the joint of the composite
over all values of 'B's private variables leaves the joint of 'A' – the semantics of 'B ◦ A'
marginalised back onto 'A's variables is the semantics of 'A'. Only 'B's kernels have to be
normalised. -/
theorem marg_joint_compose [CommSemiring R] (κA : A.Kernel R) (κB : B.Kernel R)

```

```

(hlocA : ∀ m, Local κA m) (hlocB : ∀ m, Local κB m) (hnormB : ∀ m, Normalised κB m)
(x₀ : (composeNet c).Assignment) :
marg (varsB c) (joint (composeKernel c κA κB)) x₀ = joint κA (restrictA c x₀) := by
have h := marg_joint_downstream (bn := composeNet c) (composeKernel c κA κB)
(composeNet_target_injective c) (composeTopo c) (local_compose κA κB hlocA hlocB) (varsA c)
(by
  rintro (m | m) hm
  · exact absurd (inl_mem_varsA (A.target m)) hm
  · exact fun x ⇒ hnormB m (restrictB c x))
(by
  rintro (m | m) hm p hp
  · rw [composeNet_parents_inl, Finset.mem_map] at hp
  · obtain ⟨a, _, rfl⟩ := hp
  · exact inl_mem_varsA a
  · exact absurd hm (by simp [varsA]))
(by
  rintro (a | w) hv
  · exact absurd (inl_mem_varsA a) hv
  · obtain ⟨m, hm⟩ := B.exists_mechanism w.property
  · exact ⟨.inr m, congrArg Sum.inr (Subtype.ext hm)⟩
x₀
rw [compl_varsA] at h
rw [h, filter_target_mem_varsA, Finset.prod_map]
rfl

/-- Proposition 3 (sequential compositionality, SPEC §13.2, §61). For any set 'S' of composite variables coming from 'A' and any disjoint set 'T' coming from 'B's private part, integrating 'S ∪ T' out of the joint of the composite is the same as integrating 'T' out of 'B's joint first, multiplying by 'A's joint, and then integrating 'S' out. Taking 'S' to be 'A's hidden variables together with the glued interface and 'T' to be 'B's hidden variables, this reads '[B ∘ A] = [B] ∘ [A]': the composite semantics is the sum over the interface of the product of the two open semantics. -/
theorem marg_joint_compose_split [CommSemiring R] (κA : A.Kernel R) (κB : B.Kernel R)
{S T : Finset (composeNet c).V} (hS : S ⊆ varsA c) (hT : T ⊆ varsB c)
(x : (composeNet c).Assignment) :
marg (S ∪ T) (joint (composeKernel c κA κB)) x
= marg S (fun y ⇒ joint κA (restrictA c y)
  * marg T (fun z ⇒ joint κB (restrictB c z)) y) x := by
have hjc : joint (composeKernel c κA κB)
= fun z ⇒ joint κA (restrictA c z) * joint κB (restrictB c z) :=
  funext fun z ⇒ joint_compose κA κB z
rw [marg_union_disjoint ((disjoint_varsA_varsB (c := c)).mono hS hT), hjc]
unfold marg
refine Finset.sum_congr rfl fun y _ => ?_
refine marg_mul_left fun z hz => ?_
have : restrictA c z = restrictA c y := by
  funext a
  exact mem_fibre.1 hz (Sum.inl a) fun hmem ⇒ inl_notMem_varsB a (hT hmem)
rw [this]

variable (c) in
/-- Composing a *closed* network 'A' (no inputs) with an open 'B' gives a closed network: every composite variable still has exactly one mechanism. -/
theorem closed_compose (h : A.inputs = ∅) : (composeNet c).Closed := by
  refine ⟨composeNet_target_injective c, ?_⟩
  rintro (a | w)
  · obtain ⟨m, hm⟩ := A.exists_mechanism (show a ∉ A.inputs by simp [h])
  · exact ⟨.inl m, congrArg Sum.inl hm⟩
  · obtain ⟨m, hm⟩ := B.exists_mechanism w.property
  · exact ⟨.inr m, congrArg Sum.inr (Subtype.ext hm)⟩

/-- Proposition 1a for a composite with a closed left factor: '[B ∘ A]' is a probability distribution on the composite variables. -/
theorem sum_joint_compose_eq_one [CommSemiring R] (h : A.inputs = ∅) (κA : A.Kernel R)
(κB : B.Kernel R) (hlocA : ∀ m, Local κA m) (hlocB : ∀ m, Local κB m)
(hnormA : ∀ m, Normalised κA m) (hnormB : ∀ m, Normalised κB m) :

```

```

 $\sum x, \text{joint } (\text{composeKernel } c \ \kappa A \ \kappa B) \ x = 1 :=$ 
sum_joint_eq_one _ (closed_compose c h) (composeTopo c) (local_compose  $\kappa A \ \kappa B$  hlocA hlocB)
(normalised_compose  $\kappa A \ \kappa B$  hnormA hnormB)

```

end Semantics

end Composable

end OpenFinBayesNet

end BayesianNetworksProofs

BayesianNetworksProofs.Finite.OpenSemantics

```
import BayesianNetworksProofs.Finite.Open
```

Proposition 3 in each component's own open semantics (SPEC §13.2, §61).

Finite marginal sums transfer across the two pushout legs. The proof uses one-variable updates and finite Fubini, so state-space equivalences on the glued interface need not be identities. The resulting composition theorem permits a partial interface match (glue), not only a total match (compose), and needs neither kernel locality nor normalisation: the equality is an algebraic identity of finite sums and products.

osem_compose_passthrough is the formula for arbitrary interfaces. A glued variable that stays on the composite's interface – an input of A, or passed by B straight to its outputs – keeps its visible value and is never summed; only the innerGlued variables are. It needs no hypotheses at all. With disjoint input and output sets on both networks every glued variable is inner (innerGlued_eq_glued), which gives osem_compose_glue.

```
set_option autoImplicit false
```

```
namespace BayesianNetworksProofs.OpenFinBayesNet.Composable
```

```
open FinBayesNet
```

```
variable {A B : OpenFinBayesNet} (c : Composable A B) {R : Type} [CommSemiring R]
```

```
theorem restrictA_update (x : (composeNet c).Assignment) (v : A.V) (z : A.states v) :
  restrictA c (Function.update x (.inl v) z) =
    Function.update (restrictA c x) v z := by
  funext w
  by_cases h : w = v
  · subst w
    simp [restrictA]
  · simp [restrictA, h]
```

```
theorem restrictB_update (x : (composeNet c).Assignment) (v : B.V)
  (hv : v  $\notin$  B.inputs) (z : B.states v) :
  restrictB c (Function.update x (.inr ⟨v, hv⟩) z) =
    Function.update (restrictB c x) v z := by
  funext w
  by_cases h : w = v
  · subst w
    simp [restrictB, hv]
  · by_cases hw : w  $\in$  B.inputs
    · simp [restrictB, hw, Function.update_of_ne h]
    · have hsub : (⟨w, hw⟩ : Priv B)  $\neq$  ⟨v, hv⟩ := fun he  $\Rightarrow$  h (congrArg Subtype.val he)
      simp [restrictB, hw, Function.update_of_ne h, hsub]
```

-- Transfer a finite marginal across the left pushout leg. --

```
theorem marg_restrictA (S : Finset A.V) (F : A.Assignment  $\rightarrow$  R)
  (x : (composeNet c).Assignment) :
  marg (S.map Function.Embedding.inl) (fun y  $\Rightarrow$  F (restrictA c y)) x =
  marg S F (restrictA c x) := by
  induction S using Finset.induction_on generalizing x with
```

```

| empty =>
  simp only [Finset.map_empty]
  rw [marg_empty (bn := composeNet c), marg_empty]
| @insert v S hv ih =>
  rw [Finset.map_insert, marg_insert (by simp using hv), marg_insert hv]
  exact Finset.sum_congr rfl fun z _ => by
    rw [ih]
    simp only [Function.Embedding.inl_apply]
    rw [restrictA_update]

/-- Transfer a finite marginal across the right leg when no summed variable is glued. -/
theorem marg_restrictB (S : Finset B.V) (hS : Disjoint S B.inputs)
  (F : B.Assignment -> R) (x : (composeNet c).Assignment) :
  marg (S.map c.trEmb) (fun y => F (restrictB c y)) x =
  marg S F (restrictB c x) := by
  induction S using Finset.induction_on generalizing x with
  | empty =>
    simp only [Finset.map_empty]
    rw [marg_empty (bn := composeNet c), marg_empty]
  | @insert v S hv ih =>
    have hvi : v ∉ B.inputs :=
      fun h => Finset.disjoint_left.1 hS (Finset.mem_insert_self _ _) h
    have hSi : Disjoint S B.inputs := hS.mono_left (Finset.subset_insert _ _)
    have hnot : c.trEmb v ∉ S.map c.trEmb := by
      intro hm
      obtain ⟨w, hw, he⟩ := Finset.mem_map.1 hm
      exact hv (c.trEmb.injective he ▸ hw)
    rw [trEmb_apply, tr_of_notMem hvi] at hnot
    rw [Finset.map_insert, trEmb_apply, tr_of_notMem hvi,
      marg_insert hnot, marg_insert hv]
    exact Finset.sum_congr rfl fun z _ => by rw [ih hSi, restrictB_update]

theorem glued_subset_outputs : c.glued ⊆ A.outputs := by
  intro v hv
  obtain ⟨w, _, rfl⟩ := Finset.mem_image.1 hv
  exact (c.ι w).property

theorem hidden_disjoint_inputs (O : OpenFinBayesNet) : Disjoint O.hidden O.inputs := by
  rw [Finset.disjoint_left]
  intro v hv hi
  exact (Finset.mem_compl.1 hv) (Finset.mem_union_left _ hi)

/-- An 'A'-hidden variable cannot affect the assignment seen by 'B', even at a glued input. -/
theorem restrictB_eq_of_mem_fibre_hidden (x y : (composeNet c).Assignment)
  (hy : y ∈ fibre (A.hidden.map Function.Embedding.inl) x) :
  restrictB c y = restrictB c x := by
  funext v
  by_cases hv : v ∈ B.inputs
  · simp only [restrictB, dif_pos hv]
    congr 1
    apply mem_fibre.1 hy
    simp [hidden, (c.ι ⟨v, hv⟩).property]
  · rw [restrictB_of_notMem c y hv, restrictB_of_notMem c x hv]
    exact mem_fibre.1 hy _ (by simp)

/-- The glued variables that the composite hides. A glued variable stays on the composite's
interface, and must not be summed, when it is an input of 'A' (a pass-through on 'A's side,
which remains a composite input) or when 'B' passes it straight to its outputs (a
pass-through on 'B's side, which becomes a composite output). Only the rest is internal. -/
def innerGlued : Finset A.V :=
  c.glued.filter fun a => a ∉ A.inputs ∧ Sum.inl a ∉ B.outputs.map c.trEmb

theorem innerGlued_subset_glued : c.innerGlued ⊆ c.glued := Finset.filter_subset _ _

/-- An 'A' variable that 'B's outputs reach through the right leg is glued. -/
theorem mem_glued_of_inl_mem_map (a : A.V) (h : Sum.inl a ∈ B.outputs.map c.trEmb) :

```

```

    a ∈ c.glued := by
  obtain ⟨w, _, he⟩ := Finset.mem_map.1 h
  by_cases hw : w ∈ B.inputs
  · rw [trEmb_apply, tr_of_mem hw, Sum.inl.injEq] at he
    exact Finset.mem_image.2 ⟨⟨w, hw⟩, Finset.mem_univ _, he⟩
  · rw [trEmb_apply, tr_of_notMem hw] at he
    exact absurd he (by simp)

theorem inl_notMem_map_of_disjoint (a : A.V) (S : Finset B.V) (hs : Disjoint S B.inputs) :
  Sum.inl a ∉ S.map c.trEmb := by
  intro hm
  obtain ⟨v, hv, he⟩ := Finset.mem_map.1 hm
  have hvi : v ∉ B.inputs := fun hi => Finset.disjoint_left.1 hs hv hi
  rw [trEmb_apply, tr_of_notMem hvi] at he
  exact Sum.inr_ne_inl he

/-- The hidden variables of a composite, for arbitrary interfaces: pass-through variables
on either side are allowed, and exactly the inner glued variables join the two hidden sets. -/
theorem hidden_compose_general :
  (compose c).hidden =
    (c.innerGlued.map Function.Embedding.inl ∪ A.hidden.map Function.Embedding.inl) ∪
    B.hidden.map c.trEmb := by
  ext v
  cases v with
  | inl a =>
    have hbh : Sum.inl a ∉ B.hidden.map c.trEmb :=
      inl_notMem_map_of_disjoint c a B.hidden (hidden_disjoint_inputs B)
    have hin : (Sum.inl a : A.V ⊕ Priv B) ∈ A.inputs.map Function.Embedding.inl ↔ a ∈ A.inputs :=
      Finset.mem_map' _
    have hout : (Sum.inl a : A.V ⊕ Priv B) ∈ (A.outputs \ c.glued).map Function.Embedding.inl ↔
      a ∈ A.outputs ∧ a ∉ c.glued :=
      (Finset.mem_map' _).trans Finset.mem_sdiff
    have hinner : (Sum.inl a : A.V ⊕ Priv B) ∈ c.innerGlued.map Function.Embedding.inl ↔
      a ∈ c.glued ∧ a ∉ A.inputs ∧ Sum.inl a ∉ B.outputs.map c.trEmb :=
      (Finset.mem_map' _).trans Finset.mem_filter
    have hhid : (Sum.inl a : A.V ⊕ Priv B) ∈ A.hidden.map Function.Embedding.inl ↔
      ¬(a ∈ A.inputs ∨ a ∈ A.outputs) :=
      (Finset.mem_map' _).trans (by rw [hidden, Finset.mem_compl, Finset.mem_union])
    rw [hidden, Finset.mem_compl]
    change (Sum.inl a ∉ A.inputs.map Function.Embedding.inl ∪
      ((A.outputs \ c.glued).map Function.Embedding.inl ∪ B.outputs.map c.trEmb)) ↔
      Sum.inl a ∈ (c.innerGlued.map Function.Embedding.inl ∪
        A.hidden.map Function.Embedding.inl) ∪ B.hidden.map c.trEmb
    rw [Finset.mem_union, Finset.mem_union, Finset.mem_union, Finset.mem_union, hin, hout,
      hinner, hhid]
    simp only [hbh, or_false]
    -- Propositional reasoning, spelled out: 'tauto' would unfold the memberships of the
    -- composite, whose definition is large.
  by_cases hg : a ∈ c.glued
  · have ho := glued_subset_outputs c hg
    constructor
    · intro h
      exact Or.inl ⟨hg, fun hi => h (Or.inl hi), fun hbo => h (Or.inr (Or.inr hbo))⟩
    · rintro ⟨⟨-, hi, hbo⟩ | hn⟩ (h | ⟨-, hng⟩ | h)
      · exact hi h
      · exact hng hg
      · exact hbo h
      · exact hn (Or.inl h)
      · exact hng hg
      · exact hn (Or.inr ho)
  · have hb : Sum.inl a ∉ B.outputs.map c.trEmb :=
      fun h => hg (mem_glued_of_inl_mem_map c a h)
    constructor
    · intro h
      exact Or.inr fun h' => h'.elim (fun hi => h (Or.inl hi))
        (fun ho => h (Or.inr (Or.inl ⟨ho, hg⟩)))

```

```

    • rintro ((hg', -, -) | hn) (h | (ho, -) | h)
    • exact hg hg'
    • exact hg hg'
    • exact hg hg'
    • exact hn (Or.inl h)
    • exact hn (Or.inr ho)
    • exact hb h
| inr b =>
  rw [hidden, Finset.mem_compl]
  change (Sum.inr b ∉ A.inputs.map Function.Embedding.inl ∪
    ((A.outputs \ c.glued).map Function.Embedding.inl ∪ B.outputs.map c.trEmb)) ↔
    Sum.inr b ∈ (c.innerGlued.map Function.Embedding.inl ∪
      A.hidden.map Function.Embedding.inl) ∪ B.hidden.map c.trEmb
  simp only [Finset.mem_union, inr_mem_map_trEmb (c := c)]
  simp [hidden, b.property]

/-- The inner glued variables are exactly the glued variables that the composite hides. -/
theorem innerGlued_eq_filter_hidden :
  c.innerGlued =
    c.glued.filter fun a => (Sum.inl a : A.V ⊕ Priv B) ∈ (compose c).hidden := by
  ext a
  rw [Finset.mem_filter, hidden_compose_general c]
  constructor
  • intro h
    exact ⟨innerGlued_subset_glued c h,
      Finset.mem_union_left _ (Finset.mem_union_left _ (Finset.mem_map_of_mem _ h))⟩
  • rintro ⟨hg, h⟩
    rcases Finset.mem_union.1 h with h | h
    • rcases Finset.mem_union.1 h with h | h
      • exact (Finset.mem_map' _).1 h
      • exfalse
        have hah : a ∈ A.hidden := (Finset.mem_map' _).1 h
        exact Finset.mem_compl.1 hah (Finset.mem_union_right _ (glued_subset_outputs c hg))
    • exact absurd h (inl_notMem_map_of_disjoint c a B.hidden (hidden_disjoint_inputs B))

/-- With disjoint input and output sets on both networks there is no pass-through, and every
glued variable is inner. -/
theorem innerGlued_eq_glued (hA : A.inputs ∩ A.outputs = ∅)
  (hB : B.inputs ∩ B.outputs = ∅) : c.innerGlued = c.glued := by
  refine Finset.filter_true_of_mem fun a ha => ⟨fun hi => ?_, fun hm => ?_⟩
  • have := Finset.mem_inter.2 ⟨hi, glued_subset_outputs c ha⟩
    rw [hA] at this
    exact Finset.notMem_empty _ this
  • obtain ⟨w, hwo, he⟩ := Finset.mem_map.1 hm
    by_cases hw : w ∈ B.inputs
    • have := Finset.mem_inter.2 ⟨hw, hwo⟩
      rw [hB] at this
      exact Finset.notMem_empty _ this
    • rw [trEmb_apply, tr_of_notMem hw] at he
      exact absurd he (by simp)

/-- The hidden variables of a composite, with unglued outputs retained. Total matching is
not required. -/
theorem hidden_compose (hA : A.inputs ∩ A.outputs = ∅)
  (hB : B.inputs ∩ B.outputs = ∅) :
  (compose c).hidden =
    (c.glued.map Function.Embedding.inl ∪ A.hidden.map Function.Embedding.inl) ∪
    B.hidden.map c.trEmb := by
  rw [hidden_compose_general, innerGlued_eq_glued c hA hB]

/-- **Proposition 3 with pass-through.** For arbitrary interfaces -- copied or passed-through
variables on either side, partial gluing, no locality, normalisation or matching assumptions --
the composite's open semantics is the product of the two own-variable open semantics summed
over the inner glued variables only. A glued variable that stays on the composite's interface
(an input of 'A', or passed by 'B' to its outputs) keeps its visible value: it is never summed. -/
theorem osem_compose_passthrough (κA : A.Kernel R) (κB : B.Kernel R)

```

```

(x : (compose c).Assignment) :
osem (compose c) (composeKernel c κA κB) x =
  marg (c.innerGlued.map Function.Embedding.inl)
  (fun y ⇒ osem A κA (restrictA c y) * osem B κB (restrictB c y)) x := by
have hg : Disjoint (c.innerGlued.map Function.Embedding.inl : Finset (composeNet c).V)
(A.hidden.map Function.Embedding.inl) := by
rw [Finset.disjoint_left]
intro v hv hh
obtain ⟨a, ha, rfl⟩ := Finset.mem_map.1 hv
have hah : a ∈ A.hidden := by simp using hh
exact Finset.mem_compl.1 hah (Finset.mem_union_right _
(glued_subset_outputs c (innerGlued_subset_glued c ha)))
have hS : c.innerGlued.map Function.Embedding.inl ∪ A.hidden.map Function.Embedding.inl
⊆ varsA c := by
intro v hv
rcases Finset.mem_union.1 hv with hv | hv <;>
obtain ⟨a, _, rfl⟩ := Finset.mem_map.1 hv <;> exact inl_mem_varsA a
have hT : B.hidden.map c.trEmb ⊆ varsB c := by
intro v hv
obtain ⟨b, hb, rfl⟩ := Finset.mem_map.1 hv
have hbi : b ∉ B.inputs := fun hi ⇒
Finset.disjoint_left.1 (hidden_disjoint_inputs B) hb hi
rw [trEmb_apply, tr_of_notMem hbi]
exact inr_mem_varsB _
unfold osem
rw [hidden_compose_general c, marg_joint_compose_split κA κB hS hT,
marg_union_disjoint hg]
have heq : ∀ z, marg (B.hidden.map c.trEmb)
(fun w ⇒ joint κB (restrictB c w)) z = marg B.hidden (joint κB) (restrictB c z) :=
marg_restrictB c B.hidden (hidden_disjoint_inputs B) (joint κB)
simp_rw [heq]
apply congrArg (fun F ⇒ marg (c.innerGlued.map Function.Embedding.inl) F x)
funext y
calc
_ = marg (A.hidden.map Function.Embedding.inl)
(fun z ⇒ joint κA (restrictA c z)) y *
marg B.hidden (joint κB) (restrictB c y) := by
unfold marg
rw [Finset.sum_mul]
refine Finset.sum_congr rfl fun z hz ⇒ ?_
dsimp only
rw [restrictB_eq_of_mem_fibre_hidden c y z hz]
_ = _ := by rw [marg_restrictA]

-- **Proposition 3, including partial gluing.** Sum the product of the two own-variable
-- pen semantics over the glued interface. No stochastic or locality assumptions are needed.
-- without pass-through every glued variable is inner, so this is 'osem_compose_passthrough'. -/
theorem osem_compose_glue (κA : A.Kernel R) (κB : B.Kernel R)
(hA : A.inputs ∩ A.outputs = ∅) (hB : B.inputs ∩ B.outputs = ∅)
(x : (compose c).Assignment) :
osem (compose c) (composeKernel c κA κB) x =
  marg (c.glued.map Function.Embedding.inl)
  (fun y ⇒ osem A κA (restrictA c y) * osem B κB (restrictB c y)) x := by
rw [← innerGlued_eq_glued c hA hB]
exact osem_compose_passthrough c κA κB x

-- The original Roadmap statement, with its original (now known to be stronger than
-- necessary) premises retained. 'osem_compose_glue' proves it without locality, normalisation,
-- or surjectivity of the interface map. -/
theorem osem_compose (κA : A.Kernel R) (κB : B.Kernel R)
(_hlocA : ∀ m, Local κA m) (_hlocB : ∀ m, Local κB m) (_hnormB : ∀ m, Normalised κB m)
(hA : A.inputs ∩ A.outputs = ∅) (hB : B.inputs ∩ B.outputs = ∅)
(_h1 : Function.Surjective c.1) (x : (compose c).Assignment) :
osem (compose c) (composeKernel c κA κB) x =
  marg (c.glued.map Function.Embedding.inl)
  (fun y ⇒ osem A κA (restrictA c y) * osem B κB (restrictB c y)) x := by

```

```

theorem osem_compose_glue (κA : A.Kernel R) (κB : B.Kernel R)
  (hA : A.inputs ∩ A.outputs = ∅) (hB : B.inputs ∩ B.outputs = ∅)
  (x : (compose c).Assignment) :
  osem (compose c) (composeKernel c κA κB) x =
    marg (c.glued.map Function.Embedding.inl)
      (fun y ⇒ osem A κA (restrictA c y) * osem B κB (restrictB c y)) x := by
rw [← innerGlued_eq_glued c hA hB]
exact osem_compose_passthrough c κA κB x

```

```
-- The original Roadmap statement, with its original (now known to be stronger than
necessary) premises retained. 'osem_compose_glue' proves it without locality, normalisation,
or surjectivity of the interface map. -/
```

```

theorem osem_compose (κA : A.Kernel R) (κB : B.Kernel R)
  (_hlocA : ∀ m, Local κA m) (_hlocB : ∀ m, Local κB m) (_hnormB : ∀ m, Normalised κB m)
  (hA : A.inputs ∩ A.outputs = ∅) (hB : B.inputs ∩ B.outputs = ∅)
  (_hι : Function.Surjective c.ι) (x : (compose c).Assignment) :
  osem (compose c) (composeKernel c κA κB) x =
    marg (c.glued.map Function.Embedding.inl)
    (fun y ⇒ osem A κA (restrictA c y) * osem B κB (restrictB c y)) x := by

```

```

    exact osem_compose_glue c κA κB hA hB x
end BayesianNetworksProofs.OpenFinBayesNet.Composable

```

BayesianNetworksProofs.Finite.BoundaryCases

```

import BayesianNetworksProofs.Finite.OpenSemantics
import Mathlib.Data.Finset.Card

```

Boundary counterexamples and representation limits (SPEC §13, §61 and §62.1).

The own-semantics formula `osem_compose_glue` sums over every glued variable, so it needs disjoint inputs/outputs on both networks. On the binary identity wire, the open semantics is one, while blindly summing its visible glued state gives two: `passthrough_formula_fails` refutes that formula with its side conditions removed. `osem_compose_passthrough` is the formula for arbitrary interfaces: it sums only the inner glued variables, and on the same wire it gives one (`passthrough_formula_holds_on_wire`).

There is a separate structural limitation: with no mechanisms, the typed-interface rule forces every variable to be an input. Since outputs are a subset of the apex, their number cannot exceed the input count. Thus this representation cannot express a mechanism-free one-input/two-output copying *foot* by listing an apex variable twice. General boundary maps, including noninjective output legs, are not modelled. This is not a proof that no possible category or semantic copy construction exists.

```

set_option autoImplicit false

namespace BayesianNetworksProofs.OpenFinBayesNet

open FinBayesNet

/-- In a mechanism-free network every apex variable is necessarily an input. -/
theorem inputs_eq_univ_of_no_mechanisms (O : OpenFinBayesNet) [IsEmpty O.M] :
  O.inputs = Finset.univ := by
  apply Finset.eq_univ_of_forall
  intro v
  exact O.exogenous_input v fun m => isEmptyElim m

/-- Subset feet cannot represent mechanism-free copying by duplicated output ports. -/
theorem outputs_card_le_inputs_card_of_no_mechanisms (O : OpenFinBayesNet) [IsEmpty O.M] :
  O.outputs.card ≤ O.inputs.card := by
  rw [inputs_eq_univ_of_no_mechanisms]
  exact Finset.card_le_card (Finset.subset_univ _)

/-- The binary pass-through wire, with no mechanism. -/
def binaryWire : OpenFinBayesNet where
  V := Unit
  M := Empty
  states := fun _ => Bool
  target := Empty.elim
  parents := Empty.elim
  inputs := Finset.univ
  outputs := Finset.univ
  target_inj := by intro m; exact isEmptyElim m
  input_exogenous := by intro _ _ m; exact isEmptyElim m
  exogenous_input := by intro v _; exact Finset.mem_univ v
  topo := {
    order := [[]]
    nodup := by simp
    complete := by intro v; cases v; simp
    parents_before := by simp
    no_self := by intro m; exact isEmptyElim m
  }

/-- Total matching of the binary wire with itself. -/

```

```

def binaryWireMatch : Composable binaryWire binaryWire where
  ι := id
  ι_inj := Function.injective_id
  states_equiv := fun _ => Equiv.refl _

/-- The unique empty mechanism family. -/
def binaryWireKernel : binaryWire.Kernel M := fun m => nomatch m

/-- The retained external input of the composite is fixed to 'false'. -/
def binaryWireAssignment : (Composable.compose binaryWireMatch).Assignment
| .inl _ => false
| .inr w => False.elim (w.property (Finset.mem_univ w.val))

/-- A checked counterexample to silently removing the no-pass-through side conditions. -/
theorem passthrough_formula_fails :
  osem (Composable.compose binaryWireMatch)
    (Composable.composeKernel binaryWireMatch binaryWireKernel binaryWireKernel)
    binaryWireAssignment = 1 ∧
  marg (binaryWireMatch.glued.map Function.Embedding.inl)
    (fun y => osem binaryWire binaryWireKernel (Composable.restrictA binaryWireMatch y) *
      osem binaryWire binaryWireKernel (Composable.restrictB binaryWireMatch y))
    binaryWireAssignment = 2 := by
  decide

/-- The pass-through formula 'osem_compose_passthrough' on the same wire. The glued variable is
an input of 'A' and passed straight to 'B''s outputs, so it stays on the composite's interface:
it is not inner, nothing is summed, and the formula gives the true value one. -/
theorem passthrough_formula_holds_on_wire :
  binaryWireMatch.innerGlued = 0 ∧
  marg (binaryWireMatch.innerGlued.map Function.Embedding.inl)
    (fun y => osem binaryWire binaryWireKernel (Composable.restrictA binaryWireMatch y) *
      osem binaryWire binaryWireKernel (Composable.restrictB binaryWireMatch y))
    binaryWireAssignment = 1 := by
  decide

end BayesianNetworksProofs.OpenFinBayesNet

```

BayesianNetworksProofs.Markov.Basic

```

import Mathlib.CategoryTheory.MarkovCategory.Basic
import Mathlib.CategoryTheory.CopyDiscardCategory.Deterministic

```

Generic lemmas about Mathlib's abstract CopyDiscardCategory / MarkovCategory that the MarkovCategories.jl / BayesianNetworks.jl tests mirror for the concrete finite-stochastic instance (@instance ThMarkovCategory{FiniteSpace,FiniteKernel}):

- `discard_natural` — $f \gg \varepsilon[Y] = \varepsilon[X]$: composing a kernel with delete is delete. In Julia this is the test “discard naturality holds iff the kernel is normalised”; it is the axiom that ThMarkovCategory adds on top of ThMonoidalCategoryWithDiagonals.
- `deterministic_comp` — the composite of two deterministic (copy-preserving) kernels is deterministic; in Julia, point-mass kernels compose to point-mass kernels.
- `state_discard` — a state $p : I \rightarrow X$ composed with discard is the identity of the unit (a normalised distribution has total mass one).

The concrete finite-stochastic instance is in the sibling FiniteKernels.jl proof project, module Theory/FinStoch.lean; this module only uses Mathlib's abstract classes. Mathlib's Deterministic is IsComonHom, whose composition instance already exists; the theorem below simply names it.

```

namespace BayesianNetworksProofs.Markov

open CategoryTheory MonoidalCategory CopyDiscardCategory ComonObj

universe v u

```

```

variable {C : Type u} [Category.{v} C] [MonoidalCategory.{v} C]

section Markov

variable [MarkovCategory C]

/-- Discard is natural: every morphism of a Markov category is "normalised". -/
theorem discard_natural {X Y : C} (f : X → Y) : f » ε[Y] = ε[X] :=
  MarkovCategory.discard_natural f

/-- A state has total mass one: composing 'p : I → X' with discard is the identity on 'I'. -/
theorem state_discard {X : C} (p : 1_C → X) : p » ε[X] = 1 (1_C) := by
  rw [MarkovCategory.discard_natural, discard_unit]

end Markov

section CopyDiscard

variable [CopyDiscardCategory C]

/-- The composite of two deterministic morphisms is deterministic (Mathlib's 'IsComonHom'
composition instance, named here for the Julia test 'compose(δ_a, δ_b)' is a point mass). -/
theorem deterministic_comp {X Y Z : C} (f : X → Y) (g : Y → Z) [Deterministic f]
  [Deterministic g] : Deterministic (f » g) :=
  inferInstance

/-- Identities are deterministic. -/
theorem deterministic_id (X : C) : Deterministic (1 X) :=
  inferInstance

/-- Deterministic morphisms commute with copy: 'f » Δ = Δ » (f ⊗ f)'. -/
theorem deterministic_copy {X Y : C} (f : X → Y) [Deterministic f] :
  f » Δ[Y] = Δ[X] » (f ⊗ f) :=
  Deterministic.copy_natural f

/-- The comonoid laws every object satisfies (the Julia comonoid-law tests). -/
example (X : C) : Δ[X] » (ε[X] ▷ X) = (λ_ X).inv := counit_comul X
example (X : C) : Δ[X] » (X ◁ ε[X]) = (ρ_ X).inv := comul_counit X
example (X : C) : Δ[X] » (X ◁ Δ[X]) = Δ[X] » (Δ[X] ▷ X) » (α_ X X X).hom := comul_assoc X

end CopyDiscard

end BayesianNetworksProofs.Markov

```

BayesianNetworksProofs.Finite.Probability

```

import BayesianNetworksProofs.Finite.Intervention
import BayesianNetworksProofs.Finite.Tensor
import Mathlib.Data.NNReal.Defs

```

The finite model instantiated at $\mathbb{R}_{\geq 0}$, the value semiring of `FiniteKernel{Float64}` tables in `MarkovCategories.jl` (non-negative weights). Everything in `Finite/` is proved for an arbitrary commutative semiring; these corollaries just fix `R :=  $\mathbb{R}_{\geq 0}$` so that the statements read as statements about probability distributions.

```

namespace BayesianNetworksProofs

namespace FinBayesNet

open scoped NNReal

/-- A family of non-negative real conditional kernels. -/

```

```

abbrev ProbKernel (bn : FinBayesNet) := bn.Kernel  $\mathbb{R}_{\geq 0}$ 

variable {bn : FinBayesNet}

/-- Proposition 1a over ' $\mathbb{R}_{\geq 0}$ ': the joint of a closed, acyclic network with normalised local
kernels is a probability distribution on assignments. -/
theorem sum_joint_eq_one_nnreal ( $\kappa$  : bn.ProbKernel) (hclosed : bn.Closed) (ord : bn.TopoOrder)
  (hloc :  $\forall m$ , Local  $\kappa m$ ) (hnorm :  $\forall m$ , Normalised  $\kappa m$ ) :  $\sum x$ , joint  $\kappa x = 1 :=$ 
  sum_joint_eq_one  $\kappa$  hclosed ord hloc hnorm

/-- Proposition 4 over ' $\mathbb{R}_{\geq 0}$ ': 'do(target  $m_0 = a$ )' truncates the factorisation. -/
theorem joint_intervene_nnreal ( $\kappa$  : bn.ProbKernel) ( $m_0$  : bn.M) ( $a$  : bn.states (bn.target  $m_0$ ))
  ( $x$  : bn.Assignment) :
  joint (intervene  $\kappa m_0 a$ )  $x =$ 
    (if  $x$  (bn.target  $m_0$ ) =  $a$  then 1 else 0) *
     $\prod m \in \text{Finset.univ.erase } m_0$ ,  $\kappa m x (x \text{ (bn.target } m)) :=$ 
  joint_intervene  $\kappa m_0 a x$ 

end FinBayesNet

end BayesianNetworksProofs

```

Partial assignments and explicit conditioning

```

import BayesianNetworksProofs.Finite.Evaluation
import Mathlib.Data.Real.Basic

```

Restriction and patching connect concrete tuples of retained/free variables with the existing full-assignment fibre semantics. Sums over PartialAssignment S enumerate only the coordinates in S . Explicitly fixing evidence and multiplying by its indicator give the same finite sum.

```

set_option autoImplicit false

namespace BayesianNetworksProofs.FinBayesNet

variable {bn : FinBayesNet}

abbrev PartialAssignment (bn : FinBayesNet) ( $S$  : Finset bn.V) :=
  ( $v$  : { $v$  : bn.V //  $v \in S$ })  $\rightarrow$  bn.states  $v.val$ 

def restrictTo ( $S$  : Finset bn.V) ( $x$  : bn.Assignment) : PartialAssignment bn  $S :=$ 
  fun  $v \Rightarrow x v.val$ 

def patch ( $S$  : Finset bn.V) (base : bn.Assignment) ( $x$  : PartialAssignment bn  $S$ ) : bn.Assignment :=
  fun  $v \Rightarrow$  if  $h v : v \in S$  then  $x \langle v, h v \rangle$  else base  $v$ 

theorem restrict_patch ( $S$  : Finset bn.V) (base : bn.Assignment) ( $x$  : PartialAssignment bn  $S$ ) :
  restrictTo  $S$  (patch  $S$  base  $x$ ) =  $x :=$  by
  funext  $v$ 
  simp [restrictTo, patch,  $v$ .property]

theorem patch_mem_fibre ( $S$  : Finset bn.V) (base : bn.Assignment) ( $x$  : PartialAssignment bn  $S$ ) :
  patch  $S$  base  $x \in \text{fibre } S \text{ base} :=$  by
  apply mem_fibre.2
  intro  $v hv$ 
  simp [patch,  $hv$ ]

theorem patch_restrict_of_mem { $S$  : Finset bn.V} {base  $x$  : bn.Assignment}
  ( $h : x \in \text{fibre } S \text{ base}$ ) : patch  $S$  base (restrictTo  $S x$ ) =  $x :=$  by
  funext  $v$ 
  by_cases  $h v : v \in S$ 
  · simp [patch, restrictTo,  $h v$ ]
  · simp [patch,  $h v$ , mem_fibre.1  $h v hv$ ]

```

```

/-- A finite sum over only the free coordinates equals the existing full-assignment marginal. -/
theorem sum_partial_eq_marg {R : Type} [CommSemiring R] (S : Finset bn.V)
  (F : bn.Assignment → R) (base : bn.Assignment) :
  (∑ x : PartialAssignment bn S, F (patch S base x)) = marg S F base := by
  unfold marg
  refine Finset.sum_bij (fun x _ => patch S base x) ?_ ?_ ?_ ?_
  · intro x _
    exact patch_mem_fibre S base x
  · intro x _ y _ h
    have he := congrArg (restrictTo S) h
    simp only [restrict_patch] using he
  · intro x hx
    exact ⟨restrictTo S x, Finset.mem_univ _, patch_restrict_of_mem hx⟩
  · intro x _
    rfl

def clamp (E : Finset bn.V) (observed x : bn.Assignment) : bn.Assignment :=
  fun v => if v ∈ E then observed v else x v

abbrev agrees (E : Finset bn.V) (observed x : bn.Assignment) : Prop :=
  ∀ v ∈ E, x v = observed v

def indicator (E : Finset bn.V) (observed x : bn.Assignment) : ℝ :=
  if agrees E observed x then 1 else 0

theorem clamp_agrees (E : Finset bn.V) (observed x : bn.Assignment) :
  agrees E observed (clamp E observed x) := by
  intro v hv
  simp [clamp, hv]

theorem clamp_patch {E S : Finset bn.V} (h : Disjoint E S)
  (observed base : bn.Assignment) (x : PartialAssignment bn S) :
  clamp E observed (patch S base x) = patch S (clamp E observed base) x := by
  funext v
  by_cases hv : v ∈ S
  · have he : v ∉ E := fun he => Finset.disjoint_left.1 h he hv
    simp [clamp, patch, hv, he]
  · simp [clamp, patch, hv]

theorem marg_clamp {E S : Finset bn.V} (h : Disjoint E S)
  (F : bn.Assignment → R) (observed base : bn.Assignment) :
  marg S (fun x => F (clamp E observed x)) base = marg S F (clamp E observed base) := by
  rw [← sum_partial_eq_marg, ← sum_partial_eq_marg]
  exact Finset.sum_congr rfl fun x _ => congrArg F (clamp_patch h observed base x)

theorem fibre_condition {E S : Finset bn.V} (h : Disjoint E S)
  (observed base : bn.Assignment) :
  (fibre (S ∪ E) base).filter (agrees E observed) = fibre S (clamp E observed base) := by
  ext x
  simp only [Finset.mem_filter, mem_fibre]
  constructor
  · rintro ⟨hf, he⟩ v hv
    by_cases hvE : v ∈ E
    · simp [clamp, hvE] using he v hvE
    · simp [clamp, hvE] using hf v (by simp [hv, hvE])
  · intro hf
    constructor
    · intro v hv
      have hs : v ∉ S := fun hs => hv (Finset.mem_union_left _ hs)
      have he : v ∉ E := fun he => hv (Finset.mem_union_right _ he)
      simp [clamp, he] using hf v hs
    · intro v hv
      have hs : v ∉ S := fun hs => Finset.disjoint_left.1 h hv hs
      simp [clamp, hv] using hf v hs

/-- Explicit conditioning versus an indicator factor. The summed coordinate sets are disjoint;

```

```

no normalization or positive-mass assumption is needed for this unnormalized identity. -/
theorem marg_indicator_eq_clamp {E S : Finset bn.V} (h : Disjoint E S)
  (F : bn.Assignment → ℝ) (observed base : bn.Assignment) :
  marg (S ∪ E) (fun x ⇒ F x * indicator E observed x) base =
  marg S (fun x ⇒ F (clamp E observed x)) base := by
  rw [marg_clamp h]
  unfold marg
  rw [← fibre_condition h observed base, Finset.sum_filter]
  apply Finset.sum_congr rfl
  intro x _
  simp [indicator, mul_ite]

/-- Full evidence-conditioned enumeration needs only the unobserved coordinates. -/
theorem sum_conditioned_eq_indicator (E : Finset bn.V) (F : bn.Assignment → ℝ)
  (observed : bn.Assignment) :
  (∑ x : PartialAssignment bn Eᶜ, F (patch Eᶜ observed x)) =
  ∑ x, F x * indicator E observed x := by
  rw [sum_partial_eq_marg]
  have h := marg_indicator_eq_clamp (S := Eᶜ) (E := E)
  (Finset.disjoint_left.2 fun _ he hc ⇒ Finset.mem_compl.1 hc he) F observed observed
  have he : clamp E observed observed = observed := by funext v; simp [clamp]
  have hs : Eᶜ ∪ E = Finset.univ := by
    ext v
    by_cases hv : v ∈ E <|> simp [hv]
  rw [hs, marg_univ, marg_clamp]
  (Finset.disjoint_left.2 fun _ he hc ⇒ Finset.mem_compl.1 hc he), he] at h
  exact h.symm

end BayesianNetworksProofs.FinBayesNet

```

Finite posterior distributions and exact feasibility

```

import BayesianNetworksProofs.Finite.Assignments
import BayesianNetworksProofs.Finite.VariableElimination
import Mathlib.Algebra.Order.BigOperators.GroupWithZero.Finset
import Mathlib.Algebra.Order.BigOperators.Group.Finset
import Mathlib.Data.Fintype.BigOperators

```

Normalization returns a probability distribution only at positive mass. With nonnegative weights, zero mass is exactly the absence of a positive-weight assignment. Query distributions live on partial assignments, so eliminated coordinates are not counted repeatedly.

The VE theorem uses the existing bucket algorithm and joint semantics. Explicit conditioning is modeled by clamping evidence coordinates and dropping them from factor scopes; its finite elimination result equals indicator-likelihood elimination on the original model.

```

set_option autoImplicit false

namespace BayesianNetworksProofs

noncomputable section

namespace FiniteDistribution

variable {A B : Type} [Fintype A] [Fintype B] [DecidableEq B]

def mass (w : A → ℝ) : ℝ := ∑ a, w a

structure Distribution (A : Type) [Fintype A] where
  pmf : A → ℝ
  nonneg : ∀ a, 0 ≤ pmf a
  normalized : ∑ a, pmf a = 1

def normalize (w : A → ℝ) (hw : ∀ a, 0 ≤ w a) : Option (Distribution A) :=

```

```

if h : 0 < mass w then some {
  pmf a := w a / mass w
  nonneg a := div_nonneg (hw a) h.le
  normalized := by
    simp only [div_eq_mul_inv]
    rw [← Finset.sum_mul]
    exact mul_inv_cancel₀ (ne_of_gt h)
} else none

theorem mass_nonneg (w : A → ℝ) (hw : ∀ a, 0 ≤ w a) : 0 ≤ mass w :=
  Finset.sum_nonneg fun a _ => hw a

theorem mass_zero_iff (w : A → ℝ) (hw : ∀ a, 0 ≤ w a) :
  mass w = 0 ↔ ∀ a, w a = 0 := by
  simp [mass] using Finset.sum_eq_zero_iff_of_nonneg (s := Finset.univ) (fun a _ => hw a)

theorem mass_pos_iff (w : A → ℝ) (hw : ∀ a, 0 ≤ w a) :
  0 < mass w ↔ ∃ a, 0 < w a := by
  simp [mass] using Finset.sum_pos_iff_of_nonneg (s := Finset.univ) (fun a _ => hw a)

theorem normalize_none_iff (w : A → ℝ) (hw : ∀ a, 0 ≤ w a) :
  normalize w hw = none ↔ mass w = 0 := by
  unfold normalize
  by_cases hp : 0 < mass w
  · simp [hp, ne_of_gt hp]
  · have hz := le_antisymm (le_of_not_gt hp) (mass_nonneg w hw)
    simp [hz]

theorem normalize_value (w : A → ℝ) (hw : ∀ a, 0 ≤ w a) (d : Distribution A)
  (h : normalize w hw = some d) : ∀ a, d.pmf a = w a / mass w := by
  unfold normalize at h
  split_ifs at h with hp
  · cases Option.some.inj h
    exact fun _ => rfl

def pushWeight (f : A → B) (w : A → ℝ) (b : B) : ℝ :=
  ∑ a, if f a = b then w a else 0

omit [Fintype B] in
theorem pushWeight_nonneg (f : A → B) (w : A → ℝ) (hw : ∀ a, 0 ≤ w a) :
  ∀ b, 0 ≤ pushWeight f w b := by
  intro b
  apply Finset.sum_nonneg
  intro a _
  split_ifs
  · exact hw a
  · exact le_refl 0

theorem mass_pushWeight (f : A → B) (w : A → ℝ) :
  mass (pushWeight f w) = mass w := by
  unfold mass pushWeight
  rw [Finset.sum_comm]
  apply Finset.sum_congr rfl
  intro a _
  simp

end FiniteDistribution

namespace FinBayesNet

open FiniteDistribution

variable {bn : FinBayesNet}

def likelihoodWeight (p : bn.Assignment → ℝ) (E : Finset bn.V) (observed x : bn.Assignment) : ℝ :=
  p x * indicator E observed x

```

```

theorem likelihoodWeight_nonneg (p : bn.Assignment → ℝ) (hp : ∀ x, 0 ≤ p x)
  (E : Finset bn.V) (observed : bn.Assignment) : ∀ x, 0 ≤ likelihoodWeight p E observed x := by
  intro x
  unfold likelihoodWeight indicator
  split_ifs <=> simp_all

def posterior (Q E : Finset bn.V) (observed : bn.Assignment)
  (p : bn.Assignment → ℝ) (hp : ∀ x, 0 ≤ p x) :
  Option (Distribution (PartialAssignment bn Q)) :=
  normalize (pushWeight (restrictTo Q) (likelihoodWeight p E observed))
  (pushWeight_nonneg _ _ (likelihoodWeight_nonneg p hp E observed))

/-- Impossible evidence is rejected even for an empty query or a query consisting entirely
of observed variables; feasibility concerns the whole joint, not just the requested component. -/
theorem posterior_none_iff (Q E : Finset bn.V) (observed : bn.Assignment)
  (p : bn.Assignment → ℝ) (hp : ∀ x, 0 ≤ p x) :
  posterior Q E observed p hp = none ↔ ¬∃ x, agrees E observed x ∧ 0 < p x := by
  rw [posterior, normalize_none_iff, mass_pushWeight]
  have hmass := mass_nonneg _ (likelihoodWeight_nonneg p hp E observed)
  have hz : mass (likelihoodWeight p E observed) = 0 ↔
    ¬0 < mass (likelihoodWeight p E observed) :=
    (fun he => by rw [he]; simp, fun he => le_antisymm (le_of_not_gt he) hmass)
  rw [hz, mass_pos_iff _ (likelihoodWeight_nonneg p hp E observed)]
  apply not_congr
  constructor
  · rintro ⟨x, hx⟩
    by_cases he : agrees E observed x
    · have hi : indicator E observed x = 1 := if_pos he
      exact ⟨x, he, by simp only [likelihoodWeight, hi, mul_one] using hx⟩
    · have hi : indicator E observed x = 0 := if_neg he
      simp only [likelihoodWeight, hi, mul_zero, lt_self_iff_false] at hx
  · rintro ⟨x, he, hp⟩
    have hi : indicator E observed x = 1 := if_pos he
    exact ⟨x, by simp only [likelihoodWeight, hi, mul_one] using hp⟩

theorem fibre_query (Q : Finset bn.V) (x : bn.Assignment) :
  fibre Qc x = Finset.univ.filter (fun y => restrictTo Q y = restrictTo Q x) := by
  ext y
  simp only [mem_fibre, Finset.mem_filter, Finset.mem_univ, true_and]
  constructor
  · intro h
    funext v
    exact h v.val (by simp)
  · intro h v hv
    exact congrFun h ⟨v, by simp using hv⟩

theorem marginal_eq_query_weight (Q : Finset bn.V) (p : bn.Assignment → ℝ) (x : bn.Assignment) :
  marg Qc p x = pushWeight (restrictTo Q) p (restrictTo Q x) := by
  rw [marg, fibre_query, Finset.sum_filter]
  rfl

def evidenceFactor (E : Finset bn.V) (observed : bn.Assignment) : Factor bn ℝ where
  scope := E
  value := indicator E observed
  dependsOn x y h := by
    have he : agrees E observed x ↔ agrees E observed y := by
      constructor
      · intro hx v hv
        rw [← h v hv]
        exact hx v hv
      · intro hy v hv
        rw [h v hv]
        exact hy v hv
    simp only [indicator, he]

```

```

theorem joint_nonneg (κ : bn.Kernel ℝ) (hκ : ∀ m x a, 0 ≤ κ m x a) (x : bn.Assignment) :
  0 ≤ joint κ x := Finset.prod_nonneg fun m _ => hκ m x _

/-- The already-defined VE driver, with a real evidence indicator factor, computes the
unnormalized numerator of the genuine query posterior. -/
theorem ve_posterior_numerator (κ : bn.Kernel ℝ) (hloc : ∀ m, Local κ m)
  (Q E : Finset bn.V) (observed : bn.Assignment) (vs : List bn.V)
  (hvs : vs.Nodup) (hset : vs.toFinset = Qc) (x : bn.Assignment) :
  Factor.product (Factor.eliminateAll (evidenceFactor E observed :: Factor.ofKernel κ hloc) vs) x =
    pushWeight (restrictTo Q) (likelihoodWeight (joint κ) E observed) (restrictTo Q x) := by
  rw [Factor.eliminateAll_correct _ vs hvs, hset]
  have he : Factor.product (evidenceFactor E observed :: Factor.ofKernel κ hloc) =
    likelihoodWeight (joint κ) E observed := by
    funext y
    rw [Factor.product_cons, Factor.product_ofKernel]
    exact mul_comm _ _
  rw [he, marginal_eq_query_weight]

/-- Dividing the actual VE numerator by positive evidence mass gives the normalized posterior
entry. A returned Distribution supplies nonnegativity and total mass one. -/
theorem ve_normalized_posterior (κ : bn.Kernel ℝ) (hloc : ∀ m, Local κ m)
  (hκ : ∀ m x a, 0 ≤ κ m x a) (Q E : Finset bn.V) (observed : bn.Assignment)
  (vs : List bn.V) (hvs : vs.Nodup) (hset : vs.toFinset = Qc)
  (d : Distribution (PartialAssignment bn Q))
  (hd : posterior Q E observed (joint κ) (joint_nonneg κ hκ) = some d) (x : bn.Assignment) :
  Factor.product (Factor.eliminateAll (evidenceFactor E observed :: Factor.ofKernel κ hloc) vs) x /
    mass (likelihoodWeight (joint κ) E observed) = d.pmf (restrictTo Q x) := by
  rw [ve_posterior_numerator κ hloc Q E observed vs hvs hset]
  have h := normalize_value _ _ d hd (restrictTo Q x)
  rw [mass_pushWeight] at h
  exact h.symm

namespace Factor

def condition (E : Finset bn.V) (observed : bn.Assignment) (f : Factor bn ℝ) : Factor bn ℝ where
  scope := f.scope \ E
  value x := f.value (clamp E observed x)
  dependsOn x y h := by
    apply f.dependsOn
  intro v hv
  by_cases he : v ∈ E
  · simp [clamp, he]
  · simp [clamp, he] using h v (Finset.mem_sdiff.2 ⟨hv, he⟩)

theorem product_condition (fs : List (Factor bn ℝ)) (E : Finset bn.V) (observed : bn.Assignment) :
  product (fs.map (condition E observed)) = fun x => product fs (clamp E observed x) := by
  funext x
  simp [product, condition, List.map_map, Function.comp_def]

theorem explicit_conditioning_elimination (fs : List (Factor bn ℝ)) (E : Finset bn.V)
  (observed : bn.Assignment) (vs : List bn.V) (hvs : vs.Nodup) (hd : Disjoint E vs.toFinset)
  (x : bn.Assignment) :
  product (eliminateAll (fs.map (condition E observed)) vs) x =
    marg (vs.toFinset ∪ E) (fun y => product fs y * indicator E observed y) x := by
  rw [eliminateAll_correct _ vs hvs, product_condition]
  exact (marg_indicator_eq_clamp hd (product fs) observed x).symm

end Factor
end FinBayesNet
end
end BayesianNetworksProofs

```

Checked finite records, positions and repeated input slots

```
import BayesianNetworksProofs.Finite.OrderedParents
import BayesianNetworksProofs.Finite.VariableElimination
import Mathlib.Data.Fintype.EquivFin
import Mathlib.Data.List.OfFn
import Mathlib.Data.Real.Basic
```

Raw rows carry concrete finite IDs, names, references and numeric positions. `decodeId` checks external one-based IDs before constructing those finite IDs. Position validity is only a boundedness/injectivity check on each owner fibre; finiteness derives surjectivity onto every contiguous position. Input variables need not be injective: repeated slots read the same assignment coordinate, i.e. diagonal rather than independent evaluation.

Valid consists of decidable structural facts, not a compiler-correctness certificate. The compiler derives the abstract parent sets, state spaces, closedness and topological order. Raw names/references remain attached to the records; the finite-network reduct deliberately erases metadata. This is not a theorem about the Julia compiler or its JSON parser.

```
set_option autoImplicit false

namespace BayesianNetworksProofs.Raw

open FinBayesNet

def decodeId (n external : Nat) : Option (Fin n) :=
  if h : 0 < external ∧ external ≤ n then some (external - 1, by omega) else none

theorem decodeId_none_iff (n external : Nat) :
  decodeId n external = none ↔ external = 0 ∨ n < external := by
  unfold decodeId
  split_ifs <=> simp_all <=> omega

theorem decodeId_value {n external : Nat} {i : Fin n} (h : decodeId n external = some i) :
  i.val + 1 = external := by
  unfold decodeId at h
  split_ifs at h with hb
  · cases Option.some.inj h
    change external - 1 + 1 = external
    omega

inductive Ref
| named (key : String)
| policy (key : String)
| pointMass (state : String)
| none
deriving DecidableEq

structure VariableRow where
  name : String
  spaceRef : Ref

structure StateRow (nv : Nat) where
  var : Fin nv
  position : Nat
  name : String

structure MechanismRow (nv : Nat) where
  target : Fin nv
  name : String
  kernelRef : Ref

structure InputRow (nv nm : Nat) where
  mechanism : Fin nm
  var : Fin nv
  position : Nat
```

```

structure Network where
  nv : Nat
  ns : Nat
  nm : Nat
  ni : Nat
  vars : Fin nv → VariableRow
  states : Fin ns → StateRow nv
  mechanisms : Fin nm → MechanismRow nv
  inputs : Fin ni → InputRow nv nm
  rank : Fin nv → Fin nv

def Positioned {nr no : Nat} (owner : Fin nr → Fin no) (pos : Fin nr → Nat) : Prop :=
  (∀ r, pos r < Fintype.card {s : Fin nr // owner s = owner r}) ∧
  ∀ r s, owner r = owner s → pos r = pos s → r = s

instance {nr no : Nat} (owner : Fin nr → Fin no) (pos : Fin nr → Nat) :
  Decidable (Positioned owner pos) := by unfold Positioned; infer_instance

def positionMap {nr no : Nat} (owner : Fin nr → Fin no) (pos : Fin nr → Nat)
  (h : Positioned owner pos) (o : Fin no) (r : {s : Fin nr // owner s = o}) :
  Fin (Fintype.card {s : Fin nr // owner s = o}) :=
  ⟨pos r.val, by simp only [r.property] using h.1 r.val⟩

/-- Total positional coverage follows from checked bounds, uniqueness, and equal finite
cardinalities; an ordering equivalence is not an input assumption. -/
noncomputable def positionEquiv {nr no : Nat} (owner : Fin nr → Fin no) (pos : Fin nr → Nat)
  (h : Positioned owner pos) (o : Fin no) :
  {s : Fin nr // owner s = o} ≈ Fin (Fintype.card {s : Fin nr // owner s = o}) :=
  Equiv.ofBijective (positionMap owner pos h o) ((Fintype.bijective_iff_injective_and_card _).2 ⟨
    (by
      intro r s he
      apply Subtype.ext
      exact h.2 r.val s.val (r.property.trans s.property.symm) (congrArg Fin.val he)),
    by simp))

namespace Network

def stateOwner (r : Network) (s : Fin r.ns) := (r.states s).var
def inputOwner (r : Network) (i : Fin r.ni) := (r.inputs i).mechanism
def stateCount (r : Network) (v : Fin r.nv) := Fintype.card {s : Fin r.ns // r.stateOwner s = v}
def inputCount (r : Network) (m : Fin r.nm) := Fintype.card {i : Fin r.ni // r.inputOwner i = m}

structure Valid (r : Network) : Prop where
  state_positions : Positioned r.stateOwner (fun s ⇒ (r.states s).position)
  input_positions : Positioned r.inputOwner (fun i ⇒ (r.inputs i).position)
  nonempty_states : ∀ v, 0 < r.stateCount v
  state_names : ∀ s t, (r.states s).var = (r.states t).var →
    (r.states s).name = (r.states t).name → s = t
  generators : Function.Bijective (fun m ⇒ (r.mechanisms m).target)
  rank_injective : Function.Injective r.rank
  causal_rank : ∀ i, r.rank (r.inputs i).var < r.rank (r.mechanisms (r.inputs i).mechanism).target

theorem valid_iff (r : Network) : r.Valid ↔
  Positioned r.stateOwner (fun s ⇒ (r.states s).position) ∧
  Positioned r.inputOwner (fun i ⇒ (r.inputs i).position) ∧
  (∀ v, 0 < r.stateCount v) ∧
  (∀ s t, (r.states s).var = (r.states t).var → (r.states s).name = (r.states t).name → s = t) ∧
  Function.Bijective (fun m ⇒ (r.mechanisms m).target) ∧ Function.Injective r.rank ∧
  (∀ i, r.rank (r.inputs i).var < r.rank (r.mechanisms (r.inputs i).mechanism).target) :=
  ⟨fun h ⇒ ⟨h.state_positions, h.input_positions, h.nonempty_states, h.state_names,
    h.generators, h.rank_injective, h.causal_rank⟩,
    fun h ⇒ ⟨h.1, h.2.1, h.2.2.1, h.2.2.2.1, h.2.2.2.2.1, h.2.2.2.2.2⟩⟩

instance (r : Network) : Decidable r.Valid := by
  letI : Decidable (Function.Bijective (fun m ⇒ (r.mechanisms m).target)) := by
    unfold Function.Bijective Function.Injective Function.Surjective

```

```

infer_instance
letI : Decidable (Function.Injective r.rank) := by
  unfold Function.Injective
  infer_instance
exact decidable_of_iff _ (valid_iff r).symm

def check (r : Network) : Bool := decide r.Valid

theorem check_iff (r : Network) : r.check = true ↔ r.Valid := by simp [check]

noncomputable section

@[reducible] def compile (r : Network) (h : r.Valid) : FinBayesNet where
  V := Fin r.nv
  M := Fin r.nm
  states v := Fin (r.stateCount v)
  nonemptyS v := ⟨0, h.nonempty_states v⟩
  target m := (r.mechanisms m).target
  parents m := (Finset.univ.filter fun i => (r.inputs i).mechanism = m).image
    (fun i => (r.inputs i).var)

def stateOrder (r : Network) (h : r.Valid) (v : Fin r.nv) :
  {s : Fin r.ns // (r.states s).var = v} ≈ Fin (r.stateCount v) :=
  positionEquiv r.stateOwner (fun s => (r.states s).position) h.state_positions v

def inputOrder (r : Network) (h : r.Valid) (m : Fin r.nm) :
  {i : Fin r.ni // (r.inputs i).mechanism = m} ≈ Fin (r.inputCount m) :=
  positionEquiv r.inputOwner (fun i => (r.inputs i).position) h.input_positions m

def stateLabel (r : Network) (h : r.Valid) (v : Fin r.nv) (a : Fin (r.stateCount v)) : String :=
  (r.states ((r.stateOrder h v).symm a).val).name

theorem stateLabel_injective (r : Network) (h : r.Valid) (v : Fin r.nv) :
  Function.Injective (r.stateLabel h v) := by
  intro a b he
  have hs : (r.stateOrder h v).symm a = (r.stateOrder h v).symm b := by
    apply Subtype.ext
  exact h.state_names _ _ (((r.stateOrder h v).symm a).property.trans
    ((r.stateOrder h v).symm b).property.symm) he
  exact (r.stateOrder h v).symm.injective hs

def slotVariable (r : Network) (h : r.Valid) (m : Fin r.nm) (j : Fin (r.inputCount m)) : Fin r.nv :=
  (r.inputs ((r.inputOrder h m).symm j).val).var

theorem slot_position (r : Network) (h : r.Valid) (m : Fin r.nm) (j : Fin (r.inputCount m)) :
  (r.inputs ((r.inputOrder h m).symm j).val).position = j.val :=
  congrArg Fin.val ((r.inputOrder h m).apply_symm_apply j)

theorem state_position (r : Network) (h : r.Valid) (v : Fin r.nv) (j : Fin (r.stateCount v)) :
  (r.states ((r.stateOrder h v).symm j).val).position = j.val :=
  congrArg Fin.val ((r.stateOrder h v).apply_symm_apply j)

theorem slot_image (r : Network) (h : r.Valid) (m : Fin r.nm) :
  Finset.univ.image (r.slotVariable h m) = (r.compile h).parents m := by
  ext v
  constructor
  · rintro hv
    obtain ⟨j, _, rfl⟩ := Finset.mem_image.1 hv
    exact Finset.mem_image.2 ⟨((r.inputOrder h m).symm j).val,
      Finset.mem_filter.2 ⟨Finset.mem_univ _, ((r.inputOrder h m).symm j).property⟩, rfl⟩
  · intro hv
    obtain ⟨i, hi, rfl⟩ := Finset.mem_image.1 hv
    let s : {i : Fin r.ni // (r.inputs i).mechanism = m} := ⟨i, (Finset.mem_filter.1 hi).2⟩
    refine Finset.mem_image.2 ⟨r.inputOrder h m s, Finset.mem_univ _, ?_⟩
    simp [slotVariable, s]

```

```

def rankEquiv (r : Network) (h : r.Valid) : Fin r.nv ≈ Fin r.nv :=
  Equiv.ofBijective r.rank ⟨h.rank_injective, Finite.surjective_of_injective h.rank_injective⟩

/-- A checked ranking produces the topological list, rather than assuming a compiler theorem. -/
def topological (r : Network) (h : r.Valid) : (r.compile h).TopoOrder where
  order := List.ofFn (r.rankEquiv h).symm
  nodup := List.nodup_ofFn.2 (r.rankEquiv h).symm.injective
  complete v := List.mem_ofFn.2 ⟨r.rankEquiv h v, (r.rankEquiv h).symm_apply_apply v⟩
  parents_before := by
    apply List.pairwise_ofFn.2
    intro i j hij m hm hv
    obtain ⟨slot, hs, he⟩ := Finset.mem_image.1 hv
    have heOwner := (Finset.mem_filter.1 hs).2
    have hr := h.causal_rank slot
    have hi : r.rank ((r.rankEquiv h).symm i) = i := (r.rankEquiv h).apply_symm_apply i
    have hj : r.rank ((r.rankEquiv h).symm j) = j := (r.rankEquiv h).apply_symm_apply j
    change (r.mechanisms m).target = (r.rankEquiv h).symm i at hm
    rw [heOwner, hm, he, hi, hj] at hr
    exact (not_lt_of_ge hij.le) hr
  no_self m := by
    intro hm
    obtain ⟨i, hi, he⟩ := Finset.mem_image.1 hm
    have hr := h.causal_rank i
    rw [(Finset.mem_filter.1 hi).2, he] at hr
    exact (lt_irrefl _ hr)

theorem compile_closed (r : Network) (h : r.Valid) : (r.compile h).Closed := h.generators

abbrev SlotValues (r : Network) (h : r.Valid) (m : Fin r.nm) :=
  (j : Fin (r.inputCount m)) → (r.compile h).states (r.slotVariable h m j)

def gather (r : Network) (h : r.Valid) (m : Fin r.nm) (x : (r.compile h).Assignment) :
  r.SlotValues h m := fun j ⇒ x (r.slotVariable h m j)

abbrev CPT (r : Network) (h : r.Valid) (R : Type) :=
  (m : Fin r.nm) → r.SlotValues h m → (r.compile h).states ((r.compile h).target m) → R

def kernel (r : Network) (h : r.Valid) {R : Type} (t : r.CPT h R) : (r.compile h).Kernel R :=
  fun m x ⇒ t m (r.gather h m x)

theorem kernel_local (r : Network) (h : r.Valid) {R : Type} (t : r.CPT h R) :
  ∀ m, Local (r.kernel h t) m := by
  intro m x y hx
  apply congrArg (t m)
  funext j
  apply hx
  rw [← r.slot_image h m]
  exact Finset.mem_image_of_mem _ (Finset.mem_univ j)

/-- Repeated input-variable slots are evaluated on the diagonal, not sampled independently. -/
theorem repeated_slots_diagonal (r : Network) (h : r.Valid) (m : Fin r.nm)
  (i j : Fin (r.inputCount m)) (he : r.slotVariable h m i = r.slotVariable h m j)
  (x : (r.compile h).Assignment) :
  (r.gather h m x i).val = (r.gather h m x j).val :=
  congrArg (fun v ⇒ (x v).val) he

theorem kernel_normalised (r : Network) (h : r.Valid) {R : Type} [CommSemiring R]
  (t : r.CPT h R) (ht : ∀ m z, ∑ a, t m z a = 1) :
  ∀ m, Normalised (r.kernel h t) m := fun m x ⇒ ht m (r.gather h m x)

theorem compiled_joint_normalised (r : Network) (h : r.Valid) {R : Type} [CommSemiring R]
  (t : r.CPT h R) (ht : ∀ m z, ∑ a, t m z a = 1) :
  ∑ x, joint (r.kernel h t) x = 1 :=
  sum_joint_eq_one _ (r.compile_closed h) (r.topological h)
  (r.kernel_local h t) (r.kernel_normalised h t ht)

```

```

/-- The existing valuation/factor compiler preserves the actual repeated-slot CPT product. -/
theorem factor_compilation (r : Network) (h : r.Valid) (t : r.CPT h ℝ) :
  Factor.product (Factor.ofKernel (r.kernel h t) (r.kernel_local h t)) =
    fun x => ∏ m, t m (r.gather h m x) (x ((r.compile h).target m)) := by
  rw [Factor.product_ofKernel]
  rfl

end
end Network
end BayesianNetworksProofs.Raw

```

Finite reference tables and checked resolution

```

import BayesianNetworksProofs.Finite.RawRecords
import Mathlib.Data.List.Forall2
import Mathlib.Data.Rat.Cast.Order
import Mathlib.Data.Fintype.BigOperators
import Mathlib.Algebra.BigOperators.Fin

```

Bindings are finite lists of named/policy references and rational columns. Coordinates are zero-based after checked external-ID conversion. Complete coordinate coverage, unique columns, output lengths and nonnegative entries are checked independently of compilation; exact normalization is a separate check. NoRef and missing bindings are rejected, not silently replaced by a distribution. Point masses resolve through the checked state labels.

This file specifies a finite data checker/model bridge, not the correctness of Julia or of a JSON parser. Rational values may represent stored Float64 values exactly without pretending their row sums are exactly one.

```

set_option autoImplicit false

namespace BayesianNetworksProofs.Raw

open FinBayesNet

def coordinates : List Nat → List (List Nat)
| [] => [[]]
| n :: ns => (List.range n).flatMap fun a => (coordinates ns).map (a :: ·)

theorem coordinates_ofFn (n : Nat) (sizes : Fin n → Nat) (x : (i : Fin n) → Fin (sizes i)) :
  List.ofFn (fun i => (x i).val) ∈ coordinates (List.ofFn sizes) := by
  induction n with
  | zero => simp [coordinates]
  | succ n ih =>
    simp only [List.ofFn_succ, coordinates, List.mem_flatMap, List.mem_range, List.mem_map]
    exact ⟨(x 0).val, (x 0).isLt, List.ofFn (fun i => (x i.succ).val),
      ih (fun i => sizes i.succ) (fun i => x i.succ), rfl⟩

structure Column where
  parents : List Nat
  weights : List ℚ
  deriving DecidableEq

structure Table where
  inputSizes : List Nat
  outputSize : Nat
  columns : List Column
  inputLabels : List (List String) := []
  outputLabels : List String := []
  deriving DecidableEq

def Table.column (t : Table) (ctx : List Nat) : Option Column :=
  t.columns.find? fun c => c.parents == ctx

def Table.read (t : Table) (ctx : List Nat) (a : Nat) : ℚ :=
  ((t.column ctx).getD [], []).weights.getD a 0

```

```

def Table.Ready (t : Table) : Prop :=
  (t.columns.map Column.parents).Nodup ∧
  (∀ ctx ∈ coordinates t.inputSizes, ∃ c ∈ t.columns, c.parents = ctx) ∧
  ∀ c ∈ t.columns, c.parents ∈ coordinates t.inputSizes ∧
  c.weights.length = t.outputSize ∧ ∀ q ∈ c.weights, 0 ≤ q

def Table.Normalized (t : Table) : Prop := ∀ c ∈ t.columns, c.weights.sum = 1

instance (t : Table) : Decidable t.Ready := by unfold Table.Ready; infer_instance
instance (t : Table) : Decidable t.Normalized := by unfold Table.Normalized; infer_instance

theorem find_mem {A : Type} (p : A → Bool) (xs : List A) (a : A)
  (h : xs.find? p = some a) : a ∈ xs ∧ p a = true := by
  induction xs with
  | nil ⇒ simp at h
  | cons b xs ih ⇒
    by_cases hb : p b
    · simp [List.find?, hb] at h
      subst a
      exact ⟨List.mem_cons_self .., hb⟩
    · simp [List.find?, hb] at h
      have hi := ih h
      exact ⟨List.mem_cons_of_mem b hi.1, hi.2⟩

theorem find_exists {A : Type} (p : A → Bool) (xs : List A)
  (h : ∃ a ∈ xs, p a = true) : ∃ a, xs.find? p = some a := by
  induction xs with
  | nil ⇒ simp at h
  | cons b xs ih ⇒
    by_cases hb : p b
    · exact ⟨b, by simp [List.find?, hb]⟩
    · obtain ⟨a, ha, hp⟩ := h
      have ht : a ∈ xs := (List.mem_cons.1 ha).resolve_left (fun he ⇒ hb (he ▸ hp))
      obtain ⟨c, hc⟩ := ih ⟨a, ht, hp⟩
      exact ⟨c, by simp [List.find?, hb, hc]⟩

theorem column_exists (t : Table) (h : t.Ready) (ctx : List Nat)
  (hc : ctx ∈ coordinates t.inputSizes) : ∃ c, t.column ctx = some c := by
  apply find_exists
  obtain ⟨c, hmem, he⟩ := h.2.1 ctx hc
  exact ⟨c, hmem, by simp using he⟩

theorem getD_nonnegative (xs : List ℚ) (h : ∀ q ∈ xs, 0 ≤ q) (n : Nat) :
  0 ≤ xs.getD n 0 := by
  induction xs generalizing n with
  | nil ⇒ simp
  | cons q xs ih ⇒
    cases n with
    | zero ⇒ exact h q (List.mem_cons_self ..)
    | succ n ⇒ simp using ih (fun r hr ⇒ h r (List.mem_cons_of_mem q hr)) n

theorem sum_getD (xs : List ℚ) : (∑ i : Fin xs.length, xs.getD i.val 0) = xs.sum := by
  induction xs with
  | nil ⇒ simp
  | cons q xs ih ⇒
    rw [List.length_cons, Fin.sum_univ_succ]
    simp

theorem Table.read_nonnegative (t : Table) (h : t.Ready) (ctx : List Nat)
  (hc : ctx ∈ coordinates t.inputSizes) (a : Nat) : 0 ≤ t.read ctx a := by
  obtain ⟨c, he⟩ := column_exists t h ctx hc
  have hmem := (find_mem _ t.columns c he).1
  simp [Table.read, he] using getD_nonnegative c.weights (h.2.2 c hmem).2.2 a

theorem Table.read_normalized (t : Table) (h : t.Ready) (hn : t.Normalized)

```

```

    (ctx : List Nat) (hc : ctx ∈ coordinates t.inputSizes) :
    (∑ a : Fin t.outputSize, t.read ctx a.val) = 1 := by
  obtain ⟨c, he⟩ := column_exists t h ctx hc
  have hmem := (find_mem _ t.columns c he).1
  have hlen := (h.2.2 c hmem).2.1
  simp only [Table.read, he, Option.getD_some]
  rw [← hlen, sum_getD]
  exact hn c hmem

structure Binding where
  ref : Ref
  table : Table
  deriving DecidableEq

abbrev Bank := List Binding

def resolve (bank : Bank) (ref : Ref) : Option Table :=
  (bank.find? fun b => b.ref == ref).map Binding.table

theorem resolve_has_key (bank : Bank) (ref : Ref) (t : Table) (h : resolve bank ref = some t) :
  ∃ b ∈ bank, b.ref = ref ∧ b.table = t := by
  unfold resolve at h
  obtain ⟨b, hb, ht⟩ := Option.map_eq_some_iff.1 h
  have hm := find_mem _ bank b hb
  exact ⟨b, hm.1, by simp using hm.2, ht⟩

namespace Network

noncomputable section

def slotSizes (r : Network) (h : r.Valid) (m : Fin r.nm) : List Nat :=
  List.ofFn fun j => r.stateCount (r.slotVariable h m j)

def slotIndices (r : Network) (h : r.Valid) (m : Fin r.nm) (x : r.SlotValues h m) : List Nat :=
  List.ofFn fun j => (x j).val

theorem slotIndices_valid (r : Network) (h : r.Valid) (m : Fin r.nm) (x : r.SlotValues h m) :
  r.slotIndices h m x ∈ coordinates (r.slotSizes h m) :=
  coordinates_ofFn _ _ x

def Ready (r : Network) (h : r.Valid) (bank : Bank) : Prop :=
  (bank.map Binding.ref).Nodup ∧ ∀ m,
  match (r.mechanisms m).kernelRef with
  | .none => False
  | .pointMass label => ∃ a : Fin (r.stateCount (r.mechanisms m).target),
    r.stateLabel h (r.mechanisms m).target a = label
  | ref => match resolve bank ref with
  | none => False
  | some t => t.inputSizes = r.slotSizes h m ∧
    t.outputSize = r.stateCount (r.mechanisms m).target ∧ t.Ready

def Normalized (r : Network) (bank : Bank) : Prop :=
  ∀ m t, resolve bank (r.mechanisms m).kernelRef = some t → t.Normalized

def resolvedCPT (r : Network) (h : r.Valid) (bank : Bank) : r.CPT h ℚ :=
  fun m x a =>
    match (r.mechanisms m).kernelRef with
    | .none => 0
    | .pointMass label => if r.stateLabel h (r.mechanisms m).target a = label then 1 else 0
    | ref => match resolve bank ref with
    | none => 0
    | some t => t.read (r.slotIndices h m x) a.val

theorem resolvedCPT_nonnegative (r : Network) (h : r.Valid) (bank : Bank) (hb : r.Ready h bank) :
  ∀ m x a, 0 ≤ r.resolvedCPT h bank m x a := by
  intro m x a

```

```

have hm := hb.2 m
unfold resolvedCPT
cases hr : (r.mechanisms m).kernelRef with
| none => simp [hr] at hm
| pointMass label =>
  dsimp only
  split_ifs <=> norm_num
| named key | policy key =>
  cases he : resolve bank (r.mechanisms m).kernelRef with
  | none =>
    simp only [hr] at he
    simp [hr, he] at hm
  | some t =>
    simp only [hr] at he
    have ht : t.inputSizes = r.slotSizes h m ∧
      t.outputSize = r.stateCount (r.mechanisms m).target ∧ t.Ready := by simpa [hr, he] using hm
    simpa only [he] using t.read_nonnegative ht.2.2 - (ht.1.symm ▸ r.slotIndices_valid h m x) a.val

theorem resolvedCPT_normalized (r : Network) (h : r.Valid) (bank : Bank) (hb : r.Ready h bank)
  (hn : r.Normalized bank) : ∀ m x, ∑ a, r.resolvedCPT h bank m x a = 1 := by
  intro m x
  change (∑ a : Fin (r.stateCount (r.mechanisms m).target), r.resolvedCPT h bank m x a) = 1
  have hm := hb.2 m
  cases hr : (r.mechanisms m).kernelRef with
  | none => simp [hr] at hm
  | pointMass label =>
    obtain ⟨a₀, ha⟩ : ∃ a, r.stateLabel h (r.mechanisms m).target a = label := by simpa [hr] using hm
    have he : ∀ a, r.stateLabel h (r.mechanisms m).target a = label ↔ a = a₀ := by
      intro a
      rw [← ha, (r.stateLabel_injective h _).eq_iff]
    simp [resolvedCPT, hr, he]
  | named key | policy key =>
    cases ht : resolve bank (r.mechanisms m).kernelRef with
    | none =>
      simp only [hr] at ht
      simp [hr, ht] at hm
    | some t =>
      simp only [hr] at ht
      have hv : t.inputSizes = r.slotSizes h m ∧
        t.outputSize = r.stateCount (r.mechanisms m).target ∧ t.Ready := by simpa [hr, ht] using hm
      have hn' : t.Normalized := hn m t (by simpa [hr] using ht)
      have hs := t.read_normalized hv.2.2 hn' - (hv.1.symm ▸ r.slotIndices_valid h m x)
      rw [hv.2.1] at hs
      simpa only [resolvedCPT, hr, ht] using hs

/-- Checked reference resolution plus checked column normalization yields a normalized joint;
the compiler is not assumed correct, and repeated parent slots are evaluated diagonally. -/
theorem resolved_joint_normalized (r : Network) (h : r.Valid) (bank : Bank)
  (hb : r.Ready h bank) (hn : r.Normalized bank) :
  ∑ x, joint (r.kernel h (r.resolvedCPT h bank)) x = 1 :=
  r.compiled_joint_normalised h - (r.resolvedCPT_normalized h bank hb hn)

end

/-- Executable finite search by owner and position, before any ordering equivalence is built. -/
def rawInputAt (r : Network) (m : Fin r.nm) (position : Nat) : Option (Fin r.ni) :=
  (List.finRange r.ni).find? fun i =>
    decide ((r.inputs i).mechanism = m ∧ (r.inputs i).position = position)

def rawSlotSize (r : Network) (m : Fin r.nm) (position : Nat) : Nat :=
  match r.rawInputAt m position with
  | none => 0
  | some i => r.stateCount (r.inputs i).var

def rawSlotSizes (r : Network) (m : Fin r.nm) : List Nat :=
  List.ofFn fun i : Fin (r.inputCount m) => r.rawSlotSize m i.val

```

```

theorem rawSlotSize_eq (r : Network) (h : r.Valid) (m : Fin r.nm) (i : Fin (r.inputCount m)) :
  r.rawSlotSize m i.val = r.stateCount (r.slotVariable h m i) := by
  let wanted := (r.inputOrder h m).symm i
  have hex : ∃ j, r.rawInputAt m i.val = some j := by
    apply find_exists
    refine ⟨wanted.val, by simp, ?_⟩
    simp only [decide_eq_true_eq]
    exact ⟨wanted.property, r.slot_position h m i⟩
  obtain ⟨j, hj⟩ := hex
  have hfound := (find_mem _ (List.finRange r.ni) j hj).2
  have hpair : (r.inputs j).mechanism = m ∧ (r.inputs j).position = i.val := by simp using hfound
  have he : j = wanted.val := h.input_positions.2 j wanted.val
    (hpair.1.trans wanted.property.symm) (hpair.2.trans (r.slot_position h m i).symm)
  simp only [rawSlotSize, hj, he]
  rfl

theorem rawSlotSizes_eq (r : Network) (h : r.Valid) (m : Fin r.nm) :
  r.rawSlotSizes m = r.slotSizes h m := by
  apply congrArg List.ofFn
  funext i
  exact r.rawSlotSize_eq h m i

def AxisMatch (r : Network) (m : Fin r.nm) (t : Table) : Prop :=
  t.outputLabels.length = r.stateCount (r.mechanisms m).target ∧
  (∀ s, (r.states s).var = (r.mechanisms m).target →
    t.outputLabels.getD (r.states s).position "" = (r.states s).name) ∧
  t.inputLabels.length = r.inputCount m ∧
  ∀ i, (r.inputs i).mechanism = m →
    (t.inputLabels.getD (r.inputs i).position []).length = r.stateCount (r.inputs i).var ∧
    ∀ s, (r.states s).var = (r.inputs i).var →
      (t.inputLabels.getD (r.inputs i).position []).getD (r.states s).position "" = (r.states
← s).name

instance (r : Network) (m : Fin r.nm) (t : Table) : Decidable (r.AxisMatch m t) := by
  unfold AxisMatch
  infer_instance

theorem AxisMatch.output (r : Network) (h : r.Valid) (m : Fin r.nm) (t : Table)
  (ht : r.AxisMatch m t) (a : Fin (r.stateCount (r.mechanisms m).target)) :
  t.outputLabels.getD a.val "" = r.stateLabel h (r.mechanisms m).target a := by
  have he := ht.2.1 ((r.stateOrder h _).symm a).val ((r.stateOrder h _).symm a).property
  rw [r.state_position h _ a] at he
  exact he

theorem AxisMatch.input (r : Network) (h : r.Valid) (m : Fin r.nm) (t : Table)
  (ht : r.AxisMatch m t) (j : Fin (r.inputCount m))
  (a : Fin (r.stateCount (r.slotVariable h m j))) :
  (t.inputLabels.getD j.val []).getD a.val "" = r.stateLabel h (r.slotVariable h m j) a := by
  let i := (r.inputOrder h m).symm j
  have hi := ht.2.2.2 i.val i.property
  have he := hi.2 ((r.stateOrder h _).symm a).val ((r.stateOrder h _).symm a).property
  change (t.inputLabels.getD (r.inputs i.val).position []).getD
    (r.states ((r.stateOrder h (r.slotVariable h m j)).symm a).val).position "" = _ at he
  rw [r.state_position h _ a, r.slot_position h m j] at he
  exact he

def optionReady (r : Network) (m : Fin r.nm) : Option Table → Prop
| none ⇒ False
| some t ⇒ (t.inputSizes = r.rawSlotSizes m ∧
  t.outputSize = r.stateCount (r.mechanisms m).target ∧ t.Ready) ∧ r.AxisMatch m t

instance (r : Network) (m : Fin r.nm) (o : Option Table) : Decidable (r.optionReady m o) :=
  match o with
| none ⇒ isFalse (fun h ⇒ h)
| some t ⇒ inferInstanceAs (Decidable

```

```

    ((t.inputSizes = r.rawSlotSizes m ∧ t.outputSize = r.stateCount (r.mechanisms m).target ∧
    ↪ t.Ready) ∧
    r.AxisMatch m t))

def rawReferenceReady (r : Network) (bank : Bank) (m : Fin r.nm) : Prop :=
  match (r.mechanisms m).kernelRef with
  | .none ⇒ False
  | .pointMass label ⇒ ∃ s : Fin r.ns,
    (r.states s).var = (r.mechanisms m).target ∧ (r.states s).name = label
  | .named key ⇒ r.optionReady m (resolve bank (.named key))
  | .policy key ⇒ r.optionReady m (resolve bank (.policy key))

instance (r : Network) (bank : Bank) (m : Fin r.nm) : Decidable (r.rawReferenceReady bank m) := by
  unfold rawReferenceReady
  exact match (r.mechanisms m).kernelRef with
  | .none ⇒ isFalse (fun h ⇒ h)
  | .pointMass label ⇒ inferInstanceAs (Decidable (∃ s : Fin r.ns,
    (r.states s).var = (r.mechanisms m).target ∧ (r.states s).name = label))
  | .named key ⇒ inferInstanceAs (Decidable (r.optionReady m (resolve bank (.named key))))
  | .policy key ⇒ inferInstanceAs (Decidable (r.optionReady m (resolve bank (.policy key))))

def RawReady (r : Network) (bank : Bank) : Prop :=
  (bank.map Binding.ref).Nodup ∧ ∀ m, r.rawReferenceReady bank m

def optionNormalized : Option Table → Prop
  | none ⇒ True
  | some t ⇒ t.Normalized

instance (o : Option Table) : Decidable (optionNormalized o) :=
  match o with
  | none ⇒ isTrue trivial
  | some t ⇒ inferInstanceAs (Decidable t.Normalized)

def rawReferenceNormalized (r : Network) (bank : Bank) (m : Fin r.nm) : Prop :=
  optionNormalized (resolve bank (r.mechanisms m).kernelRef)

instance (r : Network) (bank : Bank) (m : Fin r.nm) : Decidable (r.rawReferenceNormalized bank m) :=
  inferInstanceAs (Decidable (optionNormalized (resolve bank (r.mechanisms m).kernelRef)))

def RawNormalized (r : Network) (bank : Bank) : Prop :=
  ∀ m, r.rawReferenceNormalized bank m

instance (r : Network) (bank : Bank) : Decidable (r.RawReady bank) := by
  unfold RawReady
  infer_instance

instance (r : Network) (bank : Bank) : Decidable (r.RawNormalized bank) := by
  unfold RawNormalized
  infer_instance

def checkReady (r : Network) (bank : Bank) : Bool := decide (r.RawReady bank)
def checkNormalized (r : Network) (bank : Bank) : Bool := decide (r.RawNormalized bank)

theorem rawReady_sound (r : Network) (h : r.Valid) (bank : Bank) (hb : r.RawReady bank) :
  r.Ready h bank := by
  refine ⟨hb.1, ?_⟩
  intro m
  have hm := hb.2 m
  unfold rawReferenceReady at hm
  cases hr : (r.mechanisms m).kernelRef with
  | none ⇒ simp [hr] at hm
  | pointMass label ⇒
    obtain ⟨s, hs, hn⟩ : ∃ s, (r.states s).var = (r.mechanisms m).target ∧
      (r.states s).name = label := by simp [hr] using hm
    refine ⟨r.stateOrder h _ ⟨s, hs⟩, ?_⟩
    simp [stateLabel] using hn

```

```

| named key | policy key ⇒
cases ht : resolve bank (r.mechanisms m).kernelRef with
| none ⇒
simp only [hr] at ht
simp [hr, ht, optionReady] at hm
| some t ⇒
simp only [hr] at ht
have hv' : (t.inputSizes = r.rawSlotSizes m ∧
  t.outputSize = r.stateCount (r.mechanisms m).target ∧ t.Ready) ∧ r.AxisMatch m t := by
  simp [hr, ht, optionReady] using hm
have hv := hv'.1
simp only [ht] using And.intro (hv.1.trans (r.rawSlotSizes_eq h m)) hv.2

theorem rawNormalized_sound (r : Network) (bank : Bank) (hn : r.RawNormalized bank) :
  r.Normalized bank := by
intro m t ht
have hm := hn m
simp [rawReferenceNormalized, optionNormalized, ht] using hm

/-- A finite Boolean certificate check, not an assumed compiler theorem, establishes normalized
semantics for the raw reference-resolved, repeated-slot network. -/
theorem checked_reference_normalization (r : Network) (bank : Bank)
  (hs : r.check = true) (hb : r.checkReady bank = true) (hn : r.checkNormalized bank = true) :
  let h := (r.check_iff).1 hs
  ∑ x, joint (r.kernel h (r.resolvedCPT h bank)) x = 1 := by
let h := (r.check_iff).1 hs
exact r.resolved_joint_normalized h bank (r.rawReady_sound h bank (of_decide_eq_true hb))
  (r.rawNormalized_sound bank (of_decide_eq_true hn))

noncomputable def realKernel (r : Network) (h : r.Valid) (bank : Bank) : (r.compile h).Kernel ℝ :=
  r.kernel h (fun m x a ⇒ (r.resolvedCPT h bank m x a : ℝ))

theorem realKernel_local (r : Network) (h : r.Valid) (bank : Bank) :
  ∀ m, Local (r.realKernel h bank) m := r.kernel_local h _

theorem realKernel_nonnegative (r : Network) (h : r.Valid) (bank : Bank) (hb : r.RawReady bank) :
  ∀ m x a, 0 ≤ r.realKernel h bank m x a := by
intro m x a
change (0 : ℝ) ≤ (r.resolvedCPT h bank m (r.gather h m x) a : ℝ)
exact_mod_cast r.resolvedCPT_nonnegative h bank (r.rawReady_sound h bank hb) m (r.gather h m x) a

theorem realKernel_normalized (r : Network) (h : r.Valid) (bank : Bank)
  (hb : r.RawReady bank) (hn : r.RawNormalized bank) :
  ∀ m, Normalised (r.realKernel h bank) m := by
intro m x
have hs := r.resolvedCPT_normalized h bank (r.rawReady_sound h bank hb)
  (r.rawNormalized_sound bank hn) m (r.gather h m x)
change (∑ a : Fin (r.stateCount (r.mechanisms m).target),
  (r.resolvedCPT h bank m (r.gather h m x) a : ℝ)) = 1
have hr := congrArg (Rat.castHom ℝ) hs
rw [map_sum, map_one] at hr
exact hr

end Network
end BayesianNetworksProofs.Raw

```

Scoped factor marginalization

```

import BayesianNetworksProofs.Finite.Assignments
import BayesianNetworksProofs.Finite.VariableElimination
import Mathlib.Data.List.Forall2
import Mathlib.Algebra.Order.BigOperators.Group.Finset

```

project sums only the factor coordinates being removed, not the full joint state space. The private-variable

replacement law is the algebra used in junction-tree message passing. It permits arbitrary real entries; nonnegativity is needed later only for posterior semantics.

```

set_option autoImplicit false

namespace BayesianNetworksProofs.FinBayesNet.Factor

noncomputable section
variable {bn : FinBayesNet}

def unit : Factor bn ℝ where
  scope := ∅
  value _ := 1
  dependsOn := fun _ _ => rfl

def multiply (f g : Factor bn ℝ) : Factor bn ℝ where
  scope := f.scope ∪ g.scope
  value x := f.value x * g.value x
  dependsOn x y h := by
    rw [f.dependsOn x y (fun v hv => h v (Finset.mem_union_left _ hv)),
        g.dependsOn x y (fun v hv => h v (Finset.mem_union_right _ hv))]

def combine : List (Factor bn ℝ) → Factor bn ℝ
| [] => unit
| f :: fs => multiply f (combine fs)

@[ext] theorem ext {f g : Factor bn ℝ} (hs : f.scope = g.scope) (hv : f.value = g.value) :
  f = g := by
  cases f
  cases g
  cases hs
  cases hv
  rfl

theorem multiply_comm (f g : Factor bn ℝ) : multiply f g = multiply g f := by
  apply ext
  · exact Finset.union_comm _ _
  · funext x
    exact mul_comm _ _

theorem multiply_assoc (f g h : Factor bn ℝ) :
  multiply (multiply f g) h = multiply f (multiply g h) := by
  apply ext
  · exact Finset.union_assoc _ _ _
  · funext x
    exact mul_assoc _ _ _

theorem unit_multiply (f : Factor bn ℝ) : multiply unit f = f := by
  apply ext
  · simp [multiply, unit]
  · funext x
    simp [multiply, unit]

theorem multiply_unit (f : Factor bn ℝ) : multiply f unit = f := by
  rw [multiply_comm, unit_multiply]

theorem multiply_left_comm (f g h : Factor bn ℝ) :
  multiply f (multiply g h) = multiply g (multiply f h) := by
  rw [← multiply_assoc, multiply_comm f g, multiply_assoc]

instance : Std.Associative (multiply (bn := bn)) := ⟨multiply_assoc⟩
instance : Std.Commutative (multiply (bn := bn)) := ⟨multiply_comm⟩

theorem combine_value (fs : List (Factor bn ℝ)) : (combine fs).value = product fs := by
  induction fs with
  | nil => rfl
  | cons f fs ih =>

```

```

    funext x
    change f.value x * (combine fs).value x = f.value x * product fs x
    rw [ih]

theorem combine_scope (fs : List (Factor bn R)) : (combine fs).scope = jointScope fs := by
  induction fs with
  | nil => rfl
  | cons f fs ih => exact congrArg (fun S => f.scope ∪ S) ih

def project (Q : Finset bn.V) (f : Factor bn R) : Factor bn R where
  scope := f.scope ∩ Q
  value x := ∑ a : PartialAssignment bn (f.scope \ Q), f.value (patch (f.scope \ Q) x a)
  dependsOn x y h := by
    apply Finset.sum_congr rfl
    intro a _
    apply f.dependsOn
    intro v hv
    by_cases he : v ∈ f.scope \ Q
    · simp [patch, he]
    · have hq : v ∈ Q := by
        by_contra hq
        exact he (Finset.mem_sdiff.2 ⟨hv, hq⟩)
      simp [patch, he] using h v (Finset.mem_inter.2 ⟨hv, hq⟩)

theorem project_value (Q : Finset bn.V) (f : Factor bn R) (x : bn.Assignment) :
  (project Q f).value x = marg (f.scope \ Q) f.value x :=
  sum_partial_eq_marg _ _ _

theorem project_of_scope {Q : Finset bn.V} (f : Factor bn R) (h : f.scope ⊆ Q) :
  project Q f = f := by
  apply ext
  · exact Finset.inter_eq_left.2 h
  · funext x
    rw [project_value, Finset.sdiff_eq_empty_iff_subset.2 h, marg_empty]

theorem project_nonnegative (Q : Finset bn.V) (f : Factor bn R)
  (h : ∀ x, 0 ≤ f.value x) : ∀ x, 0 ≤ (project Q f).value x := by
  intro x
  exact Finset.sum_nonneg fun a _ => h _

theorem private_factor_constant (S : Finset bn.V) (g : Factor bn R)
  (h : Disjoint S g.scope) (x y : bn.Assignment) (hy : y ∈ fibre S x) :
  g.value y = g.value x :=
  g.dependsOn y x (fun v hv => mem_fibre.1 hy v
    (fun hs => Finset.disjoint_left.1 h hs hv))

def marginal (S : Finset bn.V) (f : Factor bn R) : Factor bn R where
  scope := f.scope \ S
  value x := ∑ a : PartialAssignment bn S, f.value (patch S x a)
  dependsOn x y h := by
    apply Finset.sum_congr rfl
    intro a _
    apply f.dependsOn
    intro v hv
    by_cases hs : v ∈ S
    · simp [patch, hs]
    · simp [patch, hs] using h v (Finset.mem_sdiff.2 ⟨hv, hs⟩)

theorem marginal_value (S : Finset bn.V) (f : Factor bn R) (x : bn.Assignment) :
  (marginal S f).value x = marg S f.value x := sum_partial_eq_marg _ _ _

theorem marg_product_disjoint (S T : Finset bn.V) (f g : Factor bn R)
  (hST : Disjoint S T) (hTf : Disjoint T f.scope) (hSg : Disjoint S g.scope)
  (x : bn.Assignment) :
  marg (S ∪ T) (multiply f g).value x = marg S f.value x * marg T g.value x := by
  rw [marg_union_disjoint hST]

```

```

have hinner :  $\forall y, \text{marg } T (\text{multiply } f \ g).value \ y = f.value \ y * (\text{marginal } T \ g).value \ y := \text{by}$ 
  intro y
  rw [marginal_value]
  exact marg_mul_left (fun z hz  $\Rightarrow$  private_factor_constant T f hTf y z hz)
simp_rw [hinner]
have hg : Disjoint S (marginal T g).scope := hSg.mono_right Finset.sdiff_subset
unfold marg
rw [Finset.sum_mul]
apply Finset.sum_congr rfl
intro y hy
dsimp only
rw [private_factor_constant S (marginal T g) hg x y hy, marginal_value]
rfl

/-- Sum private variables inside one factor before combining it with the remaining factors.
This is the actual message-passing replacement law, with only structural scope conditions. -/
theorem project_private (Q C : Finset bn.V) (f g : Factor bn R)
  (h : Disjoint (f.scope \ C) (g.scope  $\cup$  Q)) :
  project Q (multiply f g) = project Q (multiply (project C f) g) := by
  let E := f.scope \ C
  let T := ((f.scope  $\cap$  C)  $\cup$  g.scope) \ Q
  have he : (f.scope  $\cup$  g.scope) \ Q = T  $\cup$  E := by
    ext v
    have hd := Finset.disjoint_left.1 h
    by_cases hf : v  $\in$  f.scope <=> by_cases hg : v  $\in$  g.scope <=>
      by_cases hc : v  $\in$  C <=> by_cases hq : v  $\in$  Q <=> simp_all [T, E]
  have hdis : Disjoint T E := by
    rw [Finset.disjoint_left]
    intro v hv he
    have hf := Finset.mem_sdiff.1 he
    have ht := Finset.mem_union.1 (Finset.mem_sdiff.1 hv).1
    rcases ht with ht | ht
    · exact hf.2 (Finset.mem_inter.1 ht).2
    · exact Finset.disjoint_left.1 h he (Finset.mem_union_left _ ht)
  have hg : Disjoint E g.scope := h.mono_right Finset.subset_union_left
  apply ext
  · ext v
    have hd := Finset.disjoint_left.1 h
    by_cases hf : v  $\in$  f.scope <=> by_cases hg : v  $\in$  g.scope <=>
      by_cases hc : v  $\in$  C <=> by_cases hq : v  $\in$  Q <=>
        simp_all [project, multiply]
  · funext x
    rw [project_value, project_value]
    change marg ((f.scope  $\cup$  g.scope) \ Q) (fun y  $\Rightarrow$  f.value y * g.value y) x =
      marg T (fun y  $\Rightarrow$  (project C f).value y * g.value y) x
    rw [he, marg_union_disjoint hdis]
    apply congrArg (fun F  $\Rightarrow$  marg T F x)
    funext y
    rw [project_value]
    unfold marg
    rw [Finset.sum_mul]
    apply Finset.sum_congr rfl
    intro z hz
    dsimp only
    rw [private_factor_constant E g hg y z hz]

theorem project_project (P Q : Finset bn.V) (f : Factor bn R)
  (h : f.scope  $\cap$  P  $\subseteq$  Q) : project P (project Q f) = project P f := by
  have hd : Disjoint (f.scope \ Q) (unit.scope  $\cup$  P) := by
    apply Finset.disjoint_left.2
  intro v hv hp
  have hP : v  $\in$  P := by simp [unit] using hp
  exact (Finset.mem_sdiff.1 hv).2 (h (Finset.mem_inter.2 ((Finset.mem_sdiff.1 hv).1, hP)))
  have he := project_private P Q f unit hd
  simp only [multiply_unit] using he.symm

```

```

theorem project_inter (P Q : Finset bn.V) (f : Factor bn R) (h : f.scope ∩ P ⊆ Q) :
  project P f = project (P ∩ Q) f := by
  have hscope : f.scope ∩ P = f.scope ∩ (P ∩ Q) := by
    ext v
    constructor
    · intro hv
      exact Finset.mem_inter.2 ⟨(Finset.mem_inter.1 hv).1,
        Finset.mem_inter.2 ⟨(Finset.mem_inter.1 hv).2, h hv⟩⟩
    · intro hv
      exact Finset.mem_inter.2 ⟨(Finset.mem_inter.1 hv).1,
        (Finset.mem_inter.1 (Finset.mem_inter.1 hv).2).1⟩
  have he : f.scope \ P = f.scope \ (P ∩ Q) := by
    ext v
    constructor
    · intro hv
      exact Finset.mem_sdiff.2 ⟨(Finset.mem_sdiff.1 hv).1,
        fun hi ⇒ (Finset.mem_sdiff.1 hv).2 (Finset.mem_inter.1 hi).1⟩
    · intro hv
      refine Finset.mem_sdiff.2 ⟨(Finset.mem_sdiff.1 hv).1, ?_⟩
      intro hp
      exact (Finset.mem_sdiff.1 hv).2
        (Finset.mem_inter.2 ⟨hp, h (Finset.mem_inter.2 ⟨(Finset.mem_sdiff.1 hv).1, hp⟩)⟩)
  apply ext hscope
  funext x
  rw [project_value, project_value, he]

/-- Independent private coordinates can be summed in separate factors. Shared coordinates
must be retained, which is exactly the structural separator condition. -/
theorem project_multiply (Q : Finset bn.V) (f g : Factor bn R)
  (h : f.scope ∩ g.scope ⊆ Q) :
  project Q (multiply f g) = multiply (project Q f) (project Q g) := by
  have hf : Disjoint (f.scope \ Q) (g.scope ∪ Q) := by
    apply Finset.disjoint_left.2
    intro v hv hg
    rcases Finset.mem_union.1 hg with hg | hq
    · exact (Finset.mem_sdiff.1 hv).2 (h (Finset.mem_inter.2 ⟨(Finset.mem_sdiff.1 hv).1, hg⟩))
    · exact (Finset.mem_sdiff.1 hv).2 hq
  have hg : Disjoint (g.scope \ Q) ((project Q f).scope ∪ Q) := by
    apply Finset.disjoint_left.2
    intro v hv hf
    have hq : v ∈ Q := (Finset.mem_union.1 hf).elim (fun h ⇒ (Finset.mem_inter.1 h).2) id
    exact (Finset.mem_sdiff.1 hv).2 hq
  rw [project_private Q Q f g hf, multiply_comm (project Q f) g,
    project_private Q Q g (project Q f) hg,
    project_of_scope _ (Finset.union_subset Finset.inter_subset_right Finset.inter_subset_right),
    multiply_comm]

end
end BayesianNetworksProofs.FinBayesNet.Factor

```

Collect/distribute on a finite junction tree

```

import BayesianNetworksProofs.Finite.FactorMarginal
import BayesianNetworksProofs.Finite.Posterior
import Mathlib.Algebra.Order.BigOperators.GroupWithZero.List

```

The working representation is a binary junction tree. Arbitrary branching is represented by copying a parent bag along a binary chain with empty local factor lists; no factors or weights are duplicated. The recursive running-intersection conditions are structural: occurrences shared by two branches meet in the parent bag, and a branch's intersection with its parent is present in the branch root.

prepare computes only local products and upward separator projections. distribute uses the cached upward messages and an incoming message to send the downward messages and produce beliefs. No whole-joint enu-

meration occurs in either computation. `full` is a proof oracle. An empty virtual-root bag connects disconnected components by scalar messages.

```

set_option autoImplicit false

namespace BayesianNetworksProofs.Junction

open FinBayesNet FinBayesNet.Factor

noncomputable section
variable {bn : FinBayesNet} {I : Type}

inductive Tree (V I : Type)
| leaf (bag : Finset V) (assigned : List I)
| branch (bag : Finset V) (assigned : List I) (left right : Tree V I)

def Tree.bag : Tree bn.V I → Finset bn.V
| .leaf B _ ⇒ B
| .branch B _ _ ⇒ B

def Tree.vars : Tree bn.V I → Finset bn.V
| .leaf B _ ⇒ B
| .branch B _ l r ⇒ B ∪ (l.vars ∪ r.vars)

def Tree.indices : Tree bn.V I → List I
| .leaf _ fs ⇒ fs
| .branch _ fs l r ⇒ fs ++ l.indices ++ r.indices

def localFactor (F : I → Factor bn R) (fs : List I) := combine (fs.map F)

def full (F : I → Factor bn R) : Tree bn.V I → Factor bn R
| .leaf _ fs ⇒ localFactor F fs
| .branch _ fs l r ⇒ multiply (localFactor F fs) (multiply (full F l) (full F r))

def Good (F : I → Factor bn R) : Tree bn.V I → Prop
| .leaf B fs ⇒ (localFactor F fs).scope ⊆ B
| .branch B fs l r ⇒ (localFactor F fs).scope ⊆ B ∧ Good F l ∧ Good F r ∧
  l.vars ∩ B ⊆ l.bag ∧ r.vars ∩ B ⊆ r.bag ∧ l.vars ∩ r.vars ⊆ B

def decideGood (F : I → Factor bn R) : (t : Tree bn.V I) → Decidable (Good F t)
| .leaf B fs ⇒ inferInstanceAs (Decidable ((localFactor F fs).scope ⊆ B))
| .branch B fs l r ⇒
  let I := decideGood F l
  let I := decideGood F r
  inferInstanceAs (Decidable ((localFactor F fs).scope ⊆ B ∧ Good F l ∧ Good F r ∧
    l.vars ∩ B ⊆ l.bag ∧ r.vars ∩ B ⊆ r.bag ∧ l.vars ∩ r.vars ⊆ B))

def checkGood (F : I → Factor bn R) (t : Tree bn.V I) : Bool :=
  @decide (Good F t) (decideGood F t)

theorem checkGood_iff (F : I → Factor bn R) (t : Tree bn.V I) :
  checkGood F t = true ↔ Good F t := by
  let I := decideGood F t
  unfold checkGood
  exact decide_eq_true_iff

def checkAssignment [Fintype I] [DecidableEq I] (t : Tree bn.V I) : Bool :=
  decide (t.indices.Nodup ∧ ∀ i : I, i ∈ t.indices)

theorem checkAssignment_sound [Fintype I] [DecidableEq I] (t : Tree bn.V I)
  (h : checkAssignment t = true) : t.indices.Perm Finset.univ.toList := by
  have hv : t.indices.Nodup ∧ ∀ i : I, i ∈ t.indices := of_decide_eq_true h
  apply (List.perm_ext_iff_of_nodup hv.1 (Finset.nodup_toList _)).2
  intro i
  simp [hv.2 i]

theorem bag_subset_vars (t : Tree bn.V I) : t.bag ⊆ t.vars := by

```

```

cases t
· exact Finset.Subset.refl _
· exact Finset.subset_union_left

theorem full_scope (F : I → Factor bn R) (t : Tree bn.V I) (h : Good F t) :
  (full F t).scope ⊆ t.vars := by
induction t with
| leaf B fs ⇒ exact h
| branch B fs l r ihl ihr ⇒
  exact Finset.union_subset_union h.1 (Finset.union_subset_union (ihl h.2.1) (ihr h.2.2.1))

theorem full_value (F : I → Factor bn R) (t : Tree bn.V I) :
  (full F t).value = Factor.product (t.indices.map F) := by
induction t with
| leaf B fs ⇒ exact combine_value _
| branch B fs l r ihl ihr ⇒
  funext x
  simp [full, localFactor, multiply, combine_value, Tree.indices, Factor.product,
    List.map_append, List.map_map, ihl, ihr]

inductive Prepared (bn : FinBayesNet)
| leaf (bag : Finset bn.V) (pot : Factor bn R) (up : Factor bn R)
| branch (bag : Finset bn.V) (pot : Factor bn R) (up : Factor bn R)
  (left right : Prepared bn)

def Prepared.bag : Prepared bn → Finset bn.V
| .leaf B _ _ ⇒ B
| .branch B _ _ _ ⇒ B

def Prepared.up : Prepared bn → Factor bn R
| .leaf _ _ u ⇒ u
| .branch _ _ u _ _ ⇒ u

/-- The collect pass; projections enumerate only removed factor coordinates. -/
def prepare (F : I → Factor bn R) (parent : Finset bn.V) : Tree bn.V I → Prepared bn
| .leaf B fs ⇒
  let f := localFactor F fs
  .leaf B f (project parent f)
| .branch B fs l r ⇒
  let pl := prepare F B l
  let pr := prepare F B r
  let f := localFactor F fs
  .branch B f (project parent (multiply f (multiply pl.up pr.up))) pl pr

theorem prepare_bag (F : I → Factor bn R) (P : Finset bn.V) (t : Tree bn.V I) :
  (prepare F P t).bag = t.bag := by cases t <;> rfl

theorem project_three (B : Finset bn.V) (f l r : Factor bn R)
  (hf : f.scope ⊆ B) (hlf : l.scope ∩ r.scope ⊆ B) :
  project B (multiply f (multiply l r)) =
    multiply f (multiply (project B l) (project B r)) := by
rw [project_multiply B f _ (Finset.inter_subset_left.trans hf),
  project_of_scope f hf, project_multiply B l r hlf]

/-- Every upward message is the side-of-cut marginal of its assigned factors. -/
theorem prepare_up (F : I → Factor bn R) (t : Tree bn.V I) (h : Good F t)
  (P : Finset bn.V) (hb : t.vars ∩ P ⊆ t.bag) :
  (prepare F P t).up = project P (full F t) := by
induction t generalizing P with
| leaf B fs ⇒ rfl
| branch B fs l r ihl ihr ⇒
  change project P (multiply (localFactor F fs)
    (multiply (prepare F B l).up (prepare F B r).up)) = _
  rw [ihl h.2.1 B h.2.2.2.1, ihr h.2.2.1 B h.2.2.2.2.1]
  have hc : (full F l).scope ∩ (full F r).scope ⊆ B :=
    (Finset.inter_subset_inter (full_scope F l h.2.1) (full_scope F r h.2.2.1)).trans h.2.2.2.2

```

```

    rw [← project_three B _ _ h.1 hc]
    exact project_project P B (full F (.branch B fs l r))
      ((Finset.inter_subset_inter (full_scope F _ h) (Finset.Subset.refl P)).trans hb)

theorem prepare_up_separator (F : I → Factor bn R) (t : Tree bn.V I) (h : Good F t)
  (P : Finset bn.V) (hb : t.vars ∩ P ⊆ t.bag) :
  (prepare F P t).up = project (t.bag ∩ P) (full F t) := by
  rw [prepare_up F t h P hb, project_inter P t.bag (full F t)
    ((Finset.inter_subset_inter (full_scope F t h) (Finset.Subset.refl P)).trans hb),
    Finset.inter_comm P t.bag]

/-- The distribute pass, using the collect pass's cached upward messages. -/
def distribute (incoming : Factor bn R) : Prepared bn → List (Finset bn.V × Factor bn R)
| .leaf B f _ ⇒ [(B, multiply f incoming)]
| .branch B f _ l r ⇒
  (B, multiply f (multiply incoming (multiply l.up r.up))) ::
  (distribute (project l.bag (multiply f (multiply incoming r.up))) l ++
   distribute (project r.bag (multiply f (multiply incoming l.up))) r)

theorem frame_scope (B V : Finset bn.V) (f o g : Factor bn R)
  (hf : f.scope ⊆ B) (ho : o.scope ∩ V ⊆ B) (hg : g.scope ∩ V ⊆ B) :
  (multiply o (multiply f g)).scope ∩ V ⊆ B := by
  intro v hv
  have hvV := (Finset.mem_inter.1 hv).2
  rcases Finset.mem_union.1 (Finset.mem_inter.1 hv).1 with hv | hv
  · exact ho (Finset.mem_inter.2 ⟨hv, hvV⟩)
  · rcases Finset.mem_union.1 hv with hv | hv
    · exact hf hv
    · exact hg (Finset.mem_inter.2 ⟨hv, hvV⟩)

theorem outgoing (B C : Finset bn.V) (f o g : Factor bn R)
  (hf : f.scope ⊆ B) (hog : o.scope ∩ g.scope ⊆ B)
  (hC : (multiply o (multiply f g)).scope ∩ C ⊆ B) :
  project C (multiply f (multiply (project B o) (project B g))) =
  project C (multiply o (multiply f g)) := by
  calc
  _ = project C (project B (multiply o (multiply f g))) := by
    rw [multiply_left_comm o f g, project_three B f o g hf hog]
  _ = _ := project_project C B _ hC

/-- The collect/distribute beliefs are the global factor product projected to each bag.
The outside factor is an induction frame; the public calibration theorem supplies unit. -/
theorem distribute_correct (F : I → Factor bn R) (t : Tree bn.V I) (h : Good F t)
  (outside : Factor bn R) (hb : outside.scope ∩ t.vars ⊆ t.bag)
  (P : Finset bn.V) :
  ∀ b ∈ distribute (project t.bag outside) (prepare F P t),
  b.2 = project b.1 (multiply outside (full F t)) := by
  induction t generalizing outside P with
  | leaf B fs ⇒
    intro b hmem
    have he : b = (B, multiply (localFactor F fs) (project B outside)) := by simpa [distribute,
      ← prepare] using hmem
    subst b
    change multiply (localFactor F fs) (project B outside) = project B (multiply outside (localFactor
      F fs))
    rw [project_multiply B outside _ (Finset.inter_subset_right.trans h), project_of_scope _ h,
      multiply_comm]
  | branch B fs l r ihl ihr ⇒
    let f := localFactor F fs
    let fl := full F l
    let fr := full F r
    have hl : fl.scope ⊆ l.vars := full_scope F l h.2.1
    have hr : fr.scope ⊆ r.vars := full_scope F r h.2.2.1
    have hvl : l.vars ⊆ (Tree.branch B fs l r).vars :=
      (Finset.subset_union_left : l.vars ⊆ l.vars ∪ r.vars).trans Finset.subset_union_right
    have hvr : r.vars ⊆ (Tree.branch B fs l r).vars :=

```

```

    (Finset.subset_union_right : r.vars  $\subseteq$  l.vars  $\cup$  r.vars).trans Finset.subset_union_right
  have hoL : outside.scope  $\cap$  l.vars  $\subseteq$  B :=
    (Finset.inter_subset_inter (Finset.Subset.refl _) hvl).trans hb
  have hoR : outside.scope  $\cap$  r.vars  $\subseteq$  B :=
    (Finset.inter_subset_inter (Finset.Subset.refl _) hvr).trans hb
  have hlr : fl.scope  $\cap$  fr.scope  $\subseteq$  B :=
    (Finset.inter_subset_inter hl hr).trans h.2.2.2.2
  have hrL : fr.scope  $\cap$  l.vars  $\subseteq$  B := by
    intro v hv
    exact h.2.2.2.2 (Finset.mem_inter.2  $\langle$  (Finset.mem_inter.1 hv).2, hr (Finset.mem_inter.1
       $\hookleftarrow$  hv).1  $\rangle$ )
  have hLR : fl.scope  $\cap$  r.vars  $\subseteq$  B :=
    (Finset.inter_subset_inter hl (Finset.Subset.refl _)).trans h.2.2.2.2
  let ol := multiply outside (multiply f fr)
  let or := multiply outside (multiply f fl)
  have hoB : ol.scope  $\cap$  l.vars  $\subseteq$  B := frame_scope B l.vars f outside fr h.1 hoL hrL
  have hoR : or.scope  $\cap$  r.vars  $\subseteq$  B := frame_scope B r.vars f outside fl h.1 hoR hLR
  have hol : ol.scope  $\cap$  l.vars  $\subseteq$  l.bag := by
    intro v hv
    exact h.2.2.2.1 (Finset.mem_inter.2  $\langle$  (Finset.mem_inter.1 hv).2, hoB hv  $\rangle$ )
  have hor : or.scope  $\cap$  r.vars  $\subseteq$  r.bag := by
    intro v hv
    exact h.2.2.2.1 (Finset.mem_inter.2  $\langle$  (Finset.mem_inter.1 hv).2, hoR hv  $\rangle$ )
  have houtl : project l.bag (multiply f (multiply (project B outside) (project B fr))) = project
     $\hookleftarrow$  l.bag ol :=
    outgoing B l.bag f outside fr h.1
    ((Finset.inter_subset_inter (Finset.Subset.refl _) hr).trans hoR)
    ((Finset.inter_subset_inter (Finset.Subset.refl _) (bag_subset_vars l)).trans hoB)
  have houtr : project r.bag (multiply f (multiply (project B outside) (project B fl))) = project
     $\hookleftarrow$  r.bag or :=
    outgoing B r.bag f outside fl h.1
    ((Finset.inter_subset_inter (Finset.Subset.refl _) hl).trans hoL)
    ((Finset.inter_subset_inter (Finset.Subset.refl _) (bag_subset_vars r)).trans hoR)
  intro b hm
  simp only [prepare, distribute, prepare_bag] at hm
  rw [prepare_up F l h.2.1 B h.2.2.2.1, prepare_up F r h.2.1 B h.2.2.2.1] at hm
  change b  $\in$  (B, multiply f (multiply (project B outside) (multiply (project B fl) (project B
     $\hookleftarrow$  fr)))) ::
    (distribute (project l.bag (multiply f (multiply (project B outside) (project B fr)))) (prepare
     $\hookleftarrow$  F B l) ++
    distribute (project r.bag (multiply f (multiply (project B outside) (project B fl)))) (prepare
     $\hookleftarrow$  F B r)) at hm
  rw [houtl, houtr] at hm
  rcases List.mem_cons.1 hm with rfl | hm
  • change multiply f (multiply (project B outside) (multiply (project B fl) (project B fr))) =
    project B (multiply outside (multiply f (multiply fl fr)))
    have hf := full_scope F (.branch B fs l r) h
    change (multiply f (multiply fl fr)).scope  $\subseteq$  (Tree.branch B fs l r).vars at hf
    have hcross : outside.scope  $\cap$  (multiply f (multiply fl fr)).scope  $\subseteq$  B :=
      (Finset.inter_subset_inter (Finset.Subset.refl _) hf).trans hb
    rw [project_multiply B outside (multiply f (multiply fl fr)) hcross,
      project_three B f fl fr h.1 hlr]
    ac_rfl
  • rcases List.mem_append.1 hm with hm | hm
    • rw [ihl h.2.1 ol hoL B b hm]
      congr 1
      change multiply (multiply outside (multiply f fr)) fl = multiply outside (multiply f
     $\hookleftarrow$  (multiply fl fr))
      ac_rfl
    • rw [ihr h.2.2.1 or hor B b hm]
      congr 1
      change multiply (multiply outside (multiply f fl)) fr = multiply outside (multiply f
     $\hookleftarrow$  (multiply fl fr))
      ac_rfl

```

```

def calibrate (F : I  $\rightarrow$  Factor bn R) (t : Tree bn.V I) :=

```

```

distribute unit (prepare F  $\emptyset$  t)

theorem calibrate_correct (F : I  $\rightarrow$  Factor bn  $\mathbb{R}$ ) (t : Tree bn.V I) (h : Good F t) :
   $\forall$  b  $\in$  calibrate F t, b.2 = project b.1 (full F t) := by
  have hu : project t.bag (unit (bn := bn)) = unit := project_of_scope _ (Finset.empty_subset _)
  have hh := distribute_correct F t h unit (by intro v hv; exact False.elim (Finset.notMem_empty v
     $\hookleftarrow$  (Finset.mem_inter.1 hv).1))  $\emptyset$ 
  simpa only [hu, unit_multiply] using hh

/-- Arbitrary branching is encoded by a chain of copies of the same bag with empty local
factor lists. This is a representation transformation, not a restriction to degree three. -/
def graft (B : Finset bn.V) (fs : List I) : List (Tree bn.V I)  $\rightarrow$  Tree bn.V I
| []  $\Rightarrow$  .leaf B fs
| t :: ts  $\Rightarrow$  .branch B fs t (graft B [] ts)

def forestVars : List (Tree bn.V I)  $\rightarrow$  Finset bn.V
| []  $\Rightarrow$   $\emptyset$ 
| t :: ts  $\Rightarrow$  t.vars  $\cup$  forestVars ts

theorem graft_bag (B : Finset bn.V) (fs : List I) (ts : List (Tree bn.V I)) :
  (graft B fs ts).bag = B := by cases ts <|> rfl

theorem graft_vars (B : Finset bn.V) (fs : List I) (ts : List (Tree bn.V I)) :
  (graft B fs ts).vars = B  $\cup$  forestVars ts := by
  induction ts generalizing fs with
  | nil  $\Rightarrow$  simp [graft, Tree.vars, forestVars]
  | cons t ts ih  $\Rightarrow$ 
    simp only [graft, Tree.vars, forestVars, ih]
    ac_rfl

theorem graft_indices (B : Finset bn.V) (fs : List I) (ts : List (Tree bn.V I)) :
  (graft B fs ts).indices = fs ++ ts.flatMap Tree.indices := by
  induction ts generalizing fs with
  | nil  $\Rightarrow$  simp [graft, Tree.indices]
  | cons t ts ih  $\Rightarrow$  simp [graft, Tree.indices, ih, List.append_assoc]

theorem mem_forestVars (ts : List (Tree bn.V I)) (v : bn.V) :
  v  $\in$  forestVars ts  $\Leftrightarrow$   $\exists$  t  $\in$  ts, v  $\in$  t.vars := by
  induction ts with
  | nil  $\Rightarrow$  simp [forestVars]
  | cons t ts ih  $\Rightarrow$  simp [forestVars, ih, List.mem_cons, or_and_right, exists_or]

/-- Ordinary multi-child running intersection and factor-cover conditions generate a valid
binary working representation. No probability or message invariant occurs in the premises. -/
theorem graft_good (F : I  $\rightarrow$  Factor bn  $\mathbb{R}$ ) (B : Finset bn.V) (fs : List I)
  (ts : List (Tree bn.V I)) (hf : (localFactor F fs).scope  $\subseteq$  B)
  (hg :  $\forall$  t  $\in$  ts, Good F t) (hb :  $\forall$  t  $\in$  ts, t.vars  $\cap$  B  $\subseteq$  t.bag)
  (hpair : ts.Pairwise (fun t s  $\Rightarrow$  t.vars  $\cap$  s.vars  $\subseteq$  B)) :
  Good F (graft B fs ts) := by
  induction ts generalizing fs with
  | nil  $\Rightarrow$  exact hf
  | cons t ts ih  $\Rightarrow$ 
    have ht := (List.pairwise_cons.1 hpair)
    refine <hf, hg t (List.mem_cons_self ..),
      ih [] (Finset.empty_subset _) (fun s hs  $\Rightarrow$  hg s (List.mem_cons_of_mem t hs))
      (fun s hs  $\Rightarrow$  hb s (List.mem_cons_of_mem t hs)) ht.2,
      hb t (List.mem_cons_self ..), ?_, ?_>
    • rw [graft_bag]
      exact Finset.inter_subset_right
    • intro v hv
      have hvT := (Finset.mem_inter.1 hv).1
      rw [graft_vars] at hv
      rcases Finset.mem_union.1 (Finset.mem_inter.1 hv).2 with hB | hrest
      • exact hB
      • obtain <s, hs, hvS> := (mem_forestVars ts v).1 hrest
        exact ht.1 s hs (Finset.mem_inter.2 <hvT, hvS>)

```

```

/-- Disconnected components are joined only by empty-separator scalar messages. -/
theorem forest_good (F : I → Factor bn R) (ts : List (Tree bn.V I))
  (hg : ∀ t ∈ ts, Good F t)
  (hd : ts.Pairwise (fun t s ⇒ Disjoint t.vars s.vars)) :
  Good F (graft ∅ [] ts) := by
apply graft_good F ∅ [] ts (Finset.empty_subset _) hg
• intro t _
  simp
• exact hd.imp (by
  intro t s h
  rw [Finset.disjoint_iff_inter_eq_empty] at h
  simp [h])

theorem mem_local_scope (F : I → Factor bn R) (fs : List I) (v : bn.V) :
  v ∈ (localFactor F fs).scope ↔ ∃ i ∈ fs, v ∈ (F i).scope := by
induction fs with
| nil ⇒ simp [localFactor, combine, unit]
| cons i fs ih ⇒
  simp only [localFactor, List.map_cons, combine, multiply, Finset.mem_union] at *
  simp [ih, List.mem_cons, or_and_right, exists_or]

theorem mem_full_scope (F : I → Factor bn R) (t : Tree bn.V I) (v : bn.V) :
  v ∈ (full F t).scope ↔ ∃ i ∈ t.indices, v ∈ (F i).scope := by
induction t with
| leaf B fs ⇒ exact mem_local_scope F fs v
| branch B fs l r ihl ihr ⇒
  simp only [full, multiply, Finset.mem_union, mem_local_scope, Tree.indices,
    List.mem_append, or_and_right, exists_or, ihl, ihr, or_assoc]

theorem full_scope_univ (F : I → Factor bn R) (t : Tree bn.V I)
  (hcover : ∀ v, ∃ i ∈ t.indices, v ∈ (F i).scope) : (full F t).scope = Finset.univ := by
apply Finset.eq_univ_of_forall
intro v
exact (mem_full_scope F t v).2 (hcover v)

theorem full_product_of_perm [Fintype I] (F : I → Factor bn R) (t : Tree bn.V I)
  (hassign : t.indices.Perm Finset.univ.toList) :
  (full F t).value = Factor.product (Finset.univ.toList.map F) := by
rw [full_value]
funext x
simp only [Factor.product, List.map_map, Function.comp_def] using
(hassign.map (fun i ⇒ (F i).value x)).prod_eq

/-- Calibrated clique beliefs agree with the independently defined variable-elimination
algorithm for any duplicate-free elimination order of the other variables. -/
theorem calibrate_eq_ve [Fintype I] (F : I → Factor bn R) (t : Tree bn.V I)
  (ht : Good F t) (hassign : t.indices.Perm Finset.univ.toList)
  (hvars : ∀ v, ∃ i, v ∈ (F i).scope) (b : Finset bn.V × Factor bn R)
  (hb : b ∈ calibrate F t) (vs : List bn.V) (hvs : vs.Nodup) (hset : vs.toFinset = b.1c)
  (x : bn.Assignment) :
  b.2.value x = Factor.product (Factor.eliminateAll (Finset.univ.toList.map F) vs) x := by
have hs : (full F t).scope = Finset.univ := full_scope_univ F t (by
  intro v
  obtain ⟨i, hi⟩ := hvars v
  exact ⟨i, hassign.mem_iff.mpr (Finset.mem_toList.2 (Finset.mem_univ _)), hi⟩)
rw [calibrate_correct F t ht b hb, project_value, hs, ← Finset.compl_eq_univ_sdiff,
  full_product_of_perm F t hassign, Factor.eliminateAll_correct _ vs hvs, hset]

def beliefWeight (b : Finset bn.V × Factor bn R) (base : bn.Assignment)
  (q : PartialAssignment bn b.1) : ℝ :=
  b.2.value (patch b.1 base q)

/-- The sum of a calibrated clique belief is the GLOBAL mass, including all disconnected
components joined through the empty virtual root. Zero entries require no division. -/
theorem belief_mass [Fintype I] (F : I → Factor bn R) (t : Tree bn.V I)

```

```

(ht : Good F t) (hassign : t.indices.Perm Finset.univ.toList)
(hvars : ∀ v, ∃ i, v ∈ (F i).scope) (b : Finset bn.V × Factor bn R)
(hb : b ∈ calibrate F t) (base : bn.Assignment) :
FiniteDistribution.mass (beliefWeight b base) =
  FiniteDistribution.mass (Factor.product (Finset.univ.toList.map F)) := by
have hs : (full F t).scope = Finset.univ := full_scope_univ F t (by
  intro v
  obtain ⟨i, hi⟩ := hvars v
  exact ⟨i, hassign.mem_iff.mpr (Finset.mem_toList.2 (Finset.mem_univ _)), hi⟩)
have hw : beliefWeight b base =
  FiniteDistribution.pushWeight (restrictTo b.1) (Factor.product (Finset.univ.toList.map F)) :=
  by
  funext q
  unfold beliefWeight
  rw [calibrate_correct F t ht b hb, project_value, hs, ← Finset.compl_eq_univ_sdiff,
    full_product_of_perm F t hassign, marginal_eq_query_weight, restrict_patch]
  rw [hw, FiniteDistribution.mass_pushWeight]

def queryWeight (Q : Finset bn.V) (b : Finset bn.V × Factor bn R) (base : bn.Assignment)
  (q : PartialAssignment bn Q) : R :=
  (project Q b.2).value (patch Q base q)

theorem queryWeight_correct [Fintype I] (F : I → Factor bn R) (t : Tree bn.V I)
  (ht : Good F t) (hassign : t.indices.Perm Finset.univ.toList)
  (hvars : ∀ v, ∃ i, v ∈ (F i).scope) (b : Finset bn.V × Factor bn R)
  (hb : b ∈ calibrate F t) (Q : Finset bn.V) (hQ : Q ⊆ b.1) (base : bn.Assignment) :
  queryWeight Q b base =
    FiniteDistribution.pushWeight (restrictTo Q) (Factor.product (Finset.univ.toList.map F)) := by
have hs : (full F t).scope = Finset.univ := full_scope_univ F t (by
  intro v
  obtain ⟨i, hi⟩ := hvars v
  exact ⟨i, hassign.mem_iff.mpr (Finset.mem_toList.2 (Finset.mem_univ _)), hi⟩)
funext q
unfold queryWeight
rw [calibrate_correct F t ht b hb,
  project_project Q b.1 _ (Finset.inter_subset_right.trans hQ),
  project_value, hs, ← Finset.compl_eq_univ_sdiff, full_product_of_perm F t hassign,
  marginal_eq_query_weight, restrict_patch]

theorem factor_product_nonnegative [Fintype I] (F : I → Factor bn R)
  (hF : ∀ i x, 0 ≤ (F i).value x) : ∀ x, 0 ≤ Factor.product (Finset.univ.toList.map F) x := by
  intro x
  unfold Factor.product
  apply List.prod_nonneg
  intro a ha
  obtain ⟨f, hf, rfl⟩ := List.mem_map.1 ha
  obtain ⟨i, _, rfl⟩ := List.mem_map.1 hf
  exact hF i x

def queryPosterior [Fintype I] (F : I → Factor bn R) (t : Tree bn.V I)
  (ht : Good F t) (hassign : t.indices.Perm Finset.univ.toList)
  (hvars : ∀ v, ∃ i, v ∈ (F i).scope) (hF : ∀ i x, 0 ≤ (F i).value x)
  (b : Finset bn.V × Factor bn R) (hb : b ∈ calibrate F t)
  (Q : Finset bn.V) (hQ : Q ⊆ b.1) (base : bn.Assignment) :
  Option (FiniteDistribution.Distribution (PartialAssignment bn Q)) :=
  FiniteDistribution.normalize (queryWeight Q b base) (by
    rw [queryWeight_correct F t ht hassign hvars b hb Q hQ base]
    exact FiniteDistribution.pushWeight_nonneg _ _ (factor_product_nonnegative F hF))

/-- Exact global feasibility, including a zero-mass component disconnected from the queried
clique. The normalized result cannot silently ignore that component. -/
theorem queryPosterior_none_iff [Fintype I] (F : I → Factor bn R) (t : Tree bn.V I)
  (ht : Good F t) (hassign : t.indices.Perm Finset.univ.toList)
  (hvars : ∀ v, ∃ i, v ∈ (F i).scope) (hF : ∀ i x, 0 ≤ (F i).value x)
  (b : Finset bn.V × Factor bn R) (hb : b ∈ calibrate F t)
  (Q : Finset bn.V) (hQ : Q ⊆ b.1) (base : bn.Assignment) :

```

```

    queryPosterior F t ht hassign hvars hF b hb Q hQ base = none ↔
      ¬∃ x, 0 < Factor.product (Finset.univ.toList.map F) x := by
  rw [queryPosterior, FiniteDistribution.normalize_none_iff,
    queryWeight_correct F t ht hassign hvars b hb Q hQ base, FiniteDistribution.mass_pushWeight]
  have hn := FiniteDistribution.mass_nonneg _ (factor_product_nonnegative F hF)
  have hz : FiniteDistribution.mass (Factor.product (Finset.univ.toList.map F)) = 0 ↔
    0 < FiniteDistribution.mass (Factor.product (Finset.univ.toList.map F)) :=
    ⟨fun he ⇒ by rw [he]; simp, fun he ⇒ le_antisymm (le_of_not_gt he) hn⟩
  rw [hz, FiniteDistribution.mass_pos_iff _ (factor_product_nonnegative F hF)]

/-- A returned normalized clique/query posterior is exactly the exhaustive joint posterior. -/
theorem queryPosterior_value [Fintype I] (F : I → Factor bn R) (t : Tree bn.V I)
  (ht : Good F t) (hassign : t.indices.Perm Finset.univ.toList)
  (hvars : ∀ v, ∃ i, v ∈ (F i).scope) (hF : ∀ i x, 0 ≤ (F i).value x)
  (b : Finset bn.V × Factor bn R) (hb : b ∈ calibrate F t)
  (Q : Finset bn.V) (hQ : Q ⊆ b.1) (base : bn.Assignment)
  (d : FiniteDistribution.Distribution (PartialAssignment bn Q))
  (hd : queryPosterior F t ht hassign hvars hF b hb Q hQ base = some d)
  (q : PartialAssignment bn Q) :
  d.pmf q =
    FiniteDistribution.pushWeight (restrictTo Q) (Factor.product (Finset.univ.toList.map F)) q /
    FiniteDistribution.mass (Factor.product (Finset.univ.toList.map F)) := by
  have he := FiniteDistribution.normalize_value _ _ d hd q
  rw [queryWeight_correct F t ht hassign hvars b hb Q hQ base, FiniteDistribution.mass_pushWeight]
  ↪ at he
  exact he

end
end BayesianNetworksProofs.Junction

```

Conditional independence from separator factorization

```

import BayesianNetworksProofs.Finite.FactorMarginal
import BayesianNetworksProofs.Finite.Posterior
import Mathlib.Tactic.Ring
import Mathlib.Tactic.FieldSimp

```

Conditional independence is stated as the finite event cross-product identity. At positive conditioning mass it is the usual product of conditional probabilities. Zero conditioning mass is not divided by. This module proves the probability algebra; graphical separation is defined independently in `DSeparation.lean`.

```

set_option autoImplicit false

namespace BayesianNetworksProofs.FinBayesNet

open Factor
noncomputable section
variable {bn : FinBayesNet}

def eventMass (p : bn.Assignment → ℝ) (A B C : Finset bn.V)
  (a b c : bn.Assignment) : ℝ :=
  ∑ x, p x * indicator A a x * indicator B b x * indicator C c x

def ConditionalIndependent (p : bn.Assignment → ℝ) (A B C : Finset bn.V) : Prop :=
  ∀ a b c, eventMass p A B C a b c * eventMass p ∅ ∅ C a b c =
    eventMass p A ∅ C a b c * eventMass p ∅ B C a b c

def relativeEventMass (U : Finset bn.V) (f : bn.Assignment → ℝ) (A B C : Finset bn.V)
  (a b c : bn.Assignment) : ℝ :=
  marg U (fun x ⇒ f x * indicator A a x * indicator B b x * indicator C c x) c

theorem indicator_empty (a x : bn.Assignment) : indicator ∅ a x = 1 := by
  simp [indicator, agrees]

```

```

theorem separator_event_mass (L R C A B : Finset bn.V) (f g : Factor bn ℝ)
  (hLR : Disjoint L R) (hLC : Disjoint L C) (hRC : Disjoint R C)
  (hf : f.scope ⊆ L ∪ C) (hg : g.scope ⊆ R ∪ C)
  (hA : A ⊆ L) (hB : B ⊆ R) (a b c : bn.Assignment) :
  relativeEventMass ((L ∪ R) ∪ C) (multiply f g).value A B C a b c =
    marg L (fun x ⇒ f.value x * indicator A a x) c *
    marg R (fun x ⇒ g.value x * indicator B b x) c := by
let fl := multiply f (evidenceFactor A a)
let fr := multiply g (evidenceFactor B b)
have hfl : fl.scope ⊆ L ∪ C :=
  Finset.union_subset hf (hA.trans Finset.subset_union_left)
have hfr : fr.scope ⊆ R ∪ C :=
  Finset.union_subset hg (hB.trans Finset.subset_union_left)
have hd : Disjoint C (L ∪ R) := (Finset.disjoint_union_right.2 (hLC.symm, hRC.symm))
have he : (fun x ⇒ (multiply f g).value x * indicator A a x *
  indicator B b x * indicator C c x) =
  (fun x ⇒ (multiply fl fr).value x * indicator C c x) := by
  funext x
  dsimp [multiply, evidenceFactor, fl, fr]
  ring
unfold relativeEventMass
rw [he, marg_indicator_eq_clamp hd, marg_clamp hd]
have hc : clamp C c c = c := by funext v; simp [clamp]
rw [hc]
apply marg_product_disjoint L R fl fr hLR
· exact (Finset.disjoint_union_right.2 (hLR.symm, hRC)).mono_right hfl
· exact (Finset.disjoint_union_right.2 (hLR, hLC)).mono_right hfr

theorem separator_cross_product (L R C A B : Finset bn.V) (f g : Factor bn ℝ)
  (hLR : Disjoint L R) (hLC : Disjoint L C) (hRC : Disjoint R C)
  (hf : f.scope ⊆ L ∪ C) (hg : g.scope ⊆ R ∪ C)
  (hA : A ⊆ L) (hB : B ⊆ R) (a b c : bn.Assignment) :
  relativeEventMass ((L ∪ R) ∪ C) (multiply f g).value A B C a b c *
  relativeEventMass ((L ∪ R) ∪ C) (multiply f g).value ∅ ∅ C a b c =
  relativeEventMass ((L ∪ R) ∪ C) (multiply f g).value A ∅ C a b c *
  relativeEventMass ((L ∪ R) ∪ C) (multiply f g).value ∅ B C a b c := by
rw [separator_event_mass L R C A B f g hLR hLC hRC hf hg hA hB,
  separator_event_mass L R C ∅ ∅ f g hLR hLC hRC hf hg (Finset.empty_subset _) (Finset.empty_subset
  _),
  separator_event_mass L R C A ∅ f g hLR hLC hRC hf hg hA (Finset.empty_subset _),
  separator_event_mass L R C ∅ B f g hLR hLC hRC hf hg (Finset.empty_subset _) hB]
ring

/-- Marginalizing away variables outside an ancestral factorization commutes with an event
that depends only on retained variables. No numerical separation condition is assumed. -/
theorem eventMass_of_marginal (p q : bn.Assignment → ℝ) (U A B C : Finset bn.V)
  (hm : ∀ x, marg Uc p x = q x) (hA : A ⊆ U) (hB : B ⊆ U) (hC : C ⊆ U)
  (a b c : bn.Assignment) :
  eventMass p A B C a b c = relativeEventMass U q A B C a b c := by
let mask := multiply (evidenceFactor A a) (multiply (evidenceFactor B b) (evidenceFactor C c))
have hs : mask.scope ⊆ U := Finset.union_subset hA (Finset.union_subset hB hC)
have hinner : ∀ x, marg Uc (fun y ⇒ p y * mask.value y) x = q x * mask.value x := by
  intro x
  have he : (fun y ⇒ p y * mask.value y) = (fun y ⇒ mask.value y * p y) := by
    funext y
    ring
  rw [he, marg_mul_left (fun y hy ⇒ mask.dependsOn y x fun v hv ⇒
    mem_fibre.1 hy v (fun hc ⇒ Finset.mem_compl.1 hc (hs hv))), hm, mul_comm]
have he : (fun x ⇒ p x * indicator A a x * indicator B b x * indicator C c x) =
  (fun x ⇒ p x * mask.value x) := by
  funext x
  dsimp [mask, multiply, evidenceFactor]
  ring
unfold eventMass relativeEventMass
rw [he, ← marg_univ _ c]

```

```

have hU : U ∪ Uc = Finset.univ := Finset.union_compl U
rw [← hU, marg_union_disjoint (Finset.disjoint_left.2 fun _ hu hc ⇒ Finset.mem_compl.1 hc hu)]
simp_rw [hinner]
apply congrArg (fun f ⇒ marg U f c)
funext x
dsimp [mask, multiply, evidenceFactor]
ring

/-- Positive conditioning mass turns the cross-product identity into the usual conditional
probability factorization. The positivity boundary is explicit. -/
theorem ConditionalIndependent.conditional (p : bn.Assignment → ℝ) (A B C : Finset bn.V)
  (h : ConditionalIndependent p A B C) (a b c : bn.Assignment)
  (hz : 0 < eventMass p ∅ ∅ C a b c) :
  eventMass p A B C a b c / eventMass p ∅ ∅ C a b c =
    (eventMass p A ∅ C a b c / eventMass p ∅ ∅ C a b c) *
    (eventMass p ∅ B C a b c / eventMass p ∅ ∅ C a b c) := by
  rw [div_mul_div_comm, ← h a b c]
  field_simp [ne_of_gt hz]

end
end BayesianNetworksProofs.FinBayesNet

```

D-separation soundness on a finite DAG

```

import BayesianNetworksProofs.Finite.ConditionalIndependence
import Mathlib.Combinatorics.SimpleGraph.Connectivity.Connected
import Mathlib.Logic.Relation

```

Separation is defined solely from the directed parent graph: take the ancestors of the query and conditioning variables, moralize every retained child/parent family, delete conditioning vertices, and require no graph path between the queries. It is not defined numerically.

Reachability generates a component partition. Every mechanism scope is a clique in the moral graph, so no retained factor crosses the partition except through conditioning variables. The ancestor-normalization theorem and the separator probability algebra prove conditional independence. This is the moralized-ancestral criterion, not a verification of Julia Bayes-ball.

```

set_option autoImplicit false

namespace BayesianNetworksProofs.FinBayesNet

open Factor
noncomputable section

variable {bn : FinBayesNet}

def ParentEdge (bn : FinBayesNet) (v w : bn.V) : Prop :=
  ∃ m, bn.target m = w ∧ v ∈ bn.parents m

def ancestors (S : Finset bn.V) : Finset bn.V := by
  classical
  exact Finset.univ.filter fun v ⇒ ∃ w ∈ S, Relation.ReflTransGen (ParentEdge bn) v w

theorem subset_ancestors (S : Finset bn.V) : S ⊆ ancestors S := by
  classical
  intro v hv
  exact Finset.mem_filter.2 ⟨Finset.mem_univ _, v, hv, Relation.ReflTransGen.refl⟩

theorem ancestors_upstream (S : Finset bn.V) : UpstreamClosed (ancestors S) := by
  classical
  intro m hm v hv
  obtain ⟨w, hw, path⟩ := (Finset.mem_filter.1 hm).2
  exact Finset.mem_filter.2 ⟨Finset.mem_univ _, w, hw,

```

```

(Relation.ReflTransGen.single (m, rfl, hv)).trans path)

def family (bn : FinBayesNet) (m : bn.M) : Finset bn.V :=
  insert (bn.target m) (bn.parents m)

def moral (U : Finset bn.V) : SimpleGraph bn.V where
  Adj v w := v ≠ w ∧ ∃ m, bn.target m ∈ U ∧ v ∈ family bn m ∧ w ∈ family bn m
  symm := by
    rintro v w ⟨hne, m, hm, hv, hw⟩
    exact ⟨Ne.symm hne, m, hm, hw, hv⟩
  loopless := ⟨fun v h ⇒ h.1 rfl⟩

def cutMoral (U C : Finset bn.V) : SimpleGraph bn.V where
  Adj v w := (moral U).Adj v w ∧ v ∉ C ∧ w ∉ C
  symm := by
    rintro v w ⟨h, hv, hw⟩
    exact ⟨h.symm, hw, hv⟩
  loopless := ⟨fun v h ⇒ h.1.1 rfl⟩

def GraphSeparated (U A B C : Finset bn.V) : Prop :=
  ∀ a ∈ A, ∀ b ∈ B, ¬(cutMoral U C).Reachable a b

def DSeparated (A B C : Finset bn.V) : Prop :=
  GraphSeparated (ancestors ((A ∪ B) ∪ C)) A B C

def leftSide (U A C : Finset bn.V) : Finset bn.V := by
  classical
  exact (U \ C).filter fun v ⇒ ∃ a ∈ A, (cutMoral U C).Reachable a v

def rightSide (U A C : Finset bn.V) : Finset bn.V := U \ (leftSide U A C ∪ C)

theorem left_mem {U A C : Finset bn.V} {v : bn.V} :
  v ∈ leftSide U A C ⇔ v ∈ U ∧ v ∉ C ∧ ∃ a ∈ A, (cutMoral U C).Reachable a v := by
  simp [leftSide, and_assoc]

theorem left_subset (U A C : Finset bn.V) : leftSide U A C ⊆ U :=
  fun _ hv ⇒ (left_mem.1 hv).1

theorem left_avoids (U A C : Finset bn.V) : Disjoint (leftSide U A C) C :=
  Finset.disjoint_left.2 fun _ hv hc ⇒ (left_mem.1 hv).2.1 hc

theorem family_subset {U : Finset bn.V} (hU : UpstreamClosed U) (m : bn.M) (hm : bn.target m ∈ U) :
  family bn m ⊆ U := Finset.insert_subset_iff.2 ⟨hm, hU m hm⟩

theorem left_closed_edge {U A C : Finset bn.V} {v w : bn.V}
  (hv : v ∈ leftSide U A C) (hw : w ∈ U) (he : (cutMoral U C).Adj v w) :
  w ∈ leftSide U A C := by
  obtain ⟨_, _, a, ha, path⟩ := left_mem.1 hv
  exact left_mem.2 ⟨hw, he.2.2, a, ha, path.trans he.reachable⟩

theorem queries_in_sides {U A B C : Finset bn.V}
  (hA : A ⊆ U) (hB : B ⊆ U) (hAC : Disjoint A C) (hBC : Disjoint B C)
  (hsep : GraphSeparated U A B C) :
  A ⊆ leftSide U A C ∧ B ⊆ rightSide U A C := by
  constructor
  · intro v hv
    exact left_mem.2 ⟨hA hv, fun hc ⇒ Finset.disjoint_left.1 hAC hv hc,
      v, hv, SimpleGraph.Reachable.refl v⟩
  · intro v hv
    refine Finset.mem_sdiff.2 ⟨hB hv, ?_⟩
    intro hc
    rcases Finset.mem_union.1 hc with hl | hc
    · obtain ⟨_, _, a, ha, path⟩ := left_mem.1 hl
      exact hsep a ha v hv path
    · exact Finset.disjoint_left.1 hBC hv hc

```

```

/-- A moral clique touching the reachable side cannot also touch the other side. -/
theorem family_left {U A C : Finset bn.V} (hU : UpstreamClosed U)
  (m : bn.M) (hm : bn.target m ∈ U) (ht : (family bn m ∩ leftSide U A C).Nonempty) :
  family bn m ⊆ leftSide U A C ∪ C := by
  obtain ⟨v, hv⟩ := ht
  have hvF := (Finset.mem_inter.1 hv).1
  have hvL := (Finset.mem_inter.1 hv).2
  intro w hw
  by_cases hc : w ∈ C
  · exact Finset.mem_union_right _ hc
  · apply Finset.mem_union_left
    by_cases he : v = w
    · simp [he] using hvL
    · exact left_closed_edge hvL (family_subset hU m hm hw)
      ⟨⟨he, m, hm, hvF, hw⟩, (left_mem.1 hvL).2.1, hc⟩

theorem family_right {U A C : Finset bn.V} (hU : UpstreamClosed U)
  (m : bn.M) (hm : bn.target m ∈ U) (ht : ¬(family bn m ∩ leftSide U A C).Nonempty) :
  family bn m ⊆ rightSide U A C ∪ C := by
  intro v hv
  by_cases hc : v ∈ C
  · exact Finset.mem_union_right _ hc
  · apply Finset.mem_union_left
    refine Finset.mem_sdiff.2 ⟨family_subset hU m hm hv, ?_⟩
    intro hbad
    rcases Finset.mem_union.1 hbad with hl | hc'
    · exact ht ⟨v, Finset.mem_inter.2 ⟨hv, hl⟩⟩
    · exact hc hc'

def groupedFactor (κ : bn.Kernel R) (hloc : ∀ m, Local κ m) (S : Finset bn.V)
  (ms : Finset bn.M) (hS : ∀ m ∈ ms, family bn m ⊆ S) : Factor bn R where
  scope := S
  value x := ∏ m ∈ ms, κ m x (x (bn.target m))
  dependsOn x y h := by
    apply Finset.prod_congr rfl
    intro m hm
    rw [hloc m x y (fun v hv ⇒ h v (hS m hm (Finset.mem_insert_of_mem hv))),
      h (bn.target m) (hS m hm (Finset.mem_insert_self _ _))]

/-- **D-separation soundness.** The premise is graph path separation in the moralized ancestral
graph. The conclusion is the finite conditional-independence event identity. No numerical
independence or precomputed correct factor partition is assumed. -/
theorem d_separation_sound (κ : bn.Kernel R) (hclosed : bn.Closed) (ord : bn.TopoOrder)
  (hloc : ∀ m, Local κ m) (hnorm : ∀ m, Normalised κ m)
  (A B C : Finset bn.V) (hAC : Disjoint A C) (hBC : Disjoint B C)
  (hsep : DSeparated A B C) : ConditionalIndependent (joint κ) A B C := by
  classical
  let U := ancestors ((A ∪ B) ∪ C)
  have hU : UpstreamClosed U := ancestors_upstream _
  have hAU : A ⊆ U := (Finset.subset_union_left.trans Finset.subset_union_left).trans
    ↪ (subset_ancestors _)
  have hBU : B ⊆ U := (Finset.subset_union_right.trans Finset.subset_union_left).trans
    ↪ (subset_ancestors _)
  have hCU : C ⊆ U := Finset.subset_union_right.trans (subset_ancestors _)
  let L := leftSide U A C
  let R := rightSide U A C
  have hSides := queries_in_sides hAU hBU hAC hBC hsep
  have hLC : Disjoint L C := left_avoids U A C
  have hLR : Disjoint L R := Finset.disjoint_left.2 fun v hv hr ⇒
    (Finset.mem_sdiff.1 hr).2 (Finset.mem_union_left _ hv)
  have hRC : Disjoint R C := Finset.disjoint_left.2 fun v hr hv ⇒
    (Finset.mem_sdiff.1 hr).2 (Finset.mem_union_right _ hv)
  have hsplit : (L ∪ R) ∪ C = U := by
    have hLU : L ⊆ U := left_subset U A C
    apply Finset.Subset.antisymm
    · intro v hv

```

```

    rcases Finset.mem_union.1 hv with hv | hc
    • rcases Finset.mem_union.1 hv with hl | hr
      • exact hLU hl
      • exact (Finset.mem_sdiff.1 hr).1
    • exact hCU hc
  • intro v hv
  by_cases hc : v ∈ C
  • exact Finset.mem_union_right _ hc
  • apply Finset.mem_union_left
    by_cases hl : v ∈ L
    • exact Finset.mem_union_left _ hl
    • exact Finset.mem_union_right _ (Finset.mem_sdiff.2 (hv, by simp using And.intro hl hc))
let ms := Finset.univ.filter (fun m => bn.target m ∈ U)
let touches := fun m => (family bn m ∩ L).Nonempty
let ml := ms.filter touches
let mr := ms.filter (fun m => ¬touches m)
have hls : ∀ m ∈ ml, family bn m ⊆ L ∪ C := by
  intro m hm
  exact family_left hU m (Finset.mem_filter.1 (Finset.mem_filter.1 hm).1).2
  (Finset.mem_filter.1 hm).2
have hrs : ∀ m ∈ mr, family bn m ⊆ R ∪ C := by
  intro m hm
  exact family_right hU m (Finset.mem_filter.1 (Finset.mem_filter.1 hm).1).2
  (Finset.mem_filter.1 hm).2
let f := groupedFactor κ hloc (L ∪ C) ml hls
let g := groupedFactor κ hloc (R ∪ C) mr hrs
have hfac : (multiply f g).value = fun x => ∏ m ∈ ms, κ m x (x (bn.target m)) := by
  funext x
  exact Finset.prod_filter_mul_prod_filter_not ms touches _
have hmarg : ∀ x, marg Uc (joint κ) x = (multiply f g).value x := by
  intro x
  rw [hfac]
  exact marg_joint_upstream κ hclosed ord hloc hnorm U hU x
intro a b c
rw [eventMass_of_marginal (joint κ) (multiply f g).value U A B C hmarg hAU hBU hCU,
  eventMass_of_marginal (joint κ) (multiply f g).value U ∅ ∅ C hmarg
    (Finset.empty_subset _) (Finset.empty_subset _) hCU,
  eventMass_of_marginal (joint κ) (multiply f g).value U A ∅ C hmarg hAU (Finset.empty_subset _)
    hCU,
  eventMass_of_marginal (joint κ) (multiply f g).value U ∅ B C hmarg (Finset.empty_subset _) hBU
    hCU,
  ← hsplit]
exact separator_cross_product L R C A B f g hLR hLC hRC (Finset.Subset.refl _)
  (Finset.Subset.refl _) hSides.1 hSides.2 a b c

/-- Probability interpretation, with nonnegativity and normalization explicitly supplied. -/
theorem d_separation_sound_probability (κ : bn.Kernel ℝ) (hclosed : bn.Closed) (ord : bn.TopoOrder)
  (hloc : ∀ m, Local κ m) (hnorm : ∀ m, Normalised κ m) (hnonneg : ∀ m x a, 0 ≤ κ m x a)
  (A B C : Finset bn.V) (hAC : Disjoint A C) (hBC : Disjoint B C) (hsep : DSeparated A B C) :
  (∀ x, 0 ≤ joint κ x) ∧ (∑ x, joint κ x = 1) ∧ ConditionalIndependent (joint κ) A B C :=
  ⟨joint_nonneg κ hnonneg, sum_joint_eq_one κ hclosed ord hloc hnorm,
    d_separation_sound κ hclosed ord hloc hnorm A B C hAC hBC hsep⟩
end
end BayesianNetworksProofs.FinBayesNet

```

Conditional numerical error contracts

```

import BayesianNetworksProofs.Finite.Posterior
import Mathlib.Algebra.BigOperators.Field
import Mathlib.Tactic.FieldSimp
import Mathlib.Tactic.Linarith
import Mathlib.Tactic.Ring

```

These are quantitative hypotheses and conclusions, not a universal Float64/oracle identity. Finite product perturbations are bounded from entry errors. Normalization has an explicit positive-mass lower bound; small evidence mass amplifies error. A separate rounded-product model assumes local multiplication-error and range-preservation contracts and derives a whole-product bound. No IEEE implementation of those local contracts is assumed proved here.

```

set_option autoImplicit false

namespace BayesianNetworksProofs.Numerical

open FiniteDistribution
noncomputable section

variable {A : Type} [Fintype A]

def l1 (p q : A → ℝ) : ℝ := ∑ a, |p a - q a|

theorem l1_nonneg (p q : A → ℝ) : 0 ≤ l1 p q := Finset.sum_nonneg fun _ => abs_nonneg _

theorem mass_error (p q : A → ℝ) : |mass p - mass q| ≤ l1 p q := by
  rw [mass, mass, ← Finset.sum_sub_distrib]
  exact Finset.abs_sum_le_sum_abs _ _

theorem point_normalization_error (p q Z W : ℝ) (hq : 0 ≤ q) (hZ : 0 < Z) (hW : 0 < W) :
  |p / Z - q / W| ≤ |p - q| / Z + (q / W) * |W - Z| / Z := by
  have he : p / Z - q / W = (p - q) / Z + (q / W) * (W - Z) / Z := by
    field_simp
    ring
  rw [he]
  calc
    _ ≤ |(p - q) / Z| + |(q / W) * (W - Z) / Z| := abs_add_le _ _
    _ = _ := by
      rw [abs_div, abs_of_pos hZ, abs_div, abs_of_pos hZ, abs_mul,
        abs_of_nonneg (div_nonneg hq hW.le)]

/-- Exact normalization is L1-stable only relative to its mass. The comparison weight is
nonnegative and both masses are positive; no entrywise lower bound is required. -/
theorem normalization_l1 (p q : A → ℝ) (hq : ∀ a, 0 ≤ q a)
  (hZ : 0 < mass p) (hW : 0 < mass q) :
  l1 (fun a => p a / mass p) (fun a => q a / mass q) ≤ 2 * l1 p q / mass p := by
  have hmass : |mass q - mass p| ≤ l1 p q := by simpa only [abs_sub_comm] using mass_error p q
  have hsum : ∑ a, q a / mass q = 1 := by
    rw [← Finset.sum_div]
    exact div_self (ne_of_gt hW)
  calc
    _ ≤ ∑ a, (|p a - q a| / mass p + (q a / mass q) * |mass q - mass p| / mass p) :=
      Finset.sum_le_sum fun a => point_normalization_error _ _ _ (hq a) hZ hW
    _ = (l1 p q + |mass q - mass p|) / mass p := by
      rw [Finset.sum_add_distrib, ← Finset.sum_div, ← Finset.sum_div, ← Finset.sum_mul, hsum]
      simp [l1, add_div]
    _ ≤ (l1 p q + l1 p q) / mass p :=
      div_le_div_of_nonneg_right (add_le_add (le_refl _)) hmass hZ.le
    _ = _ := by ring

/-- A supplied absolute input error below a genuine evidence-mass floor guarantees positive
approximate mass and bounds posterior L1 error. This is not meaningful with η=0. -/
theorem posterior_error (p q : A → ℝ) (hq : ∀ a, 0 ≤ q a)
  (η ε : ℝ) (hη : 0 < η) (hfloor : η ≤ mass p)
  (herr : l1 p q ≤ ε) (hsmall : ε < η) :
  0 < mass q ∧ l1 (fun a => p a / mass p) (fun a => q a / mass q) ≤ 2 * ε / η := by
  have hZ : 0 < mass p := hη.trans_le hfloor
  have hm := (le_abs_self (mass p - mass q)).trans ((mass_error p q).trans herr)
  have hW : 0 < mass q := by linarith
  have hε : 0 ≤ ε := (l1_nonneg p q).trans herr
  refine ⟨hW, (normalization_l1 p q hq hZ hW).trans ?_⟩
  calc

```

```

- ≤ 2 * ε / mass p := div_le_div_of_nonneg_right
  (mul_le_mul_of_nonneg_left herr (by norm_num)) hZ.le
- ≤ _ := div_le_div_of_nonneg_left (mul_nonneg (by norm_num) hε) hη hfloor

theorem prod_unit_interval {I : Type} (S : Finset I) (p : I → ℝ)
  (hp : ∀ i ∈ S, 0 ≤ p i ∧ p i ≤ 1) : 0 ≤ ∏ i ∈ S, p i ∧ (∏ i ∈ S, p i) ≤ 1 := by
  classical
  induction S using Finset.induction_on with
  | empty ⇒ simp
  | @insert i S hi ih ⇒
    have hpI := hp i (Finset.mem_insert_self _ _)
    have hpS := ih (fun j hj ⇒ hp j (Finset.mem_insert_of_mem hj))
    rw [Finset.prod_insert hi]
    exact ⟨mul_nonneg hpI.1 hpS.1,
      (mul_le_mul hpI.2 hpS.2 hpS.1 zero_le_one).trans (by simp)⟩

theorem product_error {I : Type} (S : Finset I) (p q : I → ℝ)
  (hp : ∀ i ∈ S, 0 ≤ p i ∧ p i ≤ 1) (hq : ∀ i ∈ S, 0 ≤ q i ∧ q i ≤ 1) :
  |(∏ i ∈ S, p i) - ∏ i ∈ S, q i| ≤ ∑ i ∈ S, |p i - q i| := by
  classical
  induction S using Finset.induction_on with
  | empty ⇒ simp
  | @insert i S hi ih ⇒
    have pI := hp i (Finset.mem_insert_self _ _)
    have qI := hq i (Finset.mem_insert_self _ _)
    have hpS := fun j hj ⇒ hp j (Finset.mem_insert_of_mem hj)
    have hqS := fun j hj ⇒ hq j (Finset.mem_insert_of_mem hj)
    have pS := prod_unit_interval S p hpS
    rw [Finset.prod_insert hi, Finset.prod_insert hi, Finset.sum_insert hi]
    have he : p i * (∏ j ∈ S, p j) - q i * (∏ j ∈ S, q j) =
      (p i - q i) * (∏ j ∈ S, p j) + q i * ((∏ j ∈ S, p j) - ∏ j ∈ S, q j) := by ring
    rw [he]
    calc
    - ≤ |(p i - q i) * (∏ j ∈ S, p j)| + |q i * ((∏ j ∈ S, p j) - ∏ j ∈ S, q j)| :=
      abs_add_le _ _
    - = |p i - q i| * (∏ j ∈ S, p j) + q i * |(∏ j ∈ S, p j) - ∏ j ∈ S, q j| := by
      rw [abs_mul, abs_mul, abs_of_nonneg pS.1, abs_of_nonneg qI.1]
    - ≤ |p i - q i| + |(∏ j ∈ S, p j) - ∏ j ∈ S, q j| := by
      apply add_le_add
    · simp using mul_le_mul_of_nonneg_left pS.2 (abs_nonneg (p i - q i))
    · simp using mul_le_mul_of_nonneg_right qI.2
      (abs_nonneg ((∏ j ∈ S, p j) - ∏ j ∈ S, q j))
    - ≤ _ := add_le_add (le_refl _) (ih hpS hqS)

def roundedProduct (mulR : ℝ → ℝ → ℝ) : List ℝ → ℝ
| [] ⇒ 1
| x :: xs ⇒ mulR x (roundedProduct mulR xs)

def UnitInterval (x : ℝ) : Prop := 0 ≤ x ∧ x ≤ 1

theorem roundedProduct_range (mulR : ℝ → ℝ → ℝ)
  (hclosed : ∀ x y, UnitInterval x → UnitInterval y → UnitInterval (mulR x y))
  (xs : List ℝ) (hxs : ∀ x ∈ xs, UnitInterval x) : UnitInterval (roundedProduct mulR xs) := by
  induction xs with
  | nil ⇒ exact ⟨zero_le_one, le_refl⟩
  | cons x xs ih ⇒
    exact hclosed x _ (hxs x (List.mem_cons_self ..))
      (ih (fun y hy ⇒ hxs y (List.mem_cons_of_mem x hy)))

/-- An actual recursively rounded product has a linear absolute error bound, derived from
local multiplication contracts. IEEE arithmetic is not identified with mulR without proof. -/
theorem roundedProduct_error (mulR : ℝ → ℝ → ℝ) (δ : ℝ)
  (hclosed : ∀ x y, UnitInterval x → UnitInterval y → UnitInterval (mulR x y))
  (hround : ∀ x y, UnitInterval x → UnitInterval y → |mulR x y - x * y| ≤ δ)
  (xs : List ℝ) (hxs : ∀ x ∈ xs, UnitInterval x) :
  |roundedProduct mulR xs - xs.prod| ≤ xs.length * δ := by

```

```

induction xs with
| nil => simp [roundedProduct]
| cons x xs ih =>
  have hx := hxs x (List.mem_cons_self ..)
  have ht := fun y hy => hxs y (List.mem_cons_of_mem x hy)
  have hrange := roundedProduct_range mulR hclosed xs ht
  change |mulR x (roundedProduct mulR xs) - x * xs.prod| ≤ (xs.length + 1 : Nat) * δ
  calc
    _ ≤ |mulR x (roundedProduct mulR xs) - x * roundedProduct mulR xs| +
      |x * roundedProduct mulR xs - x * xs.prod| := abs_sub_le _ _
    _ ≤ δ + |roundedProduct mulR xs - xs.prod| := by
      apply add_le_add (hround x _ hx hrange)
      rw [← mul_sub, abs_mul, abs_of_nonneg hx.1]
      simp using mul_le_mul_of_nonneg_right hx.2 (abs_nonneg (roundedProduct mulR xs - xs.prod))
    _ ≤ δ + xs.length * δ := add_le_add (le_refl _) (ih ht)
    _ = _ := by push_cast; ring

theorem push_l1 {B : Type} [Fintype B] [DecidableEq B] (f : A → B) (p q : A → ℝ) :
  l1 (pushWeight f p) (pushWeight f q) ≤ l1 p q := by
  have he : ∀ b, pushWeight f p b - pushWeight f q b =
    ∑ a, if f a = b then p a - q a else 0 := by
    intro b
    unfold pushWeight
    rw [← Finset.sum_sub_distrib]
    apply Finset.sum_congr rfl
    intro a _
    split_ifs <=> simp
  unfold l1
  simp_rw [he]
  calc
    _ ≤ ∑ b, ∑ a, |if f a = b then p a - q a else 0| :=
      Finset.sum_le_sum fun b _ => Finset.abs_sum_le_sum_abs _ _
    _ = _ := by
      rw [Finset.sum_comm]
      apply Finset.sum_congr rfl
      intro a _
      change (∑ b : B, |if f a = b then p a - q a else 0|) = |p a - q a|
      simp [apply_ite]

open FinBayesNet

theorem evidence_l1 {bn : FinBayesNet} (p q : bn.Assignment → ℝ)
  (E : Finset bn.V) (observed : bn.Assignment) :
  l1 (likelihoodWeight p E observed) (likelihoodWeight q E observed) ≤ l1 p q := by
  apply Finset.sum_le_sum
  intro x _
  by_cases he : agrees E observed x
  · have hi : indicator E observed x = 1 := if_pos he
    simp only [likelihoodWeight, hi, mul_one, le_refl]
  · have hi : indicator E observed x = 0 := if_neg he
    simp only [likelihoodWeight, hi, mul_zero, sub_self, abs_zero]
  exact abs_nonneg _

/-- Entrywise CPT perturbations imply a joint-weight L1 budget. This is an explicit
worst-case finite bound, not an assertion about machine arithmetic. -/
theorem joint_l1_error {bn : FinBayesNet} (κ κ' : bn.Kernel ℝ)
  (hκ : ∀ m x a, 0 ≤ κ m x a ∧ κ m x a ≤ 1)
  (hκ' : ∀ m x a, 0 ≤ κ' m x a ∧ κ' m x a ≤ 1)
  (ε : bn.M → ℝ) (herr : ∀ m x a, |κ m x a - κ' m x a| ≤ ε m) :
  l1 (joint κ) (joint κ') ≤ (Fintype.card bn.Assignment : ℝ) * ∑ m, ε m := by
  calc
    _ ≤ ∑ _x : bn.Assignment, ∑ m, ε m := by
      apply Finset.sum_le_sum
      intro x _
      exact (product_error Finset.univ (fun m => κ m x (x (bn.target m)))
        (fun m => κ' m x (x (bn.target m))) (fun m _ => hκ m x _) (fun m _ => hκ' m x _)).trans

```

```

      (Finset.sum_le_sum fun m _ => herr m x _)
    _ = _ := by simp

/-- A complete query-posterior input-perturbation contract. The exact VE/JT results inherit
this bound through their separately proved equality to the joint/query semantics. -/
theorem kernel_query_posterior_error {bn : FinBayesNet} (κ κ' : bn.Kernel R)
  (hκ : ∀ m x a, 0 ≤ κ m x a ∧ κ m x a ≤ 1)
  (hκ' : ∀ m x a, 0 ≤ κ' m x a ∧ κ' m x a ≤ 1)
  (ε : bn.M → R) (herr : ∀ m x a, |κ m x a - κ' m x a| ≤ ε m)
  (Q E : Finset bn.V) (observed : bn.Assignment) (η : R) (hη : 0 < η)
  (hfloor : η ≤ mass (likelihoodWeight (joint κ) E observed))
  (hsmall : (Fintype.card bn.Assignment : R) * (∑ m, ε m) < η) :
  0 < mass (likelihoodWeight (joint κ') E observed) ∧
  l1
    (fun q => pushWeight (restrictTo Q) (likelihoodWeight (joint κ) E observed) q /
      mass (likelihoodWeight (joint κ) E observed))
    (fun q => pushWeight (restrictTo Q) (likelihoodWeight (joint κ') E observed) q /
      mass (likelihoodWeight (joint κ') E observed)) ≤
    2 * ((Fintype.card bn.Assignment : R) * ∑ m, ε m) / η := by
have hnonneg := likelihoodWeight_nonneg (joint κ') (joint_nonneg κ' (fun m x a => (hκ' m x a).1)) E
  ↪ observed
have herror := (push_l1 (restrictTo Q) (likelihoodWeight (joint κ) E observed)
  (likelihoodWeight (joint κ') E observed)).trans
  ((evidence_l1 (joint κ) (joint κ') E observed).trans (joint_l1_error κ κ' hκ hκ' ε herr))
have h := posterior_error _ _ (pushWeight_nonneg _ _ hnonneg) η _ hη
  (by simpa only [mass_pushWeight] using hfloor) herror hsmall
simpa only [mass_pushWeight] using h

end
end BayesianNetworksProofs.Numerical

```

Nonvacuity and boundary checks

```

import BayesianNetworksProofs.Finite.ReferenceTables
import BayesianNetworksProofs.Finite.JunctionTree
import BayesianNetworksProofs.Finite.DSeparation
import BayesianNetworksProofs.Finite.NumericalContracts
import Mathlib.Tactic.FinCases

```

The raw certificate has shuffled state/input row IDs and two repeated parent slots. The junction example has three disconnected components and impossible evidence in a component other than the query. The graphical example is a three-variable fork: its children are d-separated given their parent, by an actual edgeless cut-moral graph argument.

```

set_option autoImplicit false

namespace BayesianNetworksProofs.RefinementExamples

open FinBayesNet FinBayesNet.Factor Junction

@[reducible] def raw : Raw.Network where
  nv := 2
  ns := 4
  nm := 2
  ni := 2
  vars v := ⟨if v = 0 then "A" else "B", .none⟩
  states s := if s = 0 then ⟨0, 1, "a1"⟩ else if s = 1 then ⟨0, 0, "a0"⟩ else
    if s = 2 then ⟨1, 0, "b0"⟩ else ⟨1, 1, "b1"⟩
  mechanisms m := ⟨m, if m = 0 then "prior" else "child",
    .named (if m = 0 then "prior" else "child")⟩
  inputs i := ⟨1, 0, if i = 0 then 1 else 0⟩
  rank := id

```

```

def bank : Raw.Bank := [
  ⟨.named "prior", ⟨[], 2, [⟨[], [1/2, 1/2]⟩], [], ["a0", "a1"]⟩⟩,
  ⟨.named "child", ⟨[2,2], 2,
    [⟨[0,0], [1,0]⟩, ⟨[0,1], [0,1]⟩, ⟨[1,0], [1/2,1/2]⟩, ⟨[1,1], [0,1]⟩],
    [["a0","a1"], ["a0","a1"], ["b0","b1"]]⟩⟩]

theorem raw_structure_checked : raw.check = true := by decide
theorem raw_references_checked : raw.checkReady bank = true := by decide +kernel
theorem raw_normalization_checked : raw.checkNormalized bank = true := by decide +kernel
theorem raw_two_slots : raw.inputCount 1 = 2 := by decide

def badPositions : Raw.Network := {raw with inputs := fun _ => ⟨1, 0, 0⟩}
theorem bad_positions_rejected : badPositions.check = false := by decide +kernel

def badLabels : Raw.Bank := bank.map fun b =>
  if b.ref = .named "prior" then {b with table := {b.table with outputLabels := ["wrong", "a1"]}}
  else b
theorem bad_labels_rejected : raw.checkReady badLabels = false := by decide +kernel

def unnormalized : Raw.Bank := bank.map fun b =>
  if b.ref = .named "prior" then {b with table := {b.table with columns := [⟨[], [2/3, 2/3]⟩]}}
  else b
theorem unnormalized_data_interpretable : raw.checkReady unnormalized = true := by decide +kernel
theorem unnormalized_probability_rejected : raw.checkNormalized unnormalized = false := by decide
  ← +kernel
theorem unresolved_reference_rejected : raw.checkReady [] = false := by decide +kernel

noncomputable section

def rawValid : raw.Valid := (raw.check_iff).1 raw_structure_checked

theorem raw_diagonal (i j : Fin (raw.inputCount 1)) (x : (raw.compile rawValid).Assignment) :
  (raw.gather rawValid 1 x i).val = (raw.gather rawValid 1 x j).val := by
  apply raw.repeated_slots_diagonal rawValid
  simp [Raw.Network.slotVariable, raw]

theorem raw_compiled_probability :
  ∑ x, joint (raw.kernel rawValid (raw.resolvedCPT rawValid bank)) x = 1 :=
  raw.checked_reference_normalization bank raw_structure_checked raw_references_checked
  ← raw_normalization_checked

@[reducible] def fork : FinBayesNet where
  V := Fin 3
  M := Fin 3
  states _ := Bool
  target := id
  parents m := if m = 0 then ∅ else {0}

theorem fork_closed : fork.Closed := ⟨Function.injective_id, Function.surjective_id⟩

def fork_order : fork.TopoOrder where
  order := [0,1,2]
  nodup := by decide
  complete := by decide
  parents_before := by decide
  no_self := by decide

def forkKernel : fork.Kernel R := fun (m : Fin 3) (x : Fin 3 → Bool) (a : Bool) =>
  if m = 0 then 1/2 else if a = x 0 then 1 else 0

theorem fork_local : ∀ m, Local forkKernel m := by
  intro m x y h
  funext a
  by_cases hm : m = 0
  · simp [forkKernel, hm]
  · have h0 := h 0 (by simp [fork, hm])

```

```

    simp [forkKernel, hm, h0]

theorem fork_normalized : ∀ m, Normalised forkKernel m := by
  intro m x
  change (∑ a : Bool, forkKernel m x a) = 1
  by_cases hm : m = 0 <|> simp [forkKernel, hm]

theorem fork_nonnegative : ∀ m x a, 0 ≤ forkKernel m x a := by
  intro m x a
  unfold forkKernel
  split_ifs <|> norm_num

theorem fork_cut_edgeless (U : Finset fork.V) : cutMoral U {0} = 1 := by
  ext v w
  constructor
  · intro he
    obtain ⟨hne, m, _, hv, hw⟩ := he.1
    have hs : ∀ m : fork.M, ∀ v w, v ∈ family fork m \ {0} → w ∈ family fork m \ {0} → v = w := by
      decide
    exact hne (hs m v w (Finset.mem_sdiff.2 ⟨hv, he.2.1⟩) (Finset.mem_sdiff.2 ⟨hw, he.2.2⟩))
  · intro he
    exact False.elim he

theorem fork_graph_separated : DSeparated (bn := fork) {1} {2} {0} := by
  intro a ha b hb hpath
  have ha' : a = 1 := by simp using ha
  have hb' : b = 2 := by simp using hb
  subst a
  subst b
  rw [fork_cut_edgeless, SimpleGraph.reachable_bot] at hpath
  exact (by decide : (1 : Fin 3) ≠ 2) hpath

theorem fork_conditional_independence :
  ConditionalIndependent (joint forkKernel) {1} {2} {0} :=
  d_separation_sound forkKernel fork_closed fork_order fork_local fork_normalized
  {1} {2} {0} (by decide) (by decide) fork_graph_separated

def forkFactor (m : fork.M) : Factor fork R := Factor.mechanism forkKernel fork_local m

def connectedTree : Tree fork.V fork.M :=
  .branch {0,1} [0,1] (.leaf {0,2} [2]) (.leaf {0,1} [])

theorem connected_tree_valid : Good forkFactor connectedTree := by
  simp [Good, connectedTree, forkFactor, localFactor, combine, multiply, unit,
    Factor.mechanism, fork, Tree.vars, Tree.bag]
  decide

theorem connected_assignment : connectedTree.indices.Perm (Finset.univ : Finset fork.M).toList := by
  apply (List.perm_ext_iff_of_nodup (by decide) (Finset.nodup_toList _)).2
  intro i
  fin_cases i <|> simp [connectedTree, Tree.indices]

theorem connected_tree_check : checkGood forkFactor connectedTree = true := by decide +kernel

theorem connected_assignment_check : checkAssignment connectedTree = true := by decide +kernel

theorem connected_variables : ∀ v : fork.V, ∃ m, v ∈ (forkFactor m).scope :=
  fun v => ⟨v, Finset.mem_insert_self _ _⟩

theorem nonempty_separator_belief_exists :
  ∃ b ∈ calibrate forkFactor connectedTree, b.1 = ({0,2} : Finset fork.V) := by
  simp [calibrate, connectedTree, prepare, distribute]

theorem connected_beliefs_match_ve (b : Finset fork.V × Factor fork R)
  (hb : b ∈ calibrate forkFactor connectedTree) (vs : List fork.V)
  (hv : vs.Nodup) (hs : vs.toFinset = b.1c) (x : fork.Assignment) :

```

```

    b.2.value x = Factor.product (Factor.eliminateAll (Finset.univ.toList.map forkFactor) vs) x :=
calibrate_eq_ve forkFactor connectedTree connected_tree_valid connected_assignment
connected_variables b hb vs hv hs x

@[reducible] def roots : FinBayesNet := {fork with parents := fun _ => ∅}

def rootKernel : roots.Kernel R := fun (m : Fin 3) (l : Fin 3 → Bool) (a : Bool) =>
  if m = 1 then (if a = false then 1 else 0) else 1/2

theorem root_local : ∀ m, Local rootKernel m := fun _ _ _ => rfl

def observed : roots.Assignment := fun _ => true

def forestFactor (i : Fin 4) : Factor roots R :=
  if h : i.val < 3 then Factor.mechanism rootKernel root_local ⟨i.val, h⟩
  else evidenceFactor {1} observed

def forestTree : Tree roots.V (Fin 4) :=
  graft ∅ [] [.leaf {0} [0], .leaf {1} [1,3], .leaf {2} [2]]

theorem forest_valid : Good forestFactor forestTree := by
  apply forest_good
  · intro t ht
    simp only [List.mem_cons, List.not_mem_nil, or_false] at ht
    rcases ht with rfl | rfl | rfl <|>
    · simp [Good, localFactor, combine, multiply, unit, forestFactor, Factor.mechanism,
      evidenceFactor, roots, fork]
  · decide

theorem forest_assignment : forestTree.indices.Perm (Finset.univ : Finset (Fin 4)).toList := by
  apply (List.perm_ext_iff_of_nodup (by decide) (Finset.nodup_toList _)).2
  intro i
  fin_cases i <|> simp [forestTree, graft_indices, Tree.indices]

theorem forest_variables : ∀ v : roots.V, ∃ i, v ∈ (forestFactor i).scope := by
  intro v
  fin_cases v
  · exact ⟨0, by decide⟩
  · exact ⟨1, by decide⟩
  · exact ⟨2, by decide⟩

theorem forest_nonnegative : ∀ i x, 0 ≤ (forestFactor i).value x := by
  intro i x
  fin_cases i
  all_goals simp [forestFactor, Factor.mechanism, rootKernel, evidenceFactor, indicator, agrees]
  all_goals split_ifs <|> norm_num

theorem product_zero_of_mem {fs : List (Factor roots R)} {f : Factor roots R}
  (hf : f ∈ fs) (x : roots.Assignment) (hz : f.value x = 0) : Factor.product fs x = 0 := by
  induction fs with
  | nil => simp at hf
  | cons g fs ih =>
    rw [Factor.product_cons]
    rcases List.mem_cons.1 hf with rfl | hf
    · rw [hz, zero_mul]
    · rw [ih hf, mul_zero]

theorem forest_zero_mass (x : roots.Assignment) :
  Factor.product ((Finset.univ : Finset (Fin 4)).toList.map forestFactor) x = 0 := by
  cases hx : x 1
  · apply product_zero_of_mem (f := forestFactor 3)
    · exact List.mem_map.2 ⟨3, Finset.mem_toList.2 (Finset.mem_univ _), rfl⟩
    · simp [forestFactor, evidenceFactor, indicator, agrees, observed, hx]
  · apply product_zero_of_mem (f := forestFactor 1)
    · exact List.mem_map.2 ⟨1, Finset.mem_toList.2 (Finset.mem_univ _), rfl⟩
    · simp [forestFactor, Factor.mechanism, rootKernel, roots, fork, hx]

```

```

/-- Even a query in the other disconnected component must reject the globally impossible event. -/
theorem disconnected_query_rejects (b : Finset roots.V × Factor roots R)
  (hb : b ∈ calibrate forestFactor forestTree) (Q : Finset roots.V) (hQ : Q ⊆ b.1)
  (base : roots.Assignment) :
  queryPosterior forestFactor forestTree forest_valid forest_assignment forest_variables
    forest_nonnegative b hb Q hQ base = none := by
  rw [queryPosterior_none_iff]
  simp [forest_zero_mass]

end
end BayesianNetworksProofs.RefinementExamples

```

Correct rounding to binary64: `_nearest_binary64`, `_rational_exponent` and `_dyadic`

```

import Mathlib.Data.Int.Log
import Mathlib.Data.Rat.Floor
import Mathlib.Data.Nat.Log
import Mathlib.Algebra.Order.Field.Power
import Mathlib.Tactic.Linarith
import Mathlib.Tactic.NormNum
import Mathlib.Tactic.Positvity
import Mathlib.Tactic.FieldSimp
import Mathlib.Tactic.Ring

```

ADR 0016 makes the exact evidence-mass fallback of `BayesianNetworks` and `BayesianNetworkInference`, and the exact decision elimination of `InfluenceDiagrams`, round an exact rational once with `_nearest_binary64` from `BayesianNetworks/src/exact_rounding.jl`. This module states IEEE 754 binary64 round-to-nearest-ties-to-even of a rational, transcribes the three Julia functions over $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$, and proves the transcriptions correct.

The format. A word is a natural number below 2^{64} read as a `UInt64`: sign bit 63, an 11-bit biased exponent field (bits 52 to 62) and a 52-bit fraction. A field of 0 is a zero or subnormal, $2^{-1022} * (\text{fraction} / 2^{52})$; a field of 1 to 2046 is normal, $2^{(\text{field} - 1023)} * (1 + \text{fraction} / 2^{52})$; a field of 2047 is an infinity (fraction 0) or a NaN. `decode` maps a word to that extended value, keeping the sign of a zero.

The specification (`RoundsTo q w`, IEEE 754-2019 §4.3.1 and §7.4). A rational of magnitude at least $2^{1024} - 2^{970} (b^{\text{emax}} * (b - b^{\text{p}}) / 2)$, the midpoint above the largest finite value) rounds to the infinity of its sign. Any other rational rounds to a finite word whose value is no farther from it than the value of any finite word; when a finite word of a different value is equally near, the result's significand is even; and the result has the sign of the rational, so a result of zero is `-0.0` for a negative rational and `+0.0` otherwise.

The transcription. `nearestBinary64` follows `_nearest_binary64` line by line with the same branches and constants (1023, -1075, 52, -1074, -1022, `0x8000000000000000`, `0x7ff0000000000000`). Julia's `Rational{BigInt}` is normalized with a positive denominator, as $\mathbb{Q}$ is, so numerator and denominator are `Rat.num` and `Rat.den`. `BigInt` shifts are `<<<` on $\mathbb{N}$ (multiplication by a power of two), `divrem` of nonnegative integers is `/` and `%`, `isodd` is `% 2 = 1`, `ndigits(n; base = 2)` is `Nat.log 2 n + 1` (1 at 0, as in Julia), and the `UInt64` conversions, `<<` and `|` act on values the proofs show to be in range, so the word is a natural number below 2^{64} . `reinterpret(Float64, word)` is `decode`. `decompose` follows `Base.decompose (::Float64)` (Julia 1.12, `base/float.jl`) on the word, and `dyadic` is `_dyadic`, with `none` for the `ArgumentError` it throws on a non-finite value.

What is proved.

- `rationalExponent_eq_log`: for $n, d > 0$, `rationalExponent n d =  $\lfloor \log_2 (n / d) \rfloor$` (`Int.log 2`), with the two-sided bound in `rationalExponent_spec`.
- `nearestBinary64_roundsTo`: for every rational q , `RoundsTo q (nearestBinary64 q)`. This is the full range: zero (to `+0.0`), negative rationals, underflow to a signed zero, subnormals, the rounding of a subnormal up to the least normal, the carry into the exponent when the mantissa rounds up to 2^{53} , and overflow exactly at $2^{1024} - 2^{970}$.
- `dyadic_value`: on a finite word, `dyadic` returns the integer $\pm \text{significand}$ and exponent `max field 1 - 1075`, whose value $n * 2^p$ is the word's value; `dyadic_none_iff`: it fails exactly on the infinities and

NaNs. A negative zero gives $n = 0$.

- `nearestBinary64_value`: rounding a finite word's value returns that word, except that both zeros return $+0.0$, which the fallback's bit-exact tests rely on.

What is not proved. These are theorems about the transcribed algorithm on mathematical integers and rationals. They do not cover Julia's execution of the source, GMP's `BigInt` arithmetic, `reinterpret`, or that the transcription matches the Julia text; that last step is a reading, recorded in `docs/LEAN-JULIA-DISCREPANCIES-2026-09-30.md`. The specification does not include uniqueness or monotonicity of rounding.

```
set_option autoImplicit false
```

```
namespace BayesianNetworksProofs.Binary64
```

Binary64 words

```
-- The sign bit (bit 63) of a word. -/
def signBit (w : ℕ) : Bool := decide (w / 2 ^ 63 % 2 = 1)

-- The 11-bit biased exponent field (bits 52 to 62). -/
def field (w : ℕ) : ℕ := w / 2 ^ 52 % 2 ^ 11

-- The 52-bit fraction (trailing significand) field. -/
def fraction (w : ℕ) : ℕ := w % 2 ^ 52

-- A 64-bit word encoding a finite number: neither an infinity nor a NaN. -/
def IsFinite (w : ℕ) : Prop := w < 2 ^ 64 ∧ field w ≠ 2047

-- The magnitude of a finite word: subnormal (and zero) for a field of '0', normal otherwise. -/
def magnitude (w : ℕ) : ℚ :=
  if field w = 0 then (2 : ℚ) ^ (-1022 : ℤ) * (fraction w / 2 ^ 52)
  else (2 : ℚ) ^ ((field w : ℤ) - 1023) * (1 + fraction w / 2 ^ 52)

-- The signed value of a finite word; both zeros have value '0'. -/
def value (w : ℕ) : ℚ := if signBit w then -magnitude w else magnitude w

-- The integral significand: the fraction, with the hidden bit '2 ^ 52' for a normal word.
Its parity is the parity of the significand's last digit, which ties to even compares. -/
def significand (w : ℕ) : ℕ := if field w = 0 then fraction w else 2 ^ 52 + fraction w

-- A binary64 datum: a finite number with its sign (so '-0.0' is 'finite true 0'), an
infinity, or a NaN. -/
inductive Extended where
| finite (negative : Bool) (x : ℚ)
| infinity (negative : Bool)
| nan
deriving DecidableEq

-- 'reinterpret(Float64, w)', as the datum it encodes. -/
def decode (w : ℕ) : Extended :=
  if field w = 2047 then (if fraction w = 0 then .infinity (signBit w) else .nan)
  else .finite (signBit w) (value w)
```

The specification

```
-- '2 ^ 1024 - 2 ^ 970': the midpoint between the largest finite binary64 value
'2 ^ 53 - 1' * 2 ^ 971' and '2 ^ 1024'. A rational of at least this magnitude overflows. -/
def overflowThreshold : ℚ := 2 ^ 1024 - 2 ^ 970

-- 'w' is the binary64 word that IEEE 754 round-to-nearest, ties-to-even, delivers for the
rational 'q': overflow to the infinity of 'q's sign at or beyond 'overflowThreshold';
otherwise a finite word with 'q's sign, at least as near 'q' as every finite word, and with an
even significand when a finite word of a different value is equally near. -/
structure RoundsTo (q : ℚ) (w : ℕ) : Prop where
  word : w < 2 ^ 64
```

```

overflow : overflowThreshold ≤ |q| → decode w = .infinity (decide (q < 0))
finite : |q| < overflowThreshold → field w ≠ 2047
sign : |q| < overflowThreshold → signBit w = decide (q < 0)
nearest : |q| < overflowThreshold → ∀ w', IsFinite w' → |value w - q| ≤ |value w' - q|
tiesToEven : |q| < overflowThreshold → ∀ w', IsFinite w' →
  |value w' - q| = |value w - q| → value w' ≠ value w → Even (significand w)

```

The transcription of src/exact_rounding.jl

```

/-- Julia's 'ndigits(n; base = 2)': the number of binary digits of 'n', and '1' for '0'. -/
def ndigits2 (n : N) : N := Nat.log 2 n + 1

/-- '_rational_exponent(n, d)': -/
def rationalExponent (n d : N) : Z :=
  let exponent : Z := (ndigits2 n : Z) - ndigits2 d
  let below : Bool :=
    if exponent ≥ 0 then decide (n < d <<< exponent.toNat)
    else decide (n <<< (-exponent).toNat < d)
  if below then exponent - 1 else exponent

/-- '_nearest_binary64(value::Rational{BigInt})', returning the word that Julia reinterprets
as a 'Float64'. -/
def nearestBinary64 (value : ℚ) : N :=
  if value = 0 then 0 else
  let sign_bits : N := if value < 0 then 0x8000000000000000 else 0
  let n := value.num.natAbs
  let d := value.den
  let exponent := rationalExponent n d
  if exponent > 1023 then sign_bits ||| 0x7ff0000000000000 else
  if exponent < -1075 then sign_bits else
  let shift : Z := max (exponent - 52) (-1074)
  let numerator_ := if shift < 0 then n <<< (-shift).toNat else n
  let denominator_ := if shift > 0 then d <<< shift.toNat else d
  let mantissa := numerator_ / denominator_
  let remainder := numerator_ % denominator_
  let twice := 2 * remainder
  let mantissa :=
    if twice > denominator_ ∨ (twice = denominator_ ∧ mantissa % 2 = 1) then mantissa + 1
    else mantissa
  let hidden_bit := 1 <<< 52
  let word :=
    if mantissa < hidden_bit then mantissa
    else
    let field := max exponent (-1022) + 1023
    let (mantissa, field) :=
      if mantissa = 2 * hidden_bit then (hidden_bit, field + 1) else (mantissa, field)
    (field.toNat <<< 52) ||| (mantissa - hidden_bit)
  sign_bits ||| word

/-- 'Base.decompose(x::Float64)' on the word of 'x': '(num, pow, den)' with 'x = num * 2 ^ pow
/ den', '(0, 0, 0)' for a NaN and '(±1, 0, 0)' for an infinity. -/
def decompose (w : N) : Z × Z × Z :=
  if field w = 2047 ∧ fraction w ≠ 0 then (0, 0, 0) else
  if field w = 2047 then ((if signBit w then -1 else 1), 0, 0) else
  let s := w &&& 0x000fffffffffffff
  let e := (w &&& 0x7ff0000000000000) >>> 52
  let s := s ||| ((if e ≠ 0 then 1 else 0) <<< 52)
  let d : Z := if signBit w then -1 else 1
  ((s : Z), (e : Z) - 1075 + (if e = 0 then 1 else 0), d)

/-- '_dyadic(x)': '(BigInt(num) * sign(den), Int(pow))', and 'none' where Julia throws the
'ArgumentError' for a non-finite 'x'. -/
def dyadic (w : N) : Option (Z × Z) :=
  if field w = 2047 then none

```

```

else
  let (num, pow, den) := decompose w
  some (num * Int.sign den, pow)

```

_rational_exponent is the binary logarithm

```

theorem rationalExponent_spec (n d : ℕ) (hn : 0 < n) (hd : 0 < d) :
  (2 : ℚ) ^ rationalExponent n d ≤ (n : ℚ) / d ∧
  (n : ℚ) / d < (2 : ℚ) ^ (rationalExponent n d + 1) := by
  have hn1 := Nat.pow_log_le_self 2 hn.ne'
  have hn2 := Nat.lt_pow_succ_log_self (by norm_num : 1 < 2) n
  have hd1 := Nat.pow_log_le_self 2 hd.ne'
  have hd2 := Nat.lt_pow_succ_log_self (by norm_num : 1 < 2) d
  set a := Nat.log 2 n
  set b := Nat.log 2 d
  have hdq : (0 : ℚ) < d := by exact_mod_cast hd
  have hnq1 : (2 : ℚ) ^ a ≤ n := by exact_mod_cast hn1
  have hnq2 : (n : ℚ) < 2 ^ (a + 1) := by exact_mod_cast hn2
  have hdq1 : (2 : ℚ) ^ b ≤ d := by exact_mod_cast hd1
  have hdq2 : (d : ℚ) < 2 ^ (b + 1) := by exact_mod_cast hd2
  -- 'below' is the comparison 'n / d < 2 ^ exponent'.
  have hbelow : ∀ E : ℤ, ((if E ≥ 0 then decide (n < d <<< E.toNat)
    else decide (n <<< (-E).toNat < d)) = true) ↔ (n : ℚ) / d < (2 : ℚ) ^ E := by
    intro E
    rw [div_lt_iff₀ hdq]
    split_ifs with h
    · have hk : (2 : ℚ) ^ E.toNat = 2 ^ E := by
        rw [← zpow_natCast, Int.toNat_of_nonneg h]
      rw [decide_eq_true_iff, Nat.shiftLeft_eq, ← Nat.cast_lt (α := ℚ)]
      push_cast
      rw [hk, mul_comm]
    · have hk : (2 : ℚ) ^ (-E).toNat = 2 ^ (-E) := by
        rw [← zpow_natCast, Int.toNat_of_nonneg (by omega)]
      rw [decide_eq_true_iff, Nat.shiftLeft_eq, ← Nat.cast_lt (α := ℚ)]
      push_cast
      rw [hk]
      have h2 : (0 : ℚ) < 2 ^ (-E) := by positivity
      have h3 : (2 : ℚ) ^ E * 2 ^ (-E) = 1 := by
        rw [← zpow_add₀ two_ne_zero]; simp
      constructor
      · intro hlt
        have := mul_lt_mul_of_pos_right hlt (show (0 : ℚ) < 2 ^ E by positivity)
        calc (n : ℚ) = n * 2 ^ (-E) * 2 ^ E := by
          rw [mul_assoc, mul_comm (2 ^ (-E) : ℚ), h3, mul_one]
        _ < d * 2 ^ E := this
        _ = _ := by ring
      · intro hlt
        have := mul_lt_mul_of_pos_right hlt h2
        calc (n : ℚ) * 2 ^ (-E) < 2 ^ E * d * 2 ^ (-E) := this
        _ = d := by rw [mul_comm _ (d : ℚ), mul_assoc, h3, mul_one]
  have hlo : (2 : ℚ) ^ ((a : ℤ) - b - 1) ≤ (n : ℚ) / d := by
    rw [le_div_iff₀ hdq]
    calc (2 : ℚ) ^ ((a : ℤ) - b - 1) * d ≤ 2 ^ ((a : ℤ) - b - 1) * 2 ^ ((b : ℤ) + 1) := by
      gcongr; exact_mod_cast hdq2.le
    _ = 2 ^ a := by rw [← zpow_add₀ two_ne_zero, ← zpow_natCast]; congr 1; ring
    _ ≤ n := hnq1
  have hhi : (n : ℚ) / d < (2 : ℚ) ^ ((a : ℤ) - b + 1) := by
    rw [div_lt_iff₀ hdq]
    calc (n : ℚ) < 2 ^ (a + 1) := hnq2
    _ = 2 ^ ((a : ℤ) - b + 1) * 2 ^ (b : ℤ) := by
      rw [← zpow_add₀ two_ne_zero, ← zpow_natCast]; congr 1; push_cast; ring
    _ ≤ 2 ^ ((a : ℤ) - b + 1) * d := by
      gcongr; rw [zpow_natCast]; exact hdq1
  unfold rationalExponent ndigits2

```

```

simp only []
have hE : ((a + 1 : ℕ) : ℤ) - ((b + 1 : ℕ) : ℤ) = (a : ℤ) - b := by push_cast; ring
rw [hE]
by_cases hb : (n : ℚ) / d < (2 : ℚ) ^ ((a : ℤ) - b)
• rw [if_pos ((hbelow _).mpr hb)]
  exact ⟨hlo, by simp using hb⟩
• rw [if_neg (fun h => hb ((hbelow _).mp h))]
  exact ⟨not_lt.mp hb, hhi⟩

/-- '_rational_exponent(n, d)' is 'log₂ (n / d)' for positive 'n' and 'd'. -/
theorem rationalExponent_eq_log (n d : ℕ) (hn : 0 < n) (hd : 0 < d) :
  rationalExponent n d = Int.log 2 ((n : ℚ) / d) := by
  obtain ⟨h1, h2⟩ := rationalExponent_spec n d hn hd
  have hpos : (0 : ℚ) < n / d := by positivity
  have l1 := Int.zpow_log_le_self (b := 2) (by norm_num) hpos
  have l2 := Int.lt_zpow_succ_log_self (b := 2) (by norm_num) ((n : ℚ) / d)
  push_cast at l1 l2
  have u1 := (zpow_lt_zpow_iff_right (by norm_num : (1 : ℚ) < 2)).mp (lt_of_le_of_lt h1 l2)
  have u2 := (zpow_lt_zpow_iff_right (by norm_num : (1 : ℚ) < 2)).mp (lt_of_le_of_lt l1 h2)
  omega

```

Rounding the scaled quotient to an integer

```

/-- The integer rounding step of '_nearest_binary64': the quotient, plus one when twice the
remainder exceeds the divisor, or equals it and the quotient is odd. -/
def roundHalfEven (N D : ℕ) : ℕ :=
  let mantissa := N / D
  let remainder := N % D
  let twice := 2 * remainder
  if twice > D ∨ (twice = D ∧ mantissa % 2 = 1) then mantissa + 1 else mantissa

private theorem scaled_abs (N D : ℕ) (hD : 0 < D) (x : ℚ) :
  |x - (N : ℚ) / D| = |x * D - N| / D := by
  have hDq : (0 : ℚ) < D := by exact_mod_cast hD
  rw [show x - (N : ℚ) / D = (x * D - N) / D by field_simp, abs_div, abs_of_pos hDq]

/-- Lower bounds on the scaled distance from an integer 'k' to 'N / D', by the position of 'k'
relative to the quotient 'm'. -/
private theorem grid_bounds (D m r k : ℚ) (hD : 0 < D) (hr : 0 ≤ r) :
  (k ≤ m - 1 → D + r ≤ |k * D - (D * m + r)|) ∧
  (k ≤ m → r ≤ |k * D - (D * m + r)|) ∧
  (k = m → |k * D - (D * m + r)| = r) ∧
  (k = m + 1 → |k * D - (D * m + r)| = |D - r|) ∧
  (m + 1 ≤ k → D - r ≤ |k * D - (D * m + r)|) ∧
  (m + 2 ≤ k → 2 * D - r ≤ |k * D - (D * m + r)|) := by
  refine ⟨fun h => ?_, fun h => ?_, fun h => ?_, fun h => ?_, fun h => ?_, fun h => ?_⟩
  • have := mul_le_mul_of_nonneg_right h hD.le
    exact le_abs.mpr (Or.inr (by nlinarith))
  • have := mul_le_mul_of_nonneg_right h hD.le
    exact le_abs.mpr (Or.inr (by nlinarith))
  • rw [h, show m * D - (D * m + r) = -r by ring, abs_neg, abs_of_nonneg hr]
  • rw [h]; congr 1; ring
  • have := mul_le_mul_of_nonneg_right h hD.le
    exact le_abs.mpr (Or.inl (by nlinarith))
  • have := mul_le_mul_of_nonneg_right h hD.le
    exact le_abs.mpr (Or.inl (by nlinarith))

/-- 'roundHalfEven N D' is an integer nearest 'N / D', its tie goes to the even integer, and it
is the quotient or the quotient plus one. -/
theorem roundHalfEven_spec (N D : ℕ) (hD : 0 < D) :
  (∀ k : ℤ, |(roundHalfEven N D : ℚ) - N / D| ≤ |(k : ℚ) - N / D|) ∧
  (∀ k : ℤ, |(k : ℚ) - N / D| = |(roundHalfEven N D : ℚ) - N / D| →
    k ≠ (roundHalfEven N D : ℤ) → Even (roundHalfEven N D)) ∧
  (N / D ≤ roundHalfEven N D ∧ roundHalfEven N D ≤ N / D + 1) := by

```

```

have hDq : (0 : ℚ) < D := by exact_mod_cast hD
have hN : (N : ℚ) = D * (N / D : ℕ) + (N % D : ℕ) := by
  exact_mod_cast (Nat.div_add_mod N D).symm
have hr0 : (0 : ℚ) ≤ (N % D : ℕ) := by positivity
have hrD : ((N % D : ℕ) : ℚ) < D := by exact_mod_cast Nat.mod_lt N hD
set m := N / D with hm
set r := N % D with hr
have key : ∀ k : ℤ, |(k : ℚ) - N / D| = |(k : ℚ) * D - (D * m + r)| / D := by
  intro k; rw [scaled_abs N D hD, hN]
unfold roundHalfEven
simp only []
rw [← hm, ← hr]
split_ifs with hup
· -- rounded up: '2 r ≥ D'
  have h2r : (D : ℚ) ≤ 2 * r := by
    rcases hup with h | ⟨h, _⟩
    · exact_mod_cast h.le
    · exact_mod_cast h.symm.le
  have hM : |((m + 1 : ℕ) : ℤ) : ℚ) * D - (D * m + r)| = D - r := by
    obtain ⟨-, -, -, h4, -, -⟩ := grid_bounds D m r ((m : ℚ) + 1) hDq hr0
    push_cast; rw [h4 rfl, abs_of_nonneg (by linarith)]
  refine ⟨fun k ⇒ ?_, fun k hk hne ⇒ ?_, by omega⟩
  · have hM' := hM; push_cast at hM'
    rw [show ((m + 1 : ℕ) : ℚ) = (((m + 1 : ℕ) : ℤ) : ℚ) by push_cast; rfl, key, key]
    apply div_le_div_of_nonneg_right _ hDq.le
    rw [hM]
    obtain ⟨-, h2, -, -, h5, -⟩ := grid_bounds D m r k hDq hr0
    rcases le_or_gt k (m : ℤ) with h | h
    · have := h2 (by exact_mod_cast h); linarith
    · exact h5 (by exact_mod_cast h)
  · rw [show ((m + 1 : ℕ) : ℚ) = (((m + 1 : ℕ) : ℤ) : ℚ) by push_cast; rfl, key, key,
    hM] at hk
    have hk' : |(k : ℚ) * D - (D * m + r)| = D - r := by
      field_simp at hk; linarith [hk]
    obtain ⟨h1, -, h3, -, -, h6⟩ := grid_bounds D m r k hDq hr0
    have hpos : (0 : ℚ) < r := by linarith
    rcases (show k ≤ (m : ℤ) - 1 ∨ k = m ∨ k = m + 1 ∨ (m : ℤ) + 2 ≤ k by omega) with
      h | h | h | h
    · have := h1 (by exact_mod_cast h); linarith
    · have := h3 (by exact_mod_cast h)
    · have h2r' : 2 * r = D := by
        have : (2 * r : ℚ) = D := by linarith
        exact_mod_cast this
      rcases hup with h' | ⟨-, hodd⟩
      · omega
      · exact (Nat.even_add_one.mpr (Nat.not_even_iff_odd.mpr (Nat.odd_iff.mpr hodd)))
    · exact absurd (by push_cast [h]; ring) hne
    · have := h6 (by exact_mod_cast h); linarith
  · -- kept: '2 r ≤ D'
    have h2r : 2 * (r : ℚ) ≤ D := by
      have : 2 * r ≤ D := by omega
      exact_mod_cast this
    have hM : |((m : ℕ) : ℤ) : ℚ) * D - (D * m + r)| = r := by
      obtain ⟨-, -, h3, -, -, -⟩ := grid_bounds D m r (m : ℚ) hDq hr0
      push_cast; exact h3 rfl
    refine ⟨fun k ⇒ ?_, fun k hk hne ⇒ ?_, by omega⟩
    · rw [show ((m : ℕ) : ℚ) = (((m : ℕ) : ℤ) : ℚ) by push_cast; rfl, key, key]
      apply div_le_div_of_nonneg_right _ hDq.le
      rw [hM]
      obtain ⟨-, h2, -, -, h5, -⟩ := grid_bounds D m r k hDq hr0
      rcases le_or_gt k (m : ℤ) with h | h
      · exact h2 (by exact_mod_cast h)
      · have := h5 (by exact_mod_cast h); linarith
    · rw [show ((m : ℕ) : ℚ) = (((m : ℕ) : ℤ) : ℚ) by push_cast; rfl, key, key, hM] at hk
      have hk' : |(k : ℚ) * D - (D * m + r)| = r := by
        field_simp at hk; linarith [hk]

```

```

obtain ⟨h1, -, -, h4, -, h6⟩ := grid_bounds D m r k hDq hr0
rcases (show k ≤ (m : ℤ) - 1 ∨ k = m ∨ k = m + 1 ∨ (m : ℤ) + 2 ≤ k by omega) with
  h | h | h | h
• have := h1 (by exact_mod_cast h); linarith
• exact absurd (by rw [h]) hne
• have := h4 (by exact_mod_cast h)
  rw [hk', abs_of_nonneg (by linarith)] at this
  have h2r' : 2 * r = D := by
    have : (2 * r : ℚ) = D := by linarith
    exact_mod_cast this
  have : ¬ m % 2 = 1 := fun hodd => hup (Or.inr ⟨h2r', hodd⟩)
  exact Nat.even_iff.mpr (by omega)
• have := h6 (by exact_mod_cast h); linarith

```

_nearest_binary64 in pieces

nearestBinary64 is the single transcription; roundHalfEven, packWord and magnitudeWord name its integer rounding step, its word construction and its result before the sign bits, and nearestBinary64_eq checks that they recompose it.

```

/-- The word construction of '_nearest_binary64' from a rounded mantissa. -/
def packWord (mantissa : ℕ) (exponent : ℤ) : ℕ :=
  let hidden_bit := 1 <<< 52
  if mantissa < hidden_bit then mantissa
  else
    let field := max exponent (-1022) + 1023
    let (mantissa, field) :=
      if mantissa = 2 * hidden_bit then (hidden_bit, field + 1) else (mantissa, field)
    (field.toNat <<< 52) ||| (mantissa - hidden_bit)

/-- '_nearest_binary64' after its sign, on the magnitude 'n / d'. -/
def magnitudeWord (n d : ℕ) : ℕ :=
  let exponent := rationalExponent n d
  if exponent > 1023 then 0x7ff0000000000000 else
  if exponent < -1075 then 0 else
  let shift : ℤ := max (exponent - 52) (-1074)
  let numerator_ := if shift < 0 then n <<< (-shift).toNat else n
  let denominator_ := if shift > 0 then d <<< shift.toNat else d
  packWord (roundHalfEven numerator_ denominator_) exponent

theorem nearestBinary64_zero : nearestBinary64 0 = 0 := by simp [nearestBinary64]

theorem nearestBinary64_eq (q : ℚ) (hq : q ≠ 0) :
  nearestBinary64 q = (if q < 0 then 0x8000000000000000 else 0) |||
    magnitudeWord q.num.natAbs q.den := by
  unfold nearestBinary64 magnitudeWord
  rw [if_neg hq]
  simp only []
  by_cases h1 : rationalExponent q.num.natAbs q.den > 1023
  • simp only [if_pos h1]
  • simp only [if_neg h1]
    by_cases h2 : rationalExponent q.num.natAbs q.den < -1075
    • simp only [if_pos h2, Nat.or_zero]
    • simp only [if_neg h2]
  rfl

```

Fields of assembled words

```

theorem or_shift52 (f t : ℕ) (ht : t < 2 ^ 52) : (f <<< 52) ||| t = f * 2 ^ 52 + t := by
  rw [← Nat.shiftLeft_add_eq_or_of_lt ht, Nat.shiftLeft_eq]

theorem sign_or (w : ℕ) (hw : w < 2 ^ 63) : (0x8000000000000000 : ℕ) ||| w = 2 ^ 63 + w := by
  rw [show (0x8000000000000000 : ℕ) = 1 <<< 63 from rfl, ← Nat.shiftLeft_add_eq_or_of_lt hw,
    Nat.shiftLeft_eq, one_mul]

```

```

theorem field_pack (f t : ℕ) (hf : f < 2048) (ht : t < 2 ^ 52) : field (f * 2 ^ 52 + t) = f := by
  unfold field; norm_num at ht ⊢; omega

theorem fraction_pack (f t : ℕ) (ht : t < 2 ^ 52) : fraction (f * 2 ^ 52 + t) = t := by
  unfold fraction; norm_num at ht ⊢; omega

theorem signBit_low (w : ℕ) (hw : w < 2 ^ 63) : signBit w = false := by
  unfold signBit; norm_num at hw ⊢; omega

theorem signBit_high (w : ℕ) (hw : w < 2 ^ 63) : signBit (2 ^ 63 + w) = true := by
  unfold signBit; norm_num at hw ⊢; omega

theorem field_high (w : ℕ) : field (2 ^ 63 + w) = field w := by
  unfold field; norm_num; omega

theorem fraction_high (w : ℕ) : fraction (2 ^ 63 + w) = fraction w := by
  unfold fraction; norm_num; omega

theorem field_lt (w : ℕ) : field w < 2048 := by
  unfold field; exact Nat.mod_lt _ (by norm_num)

theorem fraction_lt (w : ℕ) : fraction w < 2 ^ 52 := by
  unfold fraction; exact Nat.mod_lt _ (by norm_num)

```

Values

```

theorem magnitude_eq (w : ℕ) :
  magnitude w = (significand w : ℚ) * 2 ^ (max (field w : ℤ) 1 - 1075) := by
  unfold magnitude significand
  split_ifs with h
  · rw [h, show max ((0 : ℕ) : ℤ) 1 - 1075 = -1022 + (-52) by norm_num,
    zpow_add₀ two_ne_zero, show (2 : ℚ) ^ (-52 : ℤ) = 1 / 2 ^ 52 by norm_num]
    ring
  · have h1 : max (field w : ℤ) 1 - 1075 = ((field w : ℤ) - 1023) + (-52) := by omega
    rw [h1, zpow_add₀ two_ne_zero, zpow_neg,
      show ((2 : ℚ) ^ (52 : ℤ)) = 2 ^ (52 : ℕ) from zpow_natCast _ _]
    push_cast
    field_simp
    norm_num

theorem magnitude_nonneg (w : ℕ) : 0 ≤ magnitude w := by
  rw [magnitude_eq]; positivity

theorem abs_value (w : ℕ) : |value w| = magnitude w := by
  unfold value; split_ifs <;> simp [abs_of_nonneg (magnitude_nonneg w)]

/-- The finite nonnegative binary64 values: 'm * 2 ^ k' with a 53-bit integer 'm' and 'k' at
least the subnormal exponent '-1074'. The largest exponent is not needed by the proofs. -/
def Representable (y : ℚ) : Prop :=
  ∃ m : ℕ, ∃ k : ℤ, m < 2 ^ 53 ∧ -1074 ≤ k ∧ y = m * 2 ^ k

theorem magnitude_representable (w : ℕ) : Representable (magnitude w) := by
  refine ⟨significand w, max (field w : ℤ) 1 - 1075, ?_, by omega, magnitude_eq w⟩
  have := fraction_lt w
  unfold significand; split_ifs <;> norm_num at this ⊢ <;> omega

```

The word construction

```

/-- The word construction: below '2 ^ 52' a subnormal word, otherwise a normal word with the
hidden bit removed, and at '2 ^ 53' the carry into the exponent field, which at 'exponent =
1023' is the infinity. Every finite result has value 'mantissa * 2 ^ shift'. -/
theorem packWord_spec (M : ℕ) (E : ℤ) (hE : E ≤ 1023) (hM : M ≤ 2 ^ 53)

```

```

(hsub : M < 2 ^ 52 → max (E - 52) (-1074) = -1074) :
packWord M E < 2 ^ 63 ∧
(M = 2 ^ 53 ∧ E = 1023 → packWord M E = 2047 * 2 ^ 52) ∧
(¬(M = 2 ^ 53 ∧ E = 1023) → field (packWord M E) ≠ 2047 ∧
  magnitude (packWord M E) = M * 2 ^ (max (E - 52) (-1074)) ∧
  (Even M → Even (significand (packWord M E)))) := by
unfold packWord
simp only [show (1 <<< 52 : ℕ) = 2 ^ 52 from rfl]
by_cases hlt : M < 2 ^ 52
· rw [if_pos hlt]
  have hf : field M = 0 := by
    have := field_pack 0 M (by norm_num) hlt; simp using this
  have hfr : fraction M = M := by
    have := fraction_pack 0 M hlt; simp using this
  have hsig : significand M = M := by simp [significand, hf, hfr]
  refine ⟨by omega, fun h ⇒ by omega, fun _ ⇒ ⟨by omega, ?_, by rw [hsig]; exact id⟩⟩
  rw [magnitude_eq, hsig, hf, hsub hlt]; norm_num
· rw [if_neg hlt]
  by_cases hc : M = 2 * 2 ^ 52
  · rw [if_pos hc]
    simp only []
    have hF : (0 : ℤ) ≤ max E (-1022) + 1023 + 1 := by omega
    set F := (max E (-1022) + 1023 + 1).toNat with hFdef
    have hFz : (F : ℤ) = max E (-1022) + 1024 := by rw [hFdef, Int.toNat_of_nonneg hF]; ring
    have hword : (F <<< 52) ||| (2 ^ 52 - 2 ^ 52) = F * 2 ^ 52 + 0 := by
      rw [Nat.sub_self]; exact or_shift52 F 0 (by norm_num)
    rw [hword]
    have hF2 : F < 2048 := by omega
    refine ⟨by norm_num; omega, fun h ⇒ by omega, fun h ⇒ ?_⟩
    have hfield := field_pack F 0 hF2 (by norm_num)
    have hfrac := fraction_pack F 0 (by norm_num)
    have hF3 : F ≠ 2047 := by omega
    have hsig : significand (F * 2 ^ 52 + 0) = 2 ^ 52 := by
      simp only [significand, hfield, hfrac]; split_ifs <=> omega
    refine ⟨by rw [hfield]; exact hF3, ?_, fun _ ⇒ by rw [hsig]; exact ⟨2 ^ 51, by norm_num⟩⟩
    rw [magnitude_eq, hsig, hfield, hc]
    rw [show max (F : ℤ) 1 - 1075 = max (E - 52) (-1074) + 1 by omega, zpow_add_one, two_ne_zero]
    push_cast; ring
  · rw [if_neg hc]
    simp only []
    have hF : (0 : ℤ) ≤ max E (-1022) + 1023 := by omega
    set F := (max E (-1022) + 1023).toNat with hFdef
    have hFz : (F : ℤ) = max E (-1022) + 1023 := by rw [hFdef, Int.toNat_of_nonneg hF]
    have hM' : M < 2 * 2 ^ 52 := by omega
    have ht : M - 2 ^ 52 < 2 ^ 52 := by omega
    rw [or_shift52 F _ ht]
    have hF2 : F < 2047 := by omega
    have hfield := field_pack F _ (by omega) ht
    have hfrac := fraction_pack F _ ht
    have hsig : significand (F * 2 ^ 52 + (M - 2 ^ 52)) = M := by
      simp only [significand, hfield, hfrac]; split_ifs <=> omega
    refine ⟨by norm_num at ht hM' ⊢; omega, fun h ⇒ by omega, fun _ ⇒ ⟨by omega, ?_, ?_⟩⟩
    · rw [magnitude_eq, hsig, hfield, show max (F : ℤ) 1 - 1075 = max (E - 52) (-1074) by omega]
    · rw [hsig]; exact id

```

Nearest on the grid $2^s \mathbb{Z}$, and below it

```

theorem scale_abs (x a : ℚ) (s : ℤ) : |x * 2 ^ s - a| = 2 ^ s * |x - a / 2 ^ s| := by
  have hp : (0 : ℚ) < 2 ^ s := by positivity
  rw [show x * 2 ^ s - a = 2 ^ s * (x - a / 2 ^ s) by field_simp, abs_mul, abs_of_pos hp]

```

*/-- If 'M' is an integer nearest 'a / 2 ^ s', ties to even, and every binary64 value not on the grid '2 ^ s ℤ' lies below '2 ^ (s + 52) ≤ a', then 'M * 2 ^ s' is a binary64 value nearest 'a', and a different binary64 value equally near forces 'M' even. -/*

```

theorem grid_nearest (a : ℚ) (s : ℤ) (M : ℕ)
  (hnear : ∀ k : ℤ, |(M : ℚ) - a / 2 ^ s| ≤ |(k : ℚ) - a / 2 ^ s|)
  (htie : ∀ k : ℤ, |(k : ℚ) - a / 2 ^ s| = |(M : ℚ) - a / 2 ^ s| → k ≠ M → Even M)
  (hoff : -1074 < s → (2 : ℚ) ^ (s + 52) ≤ a) (y : ℚ) (hy : Representable y) :
  |(M : ℚ) * 2 ^ s - a| ≤ |y - a| ∧
  (|y - a| = |(M : ℚ) * 2 ^ s - a| → y ≠ (M : ℚ) * 2 ^ s → Even M) := by
obtain ⟨m, k, hm, hk, rfl⟩ := hy
have hp : (0 : ℚ) < 2 ^ s := by positivity
by_cases hks : s ≤ k
· set K : ℤ := ((m * 2 ^ (k - s)).toNat : ℕ) : ℤ with hK
  have hyK : (m : ℚ) * 2 ^ k = (K : ℚ) * 2 ^ s := by
    rw [hK]; push_cast
  rw [← zpow_natCast, Int.toNat_of_nonneg (by omega), mul_assoc, ← zpow_add₀ two_ne_zero]
  congr 2; ring
  rw [hyK, scale_abs, scale_abs]
  refine ⟨mul_le_mul_of_nonneg_left (hnear K) hp.le, fun h hne ⇒ ?_⟩
  have h' := mul_left_cancel₀ hp.ne' h
  refine htie K h' fun hKM ⇒ hne ?_
  rw [hKM]; push_cast; rfl
· have hs : -1074 < s := by omega
  have ha := hoff hs
  have hy1 : (m : ℚ) * 2 ^ k < 2 ^ (s + 52) := by
    have hm' : (m : ℚ) < 2 ^ (53 : ℤ) := by
      rw [show (2 : ℚ) ^ (53 : ℤ) = 2 ^ (53 : ℕ) from zpow_natCast 2 53]; exact_mod_cast hm
    calc (m : ℚ) * 2 ^ k < 2 ^ (53 : ℤ) * 2 ^ k := by
      exact mul_lt_mul_of_pos_right hm' (by positivity)
    _ = 2 ^ (k + 53) := by rw [← zpow_add₀ two_ne_zero]; ring_nf
    _ ≤ 2 ^ (s + 52) := zpow_le_zpow_right₀ (by norm_num) (by omega)
  have hy0 : (0 : ℚ) ≤ (m : ℚ) * 2 ^ k := by positivity
  have hgrid := hnear (2 ^ 52 : ℕ)
  rw [← mul_le_mul_iff_of_pos_left hp, ← scale_abs, ← scale_abs] at hgrid
  have h52 : (((2 ^ 52 : ℕ) : ℤ) : ℚ) * 2 ^ s = 2 ^ (s + 52) := by
    rw [zpow_add₀ two_ne_zero]; norm_num; ring
  rw [h52, abs_of_nonpos (show (2 : ℚ) ^ (s + 52) - a ≤ 0 by linarith)] at hgrid
  have hstrict : |(M : ℚ) * 2 ^ s - a| < |(m : ℚ) * 2 ^ k - a| := by
    rw [abs_of_nonpos (by linarith : (m : ℚ) * 2 ^ k - a ≤ 0)]; linarith
  exact ⟨hstrict.le, fun h _ ⇒ absurd h hstrict.ne'⟩

```

The magnitude word

```

set_option exponentiation.threshold 1100 in
/-- The result before the sign bits, on the magnitude 'n / d': the infinity exactly from
'overflowThreshold' on, and otherwise a finite word nearest 'n / d' among the binary64
magnitudes, ties to even. -/
theorem magnitudeWord_spec (n d : ℕ) (hn : 0 < n) (hd : 0 < d) :
  magnitudeWord n d < 2 ^ 63 ∧
  (overflowThreshold ≤ (n : ℚ) / d → magnitudeWord n d = 2047 * 2 ^ 52) ∧
  ((n : ℚ) / d < overflowThreshold → field (magnitudeWord n d) ≠ 2047 ∧
  (∀ y, Representable y →
    |magnitude (magnitudeWord n d) - (n : ℚ) / d| ≤ |y - (n : ℚ) / d|) ∧
  (∀ y, Representable y →
    |y - (n : ℚ) / d| = |magnitude (magnitudeWord n d) - (n : ℚ) / d| →
    y ≠ magnitude (magnitudeWord n d) → Even (significand (magnitudeWord n d)))) := by
obtain ⟨hE1, hE2⟩ := rationalExponent_spec n d hn hd
have hapos : (0 : ℚ) < (n : ℚ) / d := by positivity
set a : ℚ := (n : ℚ) / d with ha
have hthr : overflowThreshold = (2 ^ 53 - 1 / 2) * 2 ^ (971 : ℤ) := by
  norm_num [overflowThreshold]
have hthr2 : (2 : ℚ) ^ (1023 : ℤ) < overflowThreshold := by rw [hthr]; norm_num
unfold magnitudeWord
simp only []
set E := rationalExponent n d with hEdef
by_cases h1 : E > 1023
· rw [if_pos h1]

```

```

have h1024 : (2 : ℚ) ^ (1024 : ℤ) ≤ a :=
  le_trans (zpow_le_zpow_right0 (by norm_num) (by omega)) hE1
have hlt : ¬ a < overflowThreshold := by
  rw [hthr]; norm_num at h1024; linarith
exact ⟨by norm_num, fun _ => by norm_num, fun h => absurd h hlt⟩
rw [if_neg h1]
by_cases h2 : E < -1075
· rw [if_pos h2]
  have hsmall : a < 2 ^ (-1075 : ℤ) :=
    lt_of_lt_of_le hE2 (zpow_le_zpow_right0 (by norm_num) (by omega))
  have hm0 : magnitude 0 = 0 := by rw [magnitude_eq]; simp [significand, fraction, field]
  have hlt : a < overflowThreshold := by
    have : (2 : ℚ) ^ (-1075 : ℤ) < 2 ^ (1023 : ℤ) :=
      zpow_lt_zpow_right0 (by norm_num) (by norm_num)
    exact lt_trans hsmall (lt_trans this hthr2)
  refine ⟨by norm_num, fun h => absurd h (not_le.mpr hlt), fun _ => ⟨by simp [field], ?_,
    fun _ _ => by simp [significand, field, fraction]⟩
  intro y ⟨m, k, hm, hk, hy⟩
  rw [hm0, zero_sub, abs_neg, abs_of_pos hapos]
  rcases Nat.eq_zero_or_pos m with h0 | hpos
  · rw [hy, h0]; simp [abs_of_pos hapos]
  · have hk' : (2 : ℚ) ^ (-1074 : ℤ) ≤ 2 ^ k := zpow_le_zpow_right0 (by norm_num) hk
    have hm1 : (1 : ℚ) ≤ m := by exact_mod_cast hpos
    have hy1 : (2 : ℚ) ^ (-1074 : ℤ) ≤ y := by
      rw [hy]; nlinarith [show (0 : ℚ) < 2 ^ k by positivity]
    have h2' : (2 : ℚ) ^ (-1074 : ℤ) = 2 * 2 ^ (-1075 : ℤ) := by
      rw [show (-1074 : ℤ) = 1 + (-1075) by norm_num, zpow_add0 two_ne_zero, zpow_one]
    rw [abs_of_nonneg (by linarith)]; linarith
  rw [if_neg h2]
  set s : ℤ := max (E - 52) (-1074) with hs
  set N := (if s < 0 then n <<< (-s).toNat else n) with hN
  set D := (if s > 0 then d <<< s.toNat else d) with hD
  have hDpos : 0 < D := by
    rw [hD]; split_ifs
    · rw [Nat.shiftLeft_eq]; positivity
    · exact hd
  have hp : (0 : ℚ) < 2 ^ s := by positivity
  have hdq : (0 : ℚ) < d := by exact_mod_cast hd
  have ht : (N : ℚ) / D = a / 2 ^ s := by
    rw [hN, hD, ha]
    rcases lt_trichotomy s 0 with h | h | h
    · rw [if_pos h, if_neg (by omega), Nat.shiftLeft_eq]; push_cast
      rw [← zpow_natCast, Int.toNat_of_nonneg (by omega), zpow_neg]; field_simp
    · rw [h]; simp
    · rw [if_neg (by omega), if_pos h, Nat.shiftLeft_eq]; push_cast
      rw [← zpow_natCast, Int.toNat_of_nonneg (by omega)]; field_simp
  have hDq : (0 : ℚ) < D := by exact_mod_cast hDpos
  obtain ⟨hnear, htie, hlo, hhi⟩ := roundHalfEven_spec N D hDpos
  set M := roundHalfEven N D with hMdef
  rw [ht] at hnear htie
  set t := a / 2 ^ s with htdef
  have hta : a = t * 2 ^ s := by rw [htdef]; field_simp
  -- 't < 2 ^ 53', so 'M ≤ 2 ^ 53'.
  have htlt : t < 2 ^ 53 := by
    rw [htdef, div_lt_iff0 hp]
    calc a < 2 ^ (E + 1) := hE2
      _ = 2 ^ (53 : ℤ) * 2 ^ (E - 52) := by rw [← zpow_add0 two_ne_zero]; ring_nf
      _ ≤ 2 ^ (53 : ℤ) * 2 ^ s := by
        gcongr
        · norm_num
        · omega
      _ = 2 ^ 53 * 2 ^ s := by norm_num
  have hM53 : M ≤ 2 ^ 53 := by
    have hND : N < 2 ^ 53 * D := by
      have : (N : ℚ) < 2 ^ 53 * D := by
        rw [← div_lt_iff0 hDq, ht]; exact htlt

```

```

    exact_mod_cast this
  have := (Nat.div_lt_iff_lt_mul hDpos).mpr hND
  omega
have hsub : M < 2 ^ 52 → max (E - 52) (-1074) = -1074 := by
  intro hM
  by_contra hne
  have hsE : s = E - 52 := by omega
  have ht52 : (2 : ℚ) ^ 52 ≤ t := by
    rw [htdef, le_div_iff₀ hp, hsE]
  calc (2 : ℚ) ^ 52 * 2 ^ (E - 52) = 2 ^ E := by
    rw [show (2 : ℚ) ^ 52 = 2 ^ (52 : ℤ) by norm_num, ← zpow_add₀ two_ne_zero]; ring_nf
    _ ≤ a := hE1
  have hND : 2 ^ 52 * D ≤ N := by
    have : (2 : ℚ) ^ 52 * D ≤ N := by
      rw [← le_div_iff₀ hDq, ht]; exact ht52
    exact_mod_cast this
  have := Nat.div_le_div_right (c := D) hND
  rw [Nat.mul_div_cancel _ hDpos] at this
  omega
obtain ⟨hlt63, hover, hfin⟩ := packWord_spec M E (by omega) hM53 hsub
have hiff : overflowThreshold ≤ a ↔ (M = 2 ^ 53 ∧ E = 1023) := by
  constructor
  · intro h
    have hE : E = 1023 := by
      by_contra hne
      have : (2 : ℚ) ^ (E + 1) ≤ 2 ^ (1023 : ℤ) := zpow_le_zpow_right₀ (by norm_num) (by omega)
      linarith
    have hs971 : s = 971 := by omega
    have ht1 : 2 ^ 53 - 1 / 2 ≤ t := by
      rw [htdef, le_div_iff₀ hp, hs971]; rw [hthr] at h; exact h
    refine ⟨?_, hE⟩
    by_contra hM
    have hM' : (M : ℚ) + 1 ≤ 2 ^ 53 := by
      have : M + 1 ≤ 2 ^ 53 := by omega
      exact_mod_cast this
    have hk := hnear ((2 : ℤ) ^ 53)
    simp only [Int.cast_pow, Int.cast_ofNat] at hk
    rw [abs_of_pos (show (0 : ℚ) < 2 ^ 53 - t by linarith)] at hk
    have hge : t - M ≤ |(M : ℚ) - t| := by rw [abs_sub_comm]; exact le_abs_self _
    have hMeq : M + 1 = 2 ^ 53 := by
      have : (M : ℚ) + 1 = 2 ^ 53 := le_antisymm hM' (by linarith)
      exact_mod_cast this
    have heq : |(((2 : ℤ) ^ 53 : ℤ) : ℚ) - t| = |(M : ℚ) - t| := by
      simp only [Int.cast_pow, Int.cast_ofNat]
      rw [abs_of_pos (show (0 : ℚ) < 2 ^ 53 - t by linarith)]
      apply le_antisymm _ hk
      linarith
    have hne : (2 : ℤ) ^ 53 ≠ (M : ℤ) := by
      intro h'; have : ((M : ℕ) : ℤ) + 1 = 2 ^ 53 := by exact_mod_cast hMeq
      omega
    have hev := htie _ heq hne
    have : M = 2 ^ 53 - 1 := by omega
    rw [this] at hev
    exact absurd hev (by decide)
  · rintro ⟨hM, hE⟩
    have hs971 : s = 971 := by omega
    have hk := hnear ((2 : ℤ) ^ 53 - 1)
    rw [hM] at hk
    simp only [Int.cast_sub, Int.cast_pow, Int.cast_ofNat, Int.cast_one, Nat.cast_pow,
      Nat.cast_ofNat] at hk
    rw [abs_of_pos (show (0 : ℚ) < 2 ^ 53 - t by linarith)] at hk
    have ht1 : 2 ^ 53 - 1 / 2 ≤ t := by
      rcases le_abs.mp hk with h | h
      · linarith
      · linarith
    rw [hthr, hta, hs971]

```

```

    exact mul_le_mul_of_nonneg_right ht1 (by positivity)
  refine ⟨hlt63, fun h ⇒ hover (hiff.mp h), fun h ⇒ ?_⟩
  have hno : ¬(M = 2 ^ 53 ∧ E = 1023) := fun h' ⇒ absurd (hiff.mpr h') (not_le.mpr h)
  obtain ⟨hf, hmag, heven⟩ := hfin hno
  have hoff : -1074 < s → (2 : ℚ) ^ (s + 52) ≤ a := fun h' ⇒ by
    rw [show s + 52 = E by omega]; exact hE1
  refine ⟨hf, fun y hy ⇒ ?_, fun y hy htie' hne ⇒ ?_⟩
  · rw [hmag]; exact (grid_nearest a s M hnear htie hoff y hy).1
  · rw [hmag] at htie' hne
    exact heven ((grid_nearest a s M hnear htie hoff y hy).2 htie' hne)

```

The sign

```

theorem signed_word (P : Prop) [Decidable P] (w : ℕ) (hw : w < 2 ^ 63) :
  ((if P then 0x8000000000000000 else 0) ||| w) < 2 ^ 64 ∧
  signBit ((if P then 0x8000000000000000 else 0) ||| w) = decide P ∧
  field ((if P then 0x8000000000000000 else 0) ||| w) = field w ∧
  fraction ((if P then 0x8000000000000000 else 0) ||| w) = fraction w ∧
  significand ((if P then 0x8000000000000000 else 0) ||| w) = significand w ∧
  value ((if P then 0x8000000000000000 else 0) ||| w) =
    if P then -magnitude w else magnitude w := by
  by_cases hP : P
  · rw [if_pos hP, sign_or w hw]
    have h1 := field_high w
    have h2 := fraction_high w
    refine ⟨by omega, by rw [signBit_high w hw]; simp [hP], h1, h2,
      by simp only [significand, h1, h2], ?_⟩
    simp only [value, signBit_high w hw, magnitude, h1, h2, if_pos hP, ↓reduceIte]
  · rw [if_neg hP, Nat.zero_or]
    refine ⟨by omega, by simp [signBit_low w hw, hP], rfl, rfl, rfl, ?_⟩
    simp [value, signBit_low w hw, hP]

theorem abs_num_div_den (q : ℚ) : ((q.num.natAbs : ℕ) : ℚ) / (q.den : ℚ) = |q| := by
  have h := Rat.num_div_den q
  conv_rhs ⇒ rw [← h]
  rw [abs_div, Nat.abs_cast, Nat.cast_natAbs, Int.cast_abs]

```

Correct rounding

```

/-- **Correct rounding.** For every rational 'q', the word '_nearest_binary64(q)' computes is the
IEEE 754 binary64 round-to-nearest-ties-to-even of 'q', overflow and signed zero included. -/
theorem nearestBinary64_roundsTo (q : ℚ) : RoundsTo q (nearestBinary64 q) := by
  have hthr : (0 : ℚ) < overflowThreshold :=
    sub_pos.mpr (pow_lt_pow_right₀ (by norm_num) (by norm_num))
  by_cases hq : q = 0
  · subst hq
    rw [nearestBinary64_zero]
    have hf : field 0 = 0 := by simp [field]
    have hm0 : magnitude 0 = 0 := by
      rw [magnitude_eq]
      simp only [significand, fraction, field, Nat.zero_div, Nat.zero_mod, ↓reduceIte,
        Nat.cast_zero, zero_mul]
    have hv : value 0 = 0 := by unfold value; rw [hm0]; simp
    refine ⟨by norm_num, fun h ⇒ by simp at h; linarith, fun _ ⇒ by rw [hf]; norm_num,
      fun _ ⇒ by simp [signBit], fun _ w' _ ⇒ by rw [hv]; simp, fun _ _ _ _ ⇒ ?_⟩
    simp [significand, hf, fraction]
  set n := q.num.natAbs with hn
  set d := q.den with hd
  have hn0 : 0 < n := Int.natAbs_pos.mpr (Rat.num_ne_zero.mpr hq)
  have hd0 : 0 < d := q.den_pos
  have ha : (n : ℚ) / d = |q| := abs_num_div_den q
  have hqa : 0 < |q| := abs_pos.mpr hq
  obtain ⟨hlt63, hover, hfin⟩ := magnitudeWord_spec n d hn0 hd0

```

```

rw [ha] at hover hfin
set mw := magnitudeWord n d with hmw
rw [nearestBinary64_eq q hq]
obtain ⟨hw64, hWsign, hWfield, hWfrac, hWsig, hWval⟩ := signed_word (q < 0) mw hlt63
set W := (if q < 0 then 0x8000000000000000 else 0) ||| mw with hw
have hr := magnitude_nonneg mw
-- the distance of the result, and of a value of the result's magnitude and the other sign
have hdist : |value W - q| = abs (magnitude mw - |q|) := by
  rw [hWval]
  by_cases hnég : q < 0
  · rw [if_pos hnég, abs_of_neg hnég, show -magnitude mw - q = -(magnitude mw - -q) by ring, abs_neg]
  · rw [if_neg hnég, abs_of_nonneg (not_lt.mp hnég)]
have hopp : ∀ v : ℚ, |v| = magnitude mw → v ≠ value W → |v - q| = magnitude mw + |q| := by
  intro v hv hne
  rw [hWval] at hne
  rcases (abs_eq hr).mp hv with h | h
  · by_cases hnég : q < 0
    · rw [h, abs_of_neg hnég, abs_of_nonneg (by linarith)]; ring
    · rw [if_neg hnég] at hne; exact absurd h hne
  · by_cases hnég : q < 0
    · rw [if_pos hnég] at hne; exact absurd h hne
    · have hq' : 0 < q := lt_of_le_of_ne (not_lt.mp hnég) (Ne.symm hq)
      rw [h, abs_of_pos hq', abs_of_neg (by linarith)]; ring
have hfar : ∀ w', abs (magnitude w' - |q|) ≤ |value w' - q| := by
  intro w'
  rw [← abs_value w']
  exact abs_abs_sub_abs_le_abs_sub _ _
refine ⟨hw64, fun h ⇒ ?_, fun h ⇒ ?_, fun _ ⇒ hWsign, fun h w' hw' ⇒ ?_,
  fun h w' hw' htie hne ⇒ ?_⟩
· have h1 := hover h
  have hf : field W = 2047 := by rw [hWfield, h1]; exact field_pack 2047 0 (by norm_num) (by
    ← norm_num)
  have hfr : fraction W = 0 := by rw [hWfrac, h1]; exact fraction_pack 2047 0 (by norm_num)
  simp [decode, hf, hfr, hWsign]
· rw [hWfield]; exact (hfin h).1
· rw [hdist]
  exact le_trans ((hfin h).2.1 - (magnitude_representable w')) (hfar w')
· rw [hWsig]
  obtain ⟨-, hnear, hties⟩ := hfin h
  have heq : abs (magnitude w' - |q|) = abs (magnitude mw - |q|) := by
    apply le_antisymm _ (hnear - (magnitude_representable w'))
    rw [← hdist, ← htie]; exact hfar w'
  by_cases hm : magnitude w' = magnitude mw
  · have := hopp (value w') (by rw [abs_value, hm]) hne
    rw [this, hdist] at htie
    exfalso
    rcases abs_cases (magnitude mw - |q|) with ⟨h1, -⟩ | ⟨h1, -⟩
    · rw [h1] at htie; linarith
    · rw [h1] at htie
      have hr0 : magnitude mw = 0 := by linarith
      have hv0 : value w' = 0 := by
        rw [← abs_eq_zero, abs_value, hm, hr0]
      have hW0 : value W = 0 := by
        rw [hWval, hr0]; simp
      exact hne (hv0.trans hW0.symm)
  · exact hties - (magnitude_representable w') heq hm

```

Base.decompose and _dyadic

```

theorem fraction_mask (w : ℕ) : w &&& 0x000fffffffffffff = fraction w := by
  rw [show (0x000fffffffffffff : ℕ) = 2 ^ 52 - 1 from rfl, Nat.and_two_pow_sub_one_eq_mod]; rfl

theorem field_mask (w : ℕ) : (w &&& 0x7fff000000000000) >>> 52 = field w := by

```

```

rw [Nat.shiftRight_and_distrib, show (0x7ff0000000000000 : ℕ) >>> 52 = 2 ^ 11 - 1 from rfl,
  Nat.and_two_pow_sub_one_eq_mod, Nat.shiftRight_eq_div_pow]; rfl

/-- '_dyadic' throws exactly on the infinities and NaNs. -/
theorem dyadic_none_iff (w : ℕ) : dyadic w = none ↔ field w = 2047 := by
  unfold dyadic; split_ifs with h <|> simp [h]

/-- On a finite word, '_dyadic' returns '(±significand, max field 1 - 1075)', and 'n * 2 ^ p' is
the word's value. -/
theorem dyadic_value (w : ℕ) (hw : field w ≠ 2047) :
  dyadic w = some ((significand w : ℤ) * (if signBit w then -1 else 1),
    max (field w : ℤ) 1 - 1075) ∧
  ∀ n p, dyadic w = some (n, p) → (n : ℚ) * 2 ^ p = value w := by
  have hdec : decompose w = ((significand w : ℤ), max (field w : ℤ) 1 - 1075,
    (if signBit w then -1 else 1)) := by
    unfold decompose
  rw [if_neg (fun h => hw h.1), if_neg hw]
  simp only [fraction_mask, field_mask]
  have hor : fraction w ||| ((if field w ≠ 0 then 1 else 0) <<< 52) = significand w := by
    rw [Nat.or_comm, ← Nat.shiftLeft_add_eq_or_of_lt (fraction_lt w), Nat.shiftLeft_eq]
    unfold significand
    split_ifs <|> simp_all
  rw [hor]
  congr 2
  split_ifs with h
  · simp [h]
  · simp; omega
  have hdy : dyadic w = some ((significand w : ℤ) * (if signBit w then -1 else 1),
    max (field w : ℤ) 1 - 1075) := by
    unfold dyadic
  rw [if_neg hw, hdec]
  split_ifs <|> simp
  refine ⟨hdy, fun n p h => ?_⟩
  rw [hdy] at h
  obtain ⟨rfl, rfl⟩ := Prod.mk.inj (Option.some.inj h)
  unfold value
  rw [magnitude_eq]
  split_ifs <|> push_cast <|> ring

```

Round trip

```

theorem word_eq (w : ℕ) (hw : w < 2 ^ 64) :
  w = (if signBit w then 2 ^ 63 else 0) + field w * 2 ^ 52 + fraction w := by
  unfold signBit field fraction
  norm_num at hw
  split_ifs with h <|> omega

theorem significand_lt (w : ℕ) : significand w < 2 ^ 53 := by
  have := fraction_lt w
  unfold significand; split_ifs <|> omega

theorem significand_ge (w : ℕ) (h : field w ≠ 0) : 2 ^ 52 ≤ significand w := by
  unfold significand; rw [if_neg h]; omega

theorem significand_lt_of_zero (w : ℕ) (h : field w = 0) : significand w < 2 ^ 52 := by
  unfold significand; rw [if_pos h]; exact fraction_lt w

private theorem exponent_not_lt (w₁ w₂ : ℕ)
  (h : (significand w₁ : ℚ) * 2 ^ (max (field w₁ : ℤ) 1 - 1075) =
    (significand w₂ : ℚ) * 2 ^ (max (field w₂ : ℤ) 1 - 1075)) :
  ¬ max (field w₁ : ℤ) 1 - 1075 < max (field w₂ : ℤ) 1 - 1075 := by
  intro hlt
  set e₁ := max (field w₁ : ℤ) 1 - 1075
  set e₂ := max (field w₂ : ℤ) 1 - 1075

```

```

have hf2 : field w₂ ≠ 0 := by omega
have hm2 : (2 : ℚ) ^ 52 ≤ significand w₂ := by exact_mod_cast significand_ge w₂ hf2
have hm1 : (significand w₁ : ℚ) < 2 ^ 53 := by exact_mod_cast significand_lt w₁
have hp : (0 : ℚ) < 2 ^ e₁ := by positivity
have h2 : (2 : ℚ) ≤ 2 ^ (e₂ - e₁) := by
  calc (2 : ℚ) = 2 ^ (1 : ℤ) := by norm_num
  _ ≤ 2 ^ (e₂ - e₁) := zpow_le_zpow_right₀ (by norm_num) (by omega)
have hsplit : (2 : ℚ) ^ e₂ = 2 ^ (e₂ - e₁) * 2 ^ e₁ := by
  rw [← zpow_add₀ two_ne_zero]; ring_nf
rw [hsplit, ← mul_assoc] at h
have h' := mul_right_cancel₀ hp.ne' h
have : (2 : ℚ) * 2 ^ 52 ≤ significand w₁ := by
  rw [h']; nlinarith
linarith

theorem magnitude_injective (w₁ w₂ : ℕ) (h : magnitude w₁ = magnitude w₂) :
  field w₁ = field w₂ ∧ fraction w₁ = fraction w₂ := by
  rw [magnitude_eq, magnitude_eq] at h
have he : max (field w₁ : ℤ) 1 - 1075 = max (field w₂ : ℤ) 1 - 1075 :=
  le_antisymm (not_lt.mp (exponent_not_lt w₂ w₁ h.symm)) (not_lt.mp (exponent_not_lt w₁ w₂ h))
rw [he] at h
have hs : significand w₁ = significand w₂ := by
  exact_mod_cast mul_right_cancel₀ (zpow_ne_zero _ two_ne_zero) h
have h1 := fraction_lt w₁
have h2 := fraction_lt w₂
unfold significand at hs
split_ifs at hs with hf1 hf2 hf2 <|> omega

set_option exponentiation.threshold 1100 in
theorem magnitude_lt_threshold (w : ℕ) (hw : field w ≠ 2047) :
  magnitude w < overflowThreshold := by
  have hthr : overflowThreshold = (2 ^ 53 - 1 / 2) * 2 ^ (971 : ℤ) := by
    norm_num [overflowThreshold]
  have hf := field_lt w
  have hs : (significand w : ℚ) + 1 ≤ 2 ^ 53 := by
    have := significand_lt w
    exact_mod_cast this
  have he : (2 : ℚ) ^ (max (field w : ℤ) 1 - 1075) ≤ 2 ^ (971 : ℤ) :=
    zpow_le_zpow_right₀ (by norm_num) (by omega)
  have hp : (0 : ℚ) < 2 ^ (971 : ℤ) := by positivity
  rw [magnitude_eq, hthr]
  calc (significand w : ℚ) * 2 ^ (max (field w : ℤ) 1 - 1075)
    ≤ (2 ^ 53 - 1) * 2 ^ (971 : ℤ) := by
      apply mul_le_mul (by linarith) he (by positivity) (by norm_num)
    < (2 ^ 53 - 1 / 2) * 2 ^ (971 : ℤ) := by
      apply mul_lt_mul_of_pos_right (by norm_num) hp

/-- Round trip: a finite word's value rounds back to the same word, except that '-0.0' rounds
to '+0.0' (the rational '0' has no sign). -/
theorem nearestBinary64_value (w : ℕ) (hw : IsFinite w) :
  nearestBinary64 (value w) = if value w = 0 then 0 else w := by
  split_ifs with h0
  · rw [h0, nearestBinary64_zero]
have R := nearestBinary64_roundsTo (value w)
set r := nearestBinary64 (value w) with hr
have hlt : |value w| < overflowThreshold := by
  rw [abs_value]; exact magnitude_lt_threshold w hw.2
have hv : value r = value w := by
  have := R.nearest hlt w hw
  rw [sub_self, abs_zero] at this
  exact sub_eq_zero.mp (abs_nonpos_iff.mp this)
have hm : 0 < magnitude w := by
  refine lt_of_le_of_ne (magnitude_nonneg w) (fun h => h0 ?_)
  unfold value; split_ifs <|> simp [← h]
have hsign : signBit r = signBit w := by
  rw [R.sign hlt]

```

```

    unfold value
    cases hb : signBit w
    • simp only [Bool.false_eq_true, ↓reduceIte, decide_eq_false_iff_not, not_lt]
      exact hm.le
    • simp only [↓reduceIte, decide_eq_true_eq, Left.neg_neg_iff]
      exact hm
  have hmag : magnitude r = magnitude w := by rw [← abs_value, hv, abs_value]
  obtain ⟨hf, hfr⟩ := magnitude_injective r w hmag
  rw [word_eq r R.word, word_eq w hw.1, hsign, hf, hfr]

```

end BayesianNetworksProofs.Binary64

Forward error bounds for the floating-point paths

```

import BayesianNetworksProofs.Numeric.Binary64
import BayesianNetworksProofs.Finite.Posterior
import Mathlib.Algebra.Order.BigOperators.GroupWithZero.List
import Mathlib.Data.Nat.Lattice
import Mathlib.Analysis.SpecialFunctions.Log.Basic

```

The inference engines compute in Float64. Numeric/Binary64.lean proves that the exact fallback’s final rounding is correct; this module bounds the error of the ordinary floating-point paths, as theorems about an abstract rounding model applied to the factor algebra of Finite/VariableElimination.lean and Finite/Posterior.lean.

The standard model. Rounded $u \times c$ says that c is a computed result of an operation whose exact result is x : $c = x * (1 + \delta)$ with $|\delta| \leq u$. For binary64, $u = 2^{-53}(\text{binary64Unit})$. $\gamma_n = (1 + u)^n - 1$ (gamma) bounds the relative error of n roundings, and $\text{RelWithin } e \ y \times (|x - y| \leq e * y)$ is relative error against a nonnegative exact value. Relative errors of nonnegative quantities compose without any cancellation argument: RelWithin.mul , RelWithin.add and RelWithin.round .

The bridge to binary64 (roundsTo_relative, roundsTo_subnormal, roundsTo_rounded, and nearestBinary64_relative, nearestBinary64_subnormal, nearestBinary64_rounded for the transcription of _nearest_binary64). A word w with RoundsTo $q \ w$ (IEEE 754 round-to-nearest, ties-to-even, of the rational q) satisfies $|\text{value } w - q| \leq 2^{-53} |q|$ when $2^{-1022} \leq |q| < \text{overflowThreshold}$, and $|\text{value } w - q| \leq 2^{-1075}$ when $|q| < 2^{-1022}$. The relative bound is derived from RoundsTo.nearest by exhibiting a finite word within $2^{-53} |q|$ of q . So Rounded binary64Unit holds for correct rounding of an exact result that is zero or in the normal range. That Julia’s Float64 $+$, $*$, $-$ and $/$ deliver the correctly rounded result of their exact real operation is IEEE 754 together with Julia’s and the hardware’s semantics: it is **assumed, not proved**. Julia’s exp and log are not correctly rounded; the log-sum-exp theorem takes their relative error u as a hypothesis.

What is proved.

- compProd_forward_error: a product of k nonnegative numbers computed in any order and association is within relative $\gamma \ (k - 1)$ of the exact product.
- sumRun_forward_error: a sum of n nonnegative numbers computed along any association tree t (Assoc, SumRun) is within relative $\gamma \ (\text{height } t)$, hence $\gamma \ (n - 1)$, of the exact sum. The tree is a universally quantified argument: the bound does not depend on the order.
- eliminateAll_forward_error: an approximate run of variable elimination (Run) pairs each exact factor of eliminateAll with a computed table. Each step computes every entry of the bucket’s sum-out as a sum, in any association, over the states of v of products, in any association, of the bucket’s computed tables, every operation rounded in the standard model. For nonnegative input factors fs and an elimination list vs , every entry of every computed factor, and of any computed product of the remaining factors, is within relative $\gamma \ N$ of the exact entry, with the explicit index $N = \text{veIndex } fs.\text{length } vs = |fs| + \sum_{v \in vs} |\text{states } v| - 1$. eliminateAll_marg_forward_error restates it against marg $vs.\text{toFinset } (\text{product } fs)$, and conditioned_forward_error covers evidence by slicing (condition), the shape of the Julia engine.
- normalize_forward_error and ve_posterior_forward_error: with the evidence indicator and the compiled CPT factors, $vs = Q^c$, the mass Z summed in any association and each posterior entry the rounded quotient $\text{num } q / Z$, let $K = \text{posteriorIndex } bn \ Q \ vs = (|M| + 1) + \sum_{v \in vs} |\text{states } v| - 1 + |\text{query assignments}|$. When the evidence mass is positive and $\gamma \ K < 1$, the computed mass is within relative $\gamma \ (K - 1)$ of the evidence mass and every computed posterior entry is within relative $(1 + \gamma \ K) / (1 - \gamma \ K) - 1$ (about $2 \ K \ u$) of the exact posterior entry.

- $\log\text{SumExp_forward_error}$: $m + \log \sum \exp (x_i - m)$, with the subtraction, each exp, each addition, the log and the final addition rounded with relative error u , and the spread $m - x_i \leq D$, is within $u | \text{exact} | + (1 + u) (u (\log n + \lambda) + \lambda)$ of the exact value, where $\lambda = -\log (1 - ((1 + u) ^ n \exp (u D) - 1))$.

Scope. The model has no absolute term: the VE, posterior and log-sum-exp theorems assume that every operation's result satisfies Rounded u , which for binary64 the bridge discharges only when each exact intermediate result is zero or in the normal range. **Underflow and overflow are excluded**; only the bridge's roundsTo_subnormal and roundsTo_error carry the absolute term 2^{-1075} . The Julia engines do not rely on these theorems at small mass: an evidence mass below `floatmin(Float64)` sends the query to the exact fallback (ADR 0014, ADR 0016), which Binary64.lean covers. Intermediate products that underflow while the final mass stays above `floatmin` are outside both. These are theorems about the factor algebra under an abstract rounding model, with order and association universally quantified; they are not about Julia's loop order, SIMD reassociation, array layout or execution, nor about the junction-tree, belief-propagation or brute-force paths. The run starts from the CPT entries as given (the Float64 inputs are the exact data), and the indices are worst-case bounds, not estimates.

```
set_option autoImplicit false
```

```
namespace BayesianNetworksProofs.ErrorBounds
```

The standard model of rounding

```
-- The standard model with unit roundoff 'u': 'c' is a computed result of an operation whose
exact result is 'x' when 'c = x * (1 + δ)' for some '|δ| ≤ u'. -/
def Rounded (u x c : R) : Prop := ∃ δ : R, |δ| ≤ u ∧ c = x * (1 + δ)

theorem rounded_iff {u : R} (hu : 0 ≤ u) (x c : R) : Rounded u x c ↔ |c - x| ≤ u * |x| := by
  constructor
  · rintro ⟨δ, hδ, rfl⟩
    rw [show x * (1 + δ) - x = x * δ by ring, abs_mul, mul_comm]
    exact mul_le_mul_of_nonneg_right hδ (abs_nonneg x)
  · intro h
    by_cases hx : x = 0
    · subst hx
      refine ⟨0, by simp using hu, ?_⟩
      have : |c| ≤ 0 := by simp using h
      simp using abs_nonpos_iff.mp this
    · refine ⟨(c - x) / x, ?_, by field_simp; ring⟩
      rw [abs_div]
      exact (div_le_iff₀ (abs_pos.mpr hx)).mpr h

-- An exact operation satisfies the model. -/
theorem rounded_exact {u : R} (hu : 0 ≤ u) (x : R) : Rounded u x x := ⟨0, by simp using hu, by ring⟩

-- 'γ n = (1 + u) ^ n - 1', the relative error bound after 'n' roundings. -/
def gamma (u : R) (n : ℕ) : R := (1 + u) ^ n - 1

variable {u : R}

theorem gamma_zero : gamma u 0 = 0 := by simp [gamma]

theorem one_add_gamma (n : ℕ) : 1 + gamma u n = (1 + u) ^ n := by simp [gamma]

theorem gamma_nonneg (hu : 0 ≤ u) (n : ℕ) : 0 ≤ gamma u n := by
  unfold gamma
  linarith [one_le_pow₀ (n := n) (show (1 : R) ≤ 1 + u by linarith)]

theorem gamma_mono (hu : 0 ≤ u) {m n : ℕ} (h : m ≤ n) : gamma u m ≤ gamma u n := by
  unfold gamma
  linarith [pow_le_pow_right₀ (show (1 : R) ≤ 1 + u by linarith) h]

theorem gamma_add (a b : ℕ) : (1 + gamma u a) * (1 + gamma u b) = 1 + gamma u (a + b) := by
  simp only [one_add_gamma, pow_add]
```

Relative error of nonnegative quantities

```

/-- 'x' approximates the exact value 'y': ' $|x - y| \leq e * y$ '. Used only for ' $0 \leq y$ ', where it says ' $x \in [(1 - e) y, (1 + e) y]$ '. -/
def RelWithin (e y x : R) : Prop := |x - y| ≤ e * y

theorem relWithin_refl (y : R) {e : R} (he : 0 ≤ e) (hy : 0 ≤ y) : RelWithin e y y := by
  simp [RelWithin]; positivity

theorem RelWithin.mono {e e' y x : R} (h : RelWithin e y x) (he : e ≤ e') (hy : 0 ≤ y) :
  RelWithin e' y x := h.trans (mul_le_mul_of_nonneg_right he hy)

theorem RelWithin.abs_le {e y x : R} (h : RelWithin e y x) (hy : 0 ≤ y) : |x| ≤ (1 + e) * y := by
  have h' : |x - y| ≤ e * y := h
  calc |x| = |(x - y) + y| := by ring_nf
  _ ≤ |x - y| + |y| := abs_add_le _ _
  _ ≤ e * y + y := by rw [abs_of_nonneg hy]; linarith
  _ = _ := by ring

/-- Relative errors of a product of nonnegative exact values compose multiplicatively. -/
theorem RelWithin.mul {a b y y' x x' : R} (h : RelWithin a y x) (h' : RelWithin b y' x')
  (ha : 0 ≤ a) (hy : 0 ≤ y) (hy' : 0 ≤ y') :
  RelWithin ((1 + a) * (1 + b) - 1) (y * y') (x * x') := by
  unfold RelWithin at *
  have he : x * x' - y * y' = (x - y) * (x' - y') + y * (x' - y') + y' * (x - y) := by ring
  rw [he]
  calc _ ≤ |(x - y) * (x' - y')| + |y * (x' - y')| + |y' * (x - y)| := abs_add_three _ _ _
  _ = |x - y| * |x' - y'| + y * |x' - y'| + y' * |x - y| := by
    rw [abs_mul, abs_mul, abs_mul, abs_of_nonneg hy, abs_of_nonneg hy']
  _ ≤ (a * y) * (b * y') + y * (b * y') + y' * (a * y) := by
    gcongr
  _ = _ := by ring

/-- Relative errors of a sum of nonnegative exact values do not grow beyond the larger. -/
theorem RelWithin.add {e y y' x x' : R} (h : RelWithin e y x) (h' : RelWithin e y' x') :
  RelWithin e (y + y') (x + x') := by
  unfold RelWithin at *
  calc |x + x' - (y + y')| = |(x - y) + (x' - y')| := by ring_nf
  _ ≤ |x - y| + |x' - y'| := abs_add_le _ _
  _ ≤ e * y + e * y' := add_le_add h h'
  _ = _ := by ring

/-- One rounding in the standard model multiplies the error budget by '1 + u'. -/
theorem RelWithin.round {e y x c : R} (h : RelWithin e y x) (hy : 0 ≤ y) (hu : 0 ≤ u)
  (hc : Rounded u x c) : RelWithin ((1 + u) * (1 + e) - 1) y c := by
  rw [rounded_iff hu] at hc
  have hx := h.abs_le hy
  unfold RelWithin at *
  calc |c - y| = |(c - x) + (x - y)| := by ring_nf
  _ ≤ |c - x| + |x - y| := abs_add_le _ _
  _ ≤ u * ((1 + e) * y) + e * y := add_le_add (hc.trans (mul_le_mul_of_nonneg_left hx hu)) h
  _ = _ := by ring

theorem RelWithin.mul_gamma (hu : 0 ≤ u) {a b : ℕ} {y y' x x' : R}
  (h : RelWithin (gamma u a) y x) (h' : RelWithin (gamma u b) y' x')
  (hy : 0 ≤ y) (hy' : 0 ≤ y') : RelWithin (gamma u (a + b)) (y * y') (x * x') := by
  have := h.mul h' (gamma_nonneg hu a) hy hy'
  rwa [gamma_add, add_sub_cancel_left] at this

theorem RelWithin.round_gamma (hu : 0 ≤ u) {a : ℕ} {y x c : R} (h : RelWithin (gamma u a) y x)
  (hy : 0 ≤ y) (hc : Rounded u x c) : RelWithin (gamma u (a + 1)) y c := by
  have := h.round hy hu hc
  rwa [show (1 + u) * (1 + gamma u a) - 1 = gamma u (a + 1) by
    rw [one_add_gamma]; unfold gamma; ring] at this

```

Association trees

```

/-- A binary association: the shape of a computation that combines its leaves pairwise. Any
parenthesisation of a sum or product of ' $n \geq 1$ ' terms is such a tree. -/
inductive Assoc (ι : Type) where
| leaf (i : ι)
| node (l r : Assoc ι)

namespace Assoc

variable {ι : Type}

/-- The leaves, left to right. -/
def leaves : Assoc ι → List ι
| leaf i ⇒ [i]
| node l r ⇒ leaves l ++ leaves r

/-- The number of operations along the longest root-to-leaf path. -/
def height : Assoc ι → ℕ
| leaf _ ⇒ 0
| node l r ⇒ max (height l) (height r) + 1

/-- The number of internal nodes: the number of operations the tree performs. -/
def nodes : Assoc ι → ℕ
| leaf _ ⇒ 0
| node l r ⇒ nodes l + nodes r + 1

theorem nodes_add_one (t : Assoc ι) : t.nodes + 1 = t.leaves.length := by
  induction t with
  | leaf i ⇒ rfl
  | node l r ihl ihr ⇒ simp only [nodes, leaves, List.length_append]; omega

theorem height_le_nodes (t : Assoc ι) : t.height ≤ t.nodes := by
  induction t with
  | leaf i ⇒ exact le_rfl
  | node l r ihl ihr ⇒ simp only [height, nodes]; omega

theorem height_add_one_le (t : Assoc ι) : t.height + 1 ≤ t.leaves.length := by
  have := t.nodes_add_one; have := t.height_le_nodes; omega

end Assoc

/-- 'c' is a value computed for the sum of the leaf values 'x i' of 't', combined in the
association 't' with every addition rounded in the standard model. -/
inductive SumRun (u : ℝ) {ι : Type} (x : ι → ℝ) : Assoc ι → ℝ → Prop
| leaf (i : ι) : SumRun u x (.leaf i) (x i)
| node {l r : Assoc ι} {cl cr c : ℝ} : SumRun u x l cl → SumRun u x r cr →
  Rounded u (cl + cr) c → SumRun u x (.node l r) c

/-- 'c' is a value computed for the product of the leaf values of 't', combined in the
association 't' with every multiplication rounded in the standard model. -/
inductive ProdRun (u : ℝ) {ι : Type} (x : ι → ℝ) : Assoc ι → ℝ → Prop
| leaf (i : ι) : ProdRun u x (.leaf i) (x i)
| node {l r : Assoc ι} {cl cr c : ℝ} : ProdRun u x l cl → ProdRun u x r cr →
  Rounded u (cl * cr) c → ProdRun u x (.node l r) c

variable {ι : Type}

theorem sum_map_nonneg {l : List ι} {y : ι → ℝ} (h : ∀ i ∈ l, 0 ≤ y i) : 0 ≤ (l.map y).sum :=
  List.sum_nonneg (fun _ hz ⇒ by
    obtain ⟨i, hi, rfl⟩ := List.mem_map.mp hz
    exact h i hi)

theorem prod_map_nonneg {l : List ι} {y : ι → ℝ} (h : ∀ i ∈ l, 0 ≤ y i) : 0 ≤ (l.map y).prod :=
  List.prod_nonneg (fun _ hz ⇒ by
    obtain ⟨i, hi, rfl⟩ := List.mem_map.mp hz

```

```

exact h i hi)

/-- **Computed sums.** If every leaf approximates a nonnegative exact value to relative
'γ a', a sum computed in the association 't' approximates the exact sum to relative
'γ (a + height t)'. -/
theorem SumRun.within (hu : 0 ≤ u) {x y : ι → ℝ} {a : ℕ} {t : Assoc ι} {c : ℝ}
  (h : SumRun u x t c) (hy : ∀ i ∈ t.leaves, 0 ≤ y i ∧ RelWithin (gamma u a) (y i) (x i)) :
  RelWithin (gamma u (a + t.height)) (t.leaves.map y).sum c := by
  induction h with
  | leaf i ⇒
    simp [Assoc.leaves, Assoc.height] using (hy i (by simp [Assoc.leaves])).2
  | @node l r cl cr c _ _ hc ihl ihr ⇒
    have hl := ihl (fun i hi ⇒ hy i (by simp [Assoc.leaves, hi]))
    have hr := ihr (fun i hi ⇒ hy i (by simp [Assoc.leaves, hi]))
    have nl : 0 ≤ (l.leaves.map y).sum :=
      sum_map_nonneg fun i hi ⇒ (hy i (by simp [Assoc.leaves, hi])).1
    have nr : 0 ≤ (r.leaves.map y).sum :=
      sum_map_nonneg fun i hi ⇒ (hy i (by simp [Assoc.leaves, hi])).1
    have hsum := (hl.mono (gamma_mono hu (show a + l.height ≤ a + max l.height r.height by omega))
      nl).add (hr.mono (gamma_mono hu (show a + r.height ≤ a + max l.height r.height by omega)) nr)
    have := hsum.round_gamma hu (add_nonneg nl nr) hc
    simp [Assoc.leaves, Assoc.height, List.map_append, List.sum_append, add_assoc] using this

/-- **Computed products.** If leaf 'i' approximates a nonnegative exact value to relative
'γ (e i)', a product computed in the association 't' approximates the exact product to relative
'γ (Σ e i + nodes t)', whatever the association. -/
theorem ProdRun.within (hu : 0 ≤ u) {x y : ι → ℝ} {e : ι → ℕ} {t : Assoc ι} {c : ℝ}
  (h : ProdRun u x t c) (hy : ∀ i ∈ t.leaves, 0 ≤ y i ∧ RelWithin (gamma u (e i)) (y i) (x i)) :
  RelWithin (gamma u ((t.leaves.map e).sum + t.nodes)) (t.leaves.map y).prod c := by
  induction h with
  | leaf i ⇒
    simp [Assoc.leaves, Assoc.nodes] using (hy i (by simp [Assoc.leaves])).2
  | @node l r cl cr c _ _ hc ihl ihr ⇒
    have hl := ihl (fun i hi ⇒ hy i (by simp [Assoc.leaves, hi]))
    have hr := ihr (fun i hi ⇒ hy i (by simp [Assoc.leaves, hi]))
    have nl : 0 ≤ (l.leaves.map y).prod :=
      prod_map_nonneg fun i hi ⇒ (hy i (by simp [Assoc.leaves, hi])).1
    have nr : 0 ≤ (r.leaves.map y).prod :=
      prod_map_nonneg fun i hi ⇒ (hy i (by simp [Assoc.leaves, hi])).1
    have := (hl.mul_gamma hu hr nl nr).round_gamma hu (mul_nonneg nl nr) hc
    convert this using 2
    · simp only [Assoc.leaves, Assoc.nodes, List.map_append, List.sum_append]; ring
    · simp [Assoc.leaves, List.map_append, List.prod_append]

```

Sums and products of lists, in any order and association

```

/-- 'c' is a computed sum of the values 'x i', 'i ∈ l': '0' for the empty list, and otherwise a
'SumRun' over some association of some reordering of 'l'. -/
def CompSum (u : ℝ) (x : ι → ℝ) (l : List ι) (c : ℝ) : Prop :=
  (l = [] ∧ c = 0) ∨ ∃ t : Assoc ι, t.leaves.Perm l ∧ SumRun u x t c

/-- 'c' is a computed product of the values 'x i', 'i ∈ l': '1' for the empty list, and
otherwise a 'ProdRun' over some association of some reordering of 'l'. -/
def CompProd (u : ℝ) (x : ι → ℝ) (l : List ι) (c : ℝ) : Prop :=
  (l = [] ∧ c = 1) ∨ ∃ t : Assoc ι, t.leaves.Perm l ∧ ProdRun u x t c

theorem CompSum.within (hu : 0 ≤ u) {x y : ι → ℝ} {a : ℕ} {l : List ι} {c : ℝ}
  (h : CompSum u x l c) (hy : ∀ i ∈ l, 0 ≤ y i ∧ RelWithin (gamma u a) (y i) (x i)) :
  RelWithin (gamma u (a + (l.length - 1))) (l.map y).sum c := by
  rcases h with ⟨rfl, rfl⟩ | ⟨t, hp, ht⟩
  · simp [RelWithin]
  · have h1 := ht.within hu (fun i hi ⇒ hy i (hp.subset hi))
    have hlen := t.height_add_one_le
    rw [hp.length_eq] at hlen

```

```

    rw [← (hp.map y).sum_eq]
    exact h1.mono (gamma_mono hu (by omega))
      (sum_map_nonneg fun i hi => (hy i (hp.subset hi)).1)

theorem CompProd.within (hu : 0 ≤ u) {x y : ι → ℝ} {e : ι → ℕ} {l : List ι} {c : ℝ}
  (h : CompProd u x l c) (hy : ∀ i ∈ l, 0 ≤ y i ∧ RelWithin (gamma u (e i)) (y i) (x i)) :
  RelWithin (gamma u ((l.map e).sum + (l.length - 1))) (l.map y).prod c := by
  rcases h with ⟨rfl, rfl⟩ | ⟨t, hp, ht⟩
  · simp [RelWithin, gamma_zero]
  · have h1 := ht.within hu (e := e) (fun i hi => hy i (hp.subset hi))
    have hlen := t.nodes_add_one
    rw [hp.length_eq] at hlen
    rw [← (hp.map y).prod_eq, ← (hp.map e).sum_eq]
    rwa [show t.nodes = l.length - 1 by omega] at h1

/-- Products. A product of 'k' nonnegative numbers computed in any order and association
under the standard model is within relative 'γ (k - 1)' of the exact product. -/
theorem compProd_forward_error (hu : 0 ≤ u) {y : ι → ℝ} {l : List ι} {c : ℝ}
  (hy : ∀ i ∈ l, 0 ≤ y i) (h : CompProd u y l c) :
  RelWithin (gamma u (l.length - 1)) (l.map y).prod c := by
  have := h.within hu (e := fun _ => 0)
  (fun i hi => ⟨hy i hi, relWithin_refl _ (by simp [gamma_zero]) (hy i hi)⟩)
  simp using this

/-- Sums, in any association. For every association tree 't' with 'n' leaves, a sum of
nonnegative numbers computed along 't' under the standard model is within relative 'γ (n - 1)'
of the exact sum; more sharply, within 'γ (height t)'. -/
theorem sumRun_forward_error (hu : 0 ≤ u) {y : ι → ℝ} (t : Assoc ι) {c : ℝ}
  (hy : ∀ i ∈ t.leaves, 0 ≤ y i) (h : SumRun u y t c) :
  RelWithin (gamma u t.height) (t.leaves.map y).sum c ∧
  RelWithin (gamma u (t.leaves.length - 1)) (t.leaves.map y).sum c := by
  have h1 := h.within hu (a := 0)
  (fun i hi => ⟨hy i hi, relWithin_refl _ (by simp [gamma_zero]) (hy i hi)⟩)
  rw [zero_add] at h1
  have hn : 0 ≤ (t.leaves.map y).sum := sum_map_nonneg hy
  exact ⟨h1, h1.mono (gamma_mono hu (by have := t.height_add_one_le; omega)) hn⟩

```

The bridge to binary64

```

section Bridge

open Binary64

/-- The unit roundoff of binary64, '2 ^ -53'. -/
noncomputable def binary64Unit : ℝ := 2 ^ (-53 : ℤ)

theorem binary64Unit_nonneg : 0 ≤ binary64Unit := by unfold binary64Unit; positivity

/-- A finite word with the sign of 'q' and magnitude 'M * 2 ^ (E - 52)', for a 53-bit 'M' and a
normal exponent 'E', or 'M ≤ 2 ^ 52' at the subnormal exponent 'E = -1022'. -/
private theorem finite_word (q : ℚ) (M : ℕ) (E : ℤ) (hE : -1022 ≤ E) (hE' : E ≤ 1023)
  (hM : M ≤ 2 ^ 53) (hsub : M < 2 ^ 52 → E = -1022) (hno : ¬(M = 2 ^ 53 ∧ E = 1023)) :
  ∃ W, IsFinite W ∧ |value W - q| = abs ((M : ℚ) * 2 ^ (E - 52) - |q|) := by
  have hsub' : M < 2 ^ 52 → max (E - 52) (-1074) = -1074 := fun h => by rw [hsub h]; norm_num
  obtain ⟨hlt, -, hfin⟩ := packWord_spec M E hE' hM hsub'
  obtain ⟨hf, hmag, -⟩ := hfin hno
  rw [show max (E - 52) (-1074) = E - 52 by omega] at hmag
  obtain ⟨h64, -, hfield, -, -, hval⟩ := signed_word (q < 0) (packWord M E) hlt
  refine ⟨_, ⟨h64, by rw [hfield]; exact hf⟩, ?_⟩
  rw [hval, hmag]
  by_cases hq : q < 0
  · rw [if_pos hq, abs_of_neg hq, show -((M : ℚ) * 2 ^ (E - 52)) - q =
    -((M : ℚ) * 2 ^ (E - 52) - -q) by ring, abs_neg]
  · rw [if_neg hq, abs_of_nonneg (not_lt.mp hq)]

```

```

/-- Rounding 't' to the nearest integer: the floor of 't + 1/2' is within '1/2' of 't'. -/
private theorem floor_half (t : ℚ) (ht : 0 ≤ t) :
  |((Lt + 1 / 2)⌊ : ℕ) : ℚ) - t| ≤ 1 / 2 ∧ ((Lt + 1 / 2)⌊ : ℕ) : ℚ) ≤ t + 1 / 2 := by
  have h1 := Nat.floor_le (show 0 ≤ t + 1 / 2 by linarith)
  have h2 := Nat.lt_floor_add_one (t + 1 / 2)
  exact ⟨abs_le.mpr ⟨by linarith, by linarith⟩, h1⟩

set_option exponentiation.threshold 1100 in
/-- **The bridge, normal range.** A word that IEEE 754 round-to-nearest delivers for a rational
'q' with '2 ^ -1022 ≤ |q| < overflowThreshold' has relative error at most '2 ^ -53'. The proof
exhibits a finite word within '2 ^ -53 |q|' of 'q' and applies 'RoundsTo.nearest'. -/
theorem roundsTo_relative (q : ℚ) (w : ℕ) (hw : RoundsTo q w) :
  (hlo : (2 : ℚ) ^ (-1022 : ℤ) ≤ |q|) (hhi : |q| < overflowThreshold) :
  |value w - q| ≤ 2 ^ (-53 : ℤ) * |q| := by
  set a := |q| with ha
  have hapos : 0 < a := lt_of_lt_of_le (by positivity) hlo
  set E := Int.log 2 a with hEdef
  have l1 : (2 : ℚ) ^ E ≤ a := Int.zpow_log_le_self (b := 2) (by norm_num) hapos
  have l2 : a < (2 : ℚ) ^ (E + 1) := Int.lt_zpow_succ_log_self (b := 2) (by norm_num) a
  have hthr : overflowThreshold = (2 ^ 53 - 1 / 2) * 2 ^ (971 : ℤ) := by
    norm_num [overflowThreshold]
  have hE : -1022 ≤ E := by
    by_contra h
    have : (2 : ℚ) ^ (E + 1) ≤ 2 ^ (-1022 : ℤ) := zpow_le_zpow_right0 (by norm_num) (by omega)
    linarith
  have hE' : E ≤ 1023 := by
    by_contra h
    have : (2 : ℚ) ^ (1024 : ℤ) ≤ 2 ^ E := zpow_le_zpow_right0 (by norm_num) (by omega)
    have h2 : overflowThreshold < (2 : ℚ) ^ (1024 : ℤ) := by rw [hthr]; norm_num
    linarith
  set s : ℤ := E - 52 with hs
  have hp : (0 : ℚ) < 2 ^ s := by positivity
  set t := a / 2 ^ s with ht
  have hta : a = t * 2 ^ s := by rw [ht]; field_simp
  have ht52 : (2 : ℚ) ^ 52 ≤ t := by
    rw [ht, le_div_iff0 hp]
    calc (2 : ℚ) ^ 52 * 2 ^ s = 2 ^ E := by
      rw [show (2 : ℚ) ^ 52 = 2 ^ (52 : ℤ) by norm_num, ← zpow_add0 two_ne_zero]
      congr 1; omega
    _ ≤ a := l1
  have ht53 : t < 2 ^ 53 := by
    rw [ht, div_lt_iff0 hp]
    calc a < 2 ^ (E + 1) := l2
    _ = 2 ^ 53 * 2 ^ s := by
      rw [show (2 : ℚ) ^ 53 = 2 ^ (53 : ℤ) by norm_num, ← zpow_add0 two_ne_zero]
      congr 1; omega
  obtain ⟨hM, hMle⟩ := floor_half t (by linarith)
  set M := Lt + 1 / 2)⌊ with hMdef
  have hM52 : 2 ^ 52 ≤ M := Nat.le_floor (by push_cast; linarith)
  have hM53 : M ≤ 2 ^ 53 := by
    have : (M : ℚ) < 2 ^ 53 + 1 := by linarith
    exact_mod_cast Nat.lt_succ_iff.mp (by exact_mod_cast this)
  have hno : ¬(M = 2 ^ 53 ∧ E = 1023) := by
    rintro ⟨h1, h2⟩
    have : (2 : ℚ) ^ 53 - 1 / 2 ≤ t := by
      have : ((2 ^ 53 : ℕ) : ℚ) ≤ t + 1 / 2 := h1 ▸ hMle
      push_cast at this; linarith
    have hs' : s = 971 := by omega
    rw [hs'] at hta
    have : overflowThreshold ≤ a := by
      rw [hthr, hta]; exact mul_le_mul_of_nonneg_right this (by positivity)
    linarith
  obtain ⟨W, hW, hdist⟩ := finite_word q M E hE hE' hM53 (fun h ⇒ absurd h (by omega)) hno
  refine le_trans (hw.nearest hhi W hW) ?_
  rw [hdist, ← ha, hta, ← sub_mul, abs_mul, abs_of_pos hp]

```

```

calc |(M : ℚ) - t| * 2 ^ s ≤ 1 / 2 * 2 ^ s := mul_le_mul_of_nonneg_right hM hp.le
- = 2 ^ (-53 : ℤ) * 2 ^ E := by
  rw [← zpow_add0 two_ne_zero, show (1 : ℚ) / 2 = 2 ^ (-1 : ℤ) by norm_num,
    ← zpow_add0 two_ne_zero]
  congr 1; omega
- ≤ 2 ^ (-53 : ℤ) * (t * 2 ^ s) := by rw [← hta]; gcongr

set_option exponentiation.threshold 1100 in
/-- The bridge, subnormal range. Below '2 ^ -1022' the error is at most half the subnormal
spacing, '2 ^ -1075': an absolute, not a relative, bound. -/
theorem roundsTo_subnormal (q : ℚ) (w : ℕ) (hw : RoundsTo q w)
  (hq : |q| < (2 : ℚ) ^ (-1022 : ℤ)) : |value w - q| ≤ 2 ^ (-1075 : ℤ) := by
  have hthr : (2 : ℚ) ^ (-1022 : ℤ) < overflowThreshold := by
    norm_num [overflowThreshold]
  set a := |q| with ha
  have ha0 : 0 ≤ a := abs_nonneg q
  have hp : (0 : ℚ) < 2 ^ (-1074 : ℤ) := by positivity
  set t := a / 2 ^ (-1074 : ℤ) with ht
  have hta : a = t * 2 ^ (-1074 : ℤ) := by rw [ht]; field_simp
  have ht52 : t < 2 ^ 52 := by
    rw [ht, div_lt_iff0 hp]
    calc a < 2 ^ (-1022 : ℤ) := hq
    - = 2 ^ 52 * 2 ^ (-1074 : ℤ) := by
      rw [show (2 : ℚ) ^ 52 = 2 ^ (52 : ℤ) by norm_num, ← zpow_add0 two_ne_zero]; norm_num
  obtain ⟨hM, hMle⟩ := floor_half t (div_nonneg ha0 hp.le)
  set M := Lt + 1 / 2J, with hMdef
  have hM52 : M ≤ 2 ^ 52 := by
    have : (M : ℚ) < 2 ^ 52 + 1 := by linarith
    exact_mod_cast Nat.lt_succ_iff.mp (by exact_mod_cast this)
  obtain ⟨W, hW, hdist⟩ := finite_word q M (-1022) le_rfl (by norm_num) (by omega)
  (fun _ => rfl) (by omega)
  refine le_trans (hw.nearest (lt_trans hq hthr) W hW) ?_
  rw [hdist, ← ha, hta, show (-1022 : ℤ) - 52 = -1074 by norm_num, ← sub_mul, abs_mul,
    abs_of_pos hp]
  calc |(M : ℚ) - t| * 2 ^ (-1074 : ℤ) ≤ 1 / 2 * 2 ^ (-1074 : ℤ) :=
    mul_le_mul_of_nonneg_right hM hp.le
  - = - := by
    rw [show (-1074 : ℤ) = 1 + (-1075) by norm_num, zpow_add0 two_ne_zero]; norm_num

/-- Both ranges at once: below the overflow threshold, '|value w - q| ≤ 2 ^ -53 |q| + 2 ^ -1075'. -/
theorem roundsTo_error (q : ℚ) (w : ℕ) (hw : RoundsTo q w) (hhi : |q| < overflowThreshold) :
  |value w - q| ≤ 2 ^ (-53 : ℤ) * |q| + 2 ^ (-1075 : ℤ) := by
  by_cases h : (2 : ℚ) ^ (-1022 : ℤ) ≤ |q|
  · have := roundsTo_relative q w hw h hhi
    have : (0 : ℚ) ≤ 2 ^ (-1075 : ℤ) := by positivity
    linarith
  · have := roundsTo_subnormal q w hw (not_le.mp h)
    have : (0 : ℚ) ≤ 2 ^ (-53 : ℤ) * |q| := by positivity
    linarith

set_option exponentiation.threshold 1100 in
/-- A correctly rounded zero is exactly zero. -/
theorem roundsTo_zero (w : ℕ) (hw : RoundsTo 0 w) : value w = 0 := by
  have hthr : |(0 : ℚ)| < overflowThreshold := by norm_num [overflowThreshold]
  have h0 : IsFinite 0 := ⟨by norm_num, by simp [field]⟩
  have hv0 : value 0 = 0 := by simp [value, magnitude, signBit, field, fraction]
  have := hw.nearest hthr 0 h0
  rw [hv0, sub_zero, sub_zero, abs_zero] at this
  exact abs_nonpos_iff.mp this

/-- The standard model holds for IEEE 754 binary64 rounding away from underflow and
overflow: when the exact result is zero or has magnitude in '[2 ^ -1022, overflowThreshold)',
its correctly rounded value is 'Rounded binary64Unit'. -/
theorem roundsTo_rounded (q : ℚ) (w : ℕ) (hw : RoundsTo q w)
  (hq : q = 0 ∨ ((2 : ℚ) ^ (-1022 : ℤ) ≤ |q| ∧ |q| < overflowThreshold)) :
  Rounded binary64Unit (q : ℝ) (value w : ℝ) := by

```

```

rw [rounded_iff binary64Unit_nonneg]
rcases hq with rfl | (hlo, hhi)
• simp [roundsTo_zero w hw]
• have h := roundsTo_relative q w hw hlo hhi
  have h' : ((|value w - q| : ℚ) : ℝ) ≤ ((2 ^ (-53 : ℤ) * |q| : ℚ) : ℝ) := by exact_mod_cast h
  push_cast at h'
  simp [binary64Unit] using h'

/-- The same for the transcription of '_nearest_binary64', through 'nearestBinary64_roundsTo'. -/
theorem nearestBinary64_rounded (q : ℚ)
  (hq : q = 0 ∨ ((2 : ℚ) ^ (-1022 : ℤ) ≤ |q| ∧ |q| < overflowThreshold)) :
  Rounded binary64Unit (q : ℝ) (value (nearestBinary64 q) : ℝ) :=
  roundsTo_rounded q _ (nearestBinary64_roundsTo q) hq

/-- '_nearest_binary64' in the normal range: relative error at most '2 ^ -53'. -/
theorem nearestBinary64_relative (q : ℚ) (hlo : (2 : ℚ) ^ (-1022 : ℤ) ≤ |q|)
  (hhi : |q| < overflowThreshold) :
  |value (nearestBinary64 q) - q| ≤ 2 ^ (-53 : ℤ) * |q| :=
  roundsTo_relative q _ (nearestBinary64_roundsTo q) hlo hhi

/-- '_nearest_binary64' in the subnormal range: absolute error at most '2 ^ -1075'. -/
theorem nearestBinary64_subnormal (q : ℚ) (hq : |q| < (2 : ℚ) ^ (-1022 : ℤ)) :
  |value (nearestBinary64 q) - q| ≤ 2 ^ (-1075 : ℤ) :=
  roundsTo_subnormal q _ (nearestBinary64_roundsTo q) hq

end Bridge

```

Normalisation

```

/-- **Quotients.** If 'x' approximates 'y ≥ 0' to relative 'α ≤ 1' and 'X' approximates 'Y > 0'
to relative 'β < 1', then 'x / X' approximates 'y / Y' to relative '(1 + α) / (1 - β) - 1'. -/
theorem RelWithin.div {α β y Y x X : ℝ} (h : RelWithin α y x) (H : RelWithin β Y X)
  (hα : 0 ≤ α) (hα1 : α ≤ 1) (hβ : 0 ≤ β) (hβ1 : β < 1) (hy : 0 ≤ y) (hY : 0 < Y) :
  RelWithin ((1 + α) / (1 - β) - 1) (y / Y) (x / X) := by
  unfold RelWithin at *
  obtain ⟨hx1, hx2⟩ := _root_.abs_le.mp H
  obtain ⟨hx1, hx2⟩ := _root_.abs_le.mp h
  have hb : 0 < 1 - β := by linarith
  have hXpos : 0 < X := by nlinarith
  have hr : 0 ≤ y / Y := div_nonneg hy hY.le
  have up : x / X ≤ (1 + α) / (1 - β) * (y / Y) := by
    calc x / X ≤ (1 + α) * y / X := div_le_div_of_nonneg_right (by linarith) hXpos.le
    _ ≤ (1 + α) * y / ((1 - β) * Y) :=
      div_le_div_of_nonneg_left (by positivity) (by positivity) (by linarith)
    _ = _ := by field_simp
  have key : 2 - (1 + α) / (1 - β) ≤ (1 - α) / (1 + β) := by
    have h1 : (0 : ℝ) < 1 + β := by linarith
    rw [sub_le_iff_le_add, div_add_div _ _ h1.ne' hb.ne', le_div_iff₀ (by positivity)]
    nlinarith [mul_nonneg hα hβ, sq_nonneg β]
  have lo : (2 - (1 + α) / (1 - β)) * (y / Y) ≤ x / X := by
    calc (2 - (1 + α) / (1 - β)) * (y / Y) ≤ (1 - α) / (1 + β) * (y / Y) :=
      mul_le_mul_of_nonneg_right key hr
    _ = (1 - α) * y / ((1 + β) * Y) := by field_simp
    _ ≤ (1 - α) * y / X :=
      div_le_div_of_nonneg_left (mul_nonneg (by linarith) hy) hXpos (by linarith)
    _ ≤ x / X := div_le_div_of_nonneg_right (by linarith) hXpos.le
  rw [_root_.abs_le]
  constructor <=> nlinarith

open FiniteDistribution in
/- **Normalisation.** Exact nonnegative weights 'w' with positive mass; computed weights within
relative 'γ N'; a computed mass 'Z', summed in any order and association; and each posterior
entry the rounded quotient 'wc a / Z'. With 'K = N + |A|' and 'γ K < 1', the computed mass is
within relative 'γ (K - 1)' of the exact mass, and every computed posterior entry within

```

```

relative '(1 + γ K) / (1 - γ K) - 1' of the exact one. -/
theorem normalize_forward_error {A : Type} [Fintype A] (hu : 0 ≤ u) (w wc : A → ℝ)
  (hw : ∀ a, 0 ≤ w a) (N : ℕ) (hwc : ∀ a, RelWithin (gamma u N) (w a) (wc a)) (Z : ℝ)
  (hZ : CompSum u wc Finset.univ.toList Z) (post : A → ℝ)
  (hpost : ∀ a, Rounded u (wc a / Z) (post a)) (hmass : 0 < mass w)
  (hsmall : gamma u (N + Fintype.card A) < 1) :
  RelWithin (gamma u (N + Fintype.card A - 1)) (mass w) Z ∧
  ∀ a, RelWithin ((1 + gamma u (N + Fintype.card A)) / (1 - gamma u (N + Fintype.card A)) - 1)
    (w a / mass w) (post a) := by
set K := N + Fintype.card A with hK
have hcard : 0 < Fintype.card A := by
  obtain ⟨a, -⟩ := Finset.exists_ne_zero_of_sum_ne_zero (s := Finset.univ) (f := w)
  (by rw [← mass]; exact hmass.ne')
  exact Fintype.card_pos_iff.mpr ⟨a⟩
have hZ' := hZ.within hu (y := w) (fun a _ => ⟨hw a, hwc a⟩)
rw [Finset.sum_map_toList, Finset.length_toList, Finset.card_univ] at hZ'
have hmassZ : RelWithin (gamma u (K - 1)) (mass w) Z := by
  rw [show K - 1 = N + (Fintype.card A - 1) by omega]; exact hZ'
refine ⟨hmassZ, fun a => ?_⟩
have g1 : gamma u N ≤ gamma u K := gamma_mono hu (by omega)
have g2 : gamma u (K - 1) ≤ gamma u K := gamma_mono hu (by omega)
have g3 : gamma u (N + 1) ≤ gamma u K := gamma_mono hu (by omega)
have hq := (hwc a).div hmassZ (gamma_nonneg hu N) (by linarith) (gamma_nonneg hu _)
  (by linarith) (hw a) hmass
have hr := hq.round (div_nonneg (hw a) hmass.le) hu (hpost a)
refine hr.mono ?_ (div_nonneg (hw a) hmass.le)
have hb : 0 < 1 - gamma u K := by linarith
have hb' : 1 - gamma u K ≤ 1 - gamma u (K - 1) := by linarith
have hb'' : 0 < 1 - gamma u (K - 1) := by linarith
have h1 : (1 + u) * (1 + gamma u N) = 1 + gamma u (N + 1) := by
  rw [one_add_gamma, one_add_gamma, pow_succ]; ring
rw [show (1 + u) * (1 + ((1 + gamma u N) / (1 - gamma u (K - 1)) - 1)) - 1 =
  (1 + u) * (1 + gamma u N) / (1 - gamma u (K - 1)) - 1 by ring, h1]
gcongr ?_ - 1
calc (1 + gamma u (N + 1)) / (1 - gamma u (K - 1)) ≤ (1 + gamma u K) / (1 - gamma u (K - 1)) :=
  div_le_div_of_nonneg_right (by linarith) hb''.le
_ ≤ _ := div_le_div_of_nonneg_left (by linarith [gamma_nonneg hu K]) hb hb'

```

Variable elimination under the standard model

```

open FinBayesNet FinBayesNet.Factor FiniteDistribution

variable {bn : FinBayesNet}

/-- A factor of the exact algebra paired with the table a floating-point run computed for it. -/
abbrev Computed (bn : FinBayesNet) := Factor bn ℝ × (bn.Assignment → ℝ)

/-- The computed factors in the bucket of 'v', selected by the exact scopes as in 'bucket'. -/
def cbucket (v : bn.V) (ps : List (Computed bn)) : List (Computed bn) :=
  ps.filter (fun p => decide (v ∈ p.1.scope))

/-- The computed factors outside the bucket of 'v', as in 'outside'. -/
def coutside (v : bn.V) (ps : List (Computed bn)) : List (Computed bn) :=
  ps.filter (fun p => decide (v ∉ p.1.scope))

theorem cbucket_fst (v : bn.V) (ps : List (Computed bn)) :
  (cbucket v ps).map Prod.fst = bucket v (ps.map Prod.fst) := by
  induction ps with
  | nil => rfl
  | cons p ps ih =>
    by_cases h : v ∈ p.1.scope <=> simp_all [cbucket, bucket]

theorem coutside_fst (v : bn.V) (ps : List (Computed bn)) :
  (coutside v ps).map Prod.fst = outside v (ps.map Prod.fst) := by

```

```

induction ps with
| nil ⇒ rfl
| cons p ps ih ⇒
  by_cases h : v ∈ p.1.scope <;> simp_all [coutside, outside]

/-- One computed sum-out: every entry of 'g' is a sum, computed in some order and association,
over the states 'a' of 'v' of values 'c a', each a product, computed in some order and
association, of the bucket's computed tables at 'x' updated to 'v = a'. -/
def SumOutStep (u : ℝ) (v : bn.V) (ps : List (Computed bn)) (g : bn.Assignment → ℝ) : Prop :=
  ∀ x, ∃ c : bn.states v → ℝ, CompSum u c Finset.univ.toList (g x) ∧
  ∀ a, CompProd u (fun p : Computed bn ⇒ p.2 (Function.update x v a)) (cbucket v ps) (c a)

/-- **An approximate elimination run**, step for step beside 'eliminateAll': each step replaces
the bucket of 'v' by the exact 'sumBucket' paired with a computed table 'g' satisfying
'SumOutStep', and keeps every factor outside the bucket with its table unchanged. -/
inductive Run (u : ℝ) : List (Computed bn) → List bn.V → List (Computed bn) → Prop
| nil (ps : List (Computed bn)) : Run u ps [] ps
| cons {v : bn.V} {vs : List bn.V} {ps out : List (Computed bn)} {g : bn.Assignment → ℝ} :
  SumOutStep u v ps g →
  Run u ((sumBucket v (bucket v (ps.map Prod.fst)), g) :: coutside v ps) vs out →
  Run u ps (v :: vs) out

/-- The input factors, with their exact tables: the run starts from the CPT entries as given. -/
def inputs (fs : List (Factor bn ℝ)) : List (Computed bn) := fs.map fun f ⇒ (f, f.value)

/-- The run's exact half is 'eliminateAll'. -/
theorem Run.fst {ps : List (Computed bn)} {vs : List bn.V} {out : List (Computed bn)}
(h : Run u ps vs out) : out.map Prod.fst = eliminateAll (ps.map Prod.fst) vs := by
  induction h with
  | nil ps ⇒ rfl
  | @cons v vs ps out g _ _ ih ⇒
    rw [ih, List.map_cons, coutside_fst]
    rfl

/-- The explicit index: 'm' input factors and the summed domain sizes of the eliminated
variables, less one. -/
def veIndex (m : ℕ) (vs : List bn.V) : ℕ :=
  m + (vs.map fun v ⇒ Fintype.card (bn.states v)).sum - 1

/-- A computed factor with nonnegative exact entries and some uniform relative bound. -/
def Good (u : ℝ) (p : Computed bn) : Prop :=
  (∀ x, 0 ≤ p.1.value x) ∧ ∃ n : ℕ, ∀ x, RelWithin (gamma u n) (p.1.value x) (p.2 x)

/-- The least 'n' with every computed entry within relative 'γ n'. -/
noncomputable def errIndex (u : ℝ) (p : Computed bn) : ℕ :=
  sInf {n : ℕ | ∀ x, RelWithin (gamma u n) (p.1.value x) (p.2 x)}

theorem Good.within {p : Computed bn} (h : Good u p) :
  ∀ x, RelWithin (gamma u (errIndex u p)) (p.1.value x) (p.2 x) :=
  Nat.sInf_mem (s := {n : ℕ | ∀ x, RelWithin (gamma u n) (p.1.value x) (p.2 x)}) h.2

theorem errIndex_le {p : Computed bn} {n : ℕ}
(h : ∀ x, RelWithin (gamma u n) (p.1.value x) (p.2 x)) : errIndex u p ≤ n :=
  Nat.sInf_le h

/-- The potential 'Σ (errIndex + 1)': a step raises it by at most the domain size. -/
noncomputable def potential (u : ℝ) (ps : List (Computed bn)) : ℕ :=
  (ps.map fun p ⇒ errIndex u p + 1).sum

theorem potential_eq (ps : List (Computed bn)) :
  potential u ps = (ps.map (errIndex u)).sum + ps.length := by
  induction ps with
  | nil ⇒ rfl
  | cons p ps ih ⇒ simp only [potential, List.map_cons, List.sum_cons, List.length_cons] at *; omega

theorem potential_split (v : bn.V) (ps : List (Computed bn)) :

```

```

    potential u ps = potential u (cbucket v ps) + potential u (coutside v ps) := by
induction ps with
| nil => rfl
| cons p ps ih =>
    by_cases h : v ∈ p.1.scope <;>
    simp_all [potential, cbucket, coutside] <;> omega

theorem product_eq_map (fs : List (Factor bn ℝ)) (x : bn.Assignment) :
    product fs x = (fs.map fun f => f.value x).prod := rfl

/-- One step: the new computed factor is good, and its index plus one is at most the bucket's
potential plus the domain size of 'v'. -/
theorem sumOutStep_good (hu : 0 ≤ u) {v : bn.V} {ps : List (Computed bn)}
    {g : bn.Assignment → ℝ} (hgood : ∀ p ∈ ps, Good u p) (hstep : SumOutStep u v ps g) :
    Good u (sumBucket v (bucket v (ps.map Prod.fst)), g) ∧
    errIndex u (sumBucket v (bucket v (ps.map Prod.fst)), g) + 1 ≤
    potential u (cbucket v ps) + Fintype.card (bn.states v) := by
have hb : ∀ p ∈ cbucket v ps, Good u p := fun p hp => hgood p (List.mem_filter.mp hp).1
have hexact : ∀ z, product (bucket v (ps.map Prod.fst)) z =
    ((cbucket v ps).map fun p : Computed bn => p.1.value z).prod := by
    intro z
    rw [product_eq_map, ← cbucket_fst, List.map_map]
    rfl
have hnn : ∀ z, 0 ≤ product (bucket v (ps.map Prod.fst)) z := fun z => by
    rw [hexact]; exact prod_map_nonneg fun p hp => (hb p hp).1 z
set n := ((cbucket v ps).map (errIndex u)).sum + ((cbucket v ps).length - 1) +
    (Fintype.card (bn.states v) - 1) with hn
have hwithin : ∀ x, RelWithin (gamma u n) ((sumBucket v (bucket v (ps.map Prod.fst)))) (value x)
    (g x) := by
    intro x
    obtain ⟨c, hsum, hprod⟩ := hstep x
    have hc : ∀ a, RelWithin (gamma u (((cbucket v ps).map (errIndex u)).sum +
        ((cbucket v ps).length - 1))) (product (bucket v (ps.map Prod.fst))
        (Function.update x v a)) (c a) := by
        intro a
        rw [hexact]
        exact (hprod a).within hu (y := fun p : Computed bn => p.1.value (Function.update x v a))
            (fun p hp => ⟨(hb p hp).1 _, (hb p hp).within _⟩)
    have := hsum.within hu (y := fun a => product (bucket v (ps.map Prod.fst))
        (Function.update x v a)) (fun a _ => ⟨hnn _, hc a⟩)
    rw [Finset.sum_map_toList, Finset.length_toList, Finset.card_univ] at this
    exact this
have hcard : 0 < Fintype.card (bn.states v) := Fintype.card_pos
refine ⟨⟨fun x => Finset.sum_nonneg fun a _ => hnn _, n, hwithin⟩, ?_⟩
have := errIndex_le (p := (sumBucket v (bucket v (ps.map Prod.fst)), g)) hwithin
rw [potential_eq]
omega

/-- The invariant of a run: every computed factor stays good, and the potential grows by at most
the domain size of each eliminated variable. -/
theorem Run.good (hu : 0 ≤ u) {ps : List (Computed bn)} {vs : List bn.V}
    {out : List (Computed bn)} (h : Run u ps vs out) (hgood : ∀ p ∈ ps, Good u p) :
    (∀ p ∈ out, Good u p) ∧
    potential u out ≤ potential u ps + (vs.map fun v => Fintype.card (bn.states v)).sum := by
induction h with
| nil ps => exact ⟨hgood, by simp⟩
| @cons v vs ps out g hstep _ ih =>
    obtain ⟨hnew, hle⟩ := sumOutStep_good hu hgood hstep
    obtain ⟨hout, hpot⟩ := ih (by
        intro p hp
        rcases List.mem_cons.mp hp with rfl | hp
        · exact hnew
        · exact hgood p (List.mem_filter.mp hp).1)
    refine ⟨hout, ?_⟩
    have hsplit := potential_split (u := u) v ps
    simp only [potential, List.map_cons, List.sum_cons] at hpot hle hsplit

```

omega

```

theorem inputs_good (fs : List (Factor bn R)) (hfs : ∀ f ∈ fs, ∀ x, 0 ≤ f.value x) :
  (∀ p ∈ inputs fs, Good u p) ∧ potential u (inputs fs) = fs.length := by
have h0 : ∀ f ∈ fs, ∀ x, RelWithin (gamma u 0) (f.value x) (f.value x) := fun f hf x =>
  relWithin_refl _ (by rw [gamma_zero]) (hfs f hf x)
refine ⟨fun p hp => ?_, ?_⟩
• obtain ⟨f, hf, rfl⟩ := List.mem_map.mp hp
  exact ⟨hfs f hf, 0, h0 f hf⟩
• have : ∀ f ∈ fs, errIndex u ((f, f.value) : Computed bn) = 0 := fun f hf =>
  Nat.le_zero.mp (errIndex_le (h0 f hf))
  unfold potential inputs
  rw [List.map_map]
  induction fs with
  | nil => rfl
  | cons f fs ih =>
    simp only [List.map_cons, List.sum_cons, List.length_cons, Function.comp_apply]
    rw [this f (by simp), ih (fun g hg => hfs g (by simp [hg])) (fun g hg x => h0 g (by simp [hg])
      ↪ x)
      (fun g hg => this g (by simp [hg]))]
    ring

/-- **Forward error of variable elimination.** Let 'fs' be nonnegative factors, 'vs' an
elimination order, and 'out' the result of any approximate run under the standard model with
unit roundoff 'u ≥ 0'. Then the exact half of 'out' is 'eliminateAll fs vs'; every entry of every
computed factor in 'out' is within relative 'γ N' of the exact entry; and so is any product of the
remaining computed factors, in any order and association, with
'N = veIndex fs.length vs = |fs| + Σ_{v ∈ vs} |states v| - 1'. -/
theorem eliminateAll_forward_error (hu : 0 ≤ u) (fs : List (Factor bn R))
  (hfs : ∀ f ∈ fs, ∀ x, 0 ≤ f.value x) (vs : List bn.V) (out : List (Computed bn))
  (hrun : Run u (inputs fs) vs out) :
  out.map Prod.fst = eliminateAll fs vs ∧
  (∀ p ∈ out, ∀ x, RelWithin (gamma u (veIndex fs.length vs)) (p.1.value x) (p.2 x)) ∧
  ∀ x c, CompProd u (fun p : Computed bn => p.2 x) out c →
  RelWithin (gamma u (veIndex fs.length vs)) (product (eliminateAll fs vs) x) c := by
have hfst : out.map Prod.fst = eliminateAll fs vs := by
  rw [hrun.fst]; simp [inputs, Function.comp_def]
obtain ⟨hgood0, hpot0⟩ := inputs_good (u := u) fs hfs
obtain ⟨hgood, hpot⟩ := hrun.good hu hgood0
rw [hpot0] at hpot
have hpe := potential_eq (u := u) out
refine ⟨hfst, fun p hp x => ?_, fun x c hc => ?_⟩
• have hle : errIndex u p + 1 ≤ potential u out :=
  List.le_sum_of_mem (List.mem_map.mpr ⟨p, hp, rfl⟩)
  have hidx : errIndex u p ≤ veIndex fs.length vs := by unfold veIndex; omega
  exact ((hgood p hp).within x).mono (gamma_mono hu hidx) ((hgood p hp).1 x)
• have := hc.within hu (y := fun p : Computed bn => p.1.value x) (e := errIndex u)
  (fun p hp => ⟨(hgood p hp).1 x, (hgood p hp).within x⟩)
  rw [← hfst, product_eq_map, List.map_map]
  have hidx : (out.map (errIndex u)).sum + (out.length - 1) ≤ veIndex fs.length vs := by
    have h0 : out.length = 0 → (out.map (errIndex u)).sum = 0 := fun h => by
      rw [List.length_eq_zero_iff.mp h]; rfl
    unfold veIndex; omega
  refine this.mono (gamma_mono hu hidx) ?_
  exact prod_map_nonneg fun p hp => (hgood p hp).1 x

/-- With a duplicate-free order, the computed result approximates the exact marginal
'marg vs.toFinset (product fs)' ('eliminateAll_correct'). -/
theorem eliminateAll_marg_forward_error (hu : 0 ≤ u) (fs : List (Factor bn R))
  (hfs : ∀ f ∈ fs, ∀ x, 0 ≤ f.value x) (vs : List bn.V) (hvs : vs.Nodup)
  (out : List (Computed bn)) (hrun : Run u (inputs fs) vs out) (x : bn.Assignment) (c : R)
  (hc : CompProd u (fun p : Computed bn => p.2 x) out c) :
  RelWithin (gamma u (veIndex fs.length vs)) (marg vs.toFinset (product fs) x) c := by
  rw [← eliminateAll_correct fs vs hvs x]
  exact (eliminateAll_forward_error hu fs hfs vs out hrun).2.2 x c hc

```

```

/-- Evidence by slicing, the shape of the Julia engine: eliminating over factors conditioned on
the evidence ('condition', 'explicit_conditioning_elimination') gives the same bound, with the
evidence variables neither factors nor eliminated. -/
theorem conditioned_forward_error (hu : 0 ≤ u) (fs : List (Factor bn R))
  (hfs : ∀ f ∈ fs, ∀ x, 0 ≤ f.value x) (E : Finset bn.V) (observed : bn.Assignment)
  (vs : List bn.V) (hvs : vs.Nodup) (hd : Disjoint E vs.toFinset) (out : List (Computed bn))
  (hrun : Run u (inputs (fs.map (condition E observed))) vs out) (x : bn.Assignment) (c : R)
  (hc : CompProd u (fun p : Computed bn ⇒ p.2 x) out c) :
  RelWithin (gamma u (veIndex fs.length vs))
    (marg (vs.toFinset ∪ E) (fun y ⇒ product fs y * indicator E observed y) x) c := by
rw [← explicit_conditioning_elimination fs E observed vs hvs hd x]
have h := (eliminateAll_forward_error hu (fs.map (condition E observed))
  (fun f hf y ⇒ by
    obtain ⟨g, hg, rfl⟩ := List.mem_map.mp hf
    exact hfs g hg _) vs out hrun).2.2 x c hc
rwa [List.length_map] at h

theorem length_inputs_posterior (κ : bn.Kernel R) (hloc : ∀ m, Local κ m) (E : Finset bn.V)
  (observed : bn.Assignment) :
  (evidenceFactor E observed :: ofKernel κ hloc).length = Fintype.card bn.M + 1 := by
simp [ofKernel, Finset.length_toList]

/-- The explicit index of the posterior bound: 'N = veIndex (|M| + 1) vs' for the CPT factors
and the evidence indicator, plus the number of query assignments. -/
def posteriorIndex (bn : FinBayesNet) (Q : Finset bn.V) (vs : List bn.V) : ℕ :=
  veIndex (Fintype.card bn.M + 1) vs + Fintype.card (PartialAssignment bn Q)

/-- **Forward error of the posterior.** Run variable elimination under the standard model on
the compiled CPT factors and the evidence indicator, eliminating 'vs = Qc'; compute each
query entry 'num q' as a product of the remaining tables at a query assignment, the mass 'Z'
as a sum of the 'num q' in any order and association, and each posterior entry as the rounded
quotient 'num q / Z'. Whenever the exact posterior exists (positive evidence mass) and
'γ K < 1' with 'K = posteriorIndex bn Q vs', the computed mass is within relative 'γ (K - 1)'
of the evidence mass, and every computed posterior entry is within relative
'(1 + γ K) / (1 - γ K) - 1' of the exact posterior entry. -/
theorem ve_posterior_forward_error (hu : 0 ≤ u) (κ : bn.Kernel R) (hloc : ∀ m, Local κ m)
  (hκ : ∀ m x a, 0 ≤ κ m x a) (Q E : Finset bn.V) (observed : bn.Assignment)
  (vs : List bn.V) (hvs : vs.Nodup) (hset : vs.toFinset = Qc) (out : List (Computed bn))
  (hrun : Run u (inputs (evidenceFactor E observed :: ofKernel κ hloc)) vs out)
  (base : bn.Assignment) (num : PartialAssignment bn Q → R)
  (hnum : ∀ q, CompProd u (fun p : Computed bn ⇒ p.2 (patch Q base q)) out (num q))
  (Z : R) (hZ : CompSum u num Finset.univ.toList Z)
  (post : PartialAssignment bn Q → R) (hpost : ∀ q, Rounded u (num q / Z) (post q))
  (d : Distribution (PartialAssignment bn Q))
  (hd : posterior Q E observed (joint κ) (joint_nonneg κ hκ) = some d)
  (hsmall : gamma u (posteriorIndex bn Q vs) < 1) :
  RelWithin (gamma u (posteriorIndex bn Q vs - 1)) (mass (likelihoodWeight (joint κ) E observed))
    Z ∧
    ∀ q, RelWithin ((1 + gamma u (posteriorIndex bn Q vs)) /
      (1 - gamma u (posteriorIndex bn Q vs)) - 1) (d.pmf q) (post q) := by
set fs := evidenceFactor E observed :: ofKernel κ hloc with hfsdef
have hfs : ∀ f ∈ fs, ∀ x, 0 ≤ f.value x := by
  intro f hf x
  rcases List.mem_cons.mp hf with rfl | hf
  · change 0 ≤ indicator E observed x
  unfold indicator; split_ifs <=> norm_num
  · obtain ⟨m, -, rfl⟩ := List.mem_map.mp hf
  exact hκ m x _
set w := pushWeight (restrictTo Q) (likelihoodWeight (joint κ) E observed) with hw
have hwn : ∀ q, 0 ≤ w q :=
  pushWeight_nonneg _ _ (likelihoodWeight_nonneg _ (joint_nonneg κ hκ) E observed)
have hwc : ∀ q, RelWithin (gamma u (veIndex (Fintype.card bn.M + 1) vs)) (w q) (num q) := by
  intro q
  have h := (eliminateAll_forward_error hu fs hfs vs out hrun).2.2 _ _ (hnum q)
  rw [ve_posterior_numerator κ hloc Q E observed vs hvs hset, restrict_patch] at h
  rwa [hfsdef, length_inputs_posterior] at h

```

```

have hmass : 0 < mass w := by
  have hne : posterior Q E observed (joint κ) (joint_nonneg κ hκ) ≠ none := by rw [hd]; simp
  rw [posterior, Ne, normalize_none_iff] at hne
  exact lt_of_le_of_ne (mass_nonneg _ hwn) (Ne.symm hne)
obtain ⟨hZm, hpq⟩ := normalize_forward_error hu w num hwn _ hwc Z hZ post hpost hmass hsmall
have hmw : mass w = mass (likelihoodWeight (joint κ) E observed) := mass_pushWeight _ _
refine ⟨by rw [← hmw]; exact hZm, fun q ⇒ ?_⟩
have hdq : d.pmf q = w q / mass w := by
  have := normalize_value _ _ d hd q
  rw [this]
rw [hdq]
exact hpq q

```

Log-sum-exp

section LogSumExp

open Real

```

/-- Sums with an arbitrary leaf bound 'e': along 't', the bound becomes
'(1 + u) ^ height t * (1 + e) - 1'. -/
theorem SumRun.within_rel (hu : 0 ≤ u) {ι : Type} {x y : ι → ℝ} {e : ℝ} (he : 0 ≤ e)
  {t : Assoc ι} {c : ℝ} (h : SumRun u x t c)
  (hy : ∀ i ∈ t.leaves, 0 ≤ y i ∧ RelWithin e (y i) (x i)) :
  RelWithin ((1 + u) ^ t.height * (1 + e) - 1) (t.leaves.map y).sum c := by
have h1u : (1 : ℝ) ≤ 1 + u := by linarith
induction h with
| leaf i ⇒
  simp [Assoc.leaves, Assoc.height] using (hy i (by simp [Assoc.leaves])).2
| @node l r cl cr c _ hc ihl ihr ⇒
  have hl := ihl (fun i hi ⇒ hy i (by simp [Assoc.leaves, hi]))
  have hr := ihr (fun i hi ⇒ hy i (by simp [Assoc.leaves, hi]))
  have nl : 0 ≤ (l.leaves.map y).sum :=
    sum_map_nonneg fun i hi ⇒ (hy i (by simp [Assoc.leaves, hi])).1
  have nr : 0 ≤ (r.leaves.map y).sum :=
    sum_map_nonneg fun i hi ⇒ (hy i (by simp [Assoc.leaves, hi])).1
  have mono : ∀ k, k ≤ max l.height r.height →
    (1 + u) ^ k * (1 + e) - 1 ≤ (1 + u) ^ (max l.height r.height) * (1 + e) - 1 :=
    fun k hk ⇒ by gcongr
  have hsum := (hl.mono (mono _ (le_max_left _)) nl).add (hr.mono (mono _ (le_max_right _)) nr)
  have := hsum.round (add_nonneg nl nr) hu hc
  simp only [Assoc.leaves, Assoc.height, List.map_append, List.sum_append]
  convert this using 2
  rw [pow_succ]; ring

theorem abs_exp_sub_one_le (t : ℝ) : |exp t - 1| ≤ exp |t| - 1 := by
  have h1 := add_one_le_exp t
  have h2 := add_one_le_exp (-t)
  rcases le_total 0 t with ht | ht
  · rw [abs_of_nonneg ht, abs_of_nonneg (by linarith)]
  · rw [abs_of_nonpos ht, abs_of_nonpos (by linarith [exp_le_one_iff.mpr ht])]
  have : exp t * exp (-t) = 1 := by rw [← exp_add]; simp
  nlinarith [exp_pos t, exp_pos (-t)]

/-- **Log-sum-exp.** The 'LogDomain' sum-out computes 'm + log Σ exp (x i - m)' with 'm' the
largest 'x i'. Model each 'x i - m', each 'exp', each addition of the sum (in any order and
association), the 'log' and the final 'm + _' as one rounding with relative error 'u', and let
every 'x i' lie within 'D' of 'm'. With 'n' terms, 'ρ = (1 + u) ^ n * exp (u D) - 1 < 1' and
'λ = -log (1 - ρ)', the computed value is within
'u |exact| + (1 + u) (u (log n + λ) + λ)' of the exact one: an absolute bound on the log scale,
which is a relative bound of about 'exp' of it on the probability scale. -/
theorem logSumExp_forward_error (hu : 0 ≤ u) {ι : Type} (l : List ι) (x : ι → ℝ) (m D : ℝ)
  (hm : ∀ i ∈ l, x i ≤ m) (hpeak : ∃ i ∈ l, x i = m) (hD : ∀ i ∈ l, m - x i ≤ D)
  (de : ι → ℝ) (hd : ∀ i ∈ l, Rounded u (x i - m) (d i))

```

```

(he : ∀ i ∈ l, Rounded u (exp (d i)) (e i)) (S : ℝ) (hS : CompSum u e l S)
(L : ℝ) (hL : Rounded u (log S) L) (r : ℝ) (hr : Rounded u (m + L) r)
(hp : (1 + u) ^ l.length * exp (u * D) - 1 < 1) :
|r - (m + log (l.map fun i => exp (x i - m)).sum)| ≤
  u * |m + log (l.map fun i => exp (x i - m)).sum| +
  (1 + u) * (u * (log l.length + -log (1 - ((1 + u) ^ l.length * exp (u * D) - 1))) +
    -log (1 - ((1 + u) ^ l.length * exp (u * D) - 1))) := by
set n := l.length with hn
set ρ := (1 + u) ^ n * exp (u * D) - 1 with hpdef
set lam := -log (1 - ρ) with hlam
set T := (l.map fun i => exp (x i - m)).sum with hT
obtain ⟨i0, hi0, hxi0⟩ := hpeak
have hD0 : 0 ≤ D := by have := hD i0 hi0; rw [hxi0] at this; linarith
have hne : l ≠ [] := List.ne_nil_of_mem hi0
have hn1 : 1 ≤ n := List.length_pos_iff.mpr hne
-- the exact sum lies in '[1, n]'
have hT1 : 1 ≤ T := by
  have := List.single_le_sum (l := l.map fun i => exp (x i - m))
    (fun z hz => by obtain ⟨i, -, rfl⟩ := List.mem_map.mp hz; exact (exp_pos _).le
      (exp (x i0 - m)) (List.mem_map.mpr ⟨i0, hi0, rfl⟩))
  rw [hxi0, sub_self, exp_zero] at this; exact this
have hTn : T ≤ n := by
  have := List.sum_le_card_nsmul (l.map fun i => exp (x i - m)) 1
    (fun z hz => by
      obtain ⟨i, hi, rfl⟩ := List.mem_map.mp hz
      exact exp_le_one_iff.mpr (by linarith [hm i hi]))
  simp [hn] using this
-- each computed term
have hterm : ∀ i ∈ l, RelWithin ((1 + u) * exp (u * D) - 1) (exp (x i - m)) (e i) := by
  intro i hi
  have hdi := (rounded_iff hu _).mp (hd i hi)
  have hlocal : RelWithin (exp (u * D) - 1) (exp (x i - m)) (exp (d i)) := by
    unfold RelWithin
    rw [show exp (d i) - exp (x i - m) = exp (x i - m) * (exp (d i - (x i - m)) - 1) by
      rw [mul_sub, ← exp_add]; ring_nf, abs_mul, abs_of_pos (exp_pos _), mul_comm]
    apply mul_le_mul_of_nonneg_right _ (exp_pos _).le
    refine (abs_exp_sub_one_le _).trans ?_
    have : |d i - (x i - m)| ≤ u * D := by
      refine hdi.trans (mul_le_mul_of_nonneg_left ?_ hu)
      rw [abs_of_nonpos (by linarith [hm i hi]); linarith [hD i hi]
        linarith [exp_le_exp.mpr this]
    have := hlocal.round (exp_pos _).le hu (he i hi)
    simp [add_sub_cancel] using this
have he0 : 0 ≤ (1 + u) * exp (u * D) - 1 := by
  have : 1 ≤ exp (u * D) := one_le_exp (mul_nonneg hu hD0)
  nlinarith
-- the computed sum
have hsum : RelWithin ρ T S := by
  rcases hS with ⟨rfl, -⟩ | ⟨t, hp, ht⟩
  · exact absurd rfl hne
  · have h1 := ht.within_rel hu he0 (y := fun i => exp (x i - m))
    (fun i hi => ⟨(exp_pos _).le, hterm i (hp.subset hi)⟩)
    rw [(hp.map _).sum_eq] at h1
    have hlen := t.height_add_one_le
    rw [hp.length_eq] at hlen
    refine h1.mono ?_ (by linarith)
    rw [hpdef, show (1 + u) ^ n * exp (u * D) = (1 + u) ^ (n - 1) * ((1 + u) * exp (u * D)) by
      rw [← mul_assoc, ← pow_succ, Nat.sub_add_cancel hn1]]
    have : (1 + u) ^ t.height ≤ (1 + u) ^ (n - 1) :=
      pow_le_pow_right (by linarith) (by omega)
    have h2 : 0 ≤ 1 + ((1 + u) * exp (u * D) - 1) := by linarith
    nlinarith
-- the log of the computed sum
have hp0 : 0 ≤ ρ := by
  have : 1 ≤ (1 + u) ^ n := one_le_pow (by linarith)
  have : 1 ≤ exp (u * D) := one_le_exp (mul_nonneg hu hD0)

```

```

    rw [hpdef]; nlinarith
  obtain ⟨hS1, hS2⟩ := abs_le.mp hsum
  have hSpos : 0 < S := by nlinarith
  have hlog : |log S - log T| ≤ lam := by
    have hTpos : 0 < T := by linarith
    have h1p : 0 < 1 - ρ := by linarith
    rw [← log_div hSpos.ne' hTpos.ne']
    have hlo : 1 - ρ ≤ S / T := by rw [le_div_iff hTpos]; linarith
    have hhi : S / T ≤ 1 + ρ := by rw [div_le_iff hTpos]; linarith
    have hl1 : log (1 - ρ) ≤ -ρ := by linarith [log_le_sub_one_of_pos h1p]
    rw [abs_le]
  constructor
  · rw [hlam]; linarith [log_le_log h1p hlo]
  · have := log_le_sub_one_of_pos (show 0 < S / T by positivity)
    rw [hlam]; linarith
  have hlogT : 0 ≤ log T ∧ log T ≤ log n := ⟨log_nonneg hT1, log_le_log (by linarith) hTn⟩
  have hL' := (rounded_iff hu _ _).mp hL
  have hLT : |L - log T| ≤ u * (log n + lam) + lam := by
    calc |L - log T| ≤ |L - log S| + |log S - log T| := abs_sub_le _ _ _
      _ ≤ u * |log S| + lam := add_le_add hL' hlog
      _ ≤ u * (log n + lam) + lam := by
        gcongr
        calc |log S| ≤ |log T| + |log S - log T| := by
            have := abs_add_le (log T) (log S - log T); simp using this
            _ ≤ log n + lam := by rw [abs_of_nonneg hlogT.1]; linarith [hlogT.2]
  have hr' := (rounded_iff hu _ _).mp hr
  calc |r - (m + log T)| ≤ |r - (m + L)| + |L - log T| := by
    have := abs_add_le (r - (m + L)) (L - log T); ring_nf at this; linarith
    _ ≤ u * |m + L| + |L - log T| := by linarith
    _ ≤ u * (|m + log T| + |L - log T|) + |L - log T| := by
      gcongr
      have := abs_add_le (m + log T) (L - log T); ring_nf at this; linarith
    _ = u * |m + log T| + (1 + u) * |L - log T| := by ring
    _ ≤ _ := by gcongr
end LogSumExp

end BayesianNetworksProofs.ErrorBounds

```

Roadmap

```

import BayesianNetworksProofs.Finite.OpenSemantics
import BayesianNetworksProofs.Finite.RefinementExamples

```

The former `osem_compose` hole is now proved in `Finite/OpenSemantics.lean`, in the default target and axiom audit. Its original premises are retained by a compatibility theorem. The stronger `osem_compose_glue` needs only disjoint input/output sets on both sides: locality, normalisation and total interface matching are unnecessary for this finite-sum identity.

Concrete finite records/references and repeated slots, genuine conditioned distributions, the actual Shafer-Shenoy collect/distribute computation, moralized-ancestral d-separation soundness, and conditional numerical error contracts now have checked developments. The remaining boundary is implementation translation: Julia/JSON decoding and reference binding, concrete `CliqueTrees/Bayes-ball` code, ordered-array layout and IEEE arithmetic adapters are not established merely by those finite-model theorems. Of the IEEE arithmetic, one piece is covered: `Numeric/Binary64.lean` proves that the transcription of `_nearest_binary64`, the single rounding the exact fallbacks of ADR 0016 apply, returns the binary64 round-to-nearest-ties-to-even of every rational, overflow and signed zero included. That is a theorem about the transcribed algorithm on mathematical integers. For the ordinary (non-fallback) paths, `Numeric/ErrorBounds.lean` proves forward error bounds under the standard rounding model (each operation's result is its exact result times $1 + \delta$, $|\delta| \leq u$): sums and products of nonnegative numbers in any association, variable elimination entrywise within relative γ/N for an explicit N , the normalised posterior, and log-sum-exp. It also proves that correct binary64 rounding satisfies that model with $u = 2^{-53}$ in the normal range. Underflow and overflow are excluded from those bounds, and that Julia's `Float64` operations are

correctly rounded is assumed (IEEE 754 and Julia’s semantics), not proved; Julia’s loop order and execution, and its execution of the rounding, remain outside.

The general structural open-network category, including non-injective output legs, pass-through and coherent copy/discard, is now constructed separately in `CategoricalBayesianNetworks.jl/proofs/` (ADR 0010), together with its numerical semantics: a strong braided monoidal functor into `FinStoch` preserving copy and discard (`OpenNet.Interpretation.functor`), whose composition formula `Interpretation.kernel_comp` covers copied and pass-through outputs. This project’s own `OpenFinBayesNet` now has its pass-through formula too: `osem_compose_passthrough` in `Finite/OpenSemantics.lean` sums only the glued variables the composite hides and needs no hypotheses; `osem_compose_glue` is its disjoint-interface special case. The category of stochastic kernels is a different result, provided by `FiniteKernels.jl/proofs/FiniteKernelsProofs/Theory/FinStoch.lean`.